

THE
GEOLOGY AND GEOGRAPHY
OF
NORTHERN NIGERIA



BY
J. D. FALCONER, M.A., D.Sc.
F.G.S., F.R.S.E., F.R.G.S.

With Notes by the late
ARTHUR LONGBOTTOM, B.A., F.G.S.,
and an Appendix on the Palæontology of the Cretaceous Deposits, by
HENRY WOODS, M.A., F.G.S.
University Lecturer in Palæozoology, Cambridge

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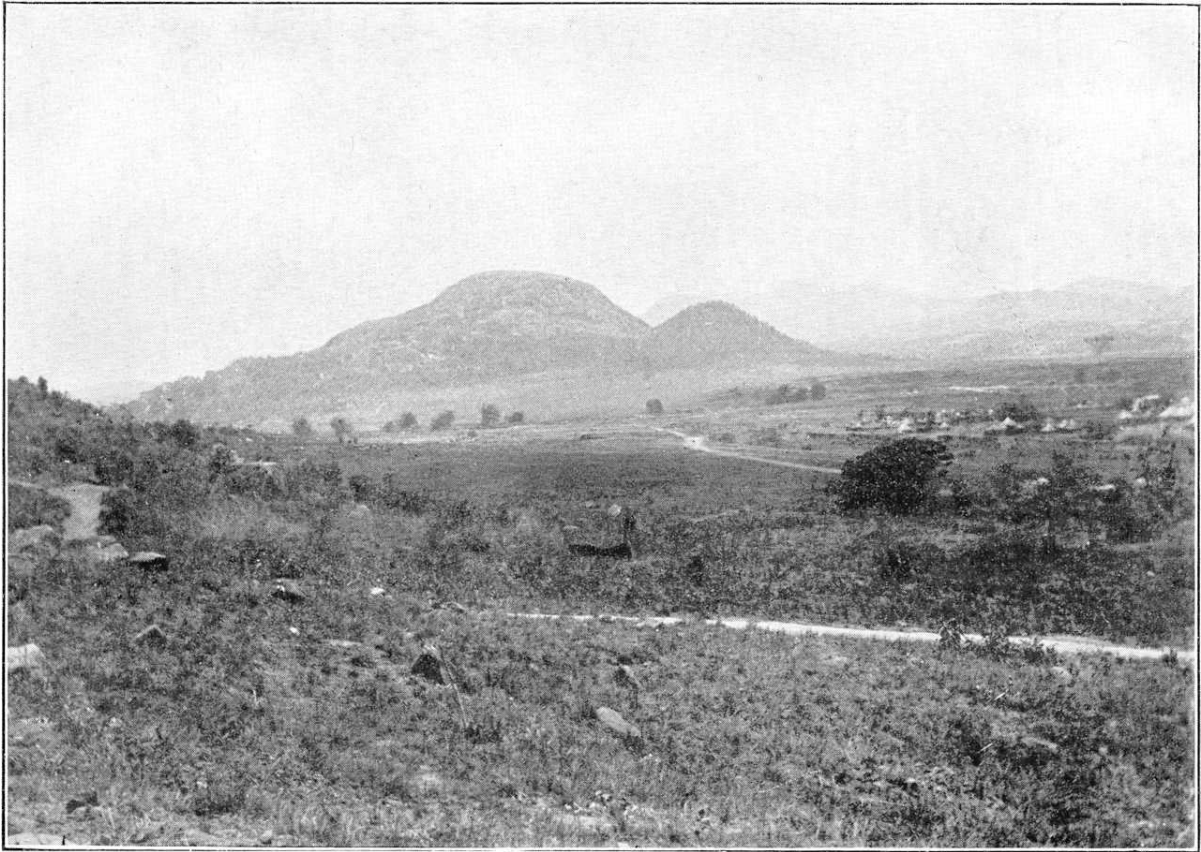
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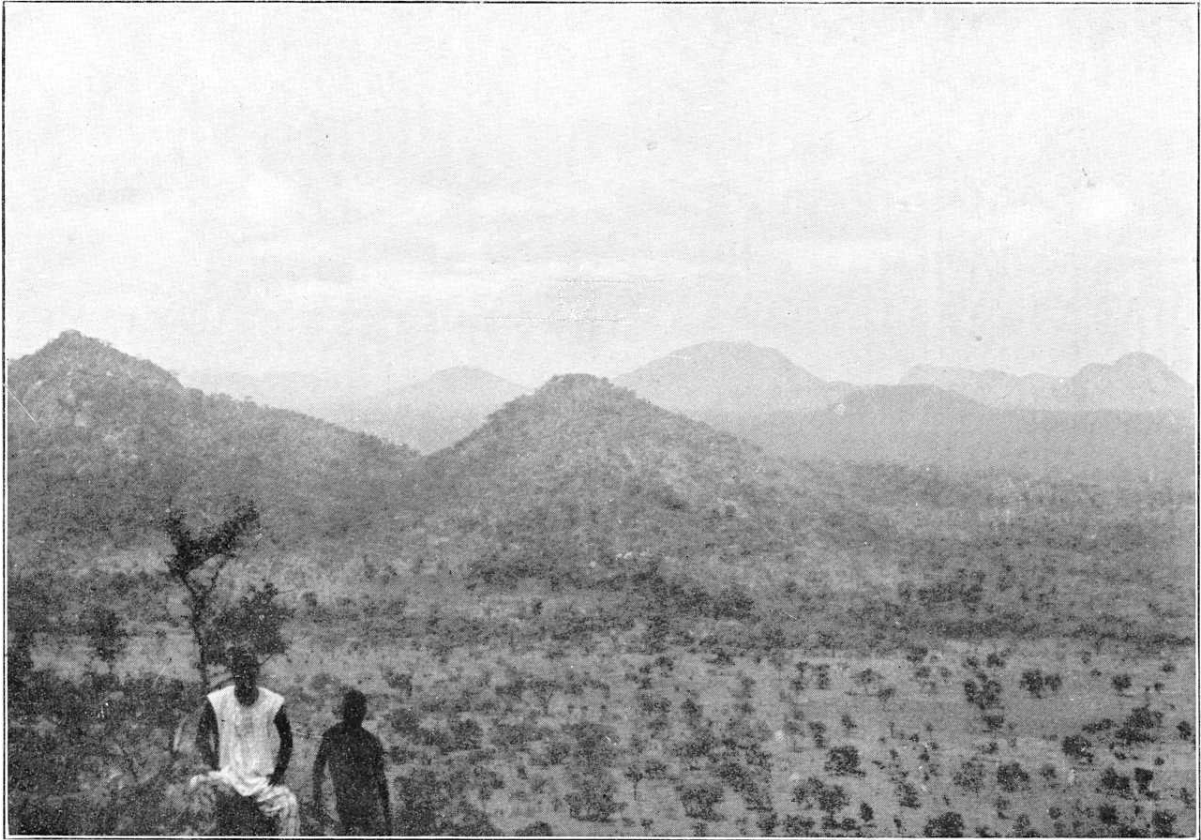
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INSELBERGE LANDSCAPE AROUND ABUJA.



INSELBERGE LANDSCAPE IN NORTHERN YOLA.

[Frontispiece.]

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WITH FIVE MAPS AND TWENTY-FOUR PLATES

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TO
MY OLD TEACHER AND FRIEND
PROFESSOR JAMES GEIKIE

PREFACE

THE observations upon which these outlines of the Geology and Geography of Northern Nigeria are based, were made in the course of the operations of the Mineral Survey of the Protectorate of which I had the honour to be the principal officer. The Survey was instituted by the Colonial Office in 1904 upon the recommendation of Professor Wyndham R. Dunstan, under whose general direction the work of investigation was carried out both in Nigeria and in the laboratories of the Imperial Institute. From 1904 to 1909 five expeditions in all were made to Northern Nigeria, in the course of which every province of the Protectorate was visited in turn. On the first two expeditions I was accompanied by Mr. Douglas R. Home, F.R.G.S., and on the remaining three by Mr. Arthur Longbottom, B.A., F.G.S. On an average seven months of each year were devoted to the actual field work, but the great extent of territory to be investigated, coupled with the difficulties of transport, rendered it impossible to do more in the time available than make a series of cross traverses through the various provinces with the object of ascertaining in a general way their geological structure and mineral resources.

In the following pages I have attempted to correlate the various observations which were made in the course of our travels and to weave a more or less consecutive story out of the disconnected contents of our notebooks. I am myself, however, only too conscious that many of my deductions may have been drawn from insufficient data and that further knowledge may overturn many of my hypotheses. In a new land someone must do the pioneering, and I shall be amply satisfied if my work affords some foundation, however insecure, for the investigations of future workers.

It is with the deepest regret that I have to record the death of my friend and former colleague, Mr. Arthur Longbottom, since this work went to press. He died of dysentery, after much suffering, in November, 1910, at

Naraguta, on the Bauchi plateau in Northern Nigeria, whither he had returned on a tour of inspection on behalf of a private syndicate. We had hoped that he would be able to continue the work which we had begun together and that he would return to England with a fresh store of observations which would increase our scanty knowledge of the geology and geography of the Protectorate. His extensive experience in various parts of Northern Nigeria would have rendered those observations of the greatest value, but his untimely death has put an end not only to a most promising career but to all hope of any further immediate advance in our knowledge of the structure of this outlying portion of the Empire.

Before he left England Mr. Longbottom contributed all the paragraphs and sections throughout the book marked with his initials (A. L.). He also supplied the photographs from which have been prepared Plates I, II, VI, VII, VIII, Fig. 1, and XVI. For Plate VIII, Fig. 2, I am indebted to the Public Works Department of Northern Nigeria. Mr. Woods has kindly contributed the Appendix on the Cretaceous Mollusca and the note upon the Eocene fossils on p. 165.

I gratefully acknowledge a Grant towards the cost of publication of this work from the Carnegie Trust for the Universities of Scotland. I am especially indebted to Sir Archibald Geikie for help and encouragement in the matter of publication. My acknowledgments are due also to various friends for assistance and advice, and more particularly to Professor James Geikie, Professor Wyndham R. Dunstan, Professor Knight, Dr. Horne, and Dr. Flett.

J. D. FALCONER.

JANUARY, 1911.

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THE GEOLOGY AND GEOGRAPHY
OF NORTHERN NIGERIA

NORTHERN NIGERIA

CHAPTER I

PHYSICAL GEOGRAPHY

Introduction. Hydrographical systems; primary watersheds. The Niger: Lokoja and Mount Patti; Jebba and the Juju Rock; the rapids of the Niger. The Benue: the great plain of Muri; sandbanks. The Niger Provinces: the tableland of central Kabba; domes and inselberge; the plain of Ilorin; Borgu; Kontagora; typical scenery of crystalline and sedimentary regions; the plains of Nupe; the sandstone plateaux of the Niger. The Benue Provinces: Bassa and Nassarawa; the river belt of sandstones; the Karshi hills and the Abuja Rock; the central plain; the northern tableland; the plains of Muri; volcanic hills of the middle Benue; the Wase Rock; the Murchison hills; the plateau of southern Yola; the lower Gongola; volcanic craters and granite inselberge. The Central Provinces: Zungeru and the gorge of the Kaduna; the plains and hills of Zaria; the Bauchi plateau; the Ningi hills; Angass and Kanna; the eastern plains; the Gongola valley; the cliffs of Tongolan; the Kudu valley; the Gateri hills; Tangale and Waja; The Peak. The Northern Provinces: southern Bornu; the Bima hills; the Barbur volcanic plateau; the Tagum hills; eastern Bornu; firki land and blown sand; Lake Chad; the central plains of Bornu; the Kerri Kerri plateau; intersecting trench-like valleys; the plains of Katagum; Kano; Mounts Dala and Kogon Dutsi; Kazaure; Katsina; Sokoto; the Gulbin Kebbi; dallols and dissected sandstone plateaux.

THE Protectorate of Northern Nigeria comprises an area of about 255,700 square miles, of which the greater part is enclosed within the obtuse angle formed by the Niger with its great tributary the Benue. The north-eastern corner is occupied by Lake Chad, whence irregular lines forming the northern and eastern boundaries between the Protectorate, the French Sudan and the Kameruns run west and south respectively to the Niger and the Benue. Within the oblique figure thus formed lie the provinces of Sokoto, Kano, Bornu, Bauchi, Zaria, Kontagora, Nupe, Nassarawa and portions of Muri and Yola. West of the Niger lie Borgu, Ilorin and Kabba, which adjoin Dahomey and Southern Nigeria, while south of the Benue lie Bassa and parts of Muri and Yola, marching with Southern Nigeria and the Kameruns. The administrative headquarters of the Protectorate are at present fixed at Zungeru on the Kaduna in western Zaria, while the subordinate headquarters of the fourteen provinces enumerated above are located respectively at Sokoto, Kano, Maidugari, Bauchi, Zaria, Kontagora, Bida, Keffi, Ibi, Yola, Kiama, Ilorin, Lokoja, and Dekina. Lokoja also, in addition to being the headquarters of Kabba province, is from its situation at the

confluence of the Niger and the Benue the natural emporium of European commerce.

Many of the more striking physical features of the Protectorate are now comparatively well known. The Niger with its flat-topped sandstone hills, the Benue and the plains of Muri, the peaks of the Murchison Hills, Lake Chad, and the sands and swamps of Bornu, have become familiar from the writings of both early and recent explorers.^[1] Equally striking, however, are the broad, flat-bottomed valleys of the north-west, in which diminutive streams now represent the mighty rivers of old; the rolling sandy plains of Hausaland^[2] with their sluggish rivers, their isolated granite domes and groups of rounded hills; and the mysterious Bauchi plateau, the home of innumerable pagan tribes, whose precipitous walls on the south and west long marked the limit between the known and the unknown.

Hydrographical Systems. ([Map. p. 4](#)).

The Bauchi plateau, which rises to a height of 4,000 to 4,500 feet above the sea, is the hydrographical centre of the Protectorate. From it the rivers radiate north, south, east, and west; but while at first they flow in such diverse directions, they become ultimately included within one or other of the two great hydrographical systems, the Niger-Benue system and the Chad or Inland system. All the rivers leaving the plateau, with the exception of those flowing directly north, belong to the Niger-Benue system. The “primary watershed,” or dividing line between the two great hydrographical systems, runs from the neighbourhood of Daura on the northern frontier of the Protectorate south-westward between Katsina and Kano to the twelfth parallel. Thence it runs south-eastward to the latitude of Zaria, and then southward to the neighbourhood of Bukuru, on the summit of the plateau. From Bukuru it strikes approximately north-eastward to Kerri Kerri, and thence south-eastward to the Yola border in the neighbourhood of Mubi. Within the Chad system the only important river is the Yo, of which the rivers Hadeija, Katagum, Delimi, and Shidya form the headwaters. The Yedseram, the only other large river of Bornu, forms for some distance the boundary line between Nigeria and the northern territories of the Kameruns.

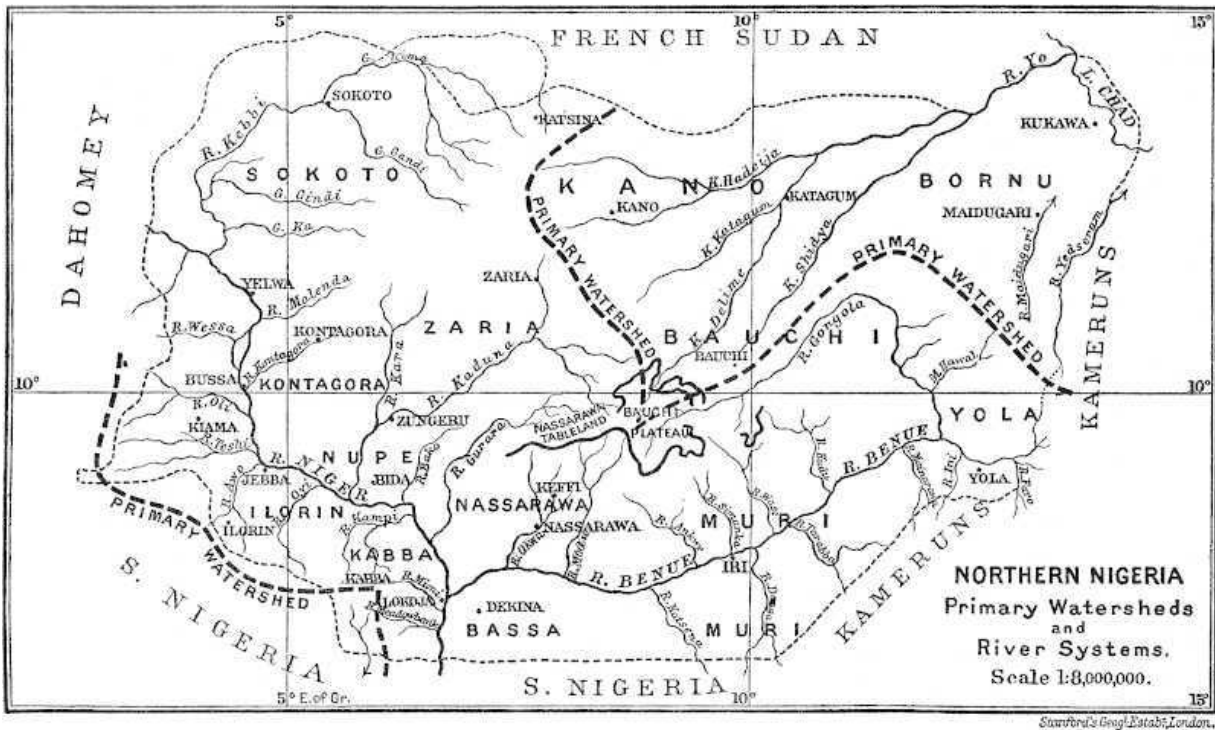
In the west the primary watershed of the Niger system coincides approximately with the boundary of the Protectorate from the eleventh parallel to the neighbourhood of Aiedi on the eighth parallel, whence it runs eastward to Kabba, and thence southward through the Egbira and Kukuruku hills into Southern Nigeria. South of the Benue, the primary watershed runs through Southern Nigeria and the Kameruns entirely beyond the southern limit of the Protectorate.

The Niger.

The Niger enters the Protectorate at Ilo in the north-west, and flows within the Protectorate for a distance of four hundred and fifty miles to Idah on the Southern Nigerian frontier. Its breadth varies in places from a few hundred yards in the dry season to several miles during the rains. The annual variation in depth may reach at Lokoja as much as thirty-five feet, and when in flood the river is navigable for small ocean-going steamers as far as Baro, the terminus of the Baro-Kano railway, and for light stern-wheelers as far as Bajibo, twenty miles north of Jebba. Its largest tributaries within the Protectorate join the Niger on the left, those on the right being comparatively short and unimportant. On the left are the Kebbi, Malenda, Kontagora, Kaduna, Bako, Gurara, and Benue; on the right the Wessa, Oli, Teshi, Awo, Oshin, Oyi, Oro, Kampi, Mimi, and Meadowbank. Of these the Gurara is navigable in flood for small stern-wheelers for a distance of thirty miles from the Niger, the Kaduna as far as Wushishi at the foot of the rapids, and the Kebbi as far as the confluence of the Jega river.

NORTHERN NIGERIA Primary Watersheds and River Systems.

MAP N^o. 1.



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Face p. 4.

The alternation of narrow gorge and open plain is a striking feature of the Niger valley within the Protectorate. Where the channel is narrow and bounded by steep cliffs on either side the seasonal variation in level gives rise to comparatively little change in the aspect of the river. The blackened rocks in the river bed and the sandy stretches on either shore may be lost to view, but the flood is still confined within its walls of rock. It is where a plain tract intervenes in the course of the river that the contrast between the Niger in flood and the Niger at low water becomes particularly striking. The sandy bars and shallows which in the dry season present such obstacles to navigation, as well as the grassy banks and islands of alluvium with their

terraced slopes of sand and clay, are buried during the rains under a flood, which may spread for miles over the plain on either side of the river.

At Idah the Niger narrows and flows swiftly between low cliffs of white sandstone which pass northward into stretches of wooded rising ground on either bank. Ten miles above Idah a group of flat-topped hills approach the right bank of the river, only to turn off abruptly inland in a line of buttresses six to seven hundred feet in height. Farther on, the low, flattened ridges on the left bank run out in a line of rounded hills: and beyond, the river broadens and a broken plain stretches inland to distant groups of lofty hills. Nearing Itobe Mount Purdy stands out, a solitary cone, in advance of the level-crested plateau beyond. From Itobe to Igbo the Niger flows through the most picturesque portion of its lower course.^[3] The hills rapidly close in, the channel becomes narrow and rocky, lofty granite domes and rounded hills separated by narrow gloomy valleys rise up on either bank, and on the left the level crest of the plateau looms behind on the sky-line (Plate I, [Fig. 1](#)). Below Lokoja the river opens out, the hills die down on either side, and from the left bank a broken hummocky plain stretches inland to the base of the Bassa hills, while at Lokoja itself flat-topped Patti (Plate I, [Fig. 2](#)), with its projecting buttresses and the red roofs of the cantonment at its base, stands sentinel over the confluence of the Niger and the Benue ([Plate II](#)).

An extensive alluvial plain annually flooded during the rains stretches away to the Benue from the left bank of the Niger for long miles above Lokoja. The right bank is occupied more or less continuously by the worn escarpment of the plateau of which Mount Patti forms the southernmost extremity. Above Koton Karifi the hills reappear on the left bank, and for more than twenty miles the Niger flows between the broken walls of what has once been a magnificent river gorge. In places for considerable distances the flat-topped hills still rise up with precipitous slopes on either side in all their ancient grandeur. More frequently, however, the escarpments are rounded off and much eroded by lateral valleys, while at times they recede to a distance, and leave between themselves and the river a more or less extensive swampy plain, with here and there a flat-topped remnant of the once continuous plateau. Before Baro is reached, however, the plateau has disappeared on the right bank, and on the left is represented only by more or less isolated flat-topped fragments. The bend of the Niger

at Egga and Kacha marks the eastward limit of an extensive alluvial flood plain, which includes the lower courses of the Kampi and the Bako, and stretches westward on the north bank as far as the Kaduna. On the south bank the pinnacle peaks of Egbom^[4] tower high above the river, while a flattened ridge extends westward as far as Pategi. Beyond the Kaduna the Niger flows through a broad and shallow valley with extensive swampy plains on either hand, and only at Shonga on the right bank and at Rabba on the left does the river approach the rising ground which elsewhere bounds the valley in the distance.

At Jebba the hills close in on either side, and the Niger issues by a double channel from a picturesque gorge which extends northward as a deeply cut and rocky valley as far as Bussa, a distance of seventy-five miles. For eight hundred yards above Jebba the gorge is narrow and bounded by vertical walls of quartzite. The Juju Rock^[5] and the other islands which interrupt the channel mount steeply from the river, the rapid current pours over sunken rocks which rise abruptly from the river bed, and mighty whirlpools bear witness to the depth of the water. Beyond Jebba the quartzite gives place to granite, the vertical walls become rounded, wooded slopes, the river broadens its rocky bed and flows in a narrow valley between two belts of broken hilly ground which lead up to the plains of Kontagora and Borgu, five hundred feet above the level of the river. In spite of the obstructions in the channel the river is navigable with caution for small stern-wheelers as far as Bajibo. Above Bajibo, however, navigation becomes precarious and dangerous even for native canoes. The current becomes more violent; rocks, cascades, and whirlpools more numerous, and small rapids more and more frequent, until above Leaba the Niger flows in a rocky gorge which leads to the base of the Great Rapid of Wuru.^[6] Here the river is formed of two branches with a rocky island between. The fall of forty feet between the summit and foot of the rapid is distributed over some thirteen hundred yards of river, giving a slope of at least 1 in 100 and a current of from fifteen to eighteen miles an hour. In the rains the granite boulders are completely submerged under a seething flood, and even in the dry season the water is thrown up by the projecting rocks in sheets of foam, and the roar of the rapid is heard for long distances on either side.

A stretch of two and a half miles of comparatively smooth water, broken only by some small rapids and cascades, separates the Great Rapid of Wuru from the rapid of Patassi, which is composed of two sets of cascades, each twelve to fifteen feet in height and separated by a straight reach of some two hundred and eighty yards. Twenty miles farther on, the third important set of rapids is reached, the rapids of Garafiri, whose cascades and whirlpools occupy a stretch of about three miles between the villages of Garafiri and Malali. At Bussa, the scene of Park's disaster, the channel is still much obstructed by projecting rocks and islands, and the small cascades and the rapid current still render navigation both difficult and dangerous. The right bank of the river is here formed of low, wooded rising ground, while on the left ranges of low hills rise in series one behind the other. Soon, however, the hills die down and the rocks disappear from the river bed, and at Warra the river is broad and smooth with an extensive flood plain on either side. From Warra to Ineku the Niger is formed of two branches, of which the more westerly preserves the calm aspect of the river at Warra, while the more easterly becomes again obstructed by rocks and rapids. Between Ineku and Yelwa the river is bounded by low, rounded hills covered with scanty forest, innumerable islands rise from the river bed supporting an abundant vegetation, while several sets of lesser rapids obstruct the channel. At Yelwa the banks become low and swampy, the hills begin to recede from the river, and five miles farther on, near the village of Sakassi, the last set of minor rapids occurs in the channel. Beyond Sakassi the whole aspect of the river changes, the channel becomes broad and sandy and obstructed by nothing more serious than sand bars. On the north bank a broad, swampy flood plain, with low rising ground in the distance, bounds the river as far as the mouth of the Kebbi, and from the Kebbi to the swamps of Ilo an open sandy plain stretches northward between the river and the base of the plateau. On the south bank the gently undulating plain which bounds the river stretches westward to the Dahomey border, broken only by some scattered groups of flat-topped hills.

The Benue.

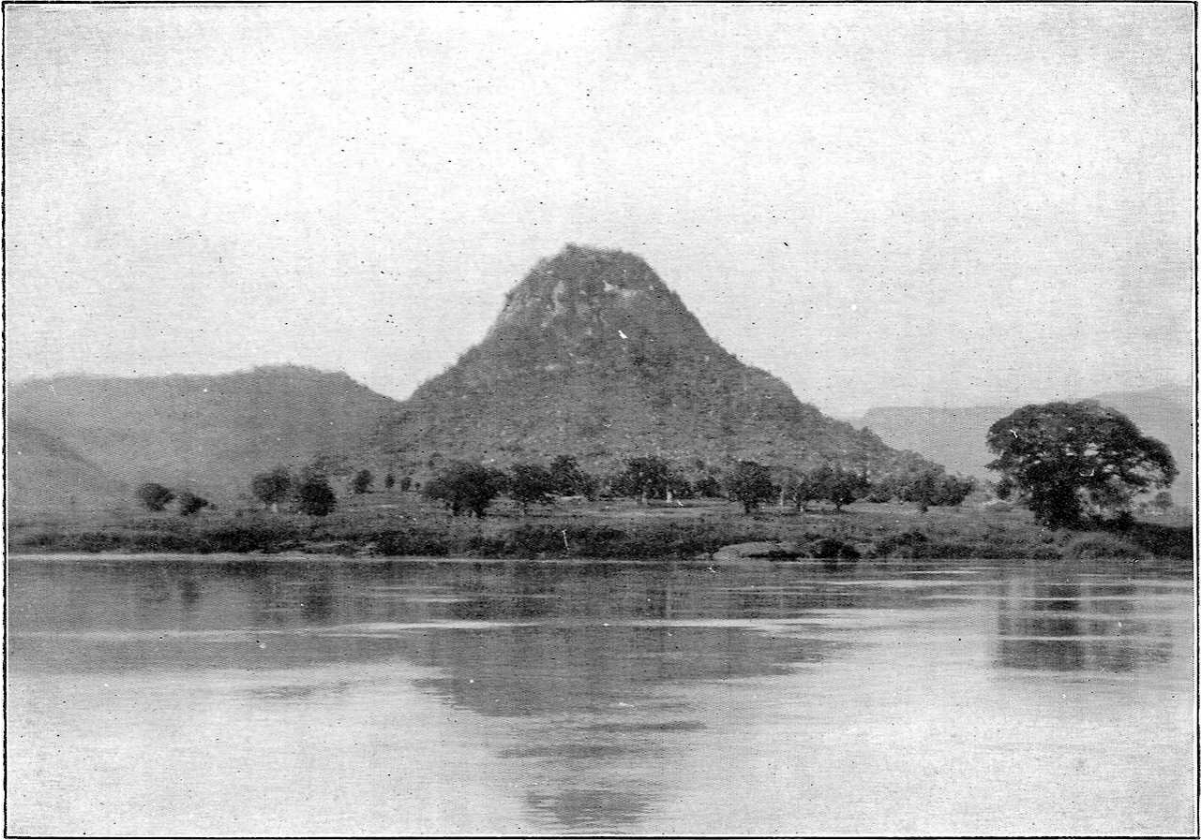
The Benue enters the Protectorate to the east of Yola, and after a course of nearly five hundred miles joins the Niger at Lokoja. Its principal tributaries are on the right bank the Gongola, Kudu, Wase, Simanka, Ankwe, Modu and Okwa; on the left bank the Faro, Ini, Mainarawa, Tarabba, Donga and Katsena. The Benue is navigable when in flood throughout its whole course for steamers of light draught. The Katsena is navigable as far as Katsena Allah, the Gongola as far as Nafada, and the Donga and Tarabba for considerable distances up stream.

In the matter of scenery the Benue presents a striking contrast to the Niger. There is on the Benue no picturesque alternation of narrow gorge and open plain; no rapids trouble its surface from Lokoja to the flat-topped hills of Yola, and even in the dry season it is only at long intervals that rocks are exposed in its sandy bed or in its steep banks of alluvium. The precipitous front of Mt. Patti and the level crest of the plateau on the right bank of the Niger are visible from the Benue for twenty miles above Lokoja, across the great alluvial plain which marks the confluence of the two mighty rivers. Above Mozum a range of flat-topped hills with occasional peaks follows the right bank for some distance, but soon turns inland and leaves a broken plain between the river and the base of the hills. A similar range, also separated from the river by a broken plain, runs parallel with the left bank a few miles inland. The escarpments of these ranges on either side of the river represent the walls of a broad flat-bottomed valley which is comparable in origin with the steep-sided gorge of the Niger above Lokoja. Here on the Benue, however, erosion has been more intense, and the hills have been cut farther back on either hand.

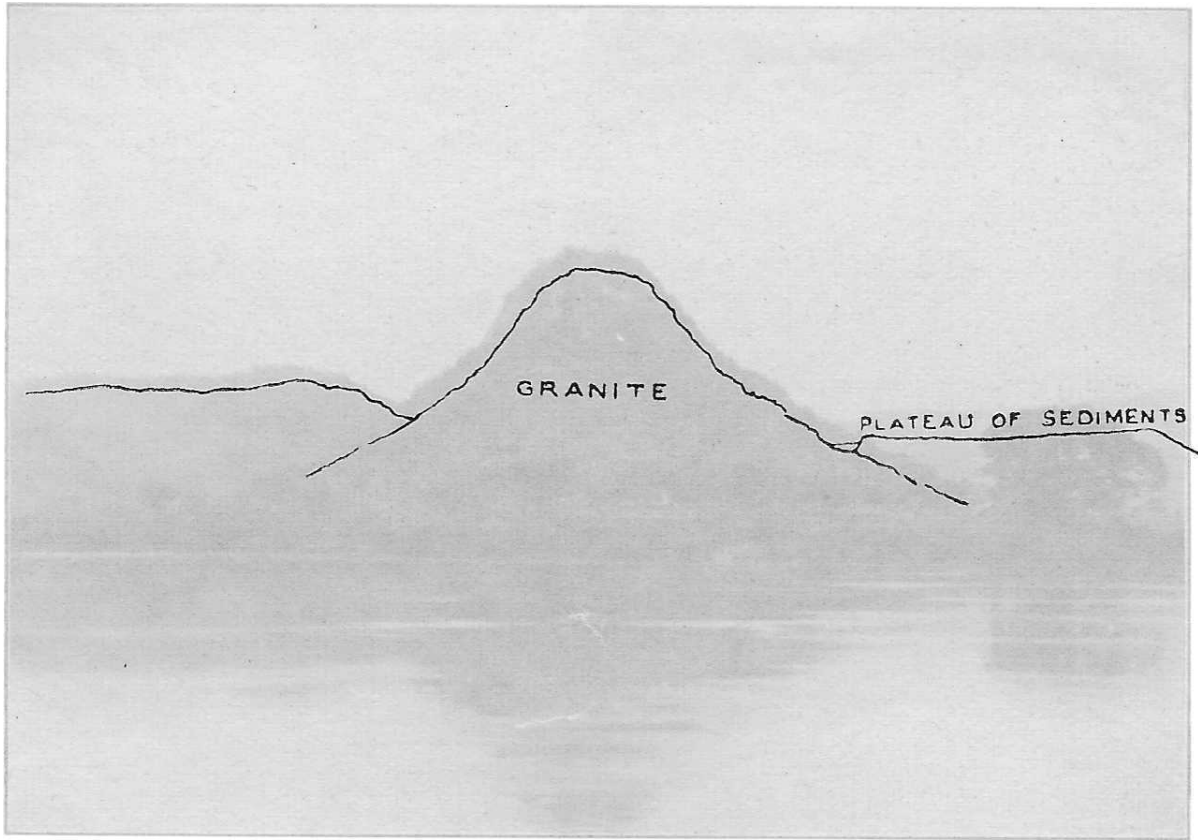
Above Umaisha a tract of broken country stretches eastward on the north bank and gives place in the neighbourhood of Loko to alluvial swamps in the foreground, with undulating sandy plains beyond. On the left bank, meanwhile, the hills approach the river at Amara and pass eastward into a hummocky plain, which is bounded to the south by a distant range of flat-topped hills. Above Loko the Doma Hills touch the river at Udeni on the right bank, but turn off abruptly inland with their low, buttressed slopes overlooking an extensive swampy plain which stretches eastward as far as Akwaneja. Beyond Akwaneja the right bank assumes a similar aspect to that of the left. The river is bounded by low, well-wooded banks, with here

and there a projecting mass of sandstone, while a hummocky plain stretches inland on either side to distant rising ground. The volcanic tract above Abinsi introduces an agreeable change into the monotony of the scenery. The worn and weathered cones and banks of lava stand out picturesquely from the wooded plains, and on the river-side the black basalt and the yellow sand offer a pleasing contrast to the eye. Soon, however, the low banks resume their familiar aspect and a featureless plain spreads out on either hand. Nearing Ibi the right bank becomes low and swampy, and the great plain of Muri opens out beyond. From Ibi to Lau the channel is broad and sandy, and the river winds about amongst everchanging sand-bars and banks of alluvium. In the rains broad stretches are flooded on either side, and for long distances on the right bank the plain beyond becomes an almost impassable swamp, while on the left long tongues of marshland run up the tributary valleys. Far to the north the peaks of the Murchison Hills^[7] tower high above the plain, while to the south the rock of Bantaji guards the swamps of the Donga.

PLATE I

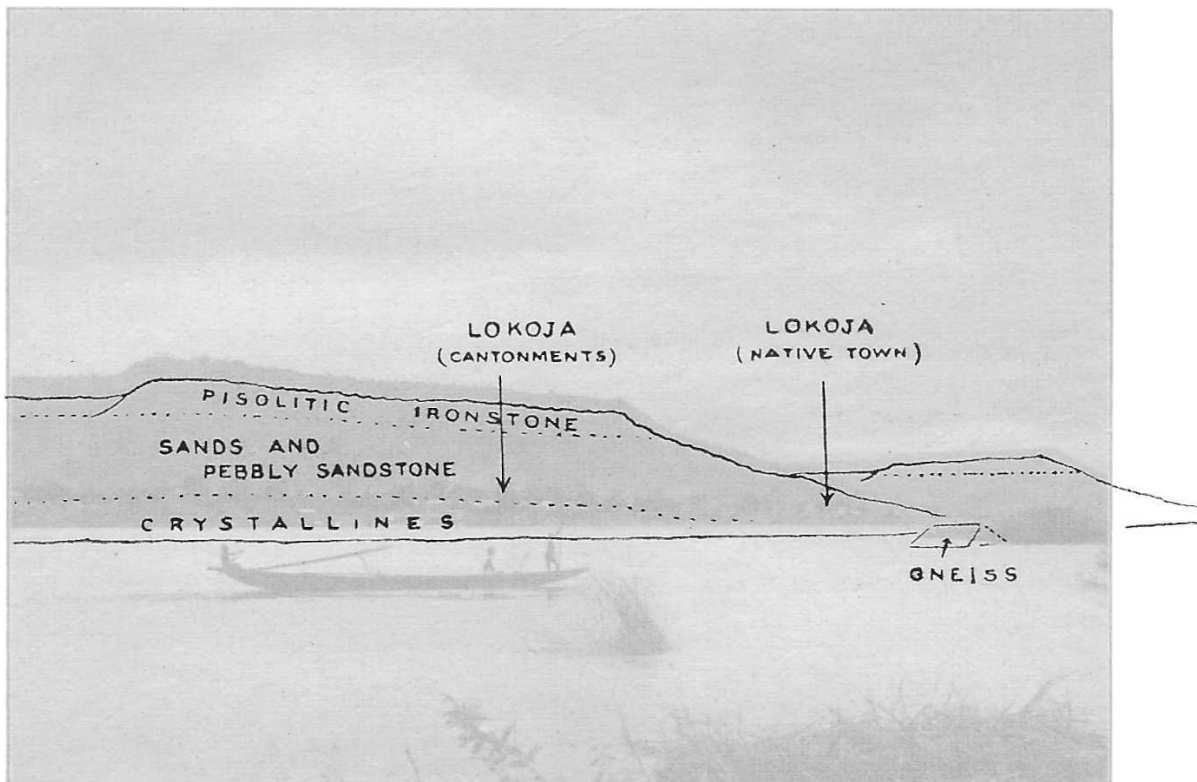


ON THE NIGER ABOVE IGBO.





LOKOJA AND MOUNT PATTI.



Farther east, and especially between Numan and Yola, the river plain is traversed by a large number of anastomosing channels, which in many cases are only connected with the main stream during the rains, and in the dry season are cut off by sand-bars into strings of shallow pools and marshes. Alluvial mud covers the rocks for some miles inland, and while in the rains travel on foot is impossible and the villages are isolated one from another, in the dry season the ground shrinks and is broken everywhere by wide cracks. This alluvial plain is perfectly level, so that the small volcanic mounds, dotted here and there, which in a more varied country would be overlooked, form conspicuous landmarks. The most striking of these hillocks is Jeme Hill, below Numan, which stands on the river bank and can be seen for twenty miles around. (A. L.)

The Benue plain, which is so extensive in Muri province, contracts near Lau, and thence eastward it is limited on both sides by plateau areas which rise suddenly and steeply at distances of fifteen miles from the river. These masses of hills with their varying outlines introduce a pleasing change into the hitherto monotonous river journey, and contrast sharply with the river plain. Here the plain of the Benue is like a broad and shallow trench, bounded by sharply rising walls, whose height is small compared with the distance between. At the same time the height is considerable in itself, for the rise to the plateau is well over a thousand feet. (A. L.)

These hills running eastward parallel to the river are breached by the Gongola plain, about ten miles wide, entering from the north, and by the plain of the Ma-Ini and Maio Belwa entering from the south. Farther on the hills come in again. To the north they are rather scattered and detached, with long tabular outlines, or, as in the Bagele Hills, with terraced sides and abrupt scarps. On the other side, though flat-topped hills do occur, the outlines are usually more varied, while south of Yola rises the typical plateau of the Vere Hills—a solid mass fronting boldly outwards upon the surrounding plains with continuous rocky faces indented by a few deeply-cut and narrow valleys. Meanwhile the plain of the Benue with its flat and marshy character continues eastward beyond Yola and the Faro river into the German Kameruns. (A. L.)

Along the whole course of the Benue from Lokoja to Yola exposures of the underlying rocks are very infrequent, and this remark applies equally to the lower courses of the tributaries. The main trunks of the Benue system have long passed the stage at which they effected any appreciable erosion, and now serve only as the channels along which detrital material brought down by the head-waters is transported to the Niger. The finer portion of the material is carried in suspension, and is responsible for the covering of alluvial mud found over those parts of the river plain which are inundated during the annual rise. The coarser portion is rolled along the bottom and gives rise to the sandbanks. These are covered in flood, but for the other part of the year they are a constant feature of the river scenery. The stream, much lessened in volume, swings first to one side and then to the other of the wide river bed, with the broad sandbanks occupying the bends. (A. L.)

The sand is rather coarse, principally of clean grains of quartz and felspar, but at times with pebbles of vein quartz or occasionally of polished limonite. Typically, a spread of sand, many acres in extent and with its level surface only a few feet above the water, ends on the down-stream side in a slope which plunges into deep water at perhaps 40° . While the river is falling, the formation of these sandbanks can be well studied. The water, covering the sandbank by only a few inches, is seen to sweep grains of sand or occasional pebbles along over the level surface till they reach the end of the bank and roll down into deep water, thus helping to form the next lamina. In this way the sandbank is built up of successive laminæ inclined down stream at a steep angle to the horizontal, and wherever a bank is cut into by a change in the direction of the current the inclined lamination is beautifully displayed. The lamination is of course only a particular instance of the "current bedding" produced whenever detritus in process of deposition is subjected to currents of air or water, but it is important to understand that the oblique lamination of the sandbanks is brought about by the river currents, and not by the wind as has been supposed. Any slight rearrangement of the surface layers by wind is obliterated by the next rise of the river. (A. L.)

The Niger Provinces.—Kabba, Ilorin, Borgu, Kontagora, Nupe.

The orographical structure of Kabba and Ilorin is somewhat complex. The primary watershed of the Niger basin which runs from Kabba to Aiedi and thence westward a few miles south of the Ilorin frontier, is not defined by any range of hills or mountains, but by the crest line of a lofty plain which, like a low, extended arch, slopes gently northward towards the Niger and southward towards the sea. In central Kabba the high plain assumes, to some extent, the character of a tableland, which is bounded to the north-east by the Lukke hills and the broken ground beyond the Mimi valley. To the east and south-east the tableland is abruptly limited by the worn escarpments of the Shokko Shokko, Agwede and Egbira hills, at whose base, 500 feet below, are spread out the broad, well-watered and well-wooded plains which stretch away eastward to the Niger, and of which passing glimpses are caught from the river below Itobe. To the south these undulating plains slope gently upward and inward from the Niger and lose themselves in the narrow valleys of the Kukuruku Hills.

Although such in brief is the orographical character of the major portion of Kabba and Ilorin, the actual conditions of the surface and the peculiar character of the scenery to some extent obscure the issue. In central Kabba the undulating surface of the tableland is in places densely wooded, in places covered with thin bush, in places open, grassy, and park-like with cultivation near the towns and villages, while above the plain at frequent intervals rise isolated rounded hills or groups of hills, or even short rocky ranges, which form secondary watersheds between the minor streams. Similarly, on the lower plains adjoining the Niger, the attention is arrested by the dome-shaped island mountains and picturesque clusters of rounded hills which rise abruptly into view. The full significance of this peculiar type of scenery can only be grasped when gazing from the summit of one of the isolated hills themselves, or from some favourable spot on the crest of the escarpment which separates the higher from the lower plains. Then all the minor irregularities disappear, and the level, wooded surface of the plain can be seen stretching up to and around and between the bases of the hills which rise like rocky islands from a verdant sea. The hills may be covered to the summit with angular *débris* and scanty forest, or the lower slopes may be littered with exfoliation fragments, amongst which a few trees find a precarious footing, while the summit may be smooth and bare, with a few

large boulders perched picturesquely in apparently impossible positions. The contours of the bare and treeless summits are frequently particularly graceful with beautiful domes and cupolas rising from gently-rounded slopes of granite or gneiss. In many cases a precipitous path leads to the village which caps the summit, while for long acres round the base the plain is cleared and cultivated. ([Plate XVII.](#))^[8]

The northern part of Kabba is more rugged, the population is more scanty and forest more abundant. The rocky valley of the Mimi is bounded to the north by the broken, hilly ground between Jakura and Akpara, which passes to the east and north under the flat-topped sandstone hills which fringe the Niger. The Jakpara Hills with their conical peaks upon rounded bases are landmarks for miles around. The white quartzite hills in the neighbourhood of Lukke alternate with belts of soft schists which support a luxuriant forest growth. Farther west by Moppa, Sanlu and Takete detached groups of granite hills rise upon an undulating plain. The thicker vegetation, however, obstructs the view, and it is only when the landscape is surveyed from the summit of one of the higher hills that the structure is seen to be very similar to that of southern Kabba. To the west, Ogga and Eri are set within a cluster of rocky hills, while to the north there is a comparatively rapid fall to the plains of the Kampi. Between the Kampi and the Niger a steep-sided range of sandstone hills capped by the pinnacle peaks of Egbom ([p. 7](#)) runs out westward in a succession of rounded ridges two hundred feet above the level of the river, and not until near Share is the river belt of sandstones again bounded by a low escarpment. The Shappa and Orissa Hills form the only prominent range in central Ilorin. The Shappa Hills are of granite, and many of the peaks are gracefully rounded and most picturesque, while the quartzites and gneisses of the Orissa hills give more rugged and irregular contours. West of the Orissa hills lies the great plain of Ilorin, thickly populated and well cultivated, in broad undulations trending generally north and south, with scattered inselberge and many kopjes of granite boulders, but with no conspicuous ranges or groups of hills. The Jebba hills, which extend southward for some distance towards Ilorin, pass westward into an extensive tract of hilly country, high lying, little known and largely uninhabited, which forms a natural boundary between Ilorin and Borgu.

The primary watershed of the Niger basin follows the frontier of the Protectorate more or less closely from the eighth parallel to the south of Ilorin as far as Okuta in the west of Borgu, whence it runs northward to the latitude of Nikki before turning off westward into Dahomey. With the watershed the high plains, along whose crest it runs, also swing round to the west of Borgu and fall off gently towards the Niger on the right, while along the river itself from Bussa to Jebba a narrow belt of broken country marks the transition from the high plains of Borgu to the gorge-like valley of the Niger.

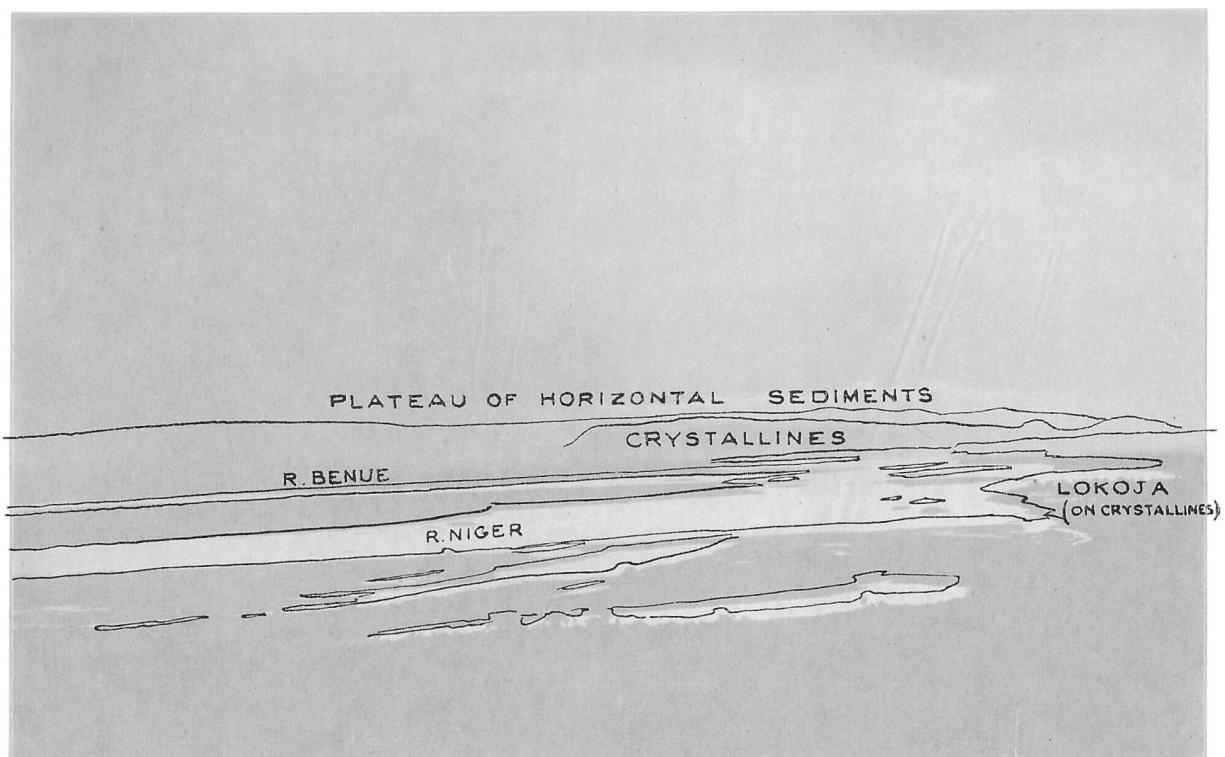
Southern Borgu is one of the least inviting districts in the Protectorate. The rocks are concealed almost everywhere by recent alluvium or by surface ironstone,^[9] which in places forms extensive platforms entirely bare of vegetation. Where rocks do emerge from the superficial accumulations they seldom form hills. The granite hills around Kiama are an exception to this rule, and along the western frontier a number of granite domes rise from two to three hundred feet high. These domes, however, are separated by wide stretches of country over which all geological processes seem to have come to a standstill. The drainage system is very poorly defined, and for the most part the water is carried off, not by stream courses with sand and boulders, but along shallow, soil-covered depressions in the plain. Such streams as there are dry up soon after the end of the rains, and the natives then resort largely to wells dug in the surface soil. The crests of the gentle undulations follow one another on the sky-line in endless succession, and the journey becomes monotonous in the extreme. The northern part of the province is diversified by a number of sandstone hills, and some swift, clear streams, fed from springs at the base of the hills, afford a welcome change to the jaded traveller. (A. L.)

In the provinces which lie to the east of the Niger the intimate relation which exists between the character of the scenery and the nature of the rocks becomes increasingly evident. In Kabba, Ilorin, and Borgu, where crystalline rocks of a granitic or gneissic character cover by far the larger part of the surface, the prevailing and most characteristic type of scenery is that of rolling sandy plains with isolated hills or groups of hills rising abruptly above the general level. On the other hand, where sedimentary rocks, such as sandstones and ironstones, cover the surface, as in the river

belt of sandstones in Kabba and Ilorin, the characteristic type of scenery is that of steep-sided and flat-topped ranges and plateaux, with detached tabular and conical hills rising from the lower plains. Over the greater part of Kontagora province, to the north of the parallel of Kontagora itself, granites and gneisses form the surface rock. In the west, near N'gashki, a group of granite hills rises abruptly from the plain which stretches away to the east, with its surface diversified in typical fashion by occasional inselberge. A belt of slaty rocks much pierced by quartz veins runs along the left bank of the Niger from N'gashki to Yelwa, and gives rise to a stretch of broken country adjoining the river. The granites and quartzites in the lower valley of the Malenda river also give rise to much rough and stony country, which culminates in the quartzite hills of Maibirro. Beyond Jindani and Yelwa the plain reappears in a more typical character, and stretches northward as a gently undulating surface covered with deep, sandy drift, from which emerge at long intervals low, rounded, or rocky hummocks of granite or gneiss.



THE CONFLUENCE OF THE NIGER AND THE BENUE.



It is in the east of the province, however, between Banaga and Kotonkoro, that the landscape assumes the typical aspect of the high plains of Hausaland. In front, behind, and for long miles on either hand, the open, sandy plain stretches away to the horizon, with here and there a low turtleback or heap of granite boulders, and at long intervals a loftier dome or a group of rounded granite hills. The rivers flow in almost imperceptible valleys in broad, sandy channels, or over smooth and bare surfaces of rock, level with the plain, while the low watersheds between the rivers are crossed almost unconsciously by the traveller.

The country around Kotonkoro and to the south-west is extremely picturesque. Numerous domes of bare rock rise from a grassy park-like country. The plain around one of these domes passes imperceptibly into a gentle slope of sandy wash, which leads the eye up to the spring of the dome. As the path winds along, first one and then another of these domes engages the attention. On approaching Mahorro two of the highest hills have flattened summits and are found to be capped by about twenty feet of stratified ironstone. To the north-east of Kontagora these outliers are more numerous, though seldom of a sufficient size to mark on the map. The country is in consequence very rough and broken and the track becomes difficult and stony. The poor agricultural prospects, however, are compensated for by good iron ore, and the crude native furnaces soon come to be a familiar sight. When these outliers are finally left behind at Libo, a belt of very rough broken country has to be crossed before descending to the Kara river. (A. L.)

The town of Kontagora is set in an open undulating sandy plain, deeply covered with surface drift, which stretches eastward to Moriga and the Kara river, with its surface diversified by a few isolated sandstone hills near Kontagora and a group of granite hills at Beri. To the south of Kontagora begins the sedimentary series which stretches southward into Nupe province and westward to within a few miles of the Niger. The change in the nature of the surface rock is accompanied by corresponding changes in the character of the scenery. Here at first are the same rolling sandy plains, if possible more deeply covered with drift than before; but here are no turtlebacks or lofty domes of granite forming conspicuous features in the landscape, no rivers flowing over smooth surfaces of rock in valleys so

shallow that they are almost indistinguishable from the plain. In place of these are low, flat-topped sandstone hills rising singly or in groups above the general level, and deep and narrow canyon-like valleys cut down, it may be, fifty or a hundred feet below the surface of the plain. A typical group of hills with their level summits of hard ferruginous grit and their slopes covered with rounded boulders of the same occurs at Madara, a few miles south of Kontagora, while at Kagara, Kaboji, and Sonko the streams run in typical canyon valleys, whose presence is unsuspected upon the plain until the chasm opens abruptly at one's feet. The larger rivers flow in broad, trench-like valleys, with their steep and wooded slopes covered with rounded pebbles and boulders of ironstone and grit. To the west the plains pass into the belt of hilly ground which leads to the Niger valley, and which in its eastern portion shows the characteristic tabular outlines of the sedimentary series, and in the west, adjoining the river, the rugged irregular contours of broken crystalline hills.

The province of Nupe is somewhat unequally divided into two parts by the Kaduna river. With the exception of the quartzite and granite hills of Jebba and Kailema and the crystalline rocks of the eastern corner of the province between Badegi and the Gurara river, the entire surface is covered by sandstones, which are continuous with the sedimentary series of southern Kontagora. West of the Kaduna, however, the high plains become much dissected and broken up into extended tabular ridges and irregular plateau-like masses, with the tracks leading over a succession of low, well-wooded hills and shallow, flat-bottomed valleys bounded by rough and stony slopes. On either side of the Kaduna rise isolated tabular hills and short ranges separated by long stretches of open plain, while to the east of Bida there open out the great rolling sandy plains of Nupe, which extend from the Bako river to Badegi and southward to the bend of the Niger. Beyond Kacha the character of the landscape again changes: scattered hills and short ranges diversify the surface and coalesce southward into the extended flat-topped ranges and plateaux, whose wooded slopes dip steeply down to the narrow ledge of crystalline rock which fringes the Niger. In the north-eastern angle of the province beyond Badegi the sandstones give place imperceptibly to granites and gneiss. The sandy plains of Nupe are continued eastward, but their drift-covered surface presents now the

characteristic features of a crystalline floor. Knobs and hummocks and low ridges of gneiss project here and there above the general level, while lofty groups of rounded granite hills, like those of Gaung, rise abruptly and picturesquely from the plain.

The sandstone plateaux which bound the Niger to the north of the Gurara river are continued southward as far as Koton Karifi, and pass eastward into Nassarawa as a dissected system of flat-topped and conical hills. To the south of the confluence of the Niger and the Benue, the scenery on the left bank of the Niger in Bassa province, as already described ([p. 6](#)) is that of a hummocky plain with groups of rounded hills in the foreground, backed and capped by lofty plateaux and conical hills. Such a type of scenery is readily explicable as that of a dissected sedimentary series resting upon an irregular crystalline floor of granite and gneiss.

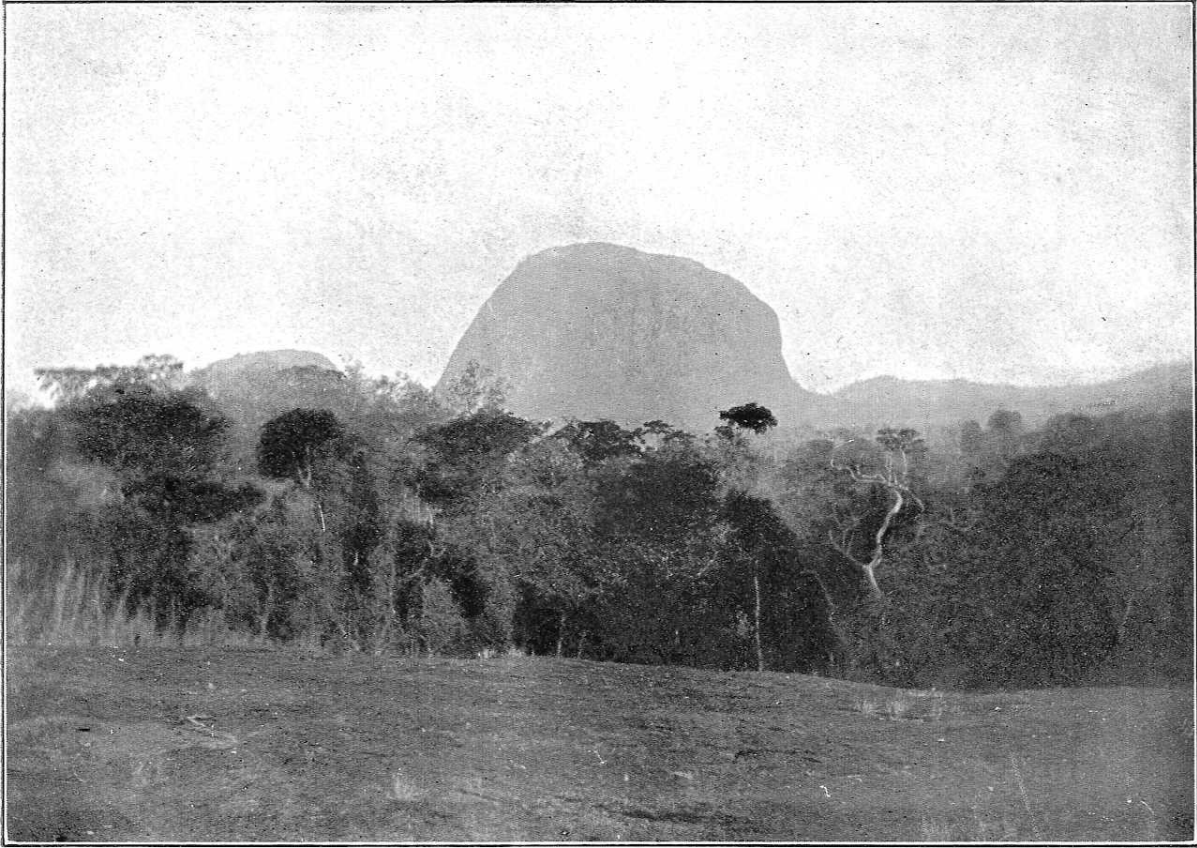
The Benue Provinces. Bassa, Nassarawa, Muri, and Yola.

With the exception of the narrow strip of crystalline rocks exposed along the Niger, the province of Bassa, so far as known, is entirely covered by the sedimentary series whose western and northern escarpments are visible respectively from the Niger and the Benue. These escarpments, however, which from a distance appear continuous, are found on nearer view to be much incised by river valleys which have been cut back for long distances into the high ground behind. A system of lofty, flat-topped and conical hills, extended ranges and plateau-like masses, separated by narrow, steep-sided valleys, has thus been dissected in the north and west out of the once continuous tableland. In the interior the relief lessens, and the surface of the tableland becomes broken up into a succession of tabular ranges and rounded ridges, separated by shallow valleys well cleared for cultivation. The hills are usually thickly wooded and the paths frequently rough and stony and covered with pebbles and broken fragments of ironstone. In the south the Akpoto country becomes in places particularly hilly and difficult, while the little known Munchi country in the east is said to open out into undulating park-like plains beyond the belt of forest which fringes the river.

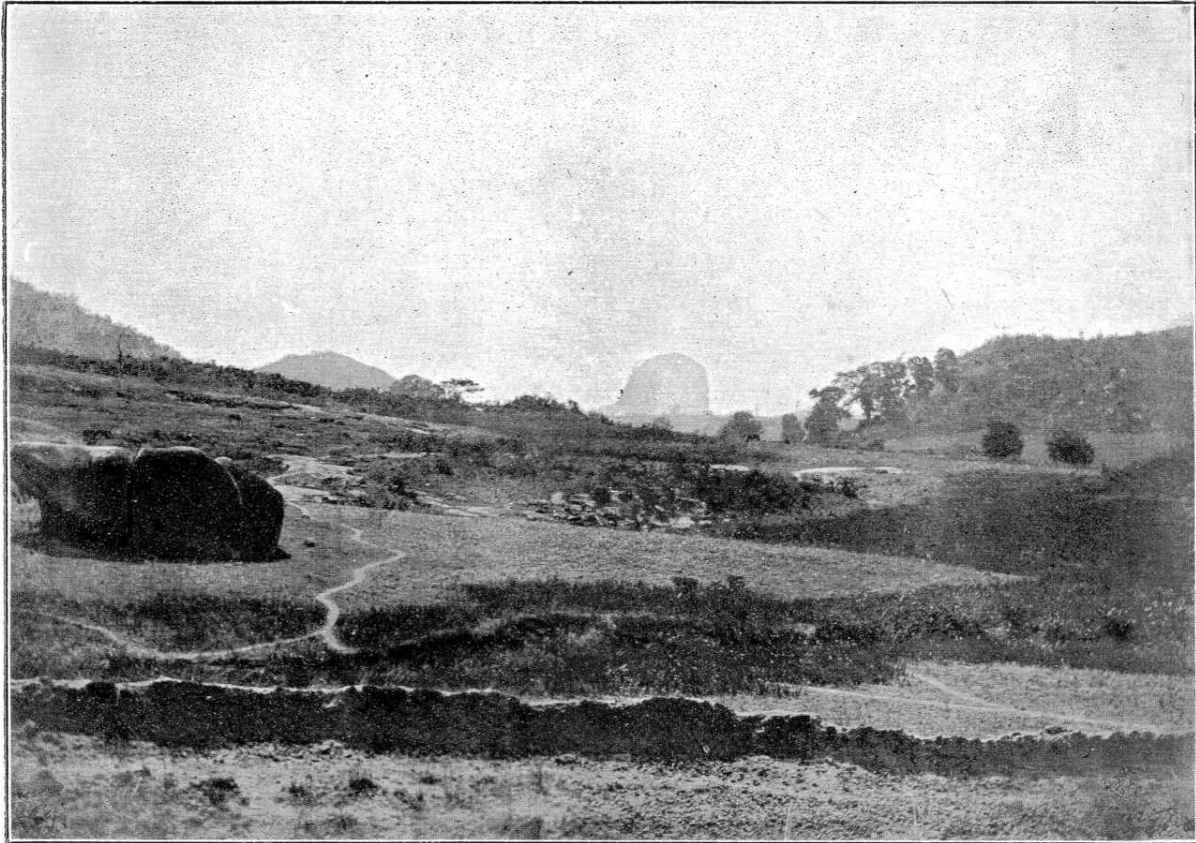
For purposes of description the province of Nassarawa may be conveniently divided into three portions, the river belt of sandstones, the central crystalline plains and hills, and the northern tableland. The Gurara river forms the greater part of the western and northern boundary and drains all but the eastern portion of the elevated plains in the north. The river belt of sandstones between the Gurara and the Okwa in the west presents the characteristic and familiar scenery of a dissected sandstone tableland. Beyond the Okwa river the flat-topped hills and ranges die down into undulating sandy plains which stretch eastward as far as the Modu river, while the detached group of hills, which strikes inland from Udeni, is the last prominent tabular range on the lower Benue. Farther east, the low-lying plains stretch into the unexplored land of the Munchis, while to the north the trade route from Loko to Ibi leads over well wooded undulating plains deeply covered with sandy drift. The solid rock is frequently exposed in the beds of the larger rivers and streams which have cut their channels through the superficial accumulations; and here and there, as at Doma and Lafia, groups of low rounded sandstone hills and projecting hummocks of grit diversify the surface of the plain.

Central Nassarawa is, in the west, particularly broken and rugged. Around Gwombe and Gwagwada and as far east as Buga a mass of low rounded hills has been carved out of a series of soft micaceous schists. The rocks are much decomposed on the surface into red gritty earth which supports an abundant vegetation and in which the rivers and streams have cut deep and narrow channels. To the north these rounded hills pass into a tract of exceedingly difficult and broken country where precipitous hills of granite and gneiss, separated by narrow rocky valleys, rise to a height of two thousand feet above the sea. This mountainous tract, of which the Karshi hills are the northern prolongation, forms the secondary watershed between the river systems of the Gurara and the Okwa. Beyond the Karshi hills a lofty and rapidly undulating plain, in places rough and stony, and much diversified by inselberge (Plate IV, [Fig. 1](#)) and short ranges of granite hills, stretches north-westward towards Abuja. The Abuja Rock, a typical granite mount, streaked and channelled by the summer rains, is a far-famed local landmark and an object of superstitious veneration to the inhabitants of the surrounding country. (Plate III, [Figs. 1](#) and [2](#).)

Eastward from Buga to Nassarawa extends a broken plain with hummocks of gneiss and detached groups of granite hills. At Kurudu, the Okwa river flows through a narrow picturesque gorge between two rounded granite hills, with the bare rock in the river bed honeycombed with potholes. Nassarawa itself is set in the midst of gently undulating park-like plains, well wooded and well cultivated, which extend northward as far as Keffi and are continuous southward with the sandy plains of the river belt. To the east they are limited by the rocky mass of the Anagoda hills and the broken hilly ground between Keffi and the Modu river. To the north of Keffi there opens out the great central plain of Nassarawa which extends from the neighbourhood of Abuja in the west to the Mada country in the east, and is limited northward by the worn escarpment of the northern tableland. Turtlebacks and kopjes of granite and broken hummocks of gneiss rise at intervals from the undulating surface of the plain, while the minor irregularities are obscured, as usual, by deep sandy drift. The larger rivers flow in broad sandy beds with long stretches of swampland on either hand, or, where the drift is thinner, in narrow rocky channels with belts of broken plain on either side from which the surface drift has been in large part removed.



ABUJA ROCK.



ABUJA ROCK FROM THE WALLS OF ABUJA.

Viewed from the central plain, the margin of the northern tableland is in places particularly picturesque. A well-marked, though much worn and rounded escarpment (Plate IV, [Fig. 2](#)), extends continuously from the neighbourhood of Kari in the west by way of Kukwi, Tataru, and Jaginde, to Darroro and Assab in the east, and the skyline is variously ornamented by domes, cupolas, conical peaks and perched blocks, with here and there, as beyond Darroro, a jagged wall of granite rising perpendicularly from the plain. The escarpment is steeper in the east than in the west, and beyond Kari it is lost in the broken hilly ground around Abuja, which falls in a succession of rocky steps to the plains of the Gurara. The tableland itself slopes gently towards the Gurara, and in the west, in the picturesque Gwari country, the numerous domes and turtlebacks of granite afford ideal sites for pagan villages. Parallel with the escarpment runs a narrow belt of broken country which represents the worn edge of the tableland, while to the north the undulating park-like plains, diversified by inselberge and groups of rounded hills, wrapped closely round with sandy drift, recall in every detail

of scenery the central plain of Nassarawa below. A belt of broken rising ground, stretching northward from Jaginde, separates the Gurara system from the Kogin Kogom which drains the grassy plains of Morrua and issues behind Jaginde from a narrow rocky valley cut backward for a considerable distance into the plateau. The flat-topped granite massif of the Kagoro hills, whose precipitous wall forms the continuation of the escarpment from Darroro to Assab, projects also on the north above the plateau and overlooks the lofty undulating plains of Morrua and Akul which stretch away northward towards the Kaduna. Far away to the south of Darroro, the peaks of the Mada hills can be seen across a broken wooded plain: but this eastern portion of the province is as yet unknown and unexplored.

The great plain, which forms the major part of Muri province, extends from the base of the Murchison hills on the north to the frontier of the Protectorate on the south, and from the Ankwe river in the west to the Wurkum and Shebshi hills in the east. With the exception of the narrow margin of crystalline rock in the north-west which runs parallel with the Murchison hills, and the broader belt in the south between Takum and Bakundi, the plain possesses throughout a floor of sedimentary rock. The flat-topped hills, which are such a conspicuous feature of the scenery of the lower Benue and the Niger, are here, however, entirely wanting. The surface of the plain slopes gently towards the Benue from either side, and into it the tributary rivers have cut a succession of shallow swampy valleys separated by tracts of gentle rising ground, which run approximately at right angles to the Benue. The undulating plain thus produced is deeply covered with surface drift and relieved only by an occasional hummock of sandstone or grit. The thin forest, which usually covers the rising ground, gives place in the neighbourhood of the rivers to extensive stretches of open grass land, which, in the rains, become almost impassable swamps.

West of the Ankwe river the surface of the plain is more frequently broken by ridges of rock, and around Awe and on either side of the Benue to the south a number of worn volcanic cones are scattered over the plain. (Plate V, [Fig. 1.](#)) Similar volcanic hills project above the surface on the northern margin of the plain between Yelua and Wase. The Wase Rock^[10] itself, a pillar of trachyte, is a landmark for miles around. (Plate V, [Fig. 2.](#)) At Namu, also, in the north-west of the province, and at Bantaji on the

Donga to the south of the Benue, solitary hills of similar origin form striking features in the landscape. Turtlebacks and kopjes of granite boulders are found only on the margins of the plain where the floor is crystalline. In the north-west the narrow belt of hummocky crystalline ground, with its inselberge and groups of granite hills, is limited by the domes and pinnacles of the little known Murchison hills,^[11] while in the south the undulating plains which stretch along the frontier of the Protectorate from the Katsena river to the Tarabba present all the characteristic features of a crystalline floor. In the north-east of the province, between Bashar and Jeb-jeb, the plains stretch northward up the Kudu valley but are limited to the east by the Ligri, Wurkum, and Jenoe hills, whose steep and rocky sandstone slopes become in places quite rugged and picturesque. The Kwana and Mumuye hills, which bound the plain to the south of the Benue, are largely granitic and present the usual rounded contours, while the foothills of the Shebshi range are of quartzite and gneiss and give rise to more rugged and irregular outlines.

The Mumuye and Shebshi hills are portions of an extensive tract of hilly country which stretches into German territory and which occupies the southern part of Yola province.^[12] The western boundary of this tract runs almost northward, past Jalingu and Kwana, as if to approach the Benue at Lau, but instead turns sharply at right angles. Running eastward from near Kwana there is a practically continuous wall of rock, and at Bamga a steep and almost precipitous climb brings the traveller from Lau to Yola on to the rim of the plateau area. From this position he looks out over the Benue plain several hundred feet below. The river is far away to the north, while beyond that again, just discernible on a clear day, is the further wall of hills stretching continuously from the Wurkum country to the Gongola. Turning south, he sees the tumbled Shebshi hills as a faint silhouette in the distance, while in the foreground, slightly below him, stretch sandy uplands with a gentle eastward slope. The track drops down upon the uplands and then proceeds just within the rim of the plateau towards Yola. For the first day this rim is made up of a chain of rounded hills. Then a number of sandstone outliers arrests the attention. (Plate VI, [Fig 1.](#)) The sandstone is massive and breaks down along vertical faces. The fallen blocks, in assuming their position of rest, smooth off the irregularities of the granite base, and from

the truncated cones thus produced, the cliff-sided outliers rise and dominate the landscape, like great fortresses. From this point the plateau descends gently to the plain of the Maio Belwa and Ma-ini rivers. (A. L.)

Opposite Lau the hilly ground to the north has similar outlines to those of the crystalline plateau to the south. Farther east, behind Wuduku, its appearance changes and the hills are composed of sandstone. Here the ridge seems to be a bold escarpment limiting the unknown Yam-yam country in precipitous cliffs, but with a dip slope to the south which plunges down below the marshy levels of the Benue. Further eastward again there is another change and the outlines of the hills become regular and flat. Where the plain of the Gongola enters the Benue valley, they turn at right angles and run northward parallel to the river. At the angle which separates the two plains the mass is seen from a distance to be made up of horizontal beds, with seven or eight terraces on its sides. These hills, which run up to the west of Kombo, are still, however, unentered and unexplored. (A. L.)

The plain of the lower Gongola is a repetition of the Benue plain on a somewhat smaller scale. The floor of the trench is about ten miles wide and the height of the walls perhaps a thousand feet. On travelling up the Gongola, Kwaba hill is conspicuous two or three days before it is actually reached. It is a massive hill of granite with numerous satellites, of which some are of granite while others are outliers of horizontal sandstones, bounded by vertical cliffs. The successive beds are exposed in their sides like string-courses of masonry, the Gongola outliers differing in this respect from those of southern Yola, which have been carved out of unbedded sandstones. (A. L.)

The plateau of the Vere hills has been already mentioned ([p. 14](#)). It is completely separated from the Karin mountains, Alantika mountains, and other hilly ground to the south by broad and level plains. The essential feature of the scenery is the contrast between the plain and the plateau masses. They rise up from one to two thousand feet higher than the plain, with no transitional foothills or slopes. The base is concealed by sandy wash, and at intervals streams, issuing from the plateau by narrow ravines, trench down through the compacted wash to form steep-sided nullahs. (A. L.)

Opposite to the Vere hills and across the Benue stand the Bagele hills. A little further north is the Hossere Furo. Both groups are of sandstone, rising up from the smooth grassy plain. The rise, however, is not immediate, as it is in the Vere hills, but through basal slopes and foothills toward the higher summits. North-eastern Yola consists of a high-level plain, elevated considerably above the plain of the Benue, but passing into it more gradually than do the plateaux to the south. There is, however, one step of about 200 feet running eastward from Song to the frontier, while the belt of country twenty miles broad to the north of it forms rather a curious contrast to the remainder of north-eastern Yola. Possibly there has been a slight local uplift. Granite is seen over almost the whole surface breaking down into coarse sand and giving rise to a hummocky country difficult to travel. The variations in level are only small, and taken as a whole it is still a plain, but the breaking down of the granite is going on actively over the whole surface. Parallel to this country and almost in alignment with it, is an interesting string of craters and flows of basalt. (Plate VI, [Fig. 2](#), VII, [Figs. 1, 2](#).) The eastern extremity of this line appears to be at Song, but westward it extends across the Mboi hills into the Yunguru country, to what distance is not yet known. The remainder of north-eastern Yola, with its rocky islets set in extensive thinly wooded plains, is similar in character to that of so many other parts of the Protectorate. Some of the inselberge, especially in the Kilba hills (Plate VIII, [Fig. 1](#)), afford excellent examples of exfoliation. (A. L.)

The Central Provinces.—Zaria and Bauchi.

The province of Zaria lies almost entirely within the drainage basin of the Kaduna river, only a narrow belt of country in the south and south-west being drained by the rivers Gurara and Bako. With the exception of the uppermost springs of the Kaduna and the lower navigable portion of its valley below Wushishi, the whole course of the river is included within the province. At Wushishi the river narrows and its channel becomes much obstructed by rocks and rapids, as it winds through the belt of hills which separate the rocky plains of western Zaria from the lower plains of Nupe. At Zungeru the Kaduna flows in the dry season in a narrow picturesque gorge

between white walls of quartzite, which in the rains are buried under a rushing foaming torrent. (Plate VIII, [Fig. 2.](#)) From the bend of the river above Girku in the east of the province, the Kaduna flows south-westward over stretches of drift-covered plain and through scattered groups of hills with its channel marked by frequent rocks and rapids, while beyond Girku the river meanders peacefully over the lofty plains which stretch eastward to the boundary of the province. The larger tributaries of the Kaduna on the north bank, the Kara, Koriga, Tubo, and Shika, take their rise upon the high plains of Hausaland, along whose crest runs the primary watershed between the systems of the Niger and Lake Chad, and, like the Kaduna itself, they flow in their middle and lower courses over broken plains and between groups of hills which separate the minor streams. The Leri river in the east rises on the high plains of the Bauchi border, while the Dunia and Serakin Pawa, the principal tributaries on the left bank, as well as the headwaters of the Gurara, take their rise amidst the scattered hills upon the lofty plains of the south-east. The Bako river rises in the hilly ground around Fuka and, with the Gurara, drains the extreme southern portion of the province.

The hills in Zaria province are of two types; (1) ranges or belts of low rounded and somewhat broken hills, composed largely of the more resistant schists, or of a mixture of granite and schist; (2) kopjes, inselberge and small detached groups of rounded granite hills, with here and there an extensive elongated mass of granite with a more or less flattened tabular summit and steep though worn and rounded slopes rising abruptly from the plains which encircle its base. To the first type belong the tract of hilly country in the west of the province which stretches from Mikonkeli to the south of the Kaduna as far as Kagara to the north, the belt of hills between Gwari and Koriga in the north of the province, the broken stony country around Fuka in the south, and the line of hills in the centre of the province from Dorka to Kaserimi. To the second type belong the solitary turtlebacks and detached masses of granite hills which so abundantly diversify the plains, while the Kudaru, Ruruma, and Kerko hills are somewhat worn examples of the steep-sided granite massifs, whose flattened though hummocky summits form the retreat of numerous pagan tribes.

Over the greater part of the province the configuration of the surface presents a very close resemblance to the characteristic scenery of the plains

of Kabba ([p. 17](#)). Between and around the island mountains and groups of hills and tracts of hilly country, spread undulating thinly-wooded drift-covered plains. The towns and villages are set picturesquely on the slopes and summits of the hills or in patches of thick forest in the midst of the plain. For long distances round the bases of the hills the plains are cleared and cultivated, and from the summit of any of the isolated mounts the peculiar character of the landscape impresses itself, as before, upon the observer. Here and there the solitary hills raise their rounded crests in lonely grandeur, while around and between them the rivers pick their sluggish course over the monotonous plain. The lofty drift-covered levels of the south-east are largely uninhabited and cultivated only by the pagans who daily descend from their rocky strongholds in the Ruruma Hills or on the margin of the Bauchi plateau. Only in the north-east between Girku, Leri, and Zaria do the plains assume the typical aspect of Hausaland ([p. 21](#)). The gently undulating surface is deeply buried in sandy drift, and in places floored with surface ironstone: extensive stretches of the grassy plains become swampland in the rains: the landscape is diversified only at long intervals by occasional turtlebacks and groups of granite hills, while well-walled towns and villages are set here and there upon the open plains or built round the bases of the solitary hills themselves, which serve as watch-towers in times of danger.

The major part of the province of Zaria may be conceived as a shallow, elongated basin, draining westward, and shut in on the north and north-east by the high plains of Hausaland and on the south-east by the Amo, Piti, and Chawai hills, the lofty plains of Katab, and the tableland of Nassarawa. The greater part of Bauchi province, on the other hand, may be looked upon as an undulating plain sloping gently towards the north-east. The south-western portion of the province is occupied by the great plateau of Bauchi, which is limited on the north and south by well-defined walls of rock. ([Map p. 40](#)). The plateau, however, is not a true tableland in the sense of an elevated plain marked off on all sides by precipitous walls from the lower plains around. Like the tablelands of Kabba, Nassarawa, and Southern Yola, which are at most inclined plains bounded by escarpments on one side only, the Bauchi plateau from its summit at Bukuru slopes gently eastward and merges gradually into the plains of the Gongola, while on the west between

the northern and southern escarpments it is continuous for a short distance between Kwole and Assab with the highest portion of the tableland of Nassarawa. The intimate relation which exists between the Bauchi plateau and the tableland of Nassarawa is further demonstrated by the continuation of the escarpment of the tableland beyond Assab to form the southern wall of the plateau. From Assab the escarpment, here 1,500 to 2,000 feet in height, runs south-eastward to the neighbourhood of the ninth parallel before swinging round to the north-east and losing itself amongst the rocks of Angass, where the single escarpment appears to be replaced by a succession of rocky steps leading down to the tract of broken country between Angass and Kanna, which is drained by the headwaters of the Wase river.

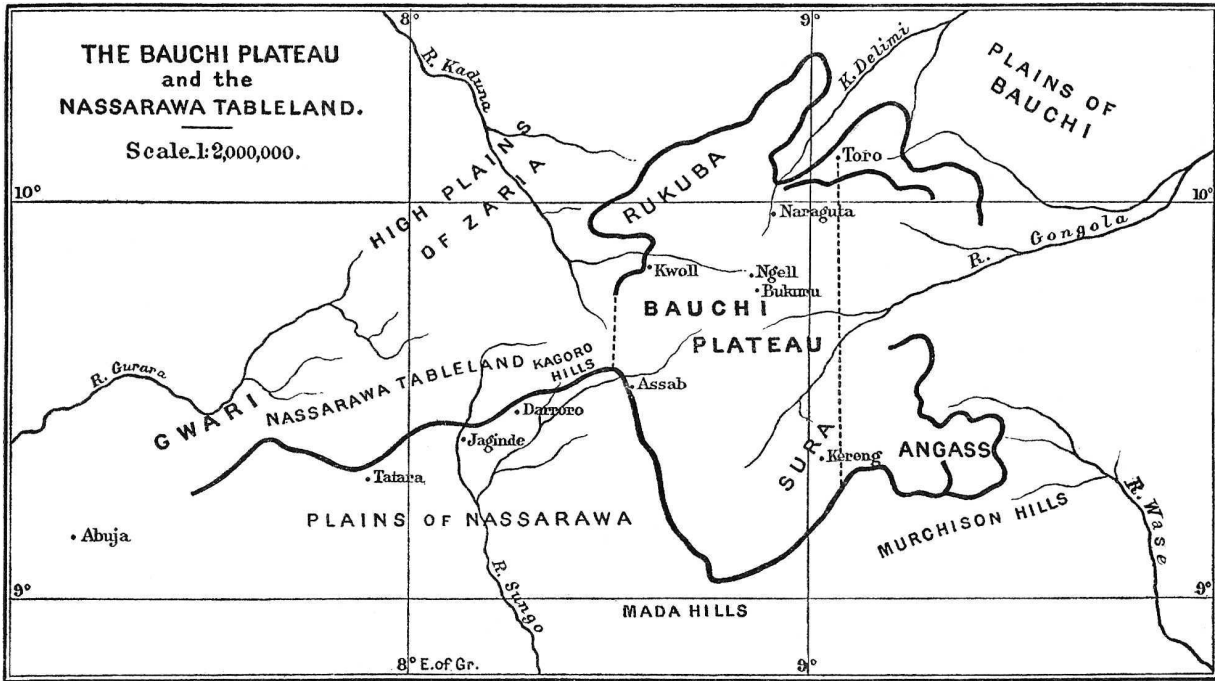
The northern margin of the plateau is more irregular than the southern and marked by projecting spurs which have now assumed the character of mountain ranges. In the west the escarpment begins in the neighbourhood of Kwole and Miango and stretches westward as a somewhat worn and rounded cliff, 1,000 feet in height, separating the lofty plains of Rukuba from the lower plains of Zaria. Beyond Chawai and Piti the escarpment is continued as the western slope of the Amo, Jengre, and Gura hills, which run out upon the broken plains to the north, while the eastern slope of the hills again assumes the character of an escarpment and bounds the valley of the Delimi as far as the gorge of the river below Naraguta. To the east of the Delimi, however, the single escarpment is replaced by at least two steps which interpose the broken plains of Toro and the summits of the Kwandokaya hills between the plateau and the northern plains, while to the south-east of Polchi the escarpments run out upon the plains of the upper Gongola in groups of picturesquely rounded hills. There is some reason also to believe that the plateau is stepped or broken in the neighbourhood of Basak in the Rukuba country, but this point must await further investigation.

The surface of the plateau reaches its highest elevation of about 4,100 feet in the neighbourhood of Ngell and Bukuru, whence the headwaters of the Delimi, Sungo, Gongola, and Kaduna radiate approximately north, south, east, and west. Of these rivers the Delimi ultimately joins the Yo and falls into Lake Chad, the Gongola and Sungo join the Benue, and the

Kaduna the Niger. Thus the few square miles in the neighbourhood of Ngell and Bukuru may rightly be claimed as the hydrographical centre of the Protectorate ([p. 3](#)), and even the most casual observer cannot fail to be impressed here with that peculiar character of the primary watershed which has already been remarked in the case of the western tributaries of the Niger and the northern tributaries of the Kaduna. There is no group of rocky mountains, no rugged peak at the hydrographical centre of the Protectorate, no lofty ranges forming the watersheds between the radiating rivers: only an undulating, swampy, alluvial plain, diversified by a few short, broken ridges and low kopjes of granite boulders. Into these plains, however, the rivers which flow north, west, and south are rapidly carving narrow rocky valleys, and their descent to the lower plains is marked in every case by picturesque waterfalls and narrow gorges cut back for considerable distances into the plateau itself. The wooded gorge of the Delimi below Naraguta, the steep waterfalls of the Kaduna between Kwooll and Miango, and the deep rocky channel of the Kogin Rim above Assab afford striking evidence of the activity of the rivers. Only in the east where the plateau inclines gently towards the Gongola do the streams flow in sandy beds between deep banks of alluvium, with broad stretches of swamp on either bank wherever the slope is insufficient to drain the summer floods.

THE BAUCHI PLATEAU and the NASSARAWA TABLELAND.

MAP N^o. 2.



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Face p. 40.

The Bauchi plateau, in the more restricted sense, may be taken to include all those portions of the elevated plains which rise above 3,500 feet. Upon this definition the plateau is bounded on the south by the continuous escarpment which stretches from Sura to Assab, and on the west by a line drawn northward from Assab to Kwole and Miango. Beyond Miango the plateau is again limited by an escarpment as far as the Jengre Hills, whence the boundary runs southward to Naraguta and thence eastward along the higher escarpment behind Toro. The eastern boundary of the plateau is formed by a line drawn approximately from south to north from Kereng to Toro. The most striking feature of the greater part of the plateau as thus defined, with the exception of the Rukuba country to the north, is the bare and treeless aspect of the surface, which assumes the character of

undulating grassy downs extensively cultivated in the neighbourhood of the towns and villages (Plate IX, [Figs. 1, 2](#)). Only amidst the clusters of rocky hills which diversify the surface do a few wind-worn trees bear witness to the former extent of the forest, while round the bases of the hills and among the boulders on their slopes are set the rectangular compounds of the pagan villages, whose dark green hedges of prickly cactus add a pleasing touch of colour to the mellow landscape. The low hills which rise from the plateau have also a peculiarity all their own, which is to be directly correlated with the greater diurnal variation in temperature. The turtlebacks and domes and rounded hills of the lower plains are here conspicuously absent, and in their place appear heaps of boulders and jagged ridges of granite, with picturesque peaks and pinnacles and perched blocks adorning their crests. Here and there, also, in the Rukuba and Kibyen country in the west and in Sura in the south-east, more or less worn and rounded volcanic cones rise above the general level of the plateau (Plate X, [Fig. 1](#)). The craters of Kereng (Plate X, [Fig. 2](#)) in the south-east, however, are remarkably well preserved. The surface of the plateau itself is, moreover, deeply covered with drifted alluvium and provided in places with a crust of surface ironstone through which the rivers and streams have in many cases cut to the rock below. Above Assab the drift has been largely removed from the surface, and between Rim and Woran numerous grassy, flat-topped hills of red, gritty earth, capped by a hard ferruginous crust, rise in places to a height of at least 300 feet above the level of the plateau margin. These hillocks recall in appearance the tabular sandstone hills of the Niger valley, and bear striking witness to the former extent and thickness of the superficial accumulations (Plate XI, [Fig. 1](#)). The broken, rocky character of the marginal belt of plateau on the north in the neighbourhood of Zammagan, Basak and Naraguta is probably also largely due to the removal of the covering of drift which formerly obscured many of the irregularities of the surface. Isolated hillocks of drift indeed are common throughout Rukuba to the west of Jos and Naraguta (Plate XI, [Fig. 2](#)), while at Basak they may be seen capping the very summit of the cliffs.

Beyond the limits of the plateau proper, a thinly wooded drift-covered plain, in places somewhat broken and hummocky and diversified in the usual manner by inselberge and scattered groups of granite hills and drained

by the headwaters of the Gongola, stretches eastward to the meridian of Bauchi. Similarly, beyond the worn and rounded escarpments and projecting spurs which form the northern margin of the plateau, an undulating wooded plain, drained by the Delimi and its tributaries, extends north-eastward towards the Kano border. The western watershed of the Delimi valley runs along the hummocky drift-covered plain which stretches northward between the Gura hills and the detached mass of the Liruei-Kano hills, and thence through the rocky hilly country round Tirkunya into the Ningi hills, which form the most striking feature of the north-western portion of the province. This elongated mass of hills, which stretches from Bura in the west to Ari in the east, is a typical instance of those steep-sided and more or less tabular granite massifs, of which examples have already been noted in the Ruruma hills of Zaria, the Kagoro hills of Nassarawa and the Vere hills of Yola. The Ningi massif reaches a higher elevation in the west than in the east, its slopes are much rounded and incised by narrow valleys, its summit is broken and hummocky with a scanty soil of disintegrated granite, while wooded drift-covered plains stretch outward from its base towards Zaria on the west, towards Kano and Katagum on the north, and towards the Delimi valley on the east and south.

The surface of these wooded plains rises gently in the north to form the low watershed between the valleys of the Delimi and the Shidya. The headwaters of the Shidya rise in the tract of broken hilly country which lies to the north and north-east of Bauchi, and forms in part the northern watershed of the Gongola basin. Around Bauchi itself domes and turtlebacks and groups of rounded hills, rising from the stony and drift-covered plains, are particularly numerous and often remarkably picturesque in their outlines, the lion couchant being one of the commonest forms which the bare granite hills assume. (Plate XII, [Figs. 1](#) and [2](#).) South of the Gongola the plains slope gently upward to the base of the Jarawa and Kanna hills, becoming more and more broken and stony as they approach the watershed of the Wase river. The granite massif of the Wadai hills between Angass and Sura is only known along the northern margin, while Angass^[13] itself is a particularly broken and rugged district separated from the Murchison Range by a series of rocky valleys still largely unexplored. Between Angass and Kanna the high drift-covered plains of Bauchi pass

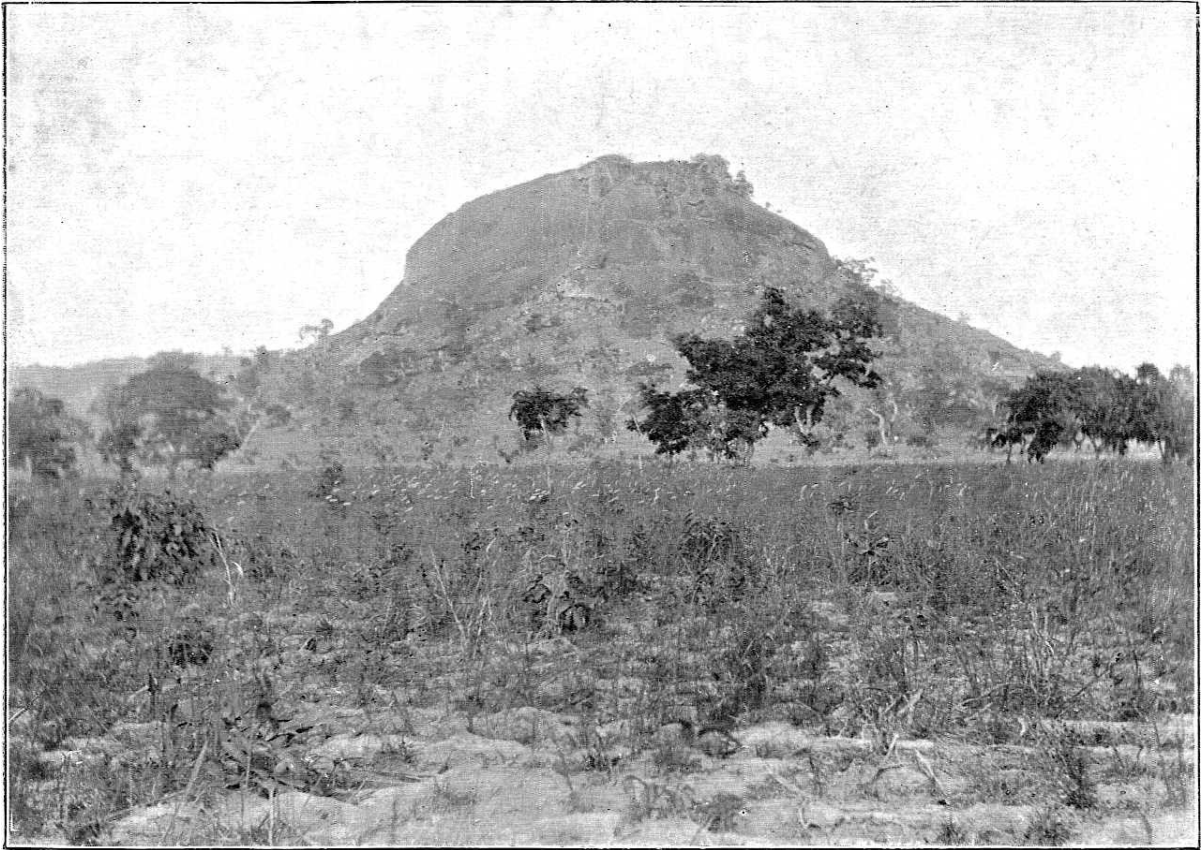
into a tract of broken country which is drained by the headwaters of the Wase river and falls rapidly to the plains of Yergum.^[14] The line of fracture to the west which forms the margin of the plateau appears to be replaced here by a simple flexure or perhaps by a succession of worn and broken steps. Farther east rise the Kanna and Kantana hills, with their projecting spur towards Kwankyam, whose steep though worn escarpments interpose an ascent of 700 to 1,000 feet between the lower and higher plains. The western hills, which are prolonged northward into the Jarawa country, are composed of crystalline rocks, and the more prominent peaks possess the usual rounded outlines of granite hills. The Kantana hills to the east, however, are sedimentary in origin, and from a distance show the familiar tabular outlines of horizontal sandstone hills, the narrow plateau on top being, however, much eroded and incised. Eastward the escarpment appears to be much stepped, and in the neighbourhood of Tukkur wheels northward to form the eastern slope of the Jarawa hills which run out upon the plain of the Gongola.

The eastern portion of the province is almost entirely covered with sedimentary rocks. In the centre, a little to the south of the parallel of Bauchi, broad plains stretch eastward from Goram and Lafiangale to Deba Habe and the Gongola, bounded on the north by the broken sandstone plateaux which are enclosed within the bend of the Gongola, and on the south by the slopes of the Kudu valley, the rounded hills of Gateri and the picturesque peaks of Tangale. In the west the plains are deeply covered with sandy drift, through which the headwaters of the Kudu have cut narrow canyon valleys into the sandstones below. Farther east the summer rains render the plains for long distances almost impassable swamps which drain imperceptibly east or west, and leave in the dry season a gently undulating surface of black, gritty clay, baked and fissured by the summer sun. To the north of the central plain and within the bend of the Gongola lies a region of detached tabular masses of elongated or irregular outline, together with groups of worn and rounded sandstone hills, separated by stretches of sandy plain, presenting all the familiar scenic features of the dissected sandstone plateaux of the Niger. The narrow valley of the Gongola, between Deu and Gombe, bounded by flat-topped ranges on either side, presents a marked contrast in scenery to the upper course of the river above Baddera, where it

threads its way over the high plains of Bauchi between turtlebacks and domes of granite. Below Gombe, however, the valley opens out and the river flows as far as Nafada through an open sandy plain, diversified by occasional flat-topped sandstone hills. Between Nafada and Ashaka the channel winds between detached hills and short tabular ranges which rise from an undulating, thinly-wooded plain, while below Ashaka the river enters the deeply drift-covered plain which stretches from the base of the Tilde and Dukul hills on the west to the foot of the escarpment of the Barbur plateau in Bornu on the east. As far as Jurara and Deba Filani the plain is thickly populated, well-cleared and cultivated; southward, however, within the Habe country, the scattered villages are separated by longer stretches of bush, while the plain is diversified by hummocks of sandstone and worn hillocks of basalt with the intervening irregularities of the surface deeply buried in drift. Beyond Deba Habe, the plain of the Gongola forms the eastward continuation of the central plain of Bauchi, whose sandy and swampy surface stretches southward to the base of the Tangale hills.

The south-eastern portion of the province is occupied in part by the upper valley of the Kudu river and in part by the extensive mass of hills which, with the exception of the outer margin around Gateri, Tangale, and Waja, is still largely unentered and unexplored. A series of narrow valleys, terminated by canyon-like gorges, have been cut by the headwaters of the Kudu into the elevated plains of Pali and Lafiangale. The difference in level between the upper plains and the river bed is greatest in Dugari in the west, where the streams flow in narrow flat-bottomed valleys, bounded by perpendicular walls of white sandstone three to five hundred feet in height. From the cliffs above Tongolan the scenery recalls in miniature the chasms, amphitheatres, and canyons of Colorado, with their promontories and projecting buttresses, steep slopes and vertical ledges, while south-eastward towards Yuli the remnants of a lower plateau mark a former pause in the activity of the river. Farther east, by Panguru and Putu, the relative height of the upper plains decreases, the sandstones become broken and inclined, and the tributary valleys are bounded, not by cliffs, but by rounded, drift-covered slopes leading upward to the higher undulating plains. The folded sandstones of Gateri and Ligri give rise to a group of rounded hills deeply incised by narrow valleys and backed by the ridges and peaks of the

unexplored Wurkum and Pero hills, which stretch north-eastward towards the Gongola. From Gateri northward to Putu the track leads over somewhat broken country, while eastward from Putu to Tangalto stretch open, undulating, swampy plains, diversified only by occasional conical stumps of basalt.



A GRANITE MOUNT NEAR ABUJA.



THE MARGIN OF THE NASSARAWA TABLELAND FROM JAGINDE.

The tract of country which lies between Tangale and the Gongola is one of the most picturesque within the Protectorate. The hills rise in detached groups from an irregular hummocky plain, and form the advanced guard of the belt of lofty highlands which intervene between the northern plains and the Benue. While the whole district may be spoken of generally as Tangale, the Tangale tribe itself occupies a very restricted area and is lodged within only the westernmost group of hills, the successive groups to the east and south being inhabited by the kindred but mutually hostile tribes of Chongwom, Ture, Awok, Tula, and Waja. Tangalto and Tall, the strongholds of the Tangale tribe, are set on the slopes of rounded granite hills, picturesquely pierced by worn pillars of trachyte and basalt, and separated by a narrow undulating plain from the neighbouring group of granite hills inhabited by the pagans of Chongwom and Kaltungo. Beyond Chongwom the character of the scenery entirely changes. In place of the rounded crystalline hills there rise precipitous and more or less tabular masses of well-bedded sedimentary rock, forming natural fortresses in the midst of an

undulating park-like plain, diversified by knobs and hummocks and standing rocks of sandstone (Plate XIII, [Fig. 1](#)). At times the summit ridges are gently curved and the hills are bounded on one side by a steep escarpment and on the other by a dip slope sinking gradually to the plains below, while the whole landscape is dominated by the Ture or Tangale Peak, a picturesque pillar of trachyte rising from a worn and flattened basal cone of sandstone (Plate XIII, [Fig. 2](#)). Eastward, around Jelengu and Reme, somewhat similar pillars of volcanic rock project here and there above the sandstone hills, while the worn escarpments of Waja appear to be continuous with those which form the western boundary of the lower Gongola valley. The country which lies to the south of Tangale and Waja is quite unknown. From the upper slopes of the Ture Peak, a second distant range of hills can be seen running parallel with the jagged ridge in the foreground, and it is probable that, instead of a mass of hills or a continuous plateau, there is within this unexplored land a succession of broken valleys and rocky ridges, forming one of the last strongholds of paganism within the Protectorate.

The Northern Provinces.—Bornu, Kano, and Sokoto.

The southern portion of Bornu, which lies immediately east of the Gongola, presents, so far as the character of its physical features is concerned, a greater affinity with the central provinces of Bauchi and Zaria than with the northern Hausa States or with the northern portion of Bornu itself. The valley of the Gongola to the east of the river is more broken and hummocky than it is to the west. The sandstones are much faulted and variously inclined and form belts of rocky plain between the swampy valleys of the tributary streams. Here and there, as at Wuyo and Gulani, a circle of jagged ridges forms a natural battlement round a pagan town, or, as at Balbea, a flattened sandstone hill forms a natural stronghold in the midst of the plain. More striking, however, are the level crests, the dip slopes and the steep escarpments of the Bima and Wadi hills, which form the landmarks of the Gongola plain from Ashaka to Wuyo. From Gulani northward to Ashaka stretches an extensive undulating plain, broken only by an occasional ridge of sandstone, with black swamp earth covering all the

minor irregularities of the surface and creeping up the slopes of the sandstone hills themselves. Eastward the broken plain of the Gongola is bounded by the worn escarpment of the Barbur volcanic plateau which becomes higher towards the north, and between Bilaraba and Dukshi is fringed with a series of flat-topped and conical outliers of basalt. To the east of Gulani and Balbea the ascent to the plateau is over basal sandstones capped by black volcanic rock. Farther north, however, in the neighbourhood of Dukshi, the escarpment is entirely composed of basalt and rises from an irregular crystalline floor. The summit of the plateau is broken and rocky in the west, deeply trenched by the tributary streams of the Gongola, and diversified by numerous worn and rounded cones of ash and lava. Eastward towards Buratai the plateau descends by a series of low rocky steps separated by drift-covered levels to the plains of Hong and Ajigin, whose deep covering of dark gritty clay, hard and fissured in the dry season, forms an almost impassable swamp in the rains. In the south-east, towards Biu and the little-known Burra country, the volcanic cones become higher and more conspicuous, while the surface of the plateau, according to native information, becomes much dissected and broken up into narrow plains and rocky valleys.

Two tracts of hilly country, separated by the swamps of Hong and Ajigin, lie respectively to the north-west and north-east of the volcanic plateau. The worn and flattened hills of sandstone and ironstone, which stretch westward from Buni and Ligdar towards Gabai, pass northward into the high, drift-covered plains of Mutwe and Gujba, which form the watershed between the Gongola and Lake Chad. In the east the Tagum and other groups of flat-topped hills around Lua, Ndufa, and Korode represent the remains of a sandstone plateau once continuous with the hills of Buni and Ligdar across the swamps of Ajigin. To the north and north-east of the Tagum hills stretch the great wooded plains of Bornu, with their gently undulating surface sloping imperceptibly towards Lake Chad, and with not a single hill nor even a hummock of boulders to relieve the monotony of the scenery. To the south-east of the Tagum hills the sandstones run out upon a crystalline floor which stretches southward into Yola province, with its surface diversified only at long intervals by island mountains and groups of granite hills.

The plains of Bornu, which stretch northward from the rocks of Chibuk and the Tagum hills to Lake Chad and the river Yo, are in the east so imperfectly drained that extensive tracts of their surface become marshy and swampy during the rainy season. The swamps reach their greatest development in the valley of the Yedseram and in the immediate neighbourhood of Chad, where the whole country is flooded after the rains. Barth^[15] has described in a sufficiently vivid manner the featureless and cheerless aspect of the swampy plains to the south and south-east of Chad. Along the caravan road from Maidugari to Kukawa transverse stretches of swamp land (firki) alternate with belts of slightly higher ground, hard and smooth on the surface, or thinly coated with loose yellow sand or dark grey loam. The character of the vegetation reflects immediately the changes in the nature of the surface soil. In the dry season, after the grass is burnt, the cracked and fissured surface of the firki land is absolutely bare and treeless, the hard and smooth surfaces of sandy clay are covered with thin mimosa bush, while the stretches of loose sandy loam support a more abundant growth of trees and scrub. The ancient dunes and the undulating sandy plains which bound the northern basin of Chad support an open forest growth, while recent accumulations of blown sand are now found only within the towns themselves, or in places which have been cleared of forest for purposes of cultivation.

The Maidugari river, which rises on the Yola border and loses itself in swamp a few miles to the north of Maidugari, has cut deeply into the superficial sandy drift in the neighbourhood of Maidugari and Maifoni and formed a comparatively narrow valley with steeply-rounded slopes. From Maifoni a gently rolling plain stretches westward to Gujba, with occasional thick-stemmed kukas rising prominently above the low growth of scrubby trees and prickly mimosa which covers the surface, and with here and there a dry “tubka” or shallow swampy hollow forming an open space within the forest. At Murguba, and again at Gujba, the plain is broken by shallow water-courses, bounded by low and ill-defined ridges of rising ground. These extended swampy hollows are flooded during the rains, and in the dry season marked by a line of wells and occasional pools separated by sandy bars. Beyond Gujba the rolling plains become again marked by a rapid alternation of sandy rising ground and clayey shallows draining

southward in the rains. Between Chumga and Abakri, however, rock begins to appear in places on the surface of the drift-covered undulations which separate the swampy valleys, while a few isolated flat-topped or rounded hills are scattered over the plains. From Abakri can be seen to the west the long flattened and rounded ridges and tabular hills, half buried in sandy drift, which introduce the Kerri Kerri plateau. Daura and Durua are set within this broken country, each at the base of a low flat-topped range of ironstone hills, while the track leads over a succession of stony ridges and shallow cultivated valleys to the continuous plateau surface which is reached about half-way between Durua and Momdo. The plateau itself is no higher than the summits of the marginal hills, which with their drift-filled valleys represent merely the dissected eastern front of the tableland. The surface of the plateau, at first thinly wooded and in places somewhat stony, passes westward into the open sandy undulating plains around Potiskum, which fall away gently to the north and west into the valley of the Kogin Shidya. Here and there a low hill or a rounded ridge of sandstone or ironstone rises above the general level, with the sandy drift which obscures all the minor irregularities of the surface deeply banked against its slopes.

The Kerri Kerri plateau, like the tableland of Nassarawa and the elevated plains of Bauchi, is not a true plateau in the strict sense of the term, but rather an inclined plane, bounded by more or less worn escarpments on the south and passing gradually northward into the plains of Ngizem. The escarpments which stretch from the neighbourhood of Daura south-westward to the provincial boundary reach their greatest elevation behind Dozi, Lewa, and Gamari to the north-west of Nafada, where precipitous cliffs of white sandstone 500 to 600 feet high rise abruptly from the sandy plains which stretch southward to the Gongola. The escarpment, however, is in no sense a continuous one, for, as already noted in the neighbourhood of Daura, the margin of the plateau has been much cut up and dissected, especially in the east, into a system of more or less detached flat-topped ranges and tabular masses separated by narrow valleys or by short stretches of open plain.

Westward where the escarpment is higher, as in the neighbourhood of Dozi and Kadi, the cliffs are furnished with the characteristic buttresses and indentations of horizontal sandstone rocks, while the margin of the plateau

is fringed with a series of detached tabular and conical outliers of varying size and elevation (Plate XIV, [Fig. 1](#)), the larger of which form natural strongholds for pagan villages.^[16] The most remarkable feature, however, about the margin of the plateau is that, behind the fringing belt of detached ranges and outlying groups of hills, the main mass of the plateau has been grooved or incised by a series of narrow flat-bottomed trench-like valleys, bounded by perpendicular walls of white sandstone and notched and indented by lateral gullies (Plate XIV, [Fig. 2](#)). For long distances the narrow valleys preserve a very constant breadth, while frequently they turn abruptly at right angles or are joined by lateral valleys of similar character which unite the whole series into an interlacing system through which it is possible to travel from Daura to Zai without once ascending to the level of the plateau above. At their upper ends the valleys rapidly contract and run out upon the plateau as steep and rocky gullies. Their floors, like the surface of the plateau above, are well cleared for cultivation and studded with scattered trees, while the villages are set aloft on the margin of the cliffs overlooking the wells in the valley below.

Beyond the western boundary of Bornu the escarpments die down, and the drift-covered surface of the plateau falls gently to the swampy valley of the Kogin Shidya. The last trace of the sandstones of Kerri Kerri is found in the low hills of conglomerate and grit around Hardawa, while twenty miles to the north according to native information the sandstones disappear in the neighbourhood of Lise under the sands and swamps of Katagum. Beyond Hardawa begin the undulating plains of Hausaland ([pp. 21, 38](#)), which stretch westward, to the north of the Ningi massif, through the heart of Kano and the north-eastern portion of Zaria into Kontagora and Sokoto. Here and there a heap of granite boulders or an isolated group of rounded hills, as at Shira, Fagam and Kila, rise out of the red sandy drift which is deeply banked against their slopes, and which elsewhere effectually buries the irregularities of the crystalline floor. Occasionally a smooth and rounded surface of gneiss or granite outcrops level with the plain, while the rivers which flow in broad, shallow and ill-defined valleys, indistinguishable at a distance from the rest of the plain, have here and there succeeded in cutting their channels through the sandy drift to the solid rock below. Where uncultivated the plains are covered with prickly mimosa bush, or with trees

well spaced or in bushy clumps with open grass around. The farm-land, however, is well cleared and comparatively bare with a few large trees scattered here and there amongst the furrows, while the fallow is thickly covered with tall aromatic grass and low scrubby trees of a few seasons' growth.

From Jemaari north-eastward to Katagum, and thence westward to the immediate neighbourhood of Kano, not a single rock is anywhere exposed, and not a single block or boulder or even a pebble of quartz is to be found in the sandy loam which covers the surface. North of Jemaari the track leads through gently undulating country presenting the usual alternations of farm-land and thinly wooded plain. Between Itas and Murmur the surface becomes in places quite hummocky with ancient dunes in the form of low rounded ridges of loose sandy drift, extended approximately east and west, rising in succession upon the level plain and bearing witness to the former prevalence of desert conditions. Around Katagum the drainage is so imperfect that long stretches of the plain are under water in the rains, while westward beyond Kwotolu the surface resumes its undulating aspect with its alternation of sandy rising ground and clayey shallows. Here and there, as at Koya, a rounded ridge of loose sand, now covered with grass and scattered trees, affords evidence of the former activity of the wind, while in places in the hollows are small sheets of fresh water forming lakes which are more or less permanent throughout the year.

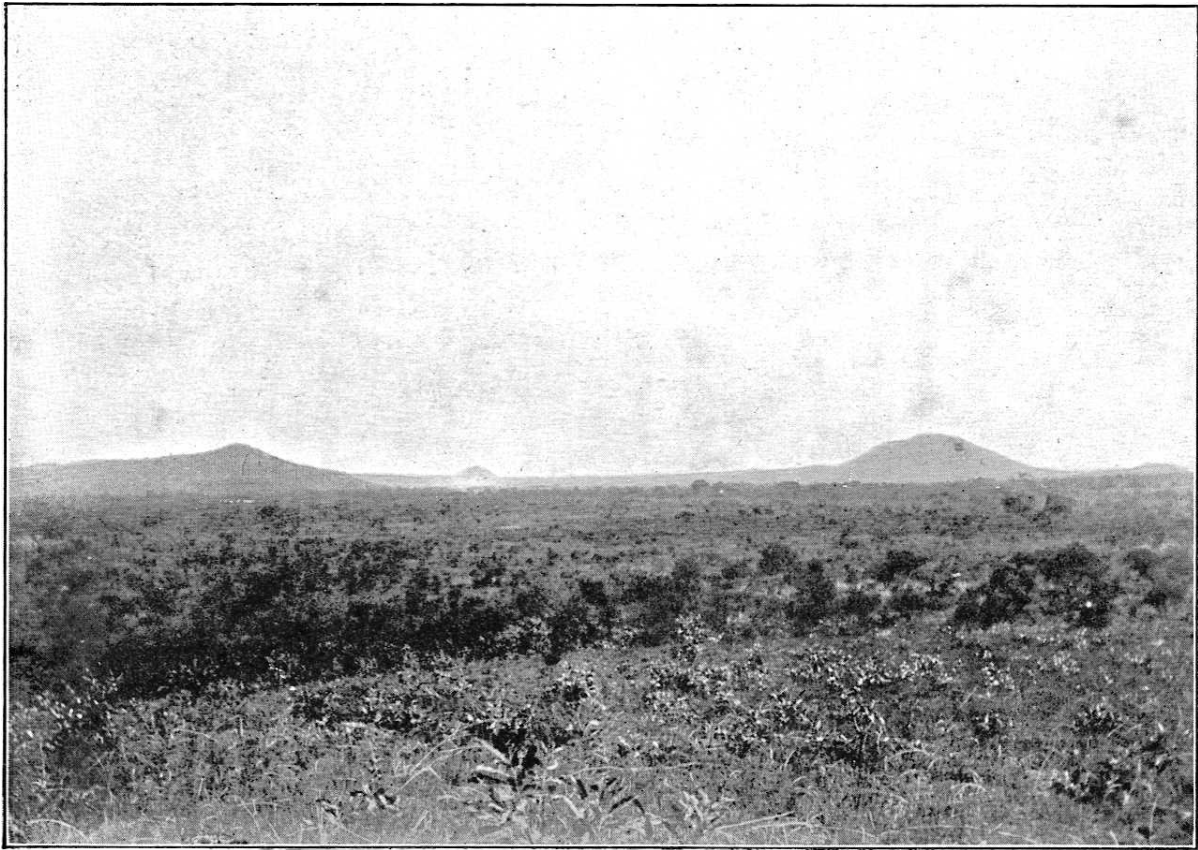
Around Kano and north-westward towards Katsina the open undulating plain presents all the characteristic features of a crystalline floor. For long distances round the towns the red sandy loam which covers the surface has been cleared of all but the larger trees and subjected for generations to intense cultivation. Here and there around Kano smooth surfaces of rock appear level with the plain, or rise into low rounded turtlebacks or heaps of granite boulders. The two prominent tabular hills of weathered diorite, Dala and Kogon Dutsi,^[17] within the walls of Kano, have a peculiar interest of their own which will be related in the sequel ([p. 201](#)). Northward between Kano and Kazaure the undulating sandy plain is diversified only by the low flat-topped hillocks of earthy grit and conglomerate in the neighbourhood of Sai and Ingogo and by the broken ridges of amethystine quartz which rise from the plain immediately south of Kunchi. Kazaure is set in the midst

of rocky ridges of blue quartzite with red sandy drift filling up the valleys and banked against the slopes of the hills. Around Rauni rise low extended flattened ridges, which appear to be composed throughout of dark red vesicular ironstone. At Tummas, half way to Ingawa, low granite knobs and heaps of boulders project above the surface, but from Ingawa north-westward to Katsina the undulating plains are featureless, with no prominent granite hills or kopjes to break the monotony of the landscape. Smooth surfaces of rock are here and there exposed level with the plain or in the channels of the streams, but otherwise the deep sandy drift which covers the surface effectually obscures all the minor irregularities.

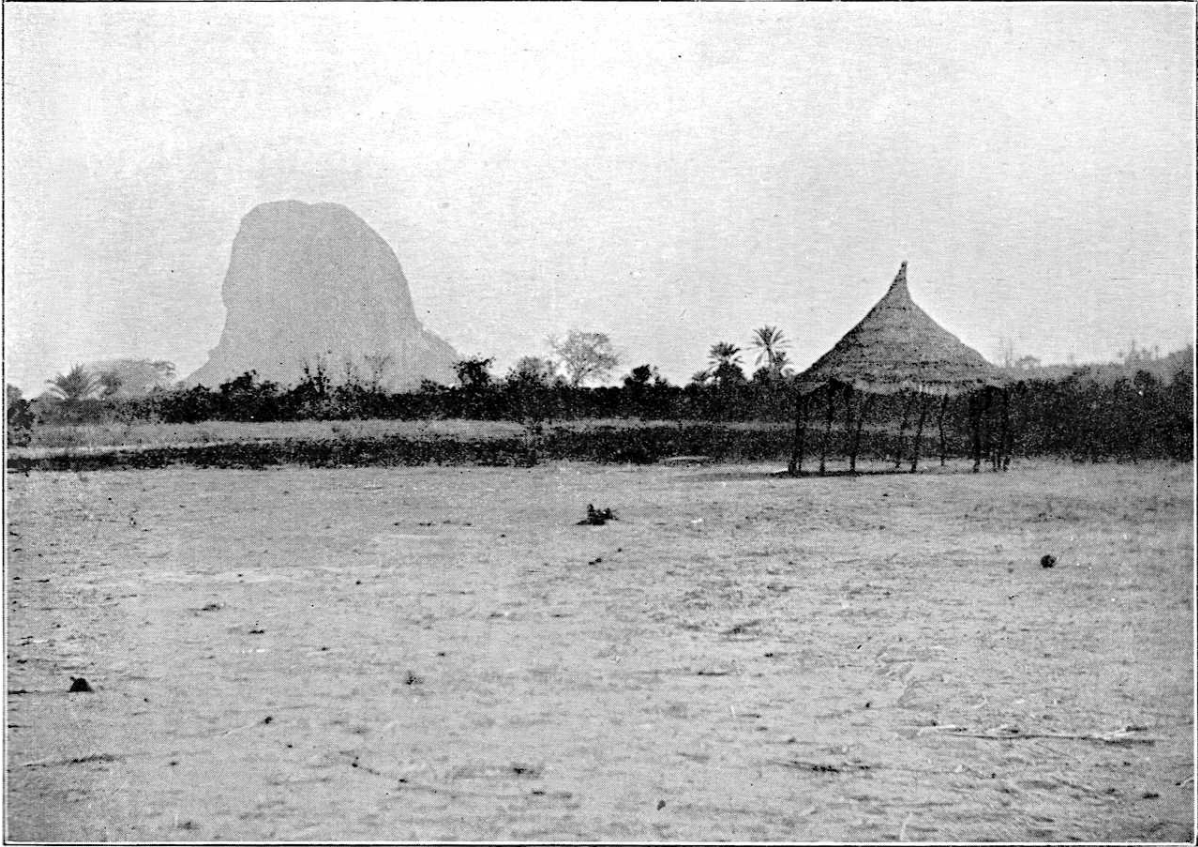
Around Katsina both sandstones and granites are exposed upon the surface of the plain, but it is not until the neighbourhood of Kurefi is reached that a decided change takes place in the character of the scenery. From Kurefi to Runka the track leads through rocky granite country which assumes its wildest aspect in the neighbourhood of Tummo, where the landscape, as far as one can see, is made up of an assemblage of low rounded hills, domes, tors, heaps of boulders, perched blocks and smooth surfaces of granite, with no particularly high or prominent peaks. On the margin of the mass near Kurefi and Runka the aspect of the country is smoother; the turtlebacks and rounded hills are separated by narrow stretches of plain, and often half-buried under the sandy drift which is banked up against their slopes. From Runka south-westward to Kotorkoshi, and thence north-westward to Karakai, the landscape assumes its familiar aspect of a gently undulating thinly-wooded surface, diversified by isolated groups of granite hills which, at Duru and Kotorkoshi, rise abruptly and picturesquely from the drift-covered plain. A belt of broken hilly country stretches N.N.E. and S.S.W. between Karakai and Maradu, parallel with the strike of the quartzites, phyllites, and gneisses out of which the hills are carved. Immediately beyond Maradu knobs and hummocks of volcanic agglomerate project from the surface of the plain, while northward the track leaves the hummocky crystalline floor in the neighbourhood of Gora and follows the level swampy plain of the Gulbin Gandi as far as Dankaiwa. At Dankaiwa, in the cliff which faces the river, is the first exposure of the red earthy sandstones of Sokoto, while at Dampo, a little farther on, the first flat-topped sandstone hill rises above the sandy plain. From Dampo the

track again runs along the swampy valley of the river as far as Rara, whence it leads across undulating park-like plains to the broken sandstone hills of Sokoto.

The northern and western portions of Sokoto province, which consist of a dissected plateau of horizontal sandstones, possess in general all the familiar scenic features of such a formation as already described from other parts of the Protectorate. The apparent differences are differences more in degree than in kind. Like the Niger above Lokoja, the Benue in its lower and upper course and the Gongola above Gombe, the Gulbin Kebbi or Sokoto river flows in a broad flat-bottomed trench-like valley between two steep and worn escarpments, which lead up to the plateau on either side. The breadth of the valley of the Gulbin Kebbi is, however, out of all proportion both to the height of the bounding walls and to the volume of the river itself. The floor of the valley is usually several miles in breadth, while the cliffs on either side rarely reach 200 feet in height. In the rainy season a sluggish flood covers the bottom of the valley, but in the dry season the river is represented by one or more diminutive streams, which disappear from time to time in stretches of swamp. The continuity of the flat-topped escarpments is much broken by lateral valleys, and here and there the vertical walls are replaced for long distances by steeply-rounded rising ground. The tributaries of the Gulbin Kebbi, where they flow through sandstone country, are likewise bounded by worn and broken escarpments. Similar broad flat-bottomed valleys with diminutive streams are common also within the French Sudan to the north of Sokoto, and are there generally known as "dallols." Of them all, however, the Gulbin Kebbi possesses the greatest volume of water, which is contributed practically entirely by its numerous southern affluents which rise on the high and well-watered plains on the borders of Kano, Zaria, Kontagora and Sokoto.



WORN VOLCANIC CONES NEAR AWE.



THE WASE ROCK.

The plateau areas between the larger valleys are themselves much dissected, especially in the east and north, by the smaller tributary streams. In the west there are some fairly extensive stretches of continuous tableland, whose gently undulating and thinly-wooded surface is diversified by occasional flat-topped and conical hills, the relics of a former plain which has now been almost entirely removed in the course of denudation. To the north and north-east of Sokoto the irregularities of the dissected plateau are softened in contour by accumulations of blown sand which covers the floors of the smaller valleys and creeps up their lateral slopes, rounding off the steeper cliffs of the flattened ridges on either side.

- [1] Mungo Park; Denham and Clapperton; R. and J. Lander; Laird and Oldfield; Allen and Thomson; Baikie; Crowther; Barth; Vogel; Passarge; Lenfant; Lugard; Wallace; Boyd-Alexander.
- [2] "Hausaland," in the general sense of the term, may be taken to include the provinces of Sokoto, Kano, Kontagora, Zaria and Bauchi.
- [3] Cf. Laird and Oldfield, *Expedition to Africa*, 1837, p. 136. The hills between Itobe and Igbo are here erroneously described as the eastward prolongation of the Kong Mountains.
- [4] Cf. Vandeleur, *Geographical Journal*, X, 1897, p. 349.
- [5] Vide Lenfant, *Le Niger*, 1905, illus., pp. 72, 73.
- [6] Lenfant, *op. cit.*, p. 78 *et seq.*
- [7] Boyd-Alexander, *From the Niger to the Nile*, 1908, p. 47.
- [8] An excellent illustration of this peculiar type of scenery (Inselberglandschaft) is given by Bornhardt, *Deutsch-Ost-Afrika—Géologie*, Bd. VII, 1900, Taf. 3, p. 30.
- [9] Erroneously designated lava by the earlier explorers.
- [10] Cf. Boyd-Alexander, *From the Niger to the Nile*, 1908, p. 86.
- [11] Cf. Boyd-Alexander, *op. cit.*, p. 46.
- [12] Cf. Whitlock, "The Yola-Cross River Boundary Commission," *G. J.*, XXXVI, 1910, p. 426.
- [13] Cf. Boyd-Alexander, *op. cit.*, p. 98.
- [14] *Ibid.*, p. 94.
- [15] Barth, *Central Africa*, 1857, II, p. 355.
- [16] Cf. Boyd-Alexander, *From the Niger to the Nile*, pp. 112, 164.
- [17] Cf. Barth, *Central Africa*, 1857, p. 96.

CHAPTER II

CRYSTALLINE ROCKS

Distribution of the crystalline rocks. Typical traverses. The harder and softer gneisses. The Jebba quartzites: the phyllites and slaty schists of the Niger valley; the sheared trachytes of Bussa; the Ngashki phyllites; replacement of the softer gneisses by the harder banded types. The gneisses and schists of Ilorin; the quartzites of the Orissa hills; the stanniferous pegmatites of Eri. The muscovite schists of Northern Kabba; the quartzites of Lukke; the crystalline limestone between Jakura and Wa; the mica schists of Sequarre; the magnetite and hæmatite schists of Eastern Kabba; the Igbo limestone. The phyllites and schists of Sokoto province; the volcanic agglomerate of Maradu; the blue quartzite of Kazaure; the diorites of Kano; Mounts Dala and Kogon Dutsi; the gneisses and schists of north-eastern Kontagora; the Zungeru belt of phyllites and schists; auriferous quartz reefs; the quartzites and schists of Eastern Zaria; the gneisses and granites of Abuja; the tourmaline rocks of Darroro; the mica schists of Gwombe; the banded gneisses of Bauchi province; the granites of the Murchison Hills; the later intrusions. The inliers of Arofu; the granites and gneisses of southern Muri; the Vere Hills; exfoliation in the Kilba Hills; the crystalline rocks of Bornu.

WHILE crystalline rocks undoubtedly everywhere form the foundation of the Protectorate, there are extensive tracts over which the later sedimentary rocks obscure the underlying granites, gneisses, and schists. The largest area over which these crystalline rocks are exposed is that which occupies the centre of the Protectorate and extends from the Niger in the west to beyond Bauchi in the east, and from Nassarawa in the south to the neighbourhood of Katsina in the north. The major portion of the provinces of Kabba, Ilorin and Borgu to the west of the Niger also possess a crystalline floor, while similar rocks cover a narrow strip on the southern border of Muri and Yola, as well as a considerable tract of country to the east of the Gongola, lying partly in Yola and partly in Bornu.

Typical Traverses.

The following typical traverses show the character of the exposures along the tracks.

I. Filelin (near Lokoja) to Ohoto:

- 9.20 A.M.— Left Filelin.
- 9.34 „ Blocks of ferruginous sandstone and grit.
- 9.38 „ Masses of biotite pegmatite; large feldspars often graphic.
- 9.40 „ Fine-grained, non-foliated reddish biotite granite with streaks and veins of graphic biotite pegmatite. The granite at places on the margin follows the foliation of the adjoining biotite gneisses, in places crosses the foliation, or is in part injected into the gneiss. The biotite gneiss is often folded and distorted; in places there is an alternation of dark bands of biotite gneiss with lighter coloured feldspathic layers. The foliation is much disturbed, but in a general way E.N.E. and vertical. The gneiss is cut by granite and pegmatite veins and dykes.
- 9.45 „ Reddish granite continued. In places the granite shows distinct banding and alternation of coarse and fine layers; sometimes banded with pegmatites.
- 9.50 „ Same pink granite; but here an irregular mixture of coarse and fine material with in places knots of coarse granite in a fine grained matrix.
- 10.0 „ Small granite knobs on right.
- 10.3 „ Sandstones.
- 10.10 „ Biotite gneiss, strike N.E.–S.W., foliation dip 40° N., cut by coarse pegmatites as before; in places interfoliated bands of segregation pegmatite.
- 10.15 „ Soft biotite gneisses.
- 10.30 „ Large blocks of pink granite and pegmatite.
- 10.40 „ Granite piercing hard white biotite gneiss with knots of amphibolite. Strike of gneisses N. 35° E., foliation dip N.W. There is a sharp junction between the granite and gneiss, but no chilled margin.
- 10.50 „ Granite.
- 11.0 „ No exposures.

- 11.30 „ Extensive mass of coarse-grained biotite gneiss, crossed by pegmatite veins and dykes of coarse red granite with large feldspars.
- 11.35 „ Biotite augen-gneiss with feldspars sometimes two inches long; crossed by veins of fine red granite or aplite.
- 11.40 „ No exposures.
- 12.30 P.M.— Bare surface of augen-gneiss level with the plain.
- 12.40 „ Dyke or sheet of fine grained reddish granite showing distinct foliation.
- 12.45 „ Fine grained and finely banded biotite gneisses crossed and injected by granite veins.
- 12.50 „ Sandstones.
- 1.15 „ Ohoto.

II. Jakura to Wa (north side of Mimi Valley).

- 6.30 A.M.— Left Jakura hill; on brow of hill quartz schist, strike N. 10° W., foliation vertical; at foot of hill fine grained pink granite with pegmatite, passing into friable pink granite with large crystals of graphic quartz and feldspar in a finer matrix.
- 6.55 „ Soft biotite gneisses, strike N.N.E., dip W.N.W.
- 6.58 „ Band of amphibolite, dip unascertainable, followed by soft biotite gneisses. Pebbles of hornblende epidote schist.
- 7.12 „ Bare exposure of red pegmatite dyke cutting fine grained felspathic gneiss, with in places a sharp junction and in others not.
- 7.20 „ Quartz schist.
- 7.35 „ Magnetite hornblende schist, strike N. 10° W., foliation dip practically horizontal.
- 7.40 „ Quartz hæmatite schist, dip unascertainable.
- 8.0 „ Massive and granular quartzite and quartz schist.
- 8.15 „ Biotite gneiss.
- 8.30 „ Quartz-muscovite schist, strike N.N.E., dip low to W.N.W.

- 8.35 „ Pebbly muscovite schist, strike N.–S., foliation vertical or highly inclined.
- 8.45 „ Pegmatites.
- 8.50 „ White marble forming rounded knobby hills cut by pegmatite veins and enclosing streaks and veins of quartz-epidote-calcite-idocrase rock on right hand; on left the marble is foliated, with folia of pyrites and knots of quartz; strike of foliation N. 10° E., dip 10° W.
- 9.5 „ Marble ends.
- 9.8 „ No exposures; grass land.
- 9.15 „ Quartzites.
- 9-35 „ Soft biotite gneiss; foliation generally vertical; strike N.N.E., dip vertical or high towards W.
- 9.50 „ Garnetiferous biotite gneiss with quartz segregations; folia crumpled and contorted; strike generally N.–S.
- 10.10 „ Garnetiferous biotite gneiss as before; folia practically horizontal.
- 10.25 „ Soft micaceous schists pierced by granite; strike N. 35° E., dip S.E.
- 10.35 „ Pink granite in places foliated; graphic pegmatites on slope of Wa hill.
- 10.45 „ Wa. On top of hill white mica schists with magnetite knots, strike N.–S., dip 20° W.; cut by quartz-muscovite-magnetite pegmatites.

III. Olle to Sura (Central Kabba).

- 6.20 A.M.— Left Olle.
- 6.25 „ Quartz schists and quartz-muscovite schists, strike N. 40° E., dip 55° W.
- 6.27 „ Soft fine-grained quartz-muscovite schists, strike N. 10° E., dip 70° W.
- 6.35 „ Muscovite schists, strike N. 30° W., foliation vertical.

- 6.40 „ Thick dyke of graphic quartz and felspar, with crystals of tourmaline.
- 7.5 „ Fine grained white granite.
- 7.25 „ Coarse white granite in river bed.
- 7.30 „ No exposures.
- 8.10 „ Micaceous gneiss, strike N. 35° E.; much folded and contorted with streaks and knots of pegmatite, probably proceeding from a thick intrusive mass of quartz-felspar pegmatite which adjoins the gneiss.
- 8.35 „ Muscovite schists, strike N.–S., foliation vertical.
- 8.40 „ Track leads along valley between hills of white quartzite; loose blocks of compact and granular quartzite.
- 9.0 „ Summit of pass; blocks of white quartzite.
- 9.5 „ Muscovite schists, strike N.–S., foliation vertical.
- 9.20 „ Lukke.
- 9.30 „ Soft muscovite schists and quartzites.
- 9.40 „ Muscovite schists, strike N. 10° W., foliation vertical.
- 9.50 „ Broken fragments of quartzite, quartz-muscovite pegmatite, and quartz-tourmaline pegmatite on road; intrusive masses of graphic felspar and quartz with crystals of muscovite, associated with exposures of coarse muscovite granite.
- 10.10 „ Biotite gneisses much crumpled and folded, exposed in bed of stream, strike N.W.
- 10.15 „ Quartzites and muscovite pegmatites.
- 10.20 „ Intrusive mass of muscovite granite with large crystals of graphic quartz and felspar in a fine grained granitic matrix; banded in places with coarse and fine layers.
- 10.40 „ Exposures of quartz-muscovite pegmatite.
- 11.0 „ Soft micaceous schist, strike N. 10° W., foliation vertical.
- 11.5 „ In bed of river and for 500 yards beyond, garnetiferous mica schists with quartz lenticles, strike N. 10° W., dip 15° W.

- 11.30 „ Schistose biotite granite, with schistose pegmatite veins forming Sura hill.
- 11.40 „ Sura.

IV. Agwede to Kabba.

- 6.30 ^{A.M.}— Left Agwede hill. On top, white granite usually showing faint foliation and frequently enclosing portions of dark mica schist; on slope, whitish ribboned gneisses, often folded and contorted, strike N.E. to E., foliation vertical.
- 6.40 „ Dyke of graphic pegmatite in gneiss; sharp junction. Frequent bands of schistose pegmatite in the pale coloured banded gneisses; in places the gneisses are hornblendic with knots of fine-grained hornblende gneiss or schist.
- 7.0 „ Ascent to plateau over contorted gneisses.
- 7.20 „ Same contorted gneisses passing into and invaded by white schistose granite; wooded plain, few exposures.
- 8.20 „ Occasional exposures of similar contorted and granitised gneiss; strike E. 10° N. to E. 10° S.
- 9.5 „ Faintly banded and indistinctly foliated biotite gneiss.
- 9.30 „ Odouape.
- 10.30 „ Level surface of composite gneiss.
- 11.30 „ Much crumpled and folded biotite gneiss; strike N.–S. Followed by garnetiferous biotite gneiss.
- 11.45 „ Composite gneiss; foliation vertical; strike N. 10° W.
- 12.0 „ Kabba.

V. Izon to Abuja (Nassarawa province).

- 6.25 ^{A.M.}— Left Izon. Open grassy or thinly-wooded plains for two hours; no exposures.
- 8.20 „ Hills; ascent of 300 feet; biotite and hornblende gneiss and amphibolite; epidote-hornblende gneisses; in places twisted and contorted and extensively invaded and injected by

masses of biotite granite; strike varies between N. 10° E. and N. 10° W.; foliation mostly vertical.

- 9.0 „ Open plain; thick granite wash.
- 9.30 „ Hills; granite invading hornblende epidote gneisses.
- 9.42 „ Abuchi. Amphibolite veined with granite.
- 10.0 „ Narrow plain; thick drift.
- 10.30 „ Hills; soft micaceous gneisses, strike N. 10° W., foliation vertical or high to E.
- 10.35 „ Hard biotite gneisses, strike N.–S., foliation linear and vertical.
- 10.45 „ Ribboned and banded biotite gneisses with quartz veins parallel to or crossing the foliation; strike N.–S.; foliation vertical.
- 11.0 „ Abuja granite with foliated margin; vertical crush lines in granite striking N.–S.
- 11.15 „ Abuja Residency.

VI. Boi to Ronn (Southern Bauchi).

- 6.30 A.M.— Left Boi; hornblende epidote gneisses broken and invaded by granite; in places much folded and crumpled.
- 6.40 „ Felspathic gneiss, strike E.–W., dip high to S.
- 7.20 „ Complex of quartz epidote rock, amphibolite, epidote hornblende schist, and gneiss, in places much granitised; general strike E. 10° N.
- 7.45 „ Hummocks of banded gneiss; open country; few exposures.
- 8.40 „ Hornblende epidote gneisses.
- 8.45 „ Goshin Duchi. Hills of biotite granite behind town.
- 9.3 „ Quartz schists, fine grained gneisses, epidote hornblende and quartz hornblende rocks, much granitised in places veins and dykes of pegmatite and basalt.
- 9.20 „ Blocks of basalt and porphyry.
- 9.35 „ Hills of fine grained biotite granite.
- 10.15 „ Granite rocks, in places schistose.

- 10.45 „ Rocky, hilly country; banded and contorted gneisses
micaceous and hornblendic; general strike E. 10° S.
- 11.10 „ Ronn.

It is unnecessary, however, to quote the details of particular traverses at greater length, especially as such descriptions serve only to show the character of the phenomena to be correlated. It is sufficiently evident that there are within the Protectorate two fairly well-defined series of crystalline rocks, a less metamorphosed series of quartzites, schists and gneisses representing an original sedimentary succession, and a more highly metamorphosed series of banded and contorted gneisses, whose relation to the less metamorphosed group will be discussed in the sequel. The less metamorphosed gneisses present in the field somewhat granular and incoherent outcrops and may be conveniently termed the “softer” gneisses, while the “harder” or more highly metamorphosed gneisses are readily distinguishable by the smooth and compact character of their exposed surfaces. The softer gneisses appear to disintegrate readily into a gritty sand under present conditions, while the harder gneisses exfoliate and pass into angular débris. Both series are extensively invaded by granitic and other igneous intrusions.

Borgu and the Niger Valley between Jebba and Yelwa.

In the province of Borgu the superficial accumulations are so extensive and the exposures so few and so widely scattered that the general character of the underlying rocks is somewhat difficult to ascertain. It is probable, however, that, like the neighbouring region of Nikki,^[18] there is a floor of well foliated and granitoid gneisses of varied origin, much pierced and invaded by veins of granite, pegmatite, quartz porphyry and quartz. Tremolite schists are known from Mori, and sheared porphyries from Kanikoko. The occasional inselberge are composed invariably of grey porphyritic granite. At Jebba the gorge of the Niger is carved out of a series of sheared quartzites and pebbly grits with the shear planes at times well spaced and the deformed pebbles still distinctly visible. Secondary mica has

been sparingly formed along the planes of movement, together with knots and streaks of tourmaline and tourmaline quartz. The predominant strike is N.–S., and the prevailing dip of the foliation 65° W. The well-jointed Juju Rock is composed of a laminated quartzite schist, which splits readily into thin curving slabs, whose surfaces are coated continuously with white mica and show a well-defined strain-slip corrugation. Westward the Jebba quartzites pass into highly sheared quartz schists, quartz mica schists and granulitic gneisses, which are followed by crumpled and contorted muscovite biotite schists and gneisses and by granite at the river Awo. The quartzites and granulitic gneisses are produced along the strike to form the range of low wooded hills which stretches southward from Jebba towards Ilorin. To the east of Jebba the quartzites are abruptly limited by hard banded and composite gneisses pierced by masses of porphyritic biotite granite.

To the north of Jebba the Kailema granite, over which the track leads from Jebba to Bajibo, is a massive hornblende granite, porphyritic with white and pink feldspars and somewhat finer-grained on the margin. The granite has evidently pierced a series of hornblende, epidote and talc schists and micaceous gneisses which outcrop imperfectly to the east. Still farther north, at Wuru, the channel of the Niger is occupied by a hard and coarsely foliated epidote hornblende gneiss, much pierced by irregular veins of schistose granite and pegmatite. The foliation planes of the gneiss are vertical and strike approximately N. 20° E. On the left bank the gneiss is succeeded by thinly bedded quartzites and quartz schists with scattered knots, lenticles and veins of quartz and finely fibrous tourmaline. Intercalations of much corrugated and wrinkled mica schist bear witness to differential movements along certain lines. The whole series, which strikes N.–S. and dips vertically or at high angles to the east or west, has apparently suffered a rapid and repeated folding. These rocks are followed by a series of fine-grained green and yellow quartzites and well-cleaved slaty schists with numerous quartz reefs, which pass eastward into a complex of hornblende gneisses, epidote hornblende gneisses, amphibolites and fine grained schistose granites which disappear at Kumboji under the sandstones of southern Kontagora.

At Bussa on the right bank of the Niger, near the Residency, there are exposures of much decomposed shaly phyllites, vertical or highly inclined to the east and striking N. 10° E. These are followed on the left bank by a thick series of little altered grey and greenish granular quartzites, which in places become pebbly and conglomeratic and enclose bands of greenish trachyte, also showing little or no sign of shearing. Further exposures of black and coarsely-sheared trachytes are found in the swamps between Bussa and Libata. Eastward the quartzites are much invaded by masses of dark green diorite and by thick veins of tourmaline quartz, the latter showing every variation between a banded tourmaline quartz rock and a felted mass of finely fibrous tourmaline. Nearer Auna these rocks are succeeded by well-foliated augen gneisses which are overlaid by the sandstones of southern Kontagora.

To the north of Bussa phyllites are again exposed in the low rounded hills to the north-west of Ngashki, where they strike N.-S., and possess a high dip of 65° to the east. Locally, however, the strike may be directed as much as 35° to the east of north, indicating that the rocks do not pursue a constantly rectilineal direction from south to north, but rather possess a zig-zag character with short and sharp deflections towards the east. The phyllites, which are mostly grey in colour, but in places mottled with large irregular whitish spots on a grey ground, pass eastward into soft grey sericitic schists with crystals of magnetite and pyrites. The whole series is pierced by quartz tourmaline veins, in which all proportions are to be found, from almost pure quartz through banded quartz tourmaline rocks to masses of pure tourmaline of the finely fibrous variety. Sills of much decomposed vesicular dolerite are common also amongst the phyllites. Farther east intercalations of micaceous gneiss become abundant within the schists, and a red fine-grained granulitic gneiss intervenes between them and the massive biotite granite of Abara, with its large crystals of felspar. To the north of Ngashki, phyllites and green slaty schists with lenticular streaks and veins of quartz and tourmaline quartz occur near Lokofe. A good exposure in the river Malenda shows pebbly lustrous schists striking N.-S., and dipping steeply towards the east. At Yelwa a narrow strip of phyllites and slaty schists on the left bank of the Niger, passes westward into knotted mica schists and biotite gneisses in the river bed and eastward into the

granitic gneiss of Bin Yauri. In the country which lies to the east of Lokofe and Yelwa, and especially in the neighbourhood of Tabachi and Maibirro, green slaty schists of varying hardness are frequently exposed, usually associated with harder gritty bands and much invaded by veins of quartz and tourmaline quartz, and irregular intrusions of granite. In places also there may be found intercalations of hard, gritty, carbonaceous and little-metamorphosed slaty beds. At Chuggu, moreover, between Tabachi and Maibirro, there is a parallel belt of biotite gneisses, hornblende gneisses, and epidote hornblende gneisses, while at Maibirro the slaty schists, here associated with thinly-bedded quartzites, quartz schists and quartz muscovite schists, striking N. 35° E., with a dip to the north-west of 65° , are succeeded in the river Malenda by well-banded biotite gneisses much veined with felspathic and granitic material. The actual junction of the banded gneisses with the quartz schists is, however, not exposed.

The rocks of the Niger valley, between Jebba and Yelwa, are thus of many and varied types. The quartzites and phyllites, the slaty schists and micaceous gneisses, and the green hornblendic and epidotic beds represent the sandstones, shales, grits and arkoses of the sedimentary series, while the sheared trachytes and porphyries, the schistose granites and the coarse-grained amphibolites represent the original igneous intrusions within the sedimentary succession. The whole series strikes fairly constantly N.-S., and forms a practically continuous belt between Jebba and Yelwa. In most places there are indications that the rocks have suffered a repeated folding into fairly sharp anticlines and synclines with vertical axial planes, the folding being apparently more rapid in the phyllites and schists than in the quartzites and gneisses. There has unfortunately been no opportunity of tracing the belt in detail along the strike, and little is known of the lateral variation or repetition of the individual members or of the original sequence of the sedimentary succession. There is no reason to believe that there has been any increase or decrease in the intensity of the metamorphism throughout the belt from south to north, and no indication has been found of the occurrence of any thermal metamorphism in the neighbourhood of the later granitic intrusions.

The most remarkable feature, however, about this belt of schistose rock within the Niger valley is the manner in which the less metamorphosed

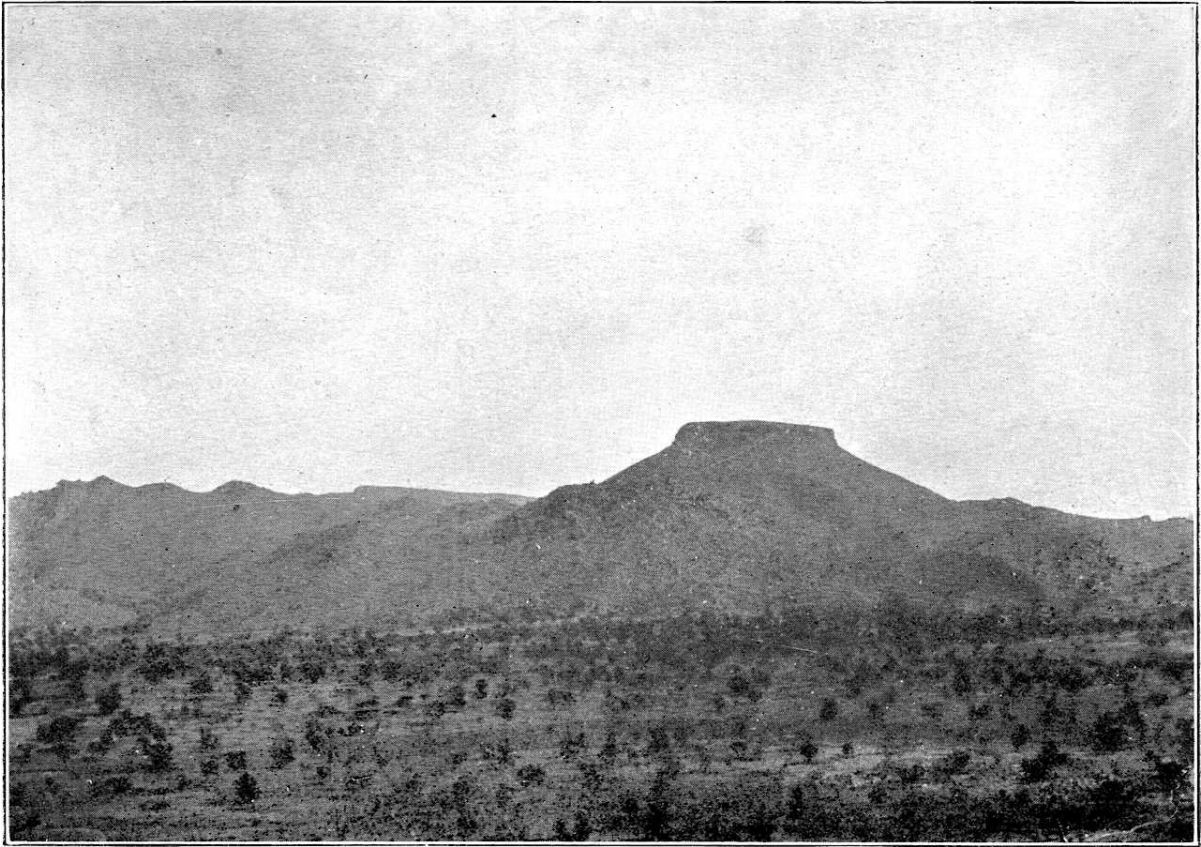
series of quartzites, phyllites and schists is bounded abruptly and locally replaced by a more highly metamorphosed and much granitised series of well-foliated and often banded gneisses. At Maibirro, for example, the schists are abruptly succeeded by banded gneisses of a type which is nowhere found between Maibirro and the Niger. Similarly at Wuru and Jebba, the quartzites give place unexpectedly to composite granitised gneisses, which must have originated under very different conditions from those which brought about the metamorphism of the phyllites and schists. This is a peculiarity of the fundamental architecture which is repeated in many other parts of the Protectorate.

Kabba and Ilorin.

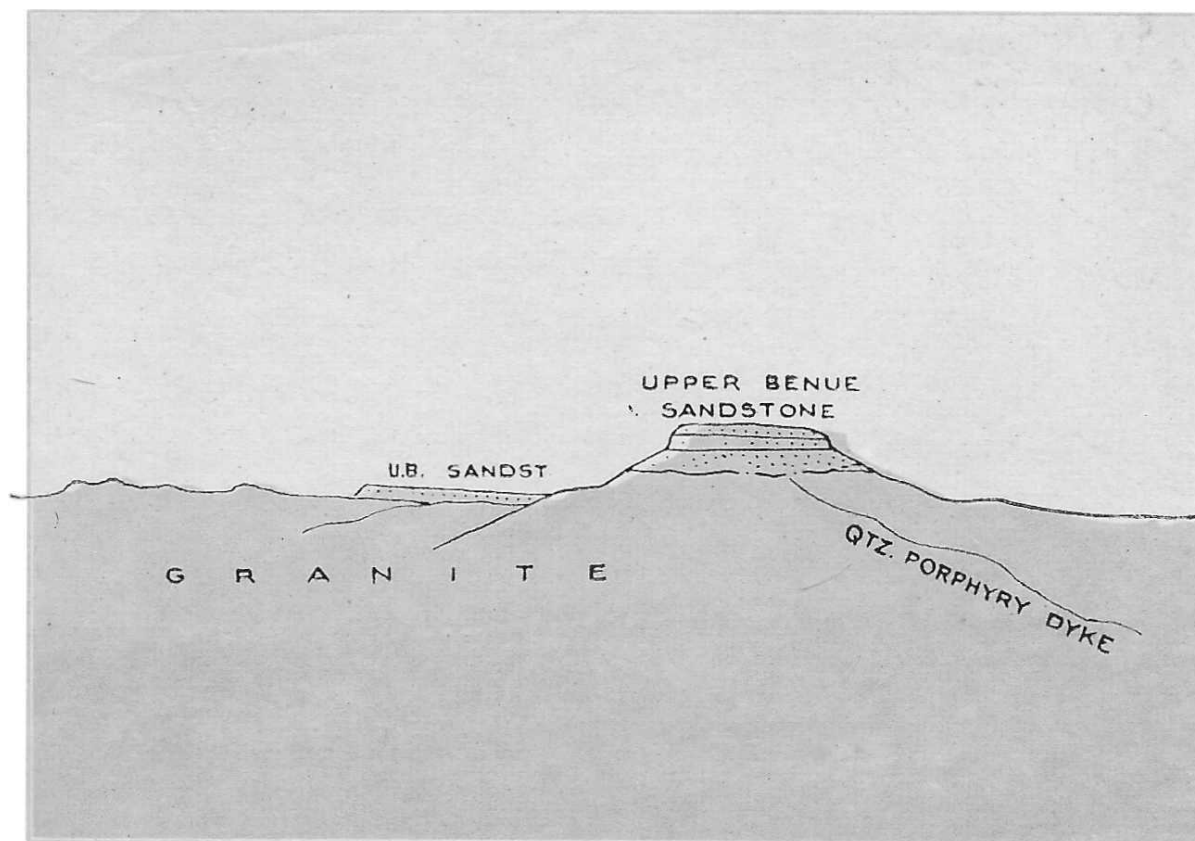
Over the greater part of the province of Ilorin the surface accumulations are so extensive that the nature of the underlying rocks can be judged only from small and scattered exposures. So far as known there are no belts of phyllites and quartzites upon the western or central plains comparable with that of the Niger valley above Jebba. The predominant rock in central Ilorin is a fairly coarse-grained, granular and feldspathic muscovite biotite gneiss, sometimes an augen-gneiss with large eyes of feldspar, which disintegrates readily into a feldspathic sand. With it are associated in places thin bands of quartzite, quartz schist, quartz muscovite schist, mica schist and gneiss, often garnetiferous, amphibolite and schistose granite, which, however, are neither sufficiently extensive nor sufficiently resistant to give rise to any prominent feature upon the surface. The whole series which strikes approximately N.-S. is much veined by muscovite and tourmaline pegmatites and invaded by masses of granite usually of a more or less foliated character. The larger masses rise into turtlebacks or heaps of boulders, and in their neighbourhood the gneisses have often been so intimately veined and traversed by the granite as to produce a composite or banded rock. At Share hard biotite gneisses with linear foliation are extensively developed and associated with hornblende gneisses and amphibolites and cut by tourmaline pegmatites.

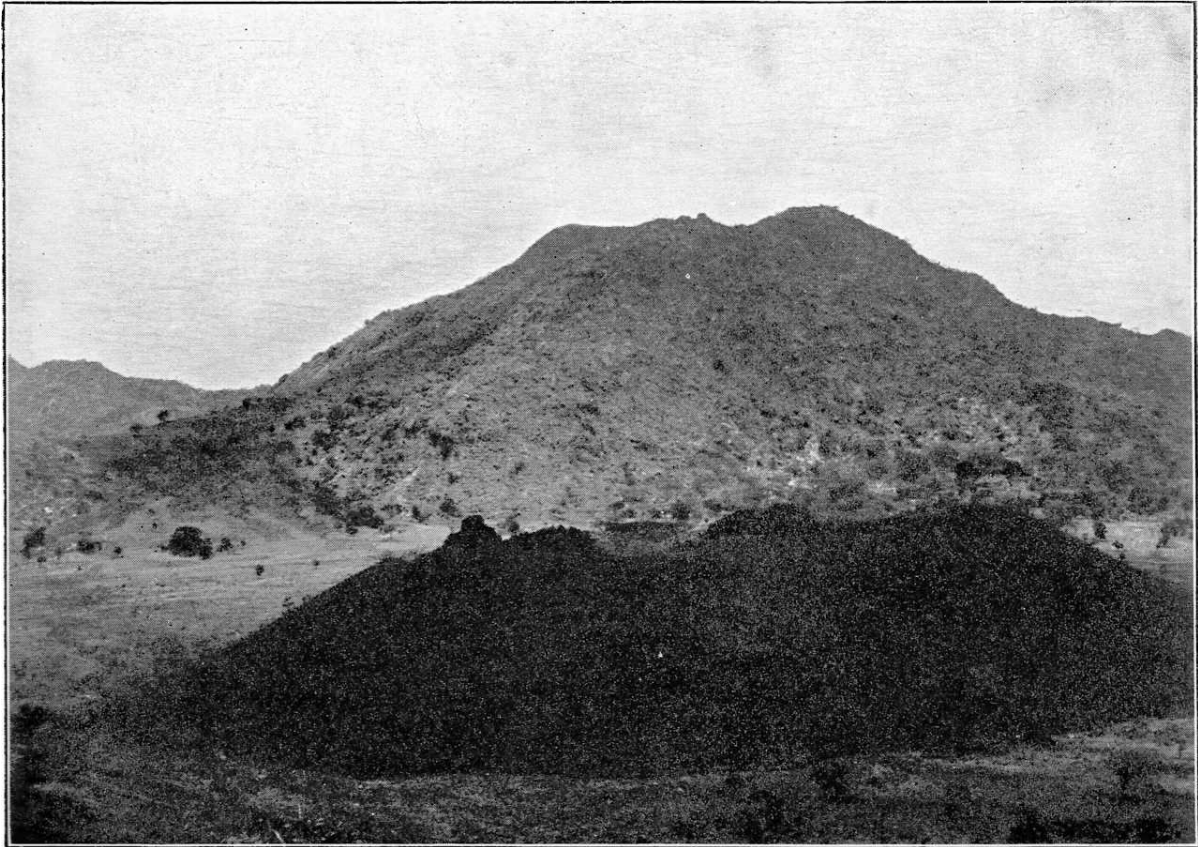
In the Orissa hills, which lie to the east of the great plain of Ilorin, the softer gneisses enclose for the first time a prominent series of quartzites,

quartz schists, quartz mica schists and granulitic gneisses which, with their general meridional strike and high dip to the west, are very similar to those which form the low hilly ground to the south and west of Jebba. The Shappa hills which succeed the Orissa hills to the south are composed in great part of biotite and hornblende granites, coarse-grained, fine-grained and porphyritic, which appear to be intrusive into a series of hard biotite and hornblende gneisses, in places well-banded and much veined with pegmatites, which occupies the south-eastern corner of the province. Around Eri the softer gneisses are again well developed and extensively exposed, the series here consisting of thin quartzites, felspathic gneisses and much contorted and occasionally garnetiferous micaceous gneisses and schists, with interbedded hornblende schists and irregular masses of coarse-grained amphibolite, probably of intrusive origin. The whole series is much pierced by veins of quartz, tourmaline quartz and muscovite pegmatite, the latter in places carrying tinstone.

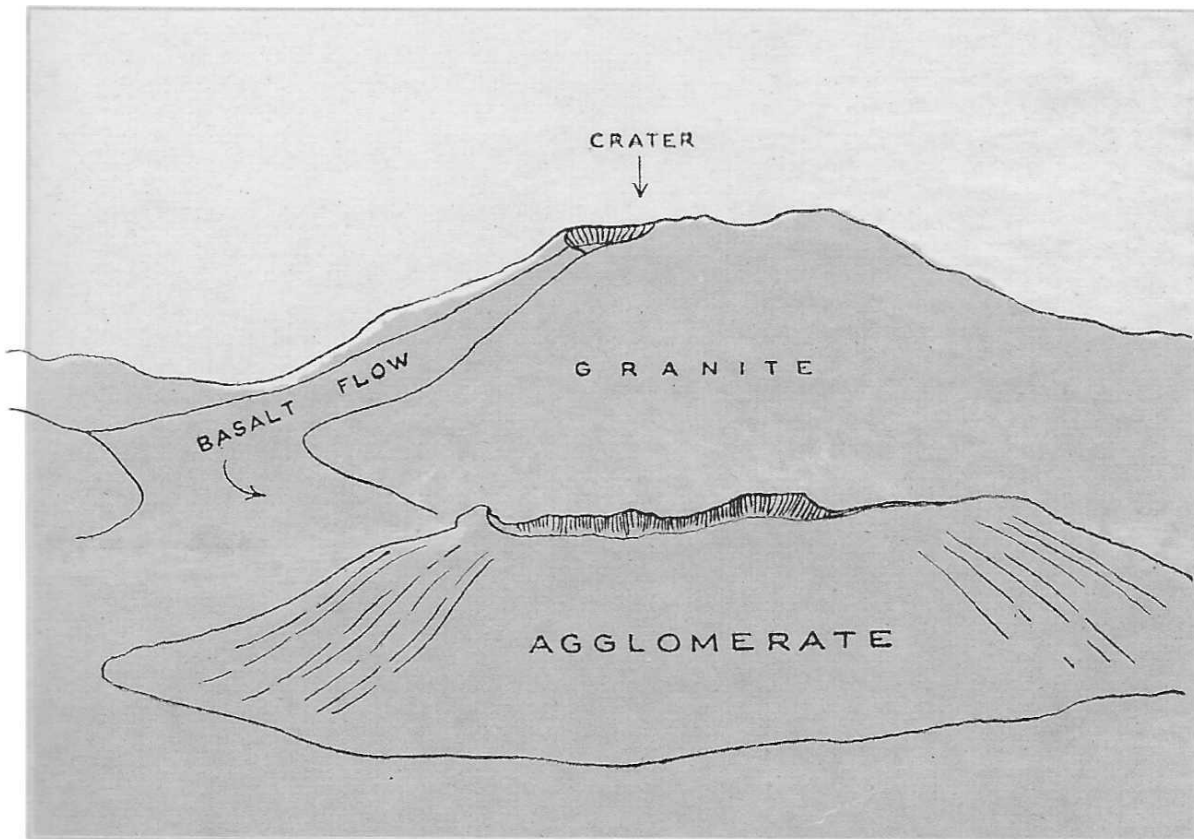


YENDAM HILL, YOLA PROVINCE.





CRATERS AT MBOL.



The softer gneisses and schists are found everywhere in northern Kabba between Eri and the Ilorin border and the sandstone plateaux which fringe the Niger. The usual succession of thinly bedded micaceous and felspathic gneisses and hornblende schists is varied by the intercalation of several belts of platy muscovite schists and quartz muscovite schists, whose deeply decomposed surface usually supports a luxuriant forest growth. Within the area mentioned there are at least four of these belts, more or less well-defined and continuous along their strike from north to south. The first runs southward from the neighbourhood of Idofin by way of Kponyon, Onore, Ote, and Omwo into southern Kabba: the second strikes southward from Okoro and Iye, skirting Sanlu to the east and Moppa to the west: the third runs from Shale and Oferre by way of Lukke, Olle, and Appe to the neighbourhood of Odouape, and the fourth is exposed along the track leading from Wa to Jakura on the north side of the Mimi valley. While the fissile quartz muscovite schist may be taken as the index rock of the various belts, each belt is in itself a complex of muscovite schists, quartz muscovite schists, quartz schists, quartzites, biotitic and hornblendic schists and other

rocks and the proportions of these rocks vary not only in the successive belts but at different points within the same belt. In the third belt for example there is along with the muscovite schist a great development of quartzites and quartz schists around Lukke, where they form prominent rocky hills (v. Traverse No. III. [p. 66](#)). Farther north several bands of chlorite and talc schist are found in the same belt in the neighbourhood of Iye, while in the fourth belt between Wa and Jakura there are intercalations of such other rocks as quartz magnetite schist, magnetite hornblende schist, and crystalline limestone (v. Traverse No. II. [p. 65](#)). The limestone is a white coarse-grained saccharoid marble which may be traced for at least 1,000 feet across the strike without affording any indication of repetition by folding. On the margin the marble is distinctly schistose with the foliation planes striking N. 10° E. and dipping at a low angle towards the west. The rocks in the neighbourhood are somewhat irregularly folded with the foliation planes as a rule parallel with the bedding. Their junction with the marble is not exposed, but if the marginal foliation of the latter be also parallel with the bedding, the actual thickness of the original limestone must have been very much less than 1000 feet. The marble is very pure, consisting almost entirely of calcite with occasional streaks and segregation bands of quartz-epidote-calcite-idocrase rock representing the original impurities within the limestone. Minute spots of graphite are abundant throughout and on the margin streaks and knots of quartz and pyrites are common along the foliation planes. In places the marble is irregularly crossed by later pegmatite veins.

The tracts which intervene between the various belts are occupied by a variety of soft micaceous and hornblendic schists and differ from the belts already described only in the comparative rarity of muscovite schists and quartzites. The absence of sufficiently detailed investigation, however, makes it impossible to say to what extent the various belts described above represent repetitions of one and the same belt, or to what extent they represent the original sequence of the sedimentary succession. While the average strike of the whole series is meridional, its undulating or zigzag character is evident from the occurrence of individual exposures showing every angle between north and east. The different members of the series, while frequently vertical, may dip at various angles to the east or west or

even in places approach the horizontal. Individual bands are frequently puckered and contorted and the whole series is invaded in places by extensive masses of biotite granite which rise into turtlebacks and groups of rounded hills. The granites are usually foliated on the margin and sometimes throughout and frequently show at their junction with the gneisses and schists a narrow zone of injected rock. Small detached exposures of the harder banded gneisses, elongated upon a parallel strike, are distributed irregularly throughout the softer series. The gneisses and schists are abundantly pierced by muscovite and tourmaline pegmatites and veins of quartz and tourmaline quartz, the latter in the neighbourhood of Iye being minutely banded as in the case of some of the tourmaline quartz veins at Bussa and Ngashki. With the pegmatites appear to be connected some small intrusive masses of non-foliated granite which become more numerous in the eastern portion of the area under consideration (v. Traverse No. I. [p. 64](#)).

The striking feature of the architecture of the high plains of central Kabba is the manner in which the softer gneisses of the north with their intercalated bands of schists and quartzes are invaded, and in part replaced, by masses of granite and of hard banded and composite gneisses. The first belt of schists can be traced southward with intervals from Omwo to the neighbourhood of Arigidi and Sequarre, where the soft micaceous schists carry much staurolite. Harder gneisses however appear to bound this softer belt irregularly to the west and east. The prolongation of the second belt of schists in northern Kabba is probably represented by two narrow strips of quartzite enclosed in the harder gneisses between Kabba and Lupa. The third belt appears to be somewhat abruptly replaced in the neighbourhood of Odouape by a complex mass of granites and hard composite and banded gneisses which extends westward across the high plains of Kabba from Agwede and Shokko Shokko to Ikinre and Omwo (v. Traverse No. IV., [p. 67](#)), and is apparently prolonged southward into the Egbira hills. The fourth belt appears to suffer much lateral variation, and is represented on the main road between Lokoja and Kabba only by some thin bands of muscovite schist enclosed within a series of micaceous and felspathic gneisses.

In southern Kabba the few facts at our disposal seem to indicate the almost complete absence of quartzites and muscovite schists and a repeated

alternation of the softer and harder gneisses, the whole series being pierced by pegmatite veins and intrusive masses of granite. Quartzites and quartz schists reappear, however, in the Kukuruku hills on the borders of Northern and Southern Nigeria. [\[19\]](#)

The lower plains which stretch eastward to the Niger from the escarpments of the Egbira and Agwede hills are occupied very largely by gneisses of the softer series. Quartzites, quartz schists and quartz-muscovite schists occur within the series, but their distribution over the plains is so irregular, and their prolongation along the strike so inconstant, that they cannot in any way be correlated with the quartzites and schists of either the third or fourth belts in northern Kabba. It would seem, indeed, that instead of the regular alternation to be found in the north, there is over these eastern plains a greater mixing of types and a greater variation in character of the individual members when traced along the strike. Quartz-magnetite and quartz-haematite schists are fairly common, typical occurrences being found in the neighbourhood of Bangede and around Okuruku and Anamu near Okeli, and on the plains to the west of the Agaiyi hills, as well as at the foot of their western slope. Amongst the granites and granitoid gneisses which form the escarpment of the Egbira hills, bands of felspathic gneiss and micaceous schist are frequently enclosed, and on the plains the gneisses and schists are crossed in the usual manner by pegmatite veins and pierced by masses of more or less foliated granite, which form the prominent isolated hills which diversify the surface of the plain. Here and there also may be found a small exposure of composite or banded gneiss, but it is only in the immediate neighbourhood of the Niger that the harder gneisses attain any extensive development. There they form a narrow belt, marked by prominent intrusions of granite, which stretches southward from Lokoja along the right bank of the river, and adds picturesqueness to the Niger valley between Igbo and Itohe ([p. 6](#)). At Lokoja the harder gneisses are exposed in the river bed, and the softer gneisses and schists at the foot of Mt. Patti. The average strike is N.E. to S.W., and the whole series is much pierced by veins of granite and graphic pegmatite, the latter containing frequent knots of magnetite.

To the east of the Niger the softer gneisses reappear in the narrow strip of crystalline rocks in Bassa province which fringes the left bank of the

river. Typical exposures are found in the neighbourhood of Igbo and Itobe, where the gneisses are associated as usual with early foliated or partially foliated granites and with later intrusions of muscovite and tourmaline granites and pegmatites, the tourmaline occurring as a rule in large and well-formed crystals. At Igbo the micaceous gneisses enclose a number of narrow bands of fine-grained, crystalline limestone, each 3 to 4 feet in thickness. The limestone varies much in colour and hardness, the purest varieties being blue, while weathering and impurities produce grey and brown tints. Much of the limestone is gritty, but the commonest accessory is a pale-coloured tremolite in rosettes and felted veins of fibrous crystals and in scattered well-formed prisms.

The crystalline rocks of Kabba and Ilorin thus present considerable resemblances in individual types to those of the Niger valley above Jebba. The most striking differences are the occurrence of such new types as crystalline limestone and magnetite and hæmatite schist in eastern Kabba and Bassa, and the apparent absence in the southern provinces of the characteristic phyllites and slaty schists of the Niger valley to the north. The place of the phyllites, however, is probably taken by the muscovite and quartz-muscovite schists, which form such prominent members of the metamorphic series in northern Kabba. If so, there would seem to have been a slight increase in the intensity of the regional metamorphism in these southern provinces as compared with the Niger valley above Jebba. Of thermal metamorphism, on the other hand, there is little trace. The schists and gneisses in the neighbourhood of Sequarre in western Kabba and in several localities in northern Kabba and Ilorin contain a considerable quantity of staurolite and other metamorphic minerals. Their distribution, however, appears to possess a linear character which is somewhat inconsistent with a contact origin, while sufficiently extensive igneous intrusions, to whose influence their local development might be ascribed, have nowhere been definitely recognised in the vicinity. The numerous intrusive masses of both foliated and unfoliated granite throughout these provinces appear to have induced little or no contact metamorphism in the surrounding gneisses and schists. An exception to this rule may perhaps be made in the case of the tourmaline granite behind Igbo on the Niger, to whose influence may possibly be ascribed the extensive development of

tremolite in the neighbouring crystalline limestones. At the same time it should be remembered that the occurrence of tremolite in metamorphic limestones cannot be taken as an invariable indication of contact action.

The irregular association of a less metamorphosed series of quartzites, schists, and gneisses with a more highly metamorphosed series of well-foliated, banded and contorted gneisses, exhibited by the crystalline rocks of the Niger valley above Jebba, forms likewise a striking feature of the crystalline rocks of Kabba and Ilorin. The banded gneisses may be either biotitic or hornblendic and the lighter and darker bands may alternate so regularly and rapidly as to give rise to a striped or ribboned appearance even in the hand specimen. They are for the most part of a medium and even grain and never porphyritic, although sometimes they enclose lenticular masses of finer grained biotite gneiss or of amphibolite. As a rule they are much invaded by granite and frequently rendered composite by thin granitic intrusions. The foliation, which may be parallel or oblique to the striation, is their least conspicuous feature. Pegmatite veins are usually abundant and frequently foliated. As already described ([p. 74](#)) these hard banded and contorted gneisses are found irregularly replacing the softer gneisses both along and across the strike.

So far as known there are no schistose rocks in Kabba and Ilorin comparable to the sheared trachytes of the Niger valley and the sheared porphyries of Borgu. Rare occurrences of thin basaltic veins have been observed, but as a rule the later intrusions in Kabba and Ilorin take the form of small masses of red or white aplitic granite or of dykes and veins of muscovite and tourmaline pegmatite or of graphic granite. These later granites have usually a peculiar granular and incoherent character and never possess chilled margins. Frequent occurrences have been noted of the granular granite acting as a kind of ground mass to large scattered porphyritic masses of graphic quartz and felspar (v. Traverses Nos. I, II, III, [pp. 64–66](#)). Frequently also the granites show an ill-defined banding caused by an alternation of coarser and finer grained layers. It is possible that the muscovite and tourmaline pegmatites, the veins of graphic granite, the quartz reefs, the banded tourmaline quartz rocks, and the small intrusive masses of acid granite may all possess a certain genetic connection. Only at Igbo, however, does there occur an extensive mass of reddish muscovite

tourmaline granite which to some extent unites the characters of all the dyke rocks and represents in a general way the original magma from which the various dykes and veins and intrusive masses have been differentiated. The abundance of tourmaline throughout Kabba and Ilorin led to the hope that tinstone would in places be associated with it, but only at Eri in eastern Ilorin have the pegmatites proved stanniferous. The quartz reefs also have nowhere been found to yield more than the minutest trace of gold.

4. The Central Crystalline Area.

The peculiarities of structure which have been remarked in the western provinces are continued throughout the larger crystalline area in the centre of the Protectorate. To the east as to the west of the Niger there has been an irregular and local replacement of a series of softer gneisses, quartzites, and schists by a complex of harder composite and often well-banded gneisses. There is also in the western part of the central area within the softer gneissic series a well-defined grouping of the quartzites and phyllites or micaceous schists into regular and definite belts which preserve their identity for considerable distances along the strike. In the east, however, this regular grouping appears to be absent, the quartzites and mica schists being distributed irregularly and in short and narrow bands amongst the felspathic and micaceous gneisses.

In the east of Sokoto province there are at least two broad and well-defined belts of phyllites and indications of a third. The first is found between Bukwium and Anka near the Kontagora border, and is well exposed in the Anka river. The strike of the phyllites varies from N.-S. to N. 20° E. Surface creep here and there makes the dips uncertain, but the inclination is usually high, and is sometimes to the west, sometimes to the east. This fact probably accounts for the great width of the belt. It is not known whether the phyllites extend to the east of Anka, but after crossing them at right angles to the strike for thirty miles to Bukwium they are still in full force, and the western limit is unknown. There are numerous quartzite bands, especially about Anka and to the south of it. The phyllites are soft, fine-grained and dark-coloured, and are traversed by abundant quartz veins, some of which carry pyrites. It is interesting to note that the

phyllites are only exposed in the neighbourhood of the river. Away from the river, on the level plain, there is no indication of them, and this fact is of significance when considering the possible extension of the phyllite belts. There is a numerous and interesting series of igneous rocks within the phyllite area, including such types as hornblende granite and syenite, pegmatite and quartz porphyry. These rocks, as a rule, show no sign of crushing, and are probably later intrusions. Some of the porphyries, however, show an imperfect foliation, and appear to have been exposed to some of the later crustal movements. Bands of amphibolite and veins of tourmaline quartz also occur within the phyllitic series. (A. L.)

The second group of phyllites and quartzites forms a belt of low hilly country striking N.N.E.–S.S.W. across the Sokoto-Kano caravan route between the villages of Maradu and Karakai, and continues northward at least as far as Bazai. The series attains across the strike a breadth of about thirteen miles, and consists of a succession of slaty schists and knotted phyllites and quartzites, much pierced by quartz veins and thin bands of schistose granite, and dipping fairly constantly in the west at 70° E.S.E., but gradually heeling over in the east to the W.N.W. Towards Karakai the quartzites predominate and contain much tourmaline, and many samples may be found showing quartz and tourmaline or quartz, slate and tourmaline in alternating ribbon-like bands. The third belt is very imperfectly exposed on the borders of the province between Woneka and Duru, and is represented only by some scattered outcrops of slaty phyllite and mica schist.

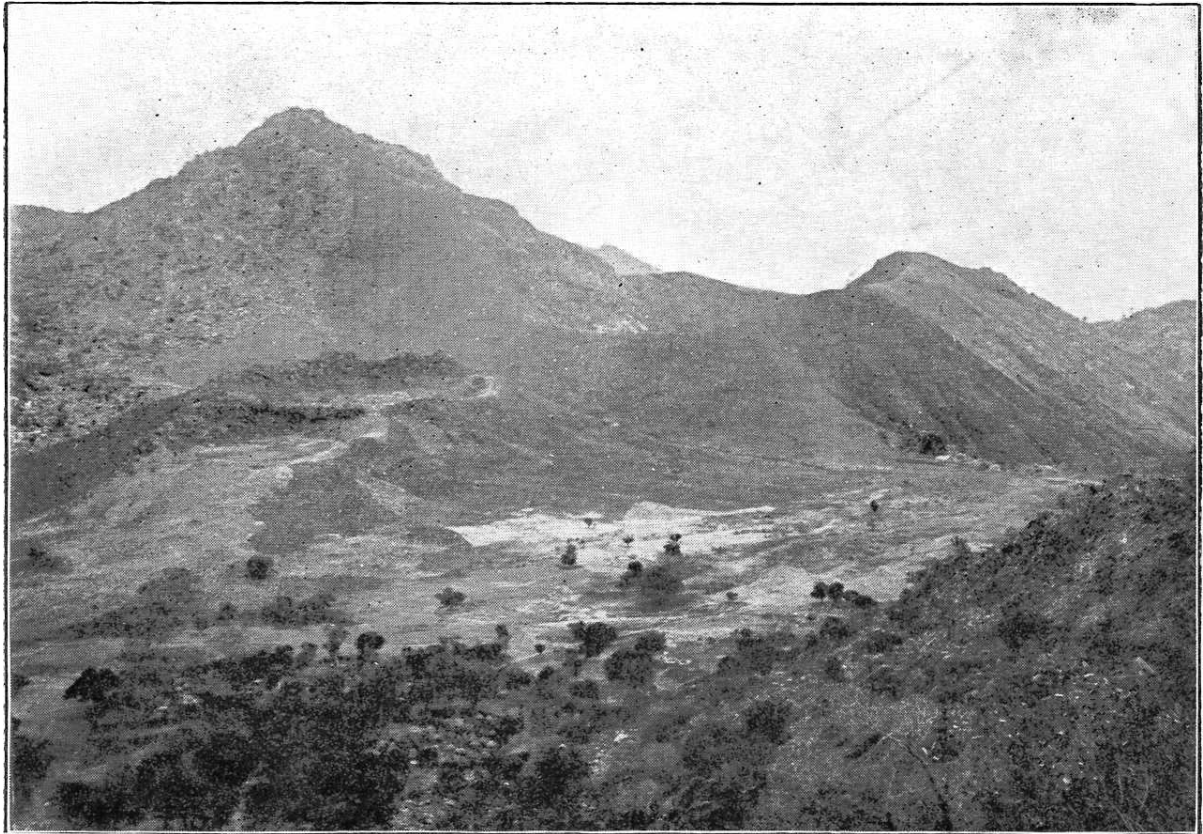
Between Maradu and the margin of the Sokoto sandstones to the west, the softer gneisses are exposed upon the sandy plains in a very fragmentary manner, and appear to consist of micaceous and hornblendic gneisses and schists, and of hornblende epidote gneisses and amphibolites with numerous small granitic intrusions. The rocky knobs and kopjes two miles west of Maradu are formed of a coarsely banded volcanic agglomerate, dipping eastward at angles of 30° – 40° . The agglomerate may be traced to within a mile of the village, where it gives place to a thinly bedded series of red ashy sandstones and ash beds. East of the rest camp, these beds strike N.N.E. and dip at 40° E.S.E. while west of the camp they strike E.N.E. and dip at 30° S.S.E. The ash is exposed in, and to the west of, the village, but

to the east its place is taken by a much broken and shattered granitic rock. The actual junction, however, was not observed. Such evidence as there is points to the former existence of a volcanic neck or vent in this locality.

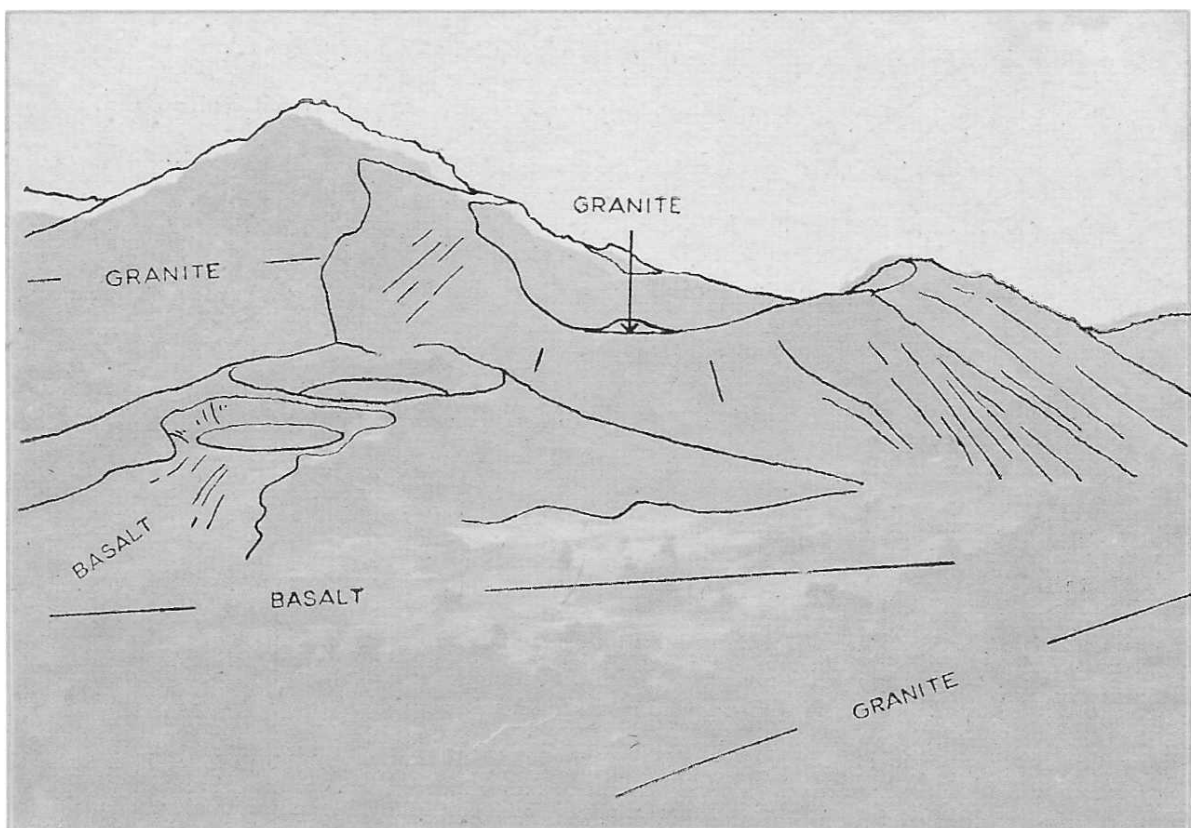
Between Karakai and Woneka the surface accumulations obscure the rock for long distances, especially to the west and east of the granite hills of Kotorkoshi. Around Bungudu, however, there are numerous exposures of a coarse-grained composite gneiss which probably extends eastward to the boundary of the province. Beyond Duru a grey biotite granite with a distinctly foliated margin forms the broken hilly country between Runka and Kurefi, but between Kurefi and Sani the granite gives place to a group of dark-coloured biotite gneisses before the crystalline rocks disappear under the sandstones of Katsina. Knobs and hummocks of granite, however, protrude through the sedimentary rocks round the walls of the town, and Barth^[20] has reported the occurrence of similar mounds and ridges immediately to the north. To the south-east of Katsina the surface rock appears to be everywhere a reddish granite as far as Duru, but beyond Duru the superficial accumulations so effectually cover up all but the occasional kopjes and boulders of granite that the character of the underlying rocks is difficult to ascertain. From the scattered fragments of quartzite, gneiss and amphibolite, and occasional granite boulders in the surface drift, it is probable, however, that for a considerable distance to the west of Kazaure the floor is formed of the sedimentary gneisses with the usual granitic intrusions. The hills around Kazaure are composed of hard blue quartzite much pierced by later veins of white quartz, and striking N.N.E.–S.S.W. with a dip of 30° to the west. This belt can be traced southward to the neighbourhood of Kasuan Kuka, where the quartzites become more micaceous and where some rocky exposures of granular amethystine quartz probably represent the intrusive quartz veins of Kazaure. From Kasuan Kuka to Kano, granites and an occasional quartz vein are the only rocks exposed, while the deep drift effectually prevents the discovery of the relationship between the surrounding rocks and the decomposed diorites of the two flat-topped hills within the walls of Kano. A few miles east of Kano the crystalline rocks disappear under the sands and clays of Katagum and Bornu. There is every reason, however, to believe that they are continuous

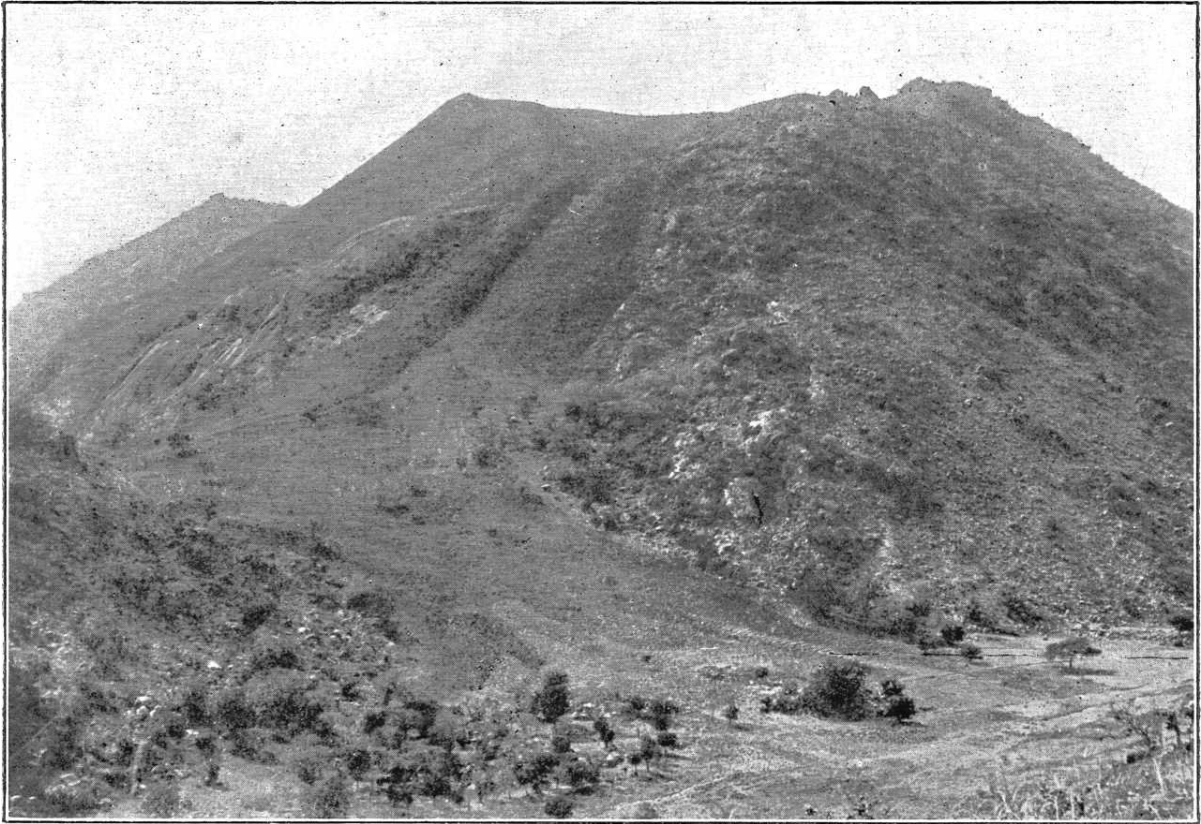
to the north-east with the granites, gneisses and quartzites of Machina, Zinder and Mounio.

The two detached hills, Dala and Kogon Dutsi, the landmarks of Kano, with their precipitous slopes and tabular summits, 150 feet above the level of the plain, present from a distance a decidedly sedimentary appearance. Clapperton,^[21] indeed, described them as consisting of “argillaceous ironstone mixed with pebbles and a rather soft kind of marl,” while Dala, the eastern mount, “is of clay-ironstone and conglomerate lying upon a bed of soft light clay apparently mixed with vegetable remains.” Closer investigation has shown, however, that both hills are of crystalline origin, Dala being composed entirely of earthy decomposition products with a gritty ferruginous cap, while Kogon Dutsi still retains cores or kernels of undecomposed diorite in the midst of the friable earthy material which forms the greater part of the hill. The felspar and hornblende of the original diorite have decomposed in the lower part of Dala into a whitish clay streaked and spotted with dark green chloritic material. In the upper part the colouration is more uniformly white or yellow, the bulk of the iron having been leached out to form the red ferruginous cap. That the latter is itself composite, however, is evident from the occurrence of gritty grains, subangular fragments of quartz and rolled pebbles of ironstone in the vacuolar and concretionary crust. Kogon Dutsi presents a similar composition, with the exception that the material becomes coarser and more granular in the neighbourhood of the kernels of undecomposed rock. The extensive decomposition of the crystalline rocks to which the earthy hills of Dala and Kogon Dutsi bear witness will be found in the sequel ([p. 201](#)) to be an important factor in the evolution of the characteristic scenery of the high plains of Hausaland.

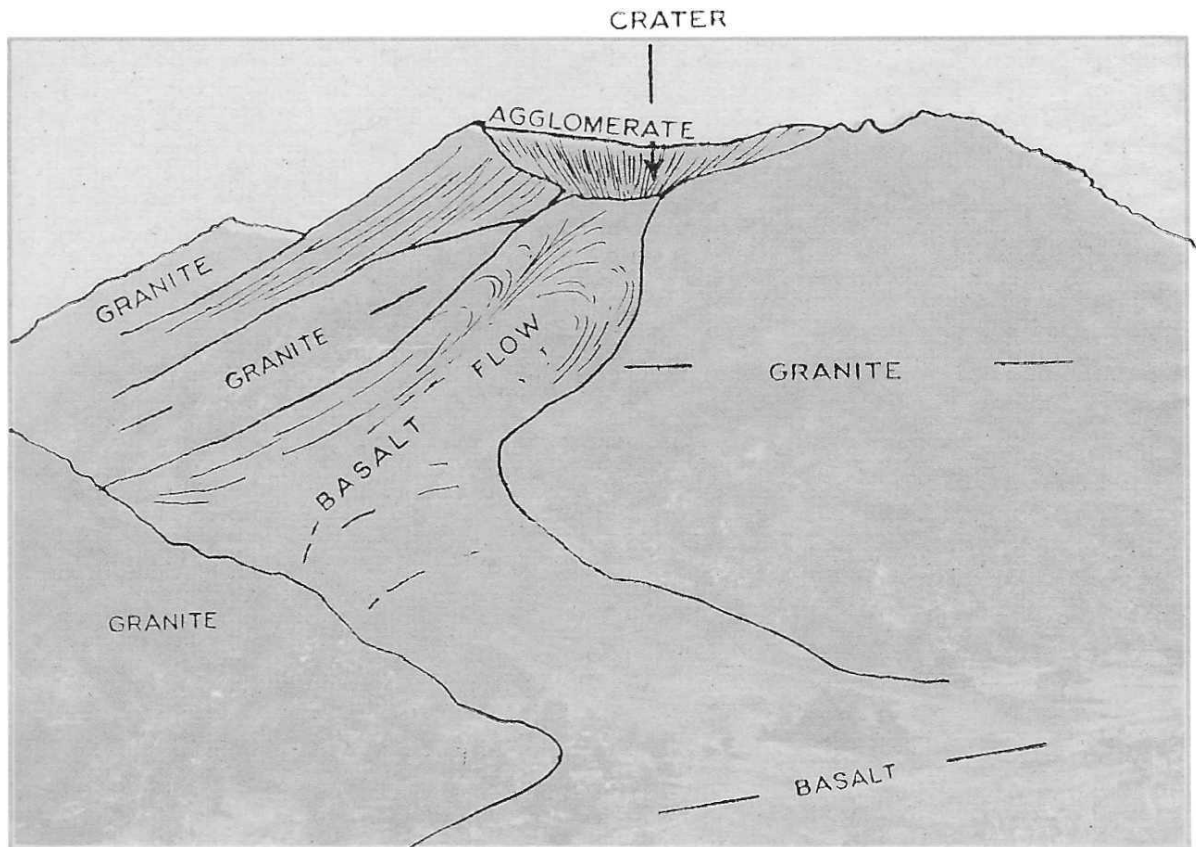


CRATERS IN THE MBOI HILLS.





A CRATER IN THE MBOI HILLS.



Around Kontagora and beyond the northern limit of the sandstones which occupy the southern portion of the province of the same name, there are found hard biotite gneisses in places well striated with pink felspathic and dark biotitic bands and crossed by numerous pegmatite veins. To the east the crystalline floor is deeply covered with drift, and outcrops are comparatively rare. In the Beri river, however, there is a good exposure of soft felspathic biotite gneisses, with knots and bands of pegmatite, while at Beri a pink granite forms a compact group of rocky hills. The granite possesses a distinctly foliated margin, and is followed to the east by hard hornblende and biotite gneisses much pierced by granite veins, which may be traced to within six miles of Bobi before they are replaced by the softer micaceous series. On account of the extent of the superficial accumulations, however, these softer gneisses are well exposed for the first time only in the rocky bed of the Kara river which forms the boundary between the provinces of Kontagora and Zaria.

Though the structure of the north-eastern part of Kontagora province is complicated in detail, it may be described in general terms as a succession

of gneiss and quartzite bands alternating with one another. They strike a few degrees east of north, and probably pass to the east of the Bukwium-Anka phyllites. The quartzites are frequently tourmaline-bearing, and one band west of Kotonkoro carries scales of graphite. This fact is interesting in connection with the graphitic schists reported from a neighbouring part of Zaria.^[22] The gneisses are usually biotite gneisses, sometimes markedly banded, at other times indistinctly so. Schistose rocks are seen in places, quartz schists being prominent to the south of Mahorro, where talc schists are also seen. The gneiss and quartzite country is traversed by pegmatite and quartz veins, but intrusions of unfoliated granite are rather uncommon. Most of the turtle-backs are foliated throughout, differing in this respect from those of many other parts of the Protectorate. Where the Sokoto road leaves the province, the boundary is marked by a high sharp ridge of amphibolite. (A. L.)

A tract of broken hilly country, almost entirely carved out of the softer gneissic series, occupies the south-western corner of Zaria province, and a diagonal traverse across the hills from Moriga on the Kara river by way of Garun Gabbas and Zungeru to Kato and Mikonkeli presents a particularly instructive section. The hills which begin about ten miles south-east of Moriga consist between Inga and Garun Gabbas of a succession of quartz schists, quartz muscovite schists, micaceous and hornblendic schists, felspathic gneisses and amphibolites, pierced by numerous quartz reefs and veins and by a few dykes of basalt and porphyry, the whole series possessing an average strike of N. 10° E. Between Garun Gabbas and the Kaduna river ridges of quartz and quartzite rise conspicuously at intervals out of the pebbly drift which covers the surface. In the railway cutting on the right bank of the Kaduna near Zungeru, quartzites and quartz schists are associated with bands of a hard and minutely foliated granulitic gneiss containing knots and lenticles of quartzo-felspathic and chloritic material. In the river bed below the railway bridge the most conspicuous rock is a well-bedded white quartzite striking N. 30° E. and dipping at an angle of 64° to the south-east (Plate VIII, [Fig. 2](#)). The quartzite varies much from a massive variety with knots of pyrites and scattered flakes of white mica through a streaky lustrous micaceous quartzite with linear foliation planes, into a well-foliated quartz muscovite biotite schist. The different varieties

are often arranged in alternating parallel bands and the whole series is pierced by occasional narrow veins of quartz. The trench which at the Kaduna bridge forms the low water channel of the river has been carved out of a band of hornblende schist intercalated along the strike of the quartzites. The schist, which in places is composed almost entirely of hornblende, but in others interfoliated and streaked with felspar, has been more easily worn away than the quartzite, and grooved and pot-holed by the action of the river. The quartzites and quartz schists, with intercalated sills of hornblende schist, continue eastward to the adjoining channel of the Dago, where they are replaced by finely striated biotite gneisses. At Zungeru micaceous and felspathic gneisses are mingled with soft, thinly bedded and repeatedly folded hornblende mica epidote gneisses or schists. The whole series is pierced in the usual manner by veins of quartz, pegmatite and granite and strikes in a general way N. 10° E. with a dip of 60° to the east. Immediately east of the railway station, however, these rocks give place to well-foliated granulitic gneisses with knots of felspar, striking N. 20° E. and dipping at 45° E. These gneisses, which probably represent more completely altered portions of the sedimentary series, are in places well striated, with the bands wavy and contorted and crossed irregularly by veins of quartz. They are followed in the vicinity of Kwarra-town by coarsely schistose grits which are among the least altered rocks within the Protectorate. Between Zungeru and Gidan Motu there are numerous exposures of green hornblendic beds similar to those at Zungeru. They become, however, more fissile and contorted towards the east. The strike oscillates between N.N.E. and N.N.W., and the dip varies between vertical and high angles to the east or west, while at Gidan Motu there is an intercalated belt of platy biotite gneisses which strike N. 60° W. and dip at 60° S.S.W. Between Gidan Motu and Kato green slaty schists and phyllites are extensively developed and steeply folded at Kato into sharp anticlines and synclines which possess an average strike of N. 20° W. The phyllites enclose much quartz in lenticles and veins and contain abundant knots and spots of pyrites. Beyond Kato the phyllites are succeeded by soft mica schists, which in turn give place to micaceous and felspathic gneisses. The latter around Mikonkeli, Minna and Paiko are replaced by hard composite and in places banded rocks and invaded by extensive masses of granite which usually show a distinct marginal foliation.

Southward the Zungeru belt of quartzites and phyllites disappears under the sandstones of Nupe, but any correlation with the quartzites and schists of Kabba cannot yet be attempted. Northward the belt has been followed as far as Birnin Gwari and from its pronounced development in the neighbourhood of that place, it is probable that it extends considerably farther north again. For twenty or thirty miles north of Tegna there are a number of granite hills strung together on a roughly north and south line, the granite having apparently been intruded along the strike of the schistose rocks. The granites become foliated locally. Between Bugai and Koriga there is a twenty mile belt of phyllites and subordinate mica schists which runs northward to Birnin Gwari, where the phyllites form a number of sharp north and south ridges and are excellently exposed, both in the sides of the ridges and in the intervening streams. The phyllites are traversed by many quartz veins and gold is widely disseminated in the streams. At Koriga there is an extensive development of hornblende gneiss. To the east are mica schists, and graphitic schists are reported from the neighbourhood.^[23] The rocks which rise above the plain in the vicinity of Zaria are all of granite. (A. L.)

So far as known, the Zungeru belt is the last prominent group of phyllites which can be traced within the Protectorate. To the east their place is taken by muscovite and biotite schists, which are to be regarded as more completely reconstructed representatives of the original shales, the metamorphic action having apparently been somewhat more intense in eastern and southern Zaria and Nassarawa. Several belts of quartzites and schists are enclosed in the usual manner within a complex of softer and harder gneisses to the east of Minna and Mikonkeli. At Kuta, for example, there is a belt of quartz muscovite schist passing on the one hand into knotted micaceous schists and on the other into quartzites and carrying much sillimanite and magnetite and large lenticles and knots of quartz and tourmaline quartz. The whole series strikes approximately N. 30° E. and dips at 45° N.W. Again between Chiri and Guni there is much soft micaceous schist rising into low rounded and well-wooded hills, striking generally N.-S., well frilled in places, with a high content in ilmenite and numerous knots and veins of tourmaline quartz. Farther east the hills which run from the neighbourhood of Dorka to Kaderri and Kaserimi are

composed very largely of quartzites, quartz schists, quartz muscovite schists and micaceous gneisses, striking N. 10° W., dipping at an angle of 60° W., and pierced by quartz veins, muscovite pegmatites and granites. Between these more prominent belts occur numerous smaller exposures of schists and quartzites and of both the softer and the harder gneisses. To the east of Kaserimi and Duchi and in the valley of the Kaduna the surface is so deeply and extensively covered with drift that little rock beyond the marginally foliated biotite granites of the isolated hills is available for investigation. There are, however, indications from occasional fragments in the drift and rare exposures in the stream courses that a considerable extent of the buried surface is composed of a softer gneissic series. To the east of the Kaduna, the Ruruma and Kerko hills which rise so prominently from the south-eastern plains are composed throughout of a coarse-grained biotite granite, locally carrying hornblende in addition and in places distinctly foliated on the margin. The junction with the gneisses is only rarely seen on account of the thick masses of drift which envelop the base of the hills. At Guri on the eastern slope of the Ruruma hills there are exposures of finely laminated hornblende epidote gneisses, while the granites and the marginal rocks are pierced by numerous dykes of basalt, dolerite and quartz porphyry.

The high plains of south-eastern Zaria are occupied by mica schists and gneisses, with granite hills in one or two localities. Along the eastern border of the province the Rukuba plateau ends in a ragged edge of biotite granite which veins the gneisses at its foot. The country around Jengre and Limoro is interesting on account of the occurrence of masses of soda granite and especially on account of numerous intrusions of dyke rocks, felsites often with flow structure, porphyries and quartz porphyries. Tinstone is frequently met with in the streams of the district. Similar dyke rocks are exposed around the base of the Kudaru hills to the north-west of Leri and there, in addition, dykes and veinlets of basalt penetrate the granite. These hills are made up of hornblende granites with syenitic modifications, and locally, towards the boundary, the granite takes on a distinctly foliated structure, with rounding of the porphyritic feldspars as in augen-gneiss. (A. L.)

In Nupe, between the Gurara river and the sandstones of the Niger valley, the crystalline series is introduced by muscovite biotite gneisses,

quartzites, hornblende and talc schists and amphibolites, pierced by extensive masses of granite around Gaung, Lefu and Paiko. In the bed of the Gurara, between Gaung and Izon, soft gneisses and hornblende epidote rocks alternate with hard and well-foliated gneisses striking a little west of north, and varying in dip from vertical to 40° or 50° E. Five miles beyond Izon, the Abuja road ascends over a crumpled and contorted series of biotite, hornblende, and epidote hornblende gneisses much injected by masses of biotite granite. Beyond Abuchi a second ascent over soft micaceous gneisses, striking N. 10° W. and vertically foliated, leads up to the striated, ribboned and banded gneisses which have been invaded by the biotite granite of Abuja. The banded gneisses strike N.–S., and are much granitised in the immediate neighbourhood of the intrusive rock, while the granite itself is marginally foliated and crossed by crush lines at the Izon gate (v. Traverse No. V, [p. 68](#)). Partially foliated granites and banded and composite gneisses continue to within five miles of Kao, whence a belt of soft micaceous gneisses and quartz muscovite schists, striking N. 20° E., stretches eastward to the neighbourhood of Kari. At Kari the escarpment of the northern tableland is formed of soft biotite gneisses, much pierced by veins of quartz and tourmaline quartz. Eastward by way of Kukwi, Anta, Tatara and Jaginde the escarpment shows much granite, associated both with banded gneisses and with softer micaceous and hornblendic rocks. The gneisses and granites are extensively invaded by muscovite and tourmaline pegmatites, and by irregular intrusions of basalt and dolerite. Between Jaginde and Darroro quartzites, quartz schists and mica schists outcrop along the escarpment, and at Darroro a similar series of thin quartzites and gneisses, striking generally N.–S., is pierced by extensive masses of quartz-muscovite-tourmaline pegmatite, and capped by the basalts of the Morrua plateau. The pegmatite is rudely banded parallel with the strike of the gneisses, the various bands being distinguished by their richness in tourmaline. All proportions may be found between quartz muscovite and quartz tourmaline rock, the tourmaline in the latter being either in large well-formed prisms, or in thin alternating bands of felted fibrous crystals. North-east of Darroro the vertical granite cliffs of the Kagoro massif form the margin of the Bauchi plateau, while in the neighbourhood of Assab the later basalts much obscure the crystalline rocks of the lower plains. At Assab the escarpment is largely composed of biotite granite, in places

porphyritic, and apparently intrusive into a series of micaceous gneisses which strike N. 10° E. and dip at an angle of 40° to the west.

The northern tableland and the central plains of Nassarawa are composed very largely of the softer gneisses with scattered outcrops of harder composite rocks. Quartz muscovite schists occur in a broad belt on the tableland in the vicinity of Aribi, and calc-silicate rocks between Kwai and Jaginde. The composite and banded gneisses reach their greatest development in the Karshi hills between Abuja and Keffi. The Mada hills and the greater part of eastern Nassarawa are still unexplored. In southern Nassarawa a broad belt of quartz muscovite schists, mica schists, hornblende schists and chlorite schists extends between Gwombe and Buga, the mica schists in the neighbourhood of Zagabutu being particularly rich in staurolite, garnet and imperfectly crystallised knots of andalusite, presumably of contact origin. The whole series has been steeply folded on a meridional axis and then coarsely corrugated transversely, the result being that individual exposures may strike at any angle between E.–W. and N.–S. The average strike is inclined around Gwombe to the north-west, and around Buga to the north-east. To the east of Buga the schists give place to the usual varied succession of softer gneisses, pierced in the usual manner by veins of quartz and tourmaline quartz, and associated from time to time with banded and composite rocks and detached masses of granite. Numerous dykes of basalt and quartz porphyry appear to be associated with the granites of the Kurudu and Anagoda hills, which lie respectively west and south-east of the town of Nassarawa.

The crystalline rocks of Bauchi province present similar characters to those of other parts of the Protectorate. The broad belts of schists and quartzites found in other provinces are here, however, conspicuously absent, such types being represented only by thin bands intercalated within the softer gneisses. Banded and composite rocks, on the other hand, appear to be present in somewhat greater proportion than elsewhere, while domes and turtlebacks of foliated or partially foliated granite (Plate XII, [Figs. 1, 2](#)) are in places particularly abundant. The softer and harder rocks are, however, so intimately mingled that a detailed description of the succession is impossible. Excellent exposures of the harder series are to be found in the Kogin Delimi below Naraguta in the west of the province, and of the softer

gneisses in the Jarawa hills to the south of Bauchi, and of both in the broken rocky country between Kanna and Angass in the southern portion of the province (v. Traverse No. VI, [p. 68](#)). The strike of the gneisses, while very variable, appears to have a greater tendency to assume an E.–W. direction in Bauchi than in other provinces, a tendency which is probably due to the greater intensity of the transverse forces which have elsewhere corrugated the prevailing meridional strike of the gneisses and schists. This feature is particularly marked in the neighbourhood of the Ningi massif, which is elongated approximately along the strike of the surrounding softer and harder gneisses.

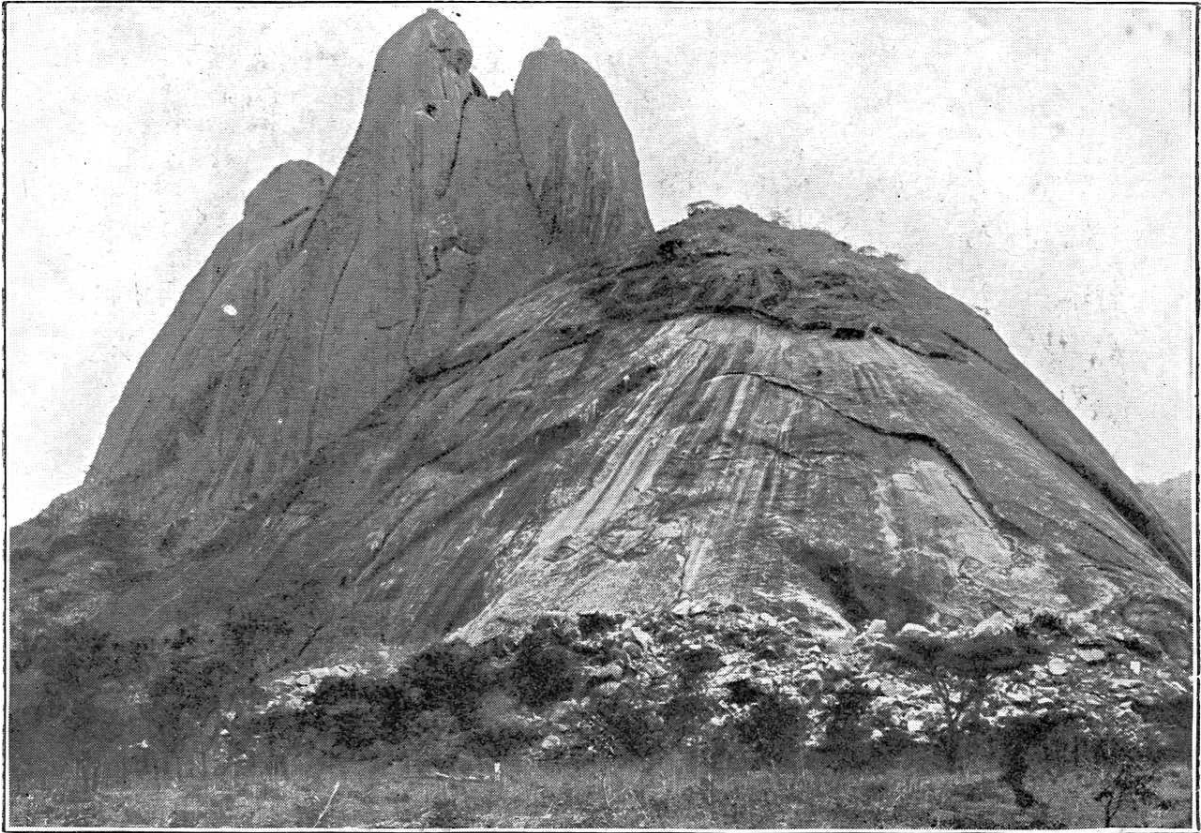
Another notable feature of the province of Bauchi is the number and variety of the later igneous intrusions. Pegmatites, quartz reefs and veins of tourmaline quartz are as abundant in Bauchi as elsewhere, but the attention is somewhat diverted from these commoner intrusive types by the presence of an abundant series of dyke rocks, and of extensive masses of non-foliated granite and syenite. The pegmatites are in places particularly rich in ilmenite, rutile and chalcedonic quartz; the dyke rocks present almost every variety between basalt and quartz porphyry; and many of the granites are especially rich in soda, and in places show chilled margins at their junction with the gneisses. The summit of the plateau around Bukuru is occupied by one of these extensive granitic intrusions, which is of peculiar interest from the fact that in places it is veined and impregnated with tinstone and other minerals. A similar granite, also carrying tinstone, forms the greater part of the Kwandokaya hills to the north of Toro. Many intrusions of syenite are found around and to the east of the town of Bauchi. The Ningi massif is largely composed of a soda granite, with a well-chilled margin, accompanied by numerous dykes of granite and porphyry, and in places carrying tinstone. Similar granites are found at Kila and Fagam on the borders of Bauchi and Kano, at Shira in Katagum to the north, and at Gurkawa on the Muri border to the south. The Murchison Hills are believed to be largely granitic, but they and the southern margin of the plateau are still practically unexplored. A small detached area of granite and gneiss,

pierced by numerous pipes of basalt and trachyte, is found around Tangalto and Chongwom in the south-east of Bauchi province.

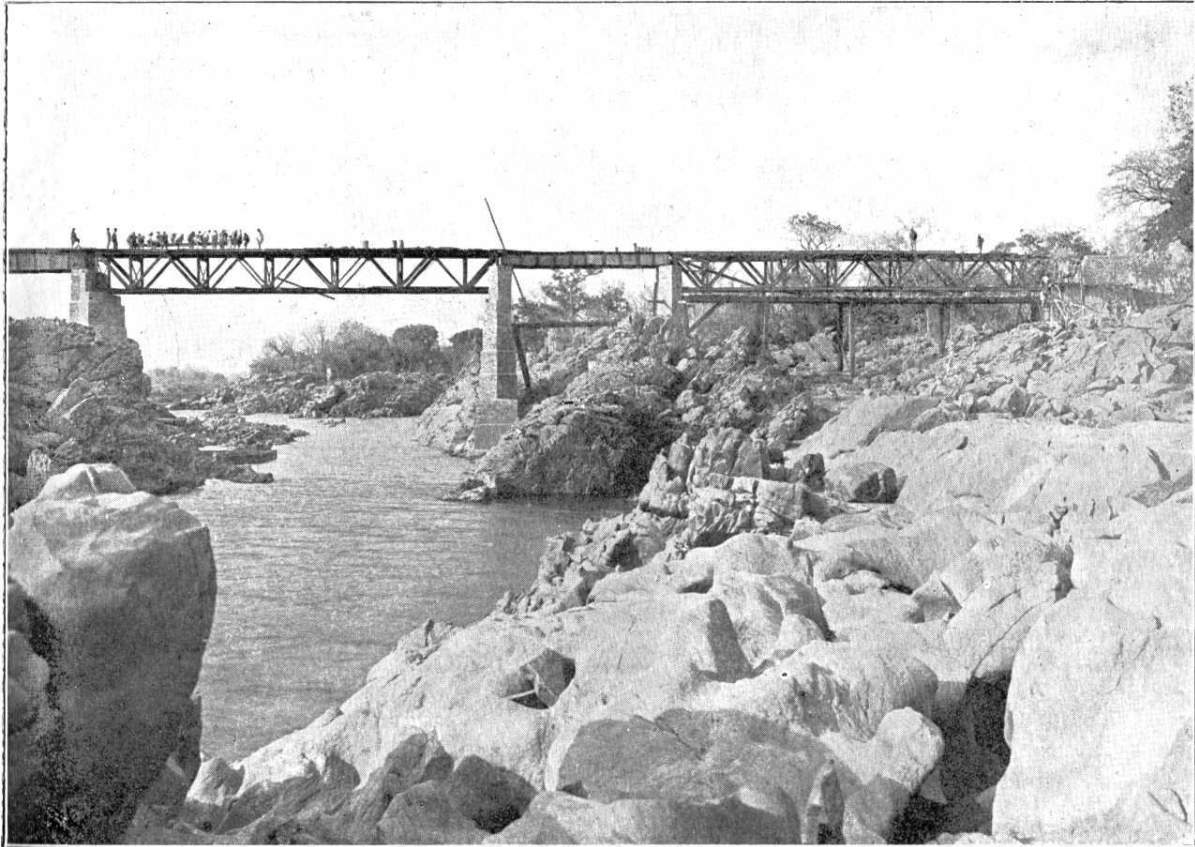
Southern Muri and Yola. (A. L.)

The strip of crystalline rocks which runs along the southern boundary of Muri and Yola has an average width of forty miles and is separated from the central area by the sedimentary rocks of the Benue plain. The crystalline floor appears at the surface, however, in a number of small inliers around Arofu in western Muri, and thus quite near to the river. Five miles east of Arofu a small area of fresh biotite-granite about fifty yards in diameter peeps from beneath the limestone. Immediately north of the town itself there is a knoll of granitoid gneisses striking north and south. On proceeding southward along this line of strike limestone is passed over for three miles, but then the gneiss appears at the surface again, and this time forms a ridge running southward for at least some miles. It may well be that other inliers of crystalline rocks occur in the unexplored Munchi country.

Farther south the transition from sedimentary to crystalline rocks is not accompanied by any feature in the drift-covered plain. Though the exposures are not good, it appears that biotite-gneisses and schists with the customary northerly strike are penetrated by granite veins. The rocky masses which lie south and south-west of Takum are probably largely or entirely of granite. The bulk of the material which forms the scattered hills around and to the north of Takum is also granite, but here it is inextricably mixed up with gneiss which is met with in patchy fashion from top to bottom of the hills. The foothills of Shebshi are schistose grits and gneisses invaded by granites in the usual manner. ^[24]



EXFOLIATION IN THE KILBA HILLS.



THE GORGE OF THE KADUNA AT ZUNGERU.

Only the northern part of the Mumuye-Manna plateau has been traversed. The rock there is almost always biotite-granite, but it is by no means always of the same variety. Occasional quartz veins are seen on the plateau and also on the lower ground between Numnai and Panteti, where the quartz carries pyrites. Intrusions of quartz-porphyry are often seen cutting through the granites, *e.g.*, east of Numnai, around Shega Mundi hill, around Pantu Baturi and at Yendam Hill. The granites only occasionally carry included blocks and fragments of gneiss, but on descending to the plain biotite-gneiss is met with striking N. 10° E., and the plain from Chukol eastward to the Vere Hills is occupied by biotite-gneiss, with some intrusions of granite. The Vere Hills themselves are made up almost entirely of granite, though to a small extent of gneiss also. Biotite-gneiss is frequently seen round the base of the plateau, generally much penetrated by granite or quartz-porphyry. Thus, speaking generally, the plains of southern Yola are mainly of gneiss and the plateau areas mainly of granite.

To the north of the Benue plain again, the country around Song, Goila and Korubi is a biotite-granite area with some conspicuous dykes of quartz-porphyry running for miles across the country. Near Zummu porphyry rises just above the soil of the plain and exhibits flow-structure, while between Zummu and Holma a felsite dyke is seen cutting through the granite. The hills around Holma are built of a well-jointed biotite-granite, and the sides of the hills are concealed by the loose blocks split off along the joints. The granite of the Kilba Hills, near Pella, is a very fresh, coarsely-crystalline biotite-granite, and locally, where the Kilba Hills rise out of the plain to the west, the granite takes on an augenstructure. In consequence of exfoliation (Plate VIII, [Fig. 1](#)) most of these hills rise up as smooth bare rocks, but some are made up of heaps of blocks, and it is not possible to explain the contrast as due to a difference in material. Between the hills around Holma and those around Pella there is a plain of gneiss about fifteen miles broad. The gneiss has been formed by the injection of a micaceous gneiss or schist with innumerable strings and veins of granite. The exposures are excellent throughout, but the gneiss never forms hills.

The country north of Pella is comparatively uninteresting. It consists of an extensive plain with isolated hills and rocks of biotite-granite. The nature of the rocks underlying the plain is practically unknown. The small area of the Kwaba Hills is composed of biotite-granite with some quartz-porphyry. (A. L.)

Crystalline rocks, largely of a gneissic character, are exposed also in southern Bornu around Kwoia to the east of the Gongola and along the base of the escarpment on the north-western margin of the Burra volcanic plateau. In eastern Bornu the granite rocks of Chibuk and Kobshi are the first prominent outcrops of crystalline rocks to the south of Maidugari. Barth,^[25] however, mentions an isolated granitic cone, Mt. Deladeba, to the N.N.E. of Isge, while Courtet,^[26] Gentil and Freydenberg^[27] note the occurrence of knobs of crystalline rock between Lake Chad and Lake Fitri. It is highly probable indeed that the thick alluvial accumulations to the east and south-east of Chad and in the lower Shari valley rest for the most part upon a crystalline floor, provided with scattered relics of Tertiary sandstone. (J. D. F.)

- [18] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 325.
- [19] Parkinson, *Q. J. G. S.* (63), 1907, p. 317.
- [20] Barth, *Central Africa*, 1857, Vol. II, pp. 41, 42.
- [21] Denham and Clapperton's *Narrative*, 1826, pp. 50, 57.
- [22] Dunstan, *Colonial Reports—Miscellaneous*, No. 26, p. 15.
- [23] Dunstan, *Colonial Reports—Miscellaneous*, No. 26, p. 15.
- [24] See also Downes, *Yola and Cross River Boundary Delimitation*, 1910, p. 35.
- [25] Barth, *Central Africa*, 1857, Vol. II, p. 369.
- [26] Courtet, *C. Rd. Ac. Sc.*, 140, p. 160.
- [27] Gentil and Freydenberg, *Bull. Soc. Geol. Fr.* (4), VIII, 1908, p. 44.

CHAPTER III

PETROGRAPHY OF THE GNEISSES AND SCHISTS AND ASSOCIATED IGNEOUS INTRUSIONS.

Subdivision of the gneisses and schists; predominant meridional strike. Gneisses and schists of sedimentary origin; variation in degree of metamorphism; phyllites and slaty schists; green schists; quartzites; quartz schists and granulitic gneisses; crystalline limestones; micaceous and felspathic gneisses; calc silicate rocks; epidote hornblende rocks; magnetite and hæmatite schists; sheared trachytes of Bussa; altered dolerites of Ngashki; amphibolites and talc schists; granulites. Petrography of the banded Archæan gneisses: relationship of the two groups of gneisses. The gneisses and schists of the Sahara and of West Africa. Age of the sedimentary gneisses. Intrusive rocks: the older granites; variation in degree of reconstruction; brecciation and injection structures; the younger granites; aplitic and tourmaline-bearing granites and pegmatites; tourmaline quartz veins; stanniferous pegmatites; schorl rocks; soda granites; biotite and riebeckite granites; tinstone associated with soda granites; biotite and augite syenites; augite diorites; riebeckite quartz porphyries; ægirine granophyre; augite syenite porphyries; alkali syenite porphyries; diorite porphyrites; mica porphyrites; hornblende porphyrites; diabases and basalts. Granites and dyke rocks of the Sahara. Age of the soda granites; successive periods of intrusion.

THE GNEISSES AND SCHISTS.

FOR descriptive purposes it has been found convenient to separate the crystalline rocks of the Protectorate into two groups, (1) a group of quartzites, phyllites, schists, and soft felspathic and micaceous gneisses, and (2) a group of hard well-foliated, much granitised and frequently banded and contorted micaceous and hornblendic gneisses. The first group occupies by far the greater area within the Protectorate, and presents in the west a well-marked alternation of belts in which quartzites, phyllites and schists predominate, with belts of which the prevailing members are felspathic and micaceous gneisses. The second group has no definite distribution within the Protectorate, but appears to replace locally and irregularly the members of the first group throughout the whole crystalline area. The members of both groups appear to be irregularly folded along axial planes, which are usually vertical or highly inclined, but it is probable that detailed investigation would reveal a regular system of major folds, accompanied by much local and minor crumpling. The folding, which is typically

meridional, has been affected by equatorial stresses in such a way as to produce a transverse twisting or corrugation of the predominant strike. The result is that individual exposures may vary in direction of strike from N.–S. to E.–W., while in places, as in Bauchi province, the strike may become regionally equatorial. A similar equatorial divergence of the prevalent meridional strike has been noted by M. Hubert^[28] in eastern Dahomey, but his suggestion that the equatorial direction becomes predominant in Borgu and the Niger valley has been found to be unsupported by local evidence. It is impossible to believe, moreover, that there can be any connection, as suggested by M. Chudeau,^[29] between this locally divergent strike of the gneisses and the distribution of the overlying Cretaceous and Eocene rocks.

The sedimentary origin of the first group is fairly certain. The pebbly quartzites, quartz schists, schistose grits, mica schists, slaty schists and phyllites represent corresponding varieties of sandstones, grits and shales, and frequently exhibit traces of their sedimentary origin in the deformed pebbles and clastic grains which they contain. The granular felspathic and micaceous gneisses on the other hand are more highly reconstructed, and afford in themselves little or no evidence of a clastic origin. Their constant association and alternation with the quartzites and schists make it probable, however, that they represent the arkoses and greywackes of the original sedimentary series. Many of the green hornblendic and epidotic beds probably also represent original gritty accumulations of chloritic and epidotic character, while the quartz magnetite schists of eastern Kabba correspond to ferruginous grits or gritty ironstones of sedimentary origin. The fine-grained crystalline limestones of Igbo, and the coarse granular marble between Jakura and Wa, similarly represent original clastic calcareous rocks within the sedimentary succession. To what extent the recurrence of belts of quartzites, phyllites and schists in the western part of the Protectorate represents the repetition of one or more belts is quite unknown. The phyllites and schists are moreover everywhere unfossiliferous and the age of the original sedimentary rocks is consequently undetermined. The foliation of the reconstructed rocks is, as a rule, parallel with the bedding, but the more micaceous schists frequently show a more or less well-defined and oblique strain-slip corrugation.

The alternation of belts of less and more highly metamorphosed rocks in the west of the Protectorate coincides to some extent with an alternation of belts in which shear structures predominate, with belts in which reconstruction is prevalent. The various belts appear to pass insensibly into each other, but it has been found impossible in the course of a rapid survey to decide either to what extent shearing may have everywhere preceded reconstruction, or to what extent the varying degree of reconstruction in alternate belts may represent a corresponding variation in the intensity of the accompanying folding. All that can be said with certainty is that the sedimentary series as a whole has been subjected to a highly complex folding accompanied by much shearing and reconstruction, and that while over the greater part of the Protectorate the stresses have been for the most part relieved by, or translated into, a molecular and mineralogical rearrangement, there are certain belts in which relief has been primarily obtained by mechanical readjustment or shearing. Moreover, as in the reconstructed areas there are certain beds which appear to be more coarsely crystalline and less distinctly foliated than others, so in the shear belts there are certain beds which have assumed the character of granulitic gneisses, while others in the immediate neighbourhood have retained to a very large extent their original clastic and gritty character. Most of the reconstructed gneisses exhibit under the microscope faint strain phenomena which, with the incipient strain-slip foliation of the more micaceous schists, are probably to be ascribed to regional stresses affecting the whole series after the period of maximum reconstruction.

Grey and purple sericitic phyllites with knots of magnetite and pyrites are extensively developed in the neighbourhood of Ngashki in the Niger valley, around Anka and Bukwium on the borders of Sokoto and Kontagora, and in the Zungeru belt in eastern Zaria from Kato to Birnin Gwari and Koriga. In Kabba and Nassarawa and southern Zaria their place is taken by fine-grained quartz muscovite schists with abundant magnetite. With the phyllites are associated in the Niger valley and Sokoto province green quartzose slaty schists (1195, 1325),^[30] whose colouration is due to recrystallised chloritic and hornblendic material. Mr. Parkinson^[31] has described similar slaty schists from Southern Nigeria. In the neighbourhood of Zungeru these green schists become softer and show under the

microscope (1563–4) ragged porous or fibrous crystals and crystal aggregates of reconstructed hornblende in a matrix of granulitic quartz, feldspar and magnetite, which encloses also scattered grains of hornblende, biotite, chlorite and epidote, as well as relics of original clastic grains of quartz and feldspar. These rocks present a considerable resemblance to the “green beds” of the Scottish Highlands.^[32] True quartzites are comparatively rare and only at Kazaure do pure siliceous rocks attain any extensive development. Elsewhere the rocks which present the characteristic appearance of quartzite in the field are usually more or less feldspathic and micaceous, and include schistose grits, micaceous quartzites, quartz muscovite schists and granulitic gneisses which show all the familiar stages in the deformation of feldspathic grits and sandstones. Some of the white and pink granulitic gneisses, such as those of Zungeru, may, however, be ultimately proved to have been originally of an eruptive character. A characteristic feature of these rocks is the development of tourmaline and sometimes of pyrites along the shear and foliation planes as the result of later pneumatolytic action. At Kuta a belt of coarse quartz muscovite schist contains abundant sillimanite in sheaf-like aggregates.

Mica schists in which biotite predominates are common throughout and frequently carry garnet, magnetite and staurolite in conspicuous crystals. In the Igbo limestones the original impurities have recrystallised as tremolite, diopside, scapolite (122), sphene, biotite, feldspar, quartz and pyrites, while the streaks of silicates in the limestone between Jakura and Wa contain diopside, epidote, idocrase, feldspar, and quartz. In the latter limestone also, scales of graphite are common throughout and pyrites on the margin. Amongst the reconstructed gneisses, which show a remarkable and rapid variation in texture, the paler-coloured feldspathic rocks are usually rich in orthoclase and microcline, and occasionally muscovite, while the darker-coloured varieties, in which biotite predominates, frequently carry garnet, sillimanite, staurolite and in places also sphene, tremolite and carbonates. Andalusite-bearing rocks are known from certain localities, but kyanite gneisses and schists have not been definitely recognised in place. The local occurrence of kyanite on the river sands, however, points to its presence within the neighbouring rocks. Calc silicate types are represented by the tremolite schists of the Nassarawa tableland (668, 689), by a diopside

tremolite schist from Borgu (1221), a diopside and wollastonite-bearing rock with sphene from Maibirro in the Niger valley (1192), a biotite tremolite schist with calcite from Olle in northern Kabba (879), and a diopside and scapolite-bearing rock with sphene and green spinel from Atuka, near Okuruku, in eastern Kabba (843). The epidote hornblende gneisses of Boi in southern Bauchi (976) contain much diopside and sphene, and are probably of sedimentary origin, while relics of clastic feldspar are common in an epidote rock from Goshin Duchi a few miles east of Boi (983). A similar clastic origin is presumed for many of the epidiosites and epidote hornblende rocks associated with the sedimentary gneisses in other parts of the Protectorate. The quartz magnetite and quartz haematite schists include coarse granular and fine-grained granulitic types (844, 1370), the latter presenting considerable resemblance to the itabirites of the United States. Their sedimentary origin is presumed throughout and only in the case of the massive quartz magnetite rock of Okuruku (148), which is associated with a magnetite-bearing epidiorite, is there any probability of an igneous origin.

Two stages of alteration can be recognised among the sheared trachytes of Bussa ([p. 72](#)). The least altered types (1164, 1167) show fracturing of the phenocrysts of feldspar and the replacement of the original ferro-magnesian minerals by fibrous urallite, chlorite, epidote and calcite. The more altered types (1166) are characterised by the production of abundant white mica in the ground-mass and the development of biotite, magnetite and carbonates from the original ferro-magnesian material. Original vesicles of quartz and calcite in these rocks have been deformed and the quartz replaced by a mosaic of angular grains. A sheared quartz-porphyry from Kanikoko (1206) exhibits deformation and granulation of the phenocrysts of quartz and feldspar and the abundant development of white mica in the ground-mass. In the altered dolerites of Ngashki (1175, 1176) the original structures are almost entirely obscured by the extensive production of granular epidote, calcite, urallite, chlorite, magnetite, feldspar and quartz. Knots of tourmaline are common in these altered rocks. The types grouped as amphibolites range from epidiorites to hornblende schists and amongst them may be recognised all the familiar stages of saussuritisation and recrystallisation of original tonalitic, dioritic and diabasic rocks.

It is probable, however, that amongst the amphibolites are included rocks which were intruded at many different periods during the metamorphism of the gneisses and that the difference of age as well as to local intensification of the metamorphism is to be ascribed the variable degree of construction which the rocks have severally undergone. The chlorite and talc schists are to be regarded as basic intrusions which had suffered considerable decomposition before being subjected to metamorphism.

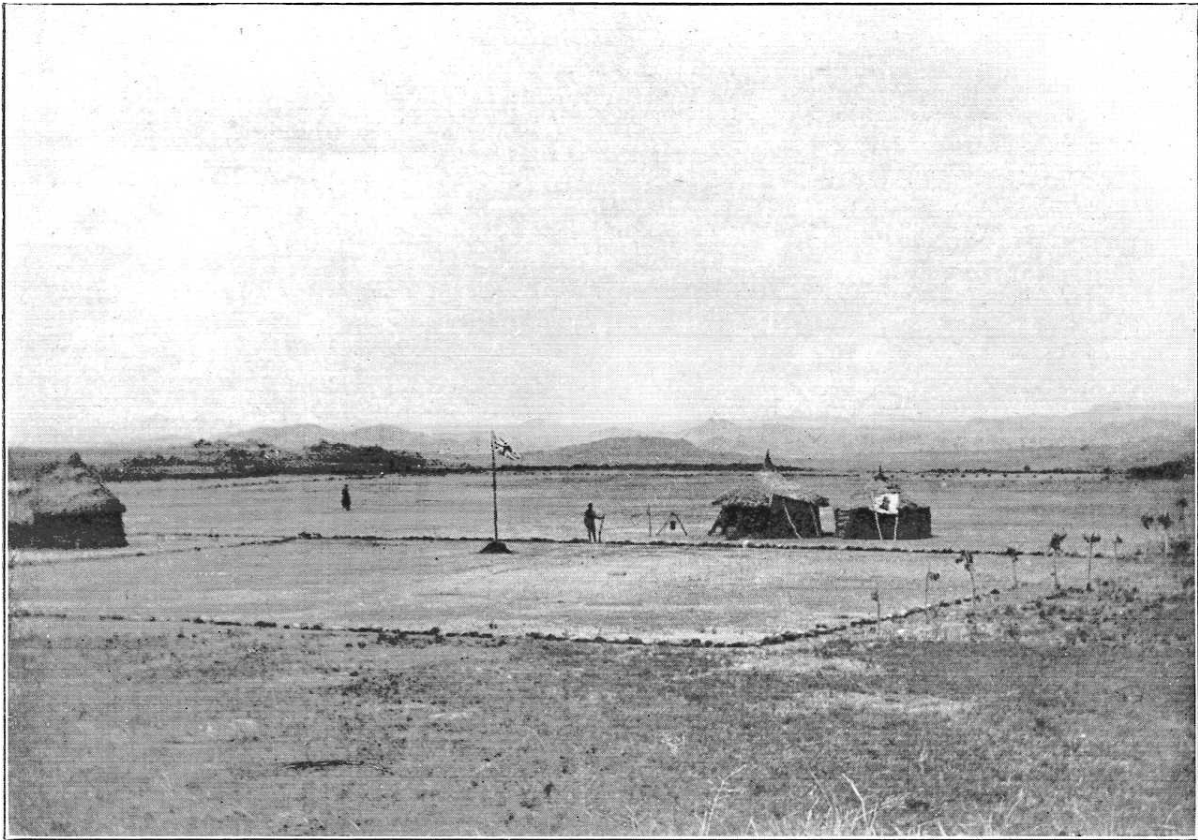
Amongst highly reconstructed rocks of doubtful derivation, associated with the sedimentary gneisses, are a garnetiferous quartz enstatite biotite granulite from Atuka near Okuruku in southern Kabba (841) and a garnet granulite from the adjoining village of Kuturu (168) composed of labradorite, quartz and garnet with a distinct centric structure. The garnet is in spongy masses associated with a small quantity of green pyroxene, sphene, apatite and iron ore. A pyroxene granulite, from Agaiyi (823), consists of a porous intergrowth of pale green pyroxene and green hornblende, the former predominating, in a matrix of clear labradorite. Sphene and ilmenite are also present in irregular intergrowths as well as a little accessory epidote and apatite. Another variety (832) contains no amphibole and consists of pale green pyroxene and garnet in a matrix of labradorite and quartz. A third type from the same locality (821) is an epidote granulite consisting of predominant epidote and garnet with subordinate green pyroxene, sphene and ilmenite in a matrix of labradorite. A pyroxene granulite from Nassarawa province (854) shows ragged crystals of diopside with quartz inclusions and rounded garnets with inclusions of both quartz and diopside in a matrix of coarsely granular quartz. A hornblende granulite from the same province (859) consists of rounded spongy garnets and radiate groups of actinolite fibres with a little accessory ilmenite in a typical fine-grained quartz-feldspathic mosaic. A remarkable biotite granulite from Eri in eastern Ilorin (1056) contains in addition to biotite and granular garnet as the predominate ingredients much sillimanite, staurolite and andalusite in a matrix of untwinned feldspar and quartz.

The harder gneisses of the second group are subordinate in amount and irregularly distributed throughout the micaceous and feldspathic gneisses of the first group. They consist of a series of much granitised medium and

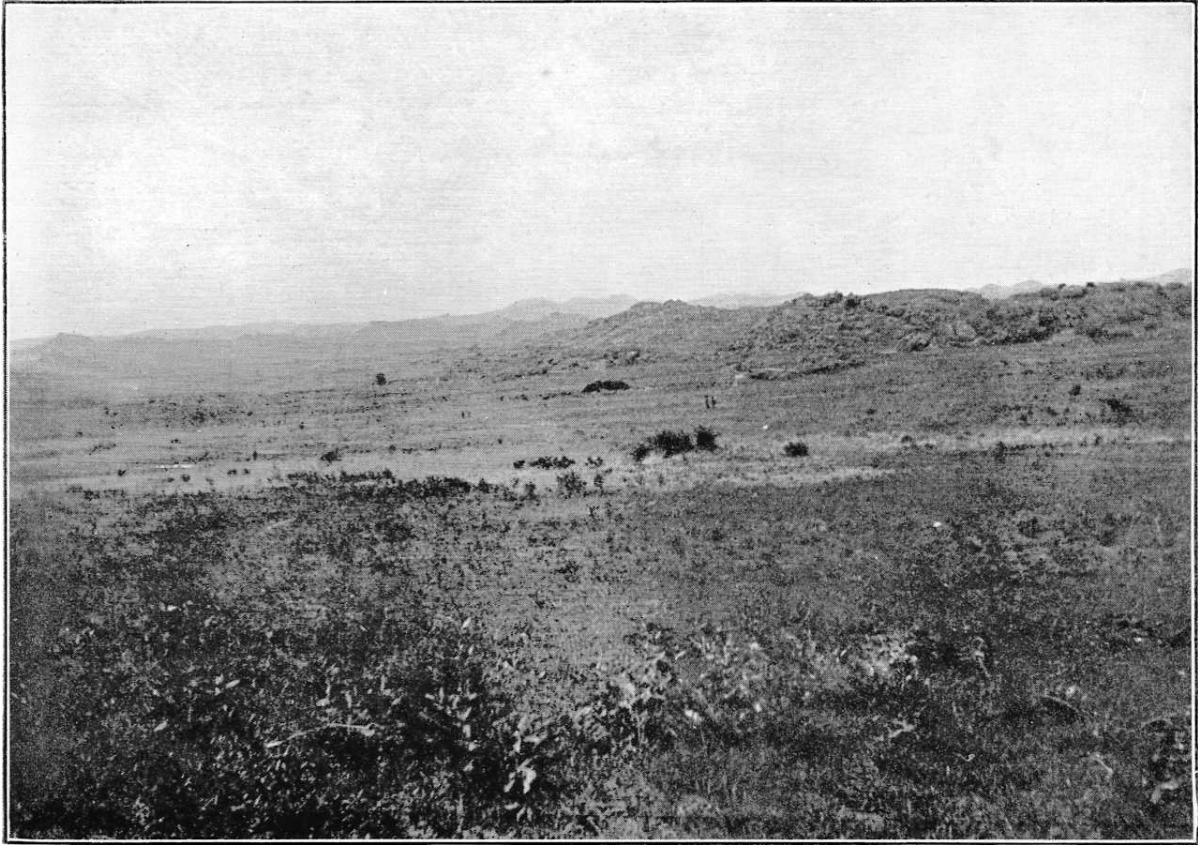
even-grained biotitic and hornblendic gneisses which usually show a lenticular or linear foliation but frequently also a ribboned or banded and contorted character. In some cases the banding is broad and only to be distinguished in the field, while hand specimens present quite a homogeneous character. All variations may be found between acid aplitic and basic hornblendic bands, and knots and lenticular masses of more basic material are common within the more acid types. As a rule the banding is much interrupted by pale coloured granitic material into which the lighter coloured bands seem to pass imperceptibly. A striking feature is the presence of a series of early pegmatite veins folded and foliated with the gneisses. There is in fact every indication that these banded rocks are of a primarily composite character and represent either an originally sedimentary or a differentiated igneous succession invaded and injected by acid granitic material and pegmatite veins in the deeper portions of the crust and subsequently subjected to a later folding, foliation and recrystallisation. The foliation is in general vertical and parallel with the banding. Under the microscope the striated and banded gneisses exhibit all the characteristic features of typical archæan rocks (801, 846). They are composed of a granular aggregate of quartz, felspar, hornblende or biotite with sphene, apatite and iron ores as accessories. The felspars are usually allotriomorphic and vary from labradorite in the more basic to orthoclase and microcline in the more acid rocks. The biotite occurs in tabular flakes singly or in aggregates with the basal plane well-developed but with ragged margins in the prism zone, while the hornblende forms spongy growths or irregular plates with embayed margins and inclusions of quartz. Strain shadows are common in the quartz and felspar but only traces of cataclastic granulisation or of secondary micropegmatite are as a rule recognisable. Van Hise^[33] is of opinion that the strain phenomena and the vermicular quartz to be observed in these thoroughly reconstituted gneisses are due to the partial lagging of reconstruction behind deformation. It is equally probable, however, that the strain effects and micropegmatitic structures have been induced after the period of maximum reconstruction by later strains of lesser magnitude.

The mutual relations of the two groups of gneisses are difficult to determine. The actual contact between the two has nowhere been found clearly exposed and its character must therefore be to a certain extent

conjectural. The apparent irregular replacement of the softer gneisses by members of the harder group when traversed across the strike and the apparent extension into the softer gneisses of tongue-like masses from the larger areas of harder rock is certainly suggestive of an intrusive character as advocated by Mr. Parkinson^[34] in Southern Nigeria and Liberia. Upon this hypothesis the harder gneisses would appear to correspond to an intimate mixture of granitic and dioritic material, whose intrusion took place during the folding and reconstruction of the sedimentary gneisses. It is difficult, however, to understand why these rocks should present in their even-grained and closely foliated and banded character such an entirely different aspect in the field from that of even the earliest recognisable granitic intrusions within the sedimentary gneisses and why they alone should possess an early set of folded and foliated pegmatite veins in addition to the later pegmatites which are common to both groups of gneisses. All the phenomena indeed are more readily explicable on the assumption that the two groups differ in age, that the sedimentary gneisses were originally deposited unconformably upon an earlier and probably archæan gneissic or igneous complex and that the latter series was refoliated and recrystallised during the reconstruction of the former. The peculiar distribution of the harder gneisses and their intimate alternation with the softer series would then be due in part to the original unconformability between the two groups and in part to the irregular character and the varying amplitude of the later folding to which both were subjected. The abrupt replacement of the sheared but little altered quartzites of the Niger valley by the harder gneisses at Jebba, Wuru and Maibirro probably indicates also a certain amount of faulting and thrusting accompanying the folding. On account of the subordinate and ill-defined character of the distribution of the harder rocks no attempt has been made to distinguish upon the map between the archæan and the later sedimentary gneisses.



ON THE SUMMIT OF THE BAUCHI PLATEAU.
The Parade Ground at Bukuru.



ON THE PLATEAU LOOKING WESTWARD FROM BUKURU.

In the Sahara to the north, M. Chudeau^[35] has described the occurrence of archæan and sedimentary gneisses whose mutual relations are similar to those which prevail in Northern Nigeria. The archæan areas are of comparatively small extent, and are primarily distinguished, according to M. Chudeau, by the presence of numerous domes and turtlebacks of granite. This method of distinguishing between the two groups, however, cannot be applied in Nigeria, where intrusive granites are equally common amongst both the earlier and the later gneisses. The metamorphic rocks of sedimentary origin attain a great development in the Sahara^[36] and include quartzites, phyllites, schists, gneisses and crystalline limestone similar in type to those of Nigeria, and folded in a similar complex manner along axes whose direction is predominantly meridional. The decrease northward of the intensity of the metamorphism, as advocated by M. Chudeau, requires much further investigation in view of the local occurrence of little altered rocks within the Protectorate. M. Chudeau believes in the original unconformability of the later gneisses upon the archæan and points out,

with M. Haug,^[37] the analogy between the system of folding of the gneisses of the Sahara and the Sudan, and the Caledonian folding of the crystalline rocks of Scotland and Scandinavia. He demonstrates the pre-Devonian age of the sedimentary gneisses, and assigns them provisionally to the Silurian on the strength of their supposed continuity with the graptolite-bearing shales of Tindesset^[38] and Hassi el Kheneg.^[39] The stratigraphical evidence available, however, seems rather to point to an unconformability between the shales and the schists, and M. Flamand^[40] prefers to assign a pre-Cambrian age to the whole series of crystalline schists and gneisses.

In Dahomey, M. Hubert^[41] has described a similar complex of archæan and sedimentary gneisses in which the latter clearly predominate. His separation as Silurian of the quartzites and schists of Atacora from the other metamorphic rocks appears, however, to be unjustifiable. Quartzites, gneisses and schists of sedimentary origin associated with granites and archæan rocks, are found also in West Africa, in French Guinea and the Fouta Djallon,^[42] in Liberia,^[43] the Ivory Coast,^[44] the Gold Coast,^[45] Togoland,^[46] and Southern Nigeria,^[47] and in Central Africa to the south and south-east of Chad,^[48] in the Kameruns,^[49] and in the French Congo.^[50] Similar rocks attain also a remarkable development to the south of the equator.^[51]

The comparison which M. Chudeau has drawn between the strike of the gneisses and schists of the Sudan on the one hand, and of the Scottish Highlands on the other, may be further extended not only to the petrographical character of the constituent groups of the gneissic complex in West Africa, but also to the respective relations of the various groups to each other, and to the intrusive rocks which they enclose.^[52] Dr. Voit^[53] has, moreover, arrived at similar conclusions in the course of his investigation into the structure of the crystalline complex of the northern Transvaal, the constitution and mutual relationships of the Swaziland system and the older granites and gneisses being closely paralleled in the rocks of Nigeria.

THE INTRUSIVE ROCKS.

Many of the thin bands of well-foliated augen and granitoid gneiss enclosed within the metamorphic series undoubtedly represent original granitic intrusions, but these are so intimately bound up with the adjoining gneisses and schists, that no discrimination can yet be attempted. Attention has been more particularly directed in the field to those granitoid rocks which form the prominent inselberge and isolated groups of hills, and amongst them it has been found possible to distinguish an older from a younger series of intrusions. The older intrusive rocks consist very largely of granite, while the younger intrusions include granitic, syenitic and diabasic types with numerous associated dyke rocks. To what extent the epidiorites, amphibolites, and granulites of the gneissic series represent more basic intrusions associated with the older granites is quite unknown.

(a.) The older Granites.

The older granites are never sharply marked off from the adjoining gneisses, and invariably show a certain amount of foliation in the field, and of cataclastic deformation and reconstruction under the microscope. They appear to have been intruded at different times within the period of the evolution of the gneisses, and the amount of reconstruction which they themselves have undergone may be taken as a general indication of their age. Vermicular micropegmatite is most extensively developed in those schistose granites which have stopped short of complete reconstruction and which still afford abundant evidence of cataclastic deformation. It may also occur, however, in small amount in thoroughly foliated and reconstructed granites, in which it is assumed to have been developed, as in the case of the banded and striated gneisses ([p. 118](#)), by minor crustal strains, after the period of maximum reconstruction.

The most highly altered types form extensive lenticular or phacolitic masses of well-foliated and fairly coarse-grained biotitic and hornblendic gneisses (849, 481), containing streaks and patches of darker-coloured and finer-grained basic material. Under the microscope they possess a coarsely granular and thoroughly reconstructed character, and in the mutual relations of their constituents present a considerable resemblance to the banded gneisses of similar composition. The quartz and felspar show much later

straining with incipient granulisation and traces of micropegmatite. These rocks have participated in all the changes which the adjoining gneisses have undergone, and were apparently intruded before the period of maximum reconstruction of the latter.

The least altered of the older granites (806, 1560) are typically distinguished by the irregular batholithic character of the intrusions and the partial and frequently only marginal character of the foliation. They sometimes have a tendency to become finer-grained towards their junction with the gneisses, and they usually enclose portions of the adjoining rocks and vein and inject with granitic material the marginal gneisses and schists. The inclusions may be of either the harder or softer gneisses according to the character of the rocks which the granite has invaded. In places a parallel arrangement of the feldspars appears to have been induced through streaming during crystallisation. The partial foliation is parallel in a general way with that of the surrounding gneisses and schists. The granites vary much in character from biotitic and hornblendic rocks with syenitic modifications to muscovite and microcline granites and aplites. The quartz and feldspar are strained throughout and cataclastic granulitisation is common on the margins of the crystals. Vermicular quartz is most abundant in the marginal and more distinctly foliated portions. These rocks would appear to have been intruded after the period of maximum reconstruction of the gneisses, but before the final cessation of stress and strain. In places, as at Abuja, crush lines cross the granites, and in their immediate neighbourhood the rocks show a higher degree of reconstruction and a greater development of micropegmatite in the matrix.

These later members of the older granites appear to have produced little or no contact metamorphism in the surrounding rocks. Brecciation and injection structures on the other hand are very frequent. The hard banded gneisses on the margin of the Abuja granite, for example, are abundantly injected with quartz and quartzo-feldspathic material. Similarly at Zwoell, in Bauchi province, a porphyritic biotite granite has brecciated a series of banded and contorted biotite and hornblende gneisses, and many of the enclosed fragments are partially absorbed or minutely injected with granitic material. The Okuruku granite in southern Kabba, on the other hand, has invaded and brecciated members of both the harder gneisses and the softer

series of biotite schists, magnetite schists and amphibolites. In the neighbourhood of the inclusions the granite becomes knotted and richer in biotite, hornblende, sphene and other minerals, while the inclusions themselves are irregularly veined and minutely penetrated by quartzofelspathic material. In every case it would appear that the rocks, which the granites have brecciated and invaded, possessed their gneissic or schistose character before the actual intrusion of the granites, and that they, as well as the larger inclusions, suffered little further change as the result of the stresses and strains which brought about the partial foliation of the granites. The gneisses as a whole show strain phenomena, but only traces of granulitisation and of secondary formation of micropegmatite, while the granites show extensive marginal formation of micropegmatite, together with intense cataclastic deformation. It would seem indeed that these granites were intruded after the period of maximum reconstruction of the gneisses, but before the final cessation of stress and strain, and that while the gneisses which had originated under conditions of maximum strain were comparatively stable under similar strains of diminishing intensity, the granites which had crystallised from igneous fusion were in a condition of unstable equilibrium with respect to the stresses and strains to which they were subjected after crystallisation. The gneisses, as it were, possessed a certain amount of elasticity and transmitted the later strains with little resultant deformation, while the rigid masses of granite arrested those strains and suffered in consequence a partial deformation. The production of a strain-slip foliation in some of the more micaceous beds is probably to be referred to the same period of differential deformation of the granites. These later strains became locally intensified along crush lines, and in the case of the Abuja granite produced granulitisation both of the granite and of the felspathised gneisses in the neighbourhood of the crush line.

In addition to the two extreme types of the older granites there are many elliptical masses of similar material, elongated in a direction parallel with the strike of the gneisses, which show a decided foliation throughout, sometimes more and sometimes less intense (48, 1223). As a rule they are indistinctly marked off from the adjoining rocks and preserve their normal texture up to the margin of the intrusion. The granites are predominantly biotite granites, more rarely muscovite biotite granites, and the feldspars are

frequently porphyritic. Under the microscope they show a higher degree of reconstruction than the least altered granites, but a less complete reconstitution than the most highly altered types. The feldspars vary from oligoclase to microcline, and the plagioclases often show basic interiors and zoned margins. Both the feldspar and the quartz show abundant strain phenomena, and the feldspars are usually partially surrounded by a secondary fringe of vermiculate micropegmatite in optically discontinuous grains. Frequently also the feldspars are corroded or irregularly penetrated by quartz or micropegmatite, while the matrix consists largely of granular and partially recrystallised feldspar, muscovite, microcline and quartz. The darker patches within these granites may be taken to represent portions of the adjoining rocks caught up during intrusion and partially absorbed, recrystallised and foliated with the granites. These rocks may possibly be correlated in the period of their intrusion with the least altered granites and regarded as the smaller and therefore more readily reconstructed representatives of the larger partially foliated masses. On the other hand, in the absence of the inclusions of adjoining gneisses which form the characteristic feature of the least altered granites, it is possible to regard them as of somewhat earlier origin and as having participated to a considerable extent in the general reconstruction of the gneisses themselves.

It seems probable indeed that the crustal movements which brought about the metamorphism of the gneisses were of a somewhat spasmodic character and spread over a very considerable interval of time and that the intrusion of granitic and other igneous material took place at intervals throughout the whole period. No hard and fast line can therefore be drawn between the highly reconstructed and gneissose granites and the imperfectly or only partially foliated types, a complete series of intermediate varieties being apparently possible between the two extreme members of the series.

(b) The Younger Intrusions.

1. Granites and pegmatites.

Amongst the later granites, which present for the most part a batholithic habit, two types may be distinguished, differing in mineralogical

composition and probably also to some extent in age. The rocks of the first type never show chilled margins, and were evidently intruded while the adjoining gneisses still possessed a comparatively high temperature. They include non-foliated aplitic granites and pegmatites usually rich in muscovite and tourmaline. The second type includes riebeckite and biotite granites which are sharply marked off from the adjoining rocks by felsitic and fine-grained margins. The rocks of the first type usually exhibit under the microscope a faint straining of the quartz and feldspar and were probably intruded before the final cessation of the crustal movements to which the reconstruction of the gneisses and the foliation of the earlier granites were due. The members of the second type, on the other hand, are practically free from cataclastic phenomena and evidently assumed their present position after the cessation of movement within the crust. There is no reason, however, to assume any great difference in age between the two types or any marked discontinuity in magmatic conditions. The same magma, of a somewhat acid and alkaline character, was probably present in the depths throughout the whole period of intrusion, and the varying character of the products may be taken as affording some clue to the nature of the differentiation within the magma itself.

The alkaline granites of the first type are pale-coloured granular rocks composed almost entirely of quartz, microcline, orthoclase, perthite, and acid plagioclase. Biotite and other ferro-magnesian minerals are comparatively rare, while muscovite and tourmaline are frequently abundant and in large primary crystals. The quartz and feldspar have a tendency to pegmatitic intergrowth, and sometimes masses of graphic pegmatite occur porphyritically in a normal quartz-feldspathic matrix. The tourmaline and muscovite may occur also as at Igbo in two generations, in large crystals and in the ground mass. Frequently also these granites possess a tendency to a coarsely banded structure, the banding being due either to variation in size of grain of the quartz and feldspar or to variation in the abundance of tourmaline and muscovite in successive bands. These granites rarely form intrusions of any size, the largest known being the tourmaline granites of Igbo and Darroro, but even they are insignificant in comparison with the alkaline granites of the second type.

The pegmatite dykes and veins which are so abundant throughout the gneissic series evidently belong to the same period of intrusion. Like the aplitic and tourmaline-bearing granites they never exhibit fine-grained margins at their junction with the gneisses. They vary much in composition, and include graphic granites, quartz-felspar pegmatites with biotite, magnetite or ilmenite, quartz-felspar pegmatites with muscovite or tourmaline or both, and quartz-muscovite pegmatites with or without tourmaline. At Eri there are stanniferous pegmatites carrying quartz, muscovite and cassiterite, but no tourmaline. The massive quartz-reefs which frequently enclose portions of the surrounding schists and carry tourmaline, pyrites and sometimes gold, are also to be referred to the same period of intrusion. Inseparable from them and from the pegmatites are the smaller veins of quartz and tourmaline quartz and the schorl rocks of a foliated or striated character which occur as a rule in association with the veins of tourmaline quartz.^[54] The schorl rocks with a gneissic habit in which the quartz is minutely granulitic and the tourmaline is in small crystals evidently recrystallised and full of quartz inclusions, are probably altered gneisses and schistose granites in which the feldspars have been replaced by quartz and tourmaline. Traces of decomposed feldspathic material are to be found in every specimen, but no pseudomorphs in tourmaline of feldspar or of other minerals have been definitely recognised. The striated schorl rocks in which the darker bands are formed of finely fibrous tourmaline, appear to arise also as modifications of the adjoining rocks in the vicinity of the tourmaline quartz veins. Sometimes a foliated rock may be traced through a striated type into a massive tourmaline rock on the margin of the vein in which the original structures have entirely disappeared, and in which the original minerals, including the quartz, have been entirely replaced by finely fibrous tourmaline. In the quartz veins, pegmatites and granites, on the other hand, the tourmaline is in well-formed crystals of primary origin. It would seem, therefore, that the pneumatolytic or metasomatic formation of the schorl rocks was brought about by emanations from the quartz veins and pegmatites before the final consolidation of the latter. The extensive development of tourmaline and pyrites in the quartzites and schists of certain areas is also to be ascribed to the circulation of mineralising vapours and solutions throughout the

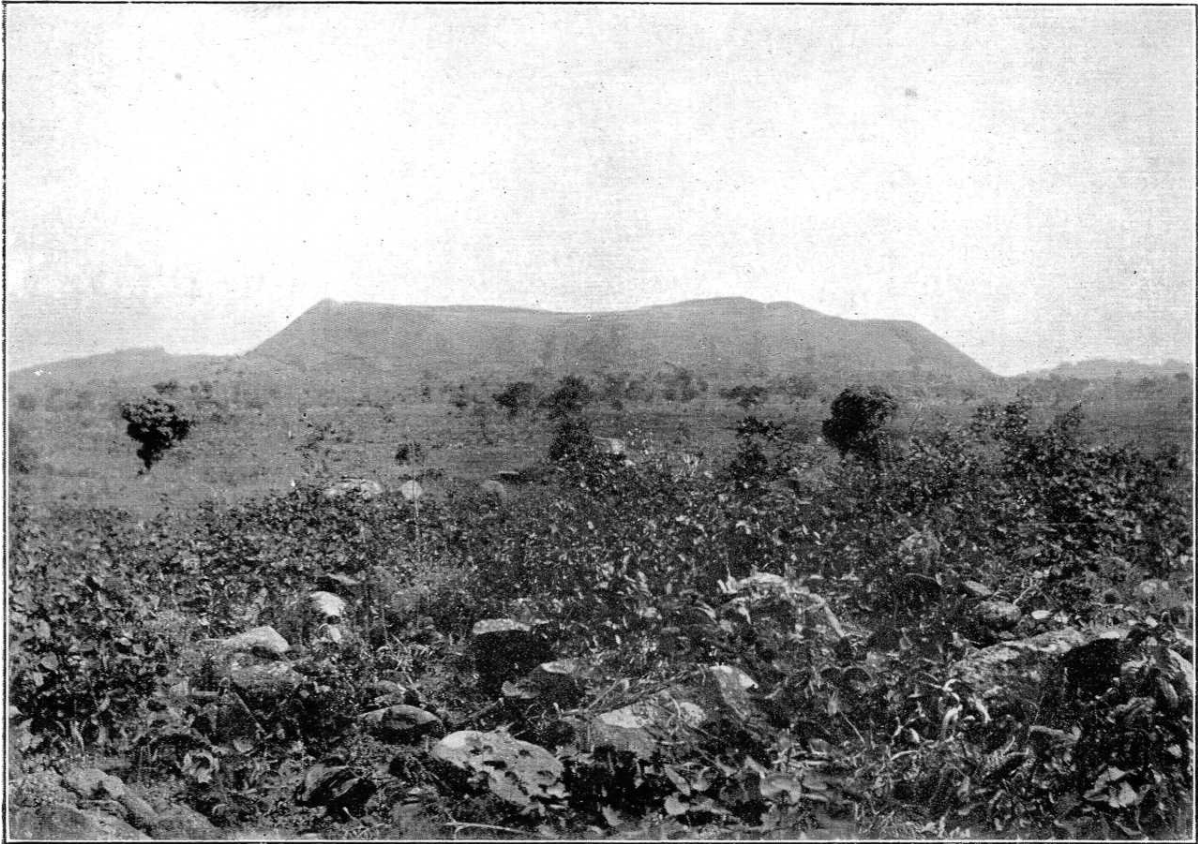
adjoining rocks during the same period. The remarkable absence of tinstone in the tourmaline pegmatites, together with its appearance in pegmatites at Eri which do not carry tourmaline, seems to point to the stanniferous pegmatites having originated at a somewhat later date than the normal tourmaline-bearing pegmatites. This suggestion is supported by the occurrence of tinstone in association with the soda granites of Bauchi province.

The alkaline granites of the second type are typically distinguished by possessing fine-grained porphyritic and felsitic margins which indicate a considerable difference in temperature between the granites and the surrounding gneisses at the period of intrusion. To this group are referred the biotite granite of the Anagoda hills in S. Nassarawa, the riebeckite granite of the Gurkawa Hills in northern Muri, the biotite granite of Bukuru and Ngell, and the riebeckite granites of the Kwandokaya Hills and of the Ningi-Burra massif in Bauchi province, the riebeckite granite of Fagam, Gadama and Galambi on the borders of Bauchi and Kano, the riebeckite granite of Shira in the sub-province of Katagum, and probably also the Kailema granite near Jebba and the biotite granite of Abara near Ngashki in western Kontagora. The alkaline character of the original magma is reflected in the nature both of the feldspars and of the ferro-magnesian. The feldspars are much kaolinised and consist chiefly of tabular crystals of untwinned orthoclase probably rich in soda, and of streaky perthitic intergrowths of orthoclase and albite. Where enclosed in quartz, the terminations of the crystals are frequently idiomorphic. Microlites of riebeckite are common inclusions and small plagioclases of an earlier growth are frequently enclosed in the later feldspars and in the quartz. The quartz forms aggregates of irregular grains showing faint strain shadows and has a great tendency to form coarsely pegmatitic intergrowths with the feldspar. Where biotite alone is present, it is of a normal greenish-brown colour. Riebeckite, however, is the most characteristic ferro-magnesian, and occurs for the most part in irregular plates moulded on the feldspars and sometimes enclosing small crystals of feldspar in poikilitic fashion. Occasionally, as in the Shira granite, the riebeckite builds elongated crystals more or less idiomorphic in the prism zone and frequently intergrown with ægirine. Crystals of riebeckite over an inch in length are common in knots

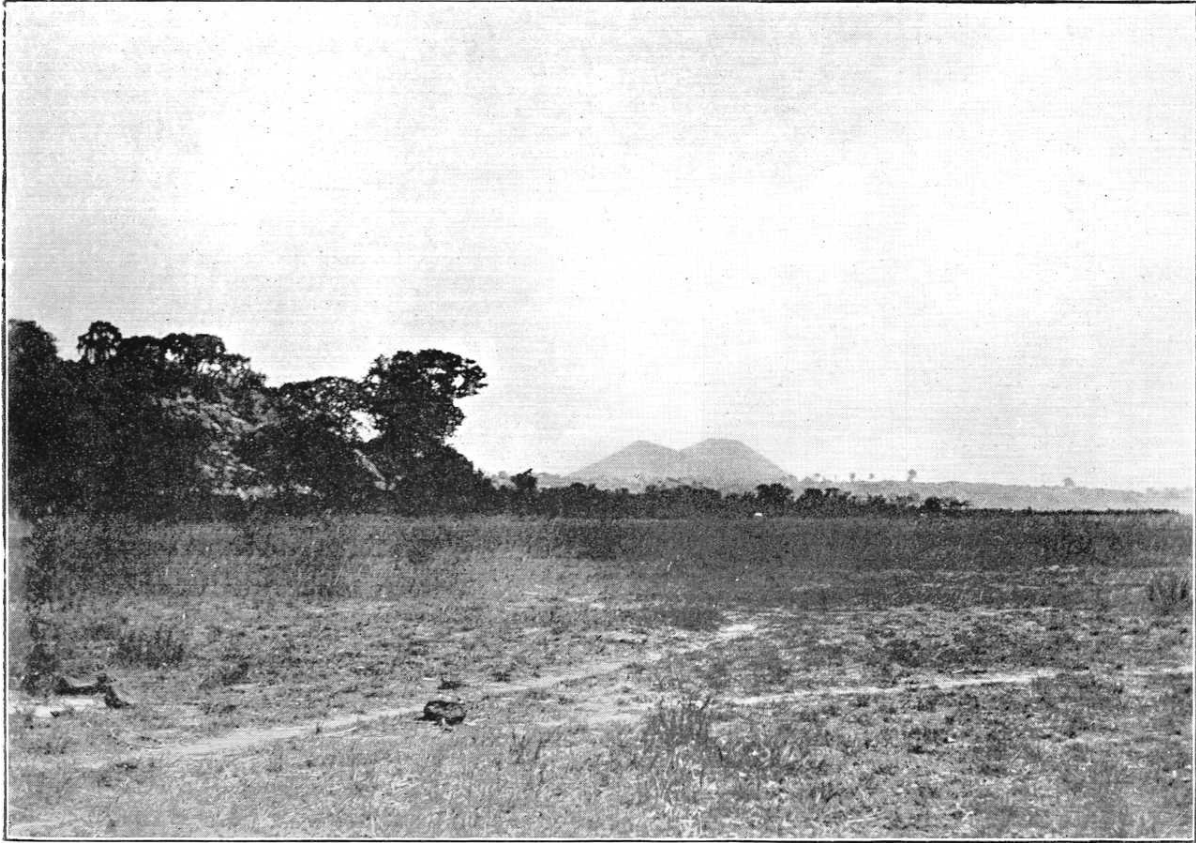
in the Kila granite, while on the other hand the margin of the Gurkawa granite is stippled with minute porous crystals of the same mineral. Associated with the riebeckite in places are other blue and brown soda amphiboles^[55] as well as green ægirine and a golden-coloured mica, while zircon, apatite and iron ores are constant but scanty accessories. Pyrites is sometimes abundant in the porphyritic and felsitic marginal rocks.

A remarkable feature of some of these later granites is the occurrence in them of tinstone and numerous sulphide ores. It would appear that in places these later granites have been broken and fissured, and that along these fissures vapours and solutions have risen which have chloritised and mineralised the adjoining rock and in part refilled the fissures with ore-bearing pegmatite and vein stuff. In the Bukuru granite the cassiterite is associated in the neighbourhood of Ngell with pyrites, chalcopyrite, tetrahedrite, blende and galena. Topaz and brown zircon are found accompanying the cassiterite in the tin-bearing alluvium at Naraguta and elsewhere, and are probably also of pneumatolytic origin. The granites of the Kwandokaya hills, of the Jengre and Limoro hills, of the Ningi-Burra massif, and of the Gadama and Fagam hills also yield in places a tin-bearing drift, and have probably suffered, like the granite of Bukuru, a local fissuring and mineralisation. Tourmaline has not been definitely recognised as a product of these later emanations.

Similar soda granites with microgranitic and felsitic margins have been described by MM. Foureau, Gentil and Lacroix^[56] from Zinder, by M. Chudeau^[57] from Mounio and Goure, by M. Garde^[58] from Machina and by M. Hubert^[59] from Dahomey, while Captain Freydenberg^[60] has noted the occurrence of a soda syenite with granitic modifications at Melfi to the south-east of Chad.



A WORN VOLCANIC HILL ON THE BAUCHI PLATEAU NEAR ZAMAGAN.



THE TWIN CONES OF KERENG FROM PANYAM

2. Syenites and Diorites

These rocks, which are by no means abundant, occur as small bosses or intrusive masses within the gneisses and schists. As a rule their junction with the surrounding rocks is not exposed, and their relationship to the later granites has not been definitely ascertained. They should probably be regarded, however, as somewhat basic modifications or differentiation products of the later granitic magma. Slight strain effects are usually apparent in the quartz and felspar.

The low hills around the town of Bauchi are composed of a coarse-grained augite syenite, in which orthoclase felspar predominates in large twinned crystals with a peculiar brownish tint and a resinous lustre. Under the microscope (49) subordinate oligoclase is present in smaller crystals, while the knots of darker minerals are composed of intergrowths of violet-coloured augite, green hornblende, brown biotite and iron ores. Quartz is

interstitial and occurs also in secondary vermicular growths on the margins of the feldspars. The accessories are zircon and apatite, and the whole rock is much stained with secondary limonite. A rock of similar appearance from Kende to the east of Bauchi (69) is somewhat more basic and may be termed a quartz augite-diorite. The predominant feldspar is a basic andesine, while the ferro-magnesian minerals are pale-coloured augite and green hornblende in irregular plates and intergrowths. The quartz is interstitial or intergrown with the feldspar in pegmatitic fashion. A similar rock from Kanna (1677) contains much biotite in addition to the augite and hornblende. The rock of Kogon Dutsi (1338) within the walls of Kano is a basic augite diorite without quartz, consisting predominantly of acid labradorite and pale-coloured augite. Both the augite and feldspar have a tendency to schillerisation and the augite is accompanied by, and in part intergrown with, green hornblende, brown biotite and iron ores, while hypersthene is a scanty accessory. An augite-diorite from eastern Kontagora (1302) contains a little interstitial quartz and feldspars varying from oligoclase to acid labradorite. The ferro-magnesian minerals are much intergrown and include pale-coloured augite, green hornblende and brown biotite.

3. Porphyries, Porphyrites, Diabases and Basalts.

The later sills and dyke rocks are most abundant in the central provinces and especially in Bauchi, eastern Zaria and northern Nassarawa, but are somewhat sparingly distributed throughout the remainder of the Protectorate. They include quartz porphyries, orthoclase porphyries, mica and hornblende porphyrites, diabases and basalts. In the central area the high content in soda which characterises the more acid types undoubtedly indicates a genetic connection between the later soda granites and the dyke rocks, although investigation has not yet proceeded far enough to allow of the recognition of the major foci to which the dyke rocks may be referred. Amongst the quartz porphyries micro-granitic, granophyric and felsophyric types are equally common. In the central provinces riebeckite is frequent both among the phenocrysts and in the ground mass, while elsewhere biotite is the commonest coloured constituent. An ægirine granophyre occurs at Chikobo (1417) on the borders of Bauchi and Zaria. Orthoclase

porphyries are commonest in the central area and possess typically a trachytic ground mass, of which riebeckite and ægirine are frequent constituents. Augite syenite porphyries are known from the Kudaru Hills in eastern Zaria (1397) and from the vicinity of Limoro on the borders of Bauchi and Zaria (1413). The latter occurrence contains hornblende and biotite in addition to the augite, as well as a small amount of quartz in the ground mass. An extensive mass of granophyric diorite porphyrite occurs in western Kontagora between Auna and Bussa (1162). The feldspars are turbid and much epidotised and surrounded by a fringe of more acid material which passes outwards into the granophyric matrix, while smaller irregular crystals of green augite and ilmenite and abundant minute plates of green mica are also enclosed within the pegmatitic material. A peculiar rock from Toro in Bauchi province (958) may be termed a quartz diorite porphyrite, and presents a certain affinity with the basic members of the charnockitic series of the Ivory Coast.^[61] The phenocrysts have a tendency to aggregate in groups and include rounded quartz grains, tabular crystals of andesine and labradorite with inclusions of granular augite, elongated crystals of hypersthene and irregular plates of biotite much corroded by the ground mass and enclosing grains of augite and apatite. The ground mass is composed of granular green augite, feldspar and quartz, with much magnetite and apatite. The feldspar of the ground mass is frequently untwinned and more acid than the phenocrysts, while the quartz forms in places micropoikilitic plates enclosing the other ingredients of the ground mass. As alkali-syenite porphyries may be grouped the bluish-grey trachytoid rocks of Kanikoko (1217–1219). The phenocrysts include tabular plates of sanidine, microcline, microperthite and microcline-microperthite and fairly well-formed crystals of diopside and enstatite, much intergrown with brown mica and bluish amphibole. The ground mass consists of microlites of alkali feldspar and granules of the ferro-magnesian minerals. Typical mica porphyrites with phenocrysts of biotite and zoned andesines occur at Barazara (1274), and between Dan Gerumfa and Gwashi (1276) in eastern Kontagora. A hornblende porphyrite between Naraguta and Tilde (956) is inconspicuously porphyritic and composed of allotriomorphic green hornblende with subordinate biotite and augite in a matrix of acid plagioclase, orthoclase and quartz. A hornblende porphyrite from Borgu

(1220) contains large poikilitic crystals of augite and hornblende in a granular matrix of labradorite, augite, hypersthene, hornblende and iron ore. A somewhat similar rock from Bauchi province (1602) contains large poikilitic hornblende crystals in a matrix of granular augite, magnetite and lath-shaped crystals of labradorite. A labradorite porphyrite from Toro in Bauchi province contains large tabular phenocrysts of acid labradorite in a ground mass of green augite, brown biotite, secondary green hornblende and lath-shaped crystals of acid plagioclase. The diabases and basalts of the central area are particularly numerous round the margin of the Bauchi plateau and the Nassarawa tableland, and probably include intrusions of many different ages. Some of the coarser-grained diabases present affinities with gabbros in the idiomorphism of their augite crystals (47, 1695). Others exhibit a typical ophitic structure (962, 985), while others are predominantly granulitic (690). A hornblende diabase with pale green augite, greenish-brown hornblende and deep red biotite occurs in the Ruruma hills (1466) in eastern Zaria. Biotite also accompanies the augite in an ophitic diabase from the Kudu hills (1390) in north-east Zaria. Uralite diabases are fairly common, but olivine diabases (747) are conspicuously rare. The basalts present few features of interest and are typically compact, non-porphyrific rocks with a holocrystalline, ophitic or granulitic structure and free from olivine.

MM. Gentil^[62] and Chudeau^[63] have described the occurrence of intrusive granites, pegmatites, porphyries, lamprophyres, diabases and gabbros, throughout the crystalline areas of the Central Sahara, and Dr. Passarge^[64] has recorded similar intrusive rocks from the northern gneissic region of Adamawa. Ægirine rhyolites and alkaline trachytes of an intrusive character are known also from Zinder, Gabana, Hadj el Hamis,^[65] and M'Burao.^[66] M. Chudeau believes that the alkaline granites of Mounio with their associated hypabyssal types pierce the sandstones of the Tegama, and would therefore refer all the known occurrences of soda granite in Central Africa to a late Cretaceous or Tertiary age. In Nigeria, however, there is little or no evidence to support this view. The soda granites are nowhere found piercing sedimentary rocks within the Protectorate, and while it is possible that they may have been originally intruded into rocks of Cretaceous age which have now been removed, it is remarkable that, with

the exception of rare sills of dolerite and the Tertiary pipes of basalt, no intrusive dykes or masses of hypabyssal or plutonic material are to be found within what remains of either Cretaceous or Tertiary rocks within the Protectorate. Such intrusive rocks, however, are particularly numerous within the crystalline areas, and especially in the neighbourhood of the larger masses of soda granite, which are themselves frequently pierced by marginal dykes of basalt and porphyry. In view of the abundance of soda types among the dyke rocks, moreover, it is hard to escape the conclusion that the later non-foliated alkaline granites of the second type ([p. 132](#)) with the associated syenites, diorites and dyke rocks of acid, intermediate and basic composition, all possess a certain genetic relationship, and should be properly ascribed to one and the same period of igneous activity. All the facts, indeed, are more readily explicable on the assumption of the pre-Cretaceous age of this period of intrusion within the Protectorate. The stratigraphy of Mounio and Koutous is somewhat doubtful, and M. Chudeau's section^[67] of the faulted margin of the Goure granite does not indicate an actual transition from the mica schists and quartzites into the sandstones and clays of the Tegama. M. Garde,^[68] moreover, while indicating the occurrence of crystalline gneisses to the north of Goure does not appear to confirm M. Chudeau's interpretation of this important section. It should be remembered, however, that even although further investigation may prove the truth of M. Chudeau's correlation, there is no reason to believe that the intrusion and effusion of igneous rocks rich in alkalis has been confined, as M. Chudeau would suggest, to one particular period in the history of the Sudan.

It is interesting to trace in a general way, and subject to the limitations indicated above, the successive changes in the character of the igneous intrusions from the earlier to the later granites and dyke rocks on the hypothesis of a continuity in magmatic conditions, or at least of a periodic renewal of activity in the same igneous focus. The successive intrusions may be tabulated as follows:—

1. Earliest well-foliated granites of intermediate composition with numerous associated amphibolites and epidiorites.

2. Partially foliated granites of both acid-alkaline and intermediate character, with partially sheared dyke rocks of acid, intermediate and basic composition.
3. Aplitic granites and pegmatites, rich in muscovite and tourmaline with associated reefs and veins of quartz and tourmaline quartz.
4. Alkaline granites rich in soda with associated syenites, diorites, and dyke rocks.

If the normal sequence of events within the period of activity of any igneous focus be taken to be the successive manifestation of volcanic, plutonic and hypabyssal activity, there may perhaps be recognised in the successive intrusions indicated above four periods of igneous activity, each of which is now represented by the products of the plutonic and hypabyssal phases only, while the products of the accompanying volcanic phases, if originally present, have been removed in the course of denudation. The tendency to the production of rocks rich in alkalis should be remarked as a notable feature of the various periods of intrusion; and in this connection it is interesting to note that Messrs. Horwood and Wade^[69] have likewise emphasized the alkaline character of many of the earlier granitic intrusions to the south of the equator.

- [28] Hubert, *Mission Scientifique au Dahomey*, 1908, p. 448.
- [29] Chudeau, *Sahara Soudanais*, 1909, p. 18.
- [30] The numbers in brackets refer to microscopic preparations in the Imperial Institute Collection, NN. F.
- [31] Parkinson, *Q. J. G. S.*, Vol. LXIII, p. 315.
- [32] *Mem. Geol. Sur. Scotland*, Expl. of Sheet 55, 1905, p. 13.
- [33] Van Hise, *Treatise on Metamorphism*, 1904, p. 696.
- [34] Parkinson, *Q. J. G. S.*, Vol. LXIII, 1907, p. 308; Vol. LXIV, 1908, p. 313.
- [35] Chudeau, *Sah. Soud.*, p. 2.
- [36] Gentil in *Doc. Sc. Miss. Sah.*, 1905, p. 721.
- [37] Haug, *C. Rd. Ac. Sc.*, 7 Aug., 1905.
- [38] Haug in *Doc. Sc. Miss. Sah.*, 1905, p. 753.
- [39] Flamand, *C. Rd. Ac. Sc.*, 140, p. 954.
- [40] Flamand in Voinot, *Bull. Com. Afr. Fr.*, 1908, p. 218.

- [41] Hubert, *C. Rd. Ac. Sc.*, 146, 3 Feb., 1908; *Miss. Sc. au Dahomey*, 1908, p. 253.
- [42] Chautard, *Le Fouta Djallon*, 1905.
- [43] Parkinson, *Q. J. G. S.*, 1908, LXIV, p. 313.
- [44] Chevalier, *La G.*, XVII, 1908, p. 201.
- [45] Lenz, *G. M.*, 1877, p. 27; Knox, *Geology of Africa*, 1905, p. 93.
- [46] Von Ammon, *Geologie von Togo, M.H. Geog. Gessell. Mün.*, 1903, I, p. 393.
- [47] Parkinson, *Q. J. G. S.*, 1907, LXIII, p. 308.
- [48] Courtet in Chevalier, *L'Afr. Cent. Fr.*, 1908, p. 646.
- [49] Passarge, *Adamaua*, 1895, p. 312; Esch, *Geol. von Kameruns*, 1904, p. 23.
- [50] Barratt, *Ann. des Mines*, VII, 1895.
- [51] Cf. Horwood and Wade, *G. M.*, V, 6, 1909, p. 455.
- [52] "The Geology of the North-West Highlands of Scotland," *Mem. Geol. Sur.*, 1907, pp. 75, 214, 595, 607.
- [53] Voit, *Trans. Geol. Soc. S. A.*, 1905, VIII, pp. 106, 141.
- [54] Cf. Flett, "Geology of Land's End District," *Mem. Geol. Sur. Eng.* 1907, p. 25.
- [55] Cf. Gentil et Freydenberg, *Bull. Soc. Geol. Fr.*, 4th Ser., VIII, 1908, p. 44.
- [56] Gentil, *C. Rd. Ac. Sc.*, 8 Aug., 1904; Foureau and Gentil, *C. Rd. Ac. Sc.*, 2 Jan., 1905; Lacroix, *C. Rd. Ac. Sc.*, 1 May, 1905; Gentil, *Doc. Sc. Miss. Sah.*, 1905, p. 697.
- [57] Chudeau, *Sah. Soud.*, 1909, p. 265; *C. Rd. Ac. Sc.* 1 July, 1907.
- [58] Garde, *C. Rd. Ac. Sc.*, 149, 5 July, 1909, p. 43.
- [59] Hubert, *C. Rd. Ac. Sc.*, 145, 1907, p. 764.
- [60] Freydenberg, *Chad et Shari*, 1908, p. 107; Gentil et Freydenberg, *Bull. Soc. Geol. Fr.* (4), VIII, 1908, p. 44.
- [61] Lacroix, *C. Rd. Ac. Sc.*, 150, 1910, p. 18.
- [62] Gentil, in Foureau's *Miss. Sahar.*, 1905, p. 720.
- [63] Chudeau, *Sah. Soud.*, 1909, p. 256.
- [64] Passarge, *Adamawa*, 1895, p. 382.
- [65] Garde, *C. Rd. Ac. Sc.*, 5 July, 1909 (149), p. 43; cf. Gentil and Freydenberg, *Bull. Soc. Geol. Fr.* (4), VIII, 1908, p. 44; cf. also Suess, *The Face of the Earth*, Vol. IV, *Oxford Trans.*, 1909, p. 283.
- [66] Hubert, *C. Rd. Ac. Sc.*, 1904 (139), p. 378.
- [67] Chudeau, *Sah. Soud.*, 1909, p. 267.
- [68] Garde, *C. Rd. Ac. Sc.*, 149, 5 July, 1909, p. 43.
- [69] Horwood and Wade, *Geol. Mag.* V, 6, 1909, p. 455.

CHAPTER IV.

CRETACEOUS ROCKS.

Distribution: relation to crystalline rocks. The Cretaceous rocks of the Benue valley: the succession around Awe; the position of the salt-bearing beds; the Arofu and Akwana limestones and galena veins; the limestone-shales and brine springs of eastern Muri; the Muri sandstones; the Gateri grits; the limestones and galena veins of the Kudu valley. The Cretaceous rocks of Yola and of the Gongola valley: the Bima sandstones; the Gongila limestone; the rocks of Gulani and Wuyo; the upper sandstones of Ture and Waja. Classification of the Cretaceous rocks. The Cretaceous rocks of the French Sudan, Southern Nigeria and the Kameruns.

No rocks of Devonian or Carboniferous age or of any period preceding the Cretaceous^[70] have been discovered within the Protectorate. The Cretaceous rocks, moreover, are themselves represented only by the upper members of the series, no deposits of earlier age than the Turonian having been anywhere definitely recognised. The Upper Cretaceous rocks are confined within the Protectorate to the Benue valley between long. $8^{\circ}30'$ E. and long. $12^{\circ}15'$ E., and to the valley of the Gongola between its confluence with the Benue and lat. $11^{\circ}15'$ N. They form the greater part of the plains of Muri and Yola, and of the narrow plain of the Gongola in Bauchi and Bornu, but only around Ligri and Gateri and in Wurkum, Tangale and Waja do they rise into ranges of rounded or tabular hills. The rocks consist of a well-bedded series of red and white sandstones and grits, carbonaceous and calcareous shales, ironstones and limestones, usually somewhat faulted and gently folded into low anticlines and shallow synclines, and in places pierced by intrusive masses of dolerite or basalt. The original thickness is unascertainable, partly on account of the folding and faulting to which the rocks have been subjected and partly on account of the thick covering of drift which obscures the rocks for long distances and renders difficult or impossible the correlation of the various exposures. In the west the Cretaceous rocks are believed to pass unconformably under the sandstones of the Lokoja series in the neighbourhood of Udeni, but the actual junction has nowhere been observed. How far westward they extend under the Lokoja sandstones is also undetermined. An exposure of white sandstones and carbonaceous shales in the bed of a stream near Akwacha in

Bassa province seems to indicate that they extend for a considerable distance west of Loko, but that they do not reach the Niger is abundantly evident from the absence of any trace of them in the neighbourhood of the river below Lokoja.

To the north and to the south of the plains of Muri, the Cretaceous sandstones give place imperceptibly to a crystalline floor. The actual contact, however, has been found wherever examined to be so much obscured by drift that it is impossible to say whether any part of the margin is bounded by faults or not. In the east the Cretaceous rocks pass under the upper Benue sandstones and, so far as known, do not reappear anywhere east of Yola. Similarly in the Kudu valley they disappear under the sandstones of Duguri and in the Gongola valley under the Gombe sandstones to the west and north. To the east of the Gongola between Bilaraba and Dusua they rest upon a hummocky crystalline surface, while farther south they pass behind Balbea under the Barbur basalt plateau. To the west of Kwoia, however, their junction with the crystalline rocks, although somewhat badly exposed, has every appearance of being fractured, while on the lower Gongola the relations between the Cretaceous and the crystalline rocks to the east are such as to suggest most forcibly an extensive faulting along the contact. Within the area covered by the Cretaceous rocks inliers of granite and gneiss are exposed around Arofu on the middle Benue, around Tangalto and Chongwom in southern Bauchi and in various places upon the narrow plain of the Gongola and, while in some cases the crystalline rocks exposed upon the plain of the Gongola have undoubtedly been brought to the surface by faulting, it is at the same time fairly certain that the earlier Cretaceous rocks were deposited upon a highly irregular crystalline floor, the more elevated portions of which have been exposed at the surface in the course of denudation. For the same reason also the apparent base of the Cretaceous in the neighbourhood of these inliers of crystalline rock cannot be regarded as the actual base of the series, which is at most very imperfectly known.

Muri Province.

In Muri province a series of limestones and shales definitely known to be of Upper Cretaceous and Turonian age^[71] is exposed in western Muri around Awe to the north and around Arofu and Akwana to the south of the Benue. In eastern Muri, the same series is found at Kumberi on the right bank of the Benue a few miles above Amar. At Awe a sharp little anticline runs E.N.E. along the north side of the town and brings up coarse white felspathic grits and sandstones with two bands of carbonaceous shales (Plate XV, [Fig. 1](#)). It begins to the north-east of the town as a gentle arch with its axis dipping gently westward. The intensity of the folding increases rapidly in that direction until the anticline passes into a fault with the downthrow towards the north, but farther west the fold reappears and speedily flattens out. The northern limb of the fold is much more steeply inclined than the southern and the beds are even in some places overturned. At the same time the sandstone outcrop takes the form of a bow with the concave side to the north, the whole structure being evidently produced by a local intensification of the folding. (A. L.)

To the north and south of Awe, there are synclinal areas in which rocks higher in the series have been preserved. From the various exposures in the neighbourhood the following approximate sequence can be made out:

Soft white quartzose sandstones, often with calcareous cement.

Thinly-bedded light-coloured calcareous shales and sandstones with bands of impure shelly limestone.

Greenish shales with ironstone bands and nodules.

White grits and sandstones and carbonaceous shales with brine springs.

Base not exposed.

(A. L.)

Grits and sandstones similar to the lowest beds seen at Awe, with brine springs but without carbonaceous shales, come up to the surface again along a line running from Akiri to Azara and Ribbi, but north of that again comes another synclinal area, for between Azara and the Ankwe river exposures of shales and limestones are met with. Over the whole of this part of Muri indeed the beds appear to be thrown into gentle anticlines and synclines whose axes run approximately E.N.E. A lower and essentially

sandy division, often yielding brine, appears along the cores of the anticlines as at Azara, Akiri and Awe, while an upper, essentially shaly division, is preserved in the intervening synclinal areas. (A. L.)

Limestones are known to occur in the volcanic tract on the Benue, south-west of Awe, while at Tunga wharf, immediately south of Awe, a river section exposes the lower sandstones striking E.N.E., and dipping northwards at 27° . At Arofu a small inlier of granite to the east of the town is seen to be five-sixths surrounded by silicified limestone with the remainder of its perimeter concealed by marshy ground. An excavation proved the limestone to rest directly on the granite and we are therefore compelled to conclude that this granite forms a knoll surrounded and at one time overlaid by the limestone. Whether similar relations hold good between the other inliers of crystalline rocks in the neighbourhood and the limestone is not certain. The outcrop of the limestones is a broad one, about half a mile across in the vicinity of Arofu and still more to the south of Akwana. The broad outcrop is accounted for partly by the gentle dip, partly perhaps by gentle undulations of the strata, but partly also by the thickness of the limestone. What this is it is impossible to say in the present state of our knowledge, but it is certainly much greater than that of the limestone bands known in the neighbouring parts of the province. (A. L.)

The saltings of Arofu and Akwana bear witness to a salt-bearing series underneath the limestones although that series itself is nowhere exposed. Above the limestones comes a series of shales, shaly sandstones and ironstones with a few thin intercalated limestone bands. This shaly series outcrops between Arofu and the river and runs a little south of west to beyond Akwana. The dip is about 10° and almost north. At Shaii however, twelve miles east of Arofu, similar shales dip at 25° towards N. 40° E., so that the folding is somewhat variable. Around Arofu the limestones are seldom seen in an unaltered condition, for almost all the rock exposed is silicified. Thus the country is crossed by innumerable dyke-like masses of silicified limestone standing up several feet above the soil. In the rising ground south of Akwana the silicified bands give rise to several small cascades. Knots of galena are common in the silicified limestone and in places as at Arofu there are veins of galena and fluorspar. The original structures of the limestone which was partly foraminiferal but very largely

oolitic and pisolitic have been excellently preserved during the change. At Arofu, however, a compact unaltered variety contains imperfect remains of ammonites. It is possible that these limestones which lie to the south of the Benue may be on a somewhat different horizon from those of Awe, but the presence of the underlying salt-bearing series indicates that the general succession is the same. (A. L.)

At Kumberi some miles above Amar there occurs the following excellent riverside section of the limestone-shale series, in which the subordinate character of the limestone bands is very evident:

Occasional sandstone seen in the river bank:—

About 100 ft.—Fine-grained sandstone, mudstone and shales alternating as below.

- | | | |
|-----------|---|--|
| 12 | „ | Sandstone. |
| About 50 | „ | Shales. |
| 2 | „ | Good white limestone with numerous teeth, bones, and phosphatic casts of lamellibranch and gasteropod shells. Ammonites. |
| 6 | „ | Shales. |
| 1½ | „ | Shelly limestone. |
| 8 | „ | Shales. |
| 2 | „ | Friable sandstone. |
| 10 | „ | Shales with mudstone bands. |
| 15 | „ | Blue-gray shales. |
| 20 | „ | Shales with mudstone bands. |
| 2 | „ | Mudstone with large calcareous concretions round ammonite shells. |
| 5 | „ | Shaly sandstone. |
| 10 | „ | Friable sandstone. |
| 2 | „ | Sandy shales. |
| 10 | „ | Soft, friable sandstones with shaly partings. |
| About 200 | „ | Fine-grained and soft sandstones, mudstones, and shales alternating as above. |

Occasional sandstones in river bank.

Total: 450 to 500 ft. Limestone : 3½ ft.

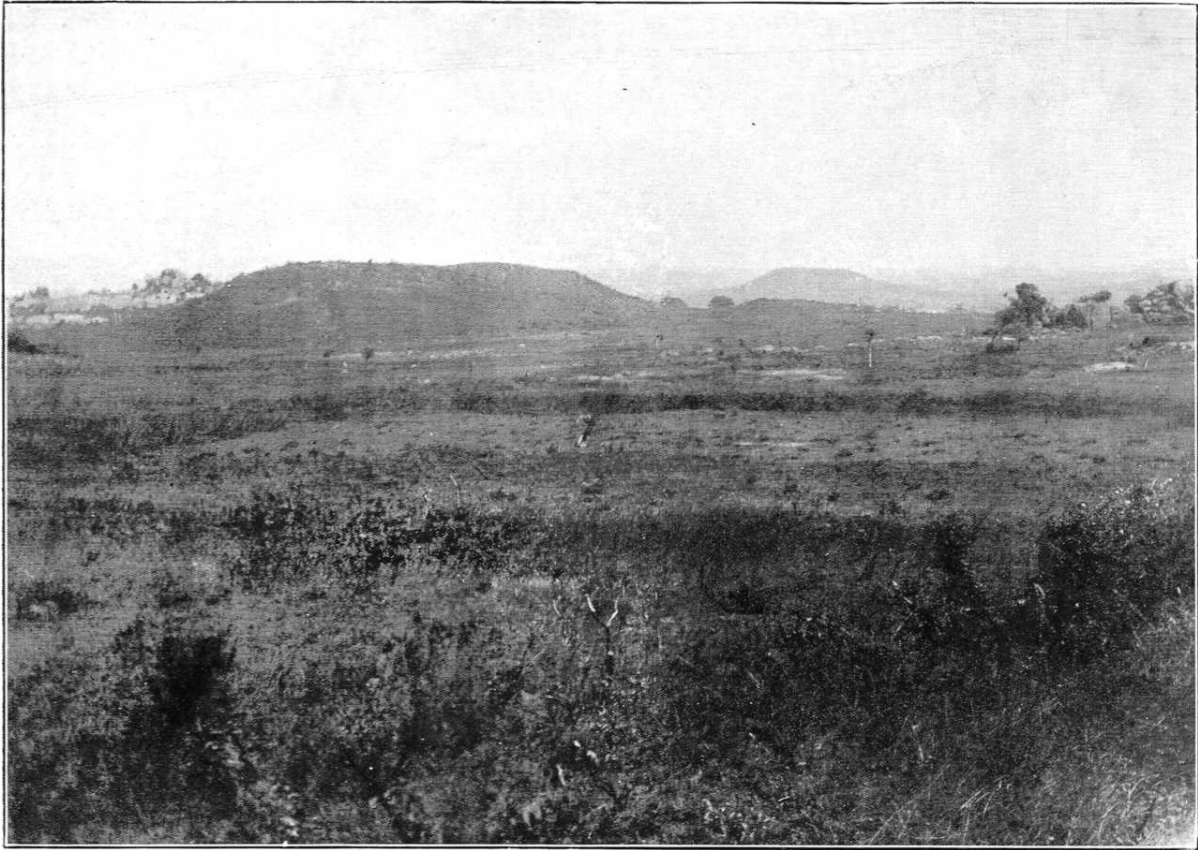
Strike uniformly N. 50° E.; dip, 30°–35° S.E.

(A. L.)

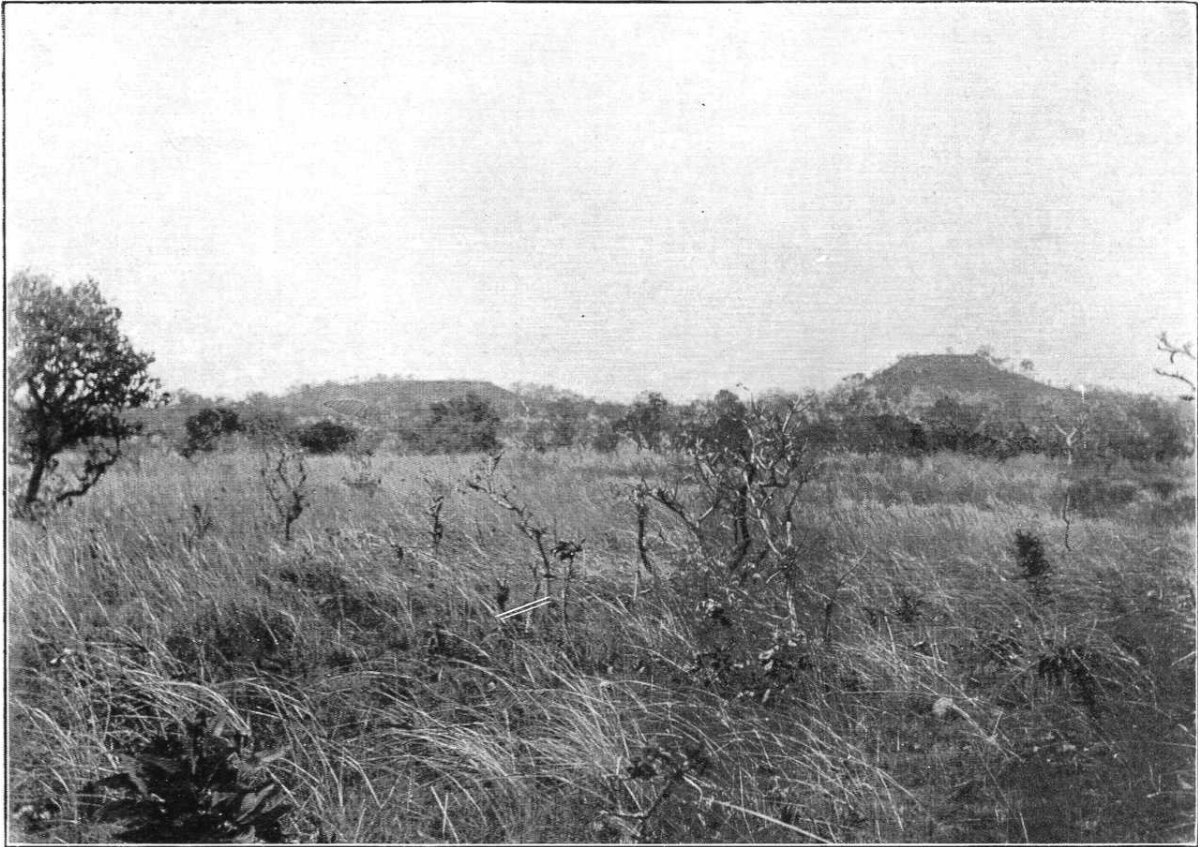
Farther east again scattered fragments of limestone indicate the presence of the limestone-shale series in a synclinal between Bomanda and Muri. An anticlinal axis, running a few hundred yards south of Bomanda, brings up a lower series of soft thinly-bedded calcareous grits with partings of green calcareous marl along the crest of the arch. Brine springs like those of Awe rise through the northern limb. At Muri the lower reddish grits are seen in full force and dip southwards at about 60°. A careful estimate of the thickness exposed worked out at over 1,500 ft. though neither the base nor the upper limit was seen. To the north fragments of limestone are found in the swamps of the Kudu valley to the west of Jebjeb and Gateri but the associated rocks have nowhere been observed. (A. L.)

The white grits and sandstones and carbonaceous shales of Awe and the calcareous grits and marls of Bomanda represent the passage beds between the upper limestone-shale series and the lower sandstones exposed at Akiri, Ribí and Tunga wharf in the west and at Muri town in the east of the province. The weak brine which rises in springs through the passage beds at Awe and Bomanda and even through the limestones at Akwana appears to take its origin in the upper part of the lower sandstone series. Brine springs and surface incrustations are found also where the lower sandstones outcrop at Jebjeb, Jebero, Jenoe, and Langa in eastern Muri, at Akiri, Azara, Ribí, Kanje and Abune in western Muri, and at Doya and Keana on the borders of Nassarawa. These localities occur immediately round the margins of the more marked synclinal areas containing the remains of the limestone-shale series, a fact which warrants the assumption that the upper part of the sandstones underlying the limestones is impregnated with salt and was originally accumulated under arid conditions. Indications of the salt-bearing strata are also found at Lafin Gisheri between Tunga and Ibi and at a place of the same name between Shishi and Wurio to the east of Ibi. In these localities we are justified in assuming that the rocks immediately below the surface belong also to the upper part of the lower sandstone series.

To the north of the salt area in western Muri well bedded sandstones are exposed in many places between Lafia in Nassarawa province, and Wase and Bashar to the north of Ibi. The prevailing type of rock is a reddish grit or sandstone, frequently brick red, with knots and spots of white felspathic material, everywhere unfossiliferous, and nowhere yielding any indication of salt impregnations. At Bashar thin beds of white and red clay are interstratified with the sandstones. The rocks are usually inclined at a low angle to the horizontal and appear to be thrown into a series of gentle undulations. From their position with respect to the salt-bearing beds and the limestone-shale series, it may be assumed, although continuous exposures are wanting, that these rocks lie conformably at or near the base of the sedimentary succession in western Muri. Their thickness is unknown, but if, as seems probable, they are to be correlated with the sandstones of Muri town which underlie the limestones in the eastern part of the province, they must be, according to Mr. Longbottom's determination ([p. 151](#)), at least 1,500 ft. thick, and they may well be more. The whole series underlying the passage beds of Awe and Bomanda and impregnated with salt in the upper part may conveniently be termed the "Muri sandstones."



HILLOCKS OF DRIFT ON THE BAUCHI PLATEAU NEAR VOM.



HILLOCKS OF DRIFT TO THE WEST OF NARAGUTA.

Upon the drift-covered plains of southern Muri are found occasional detached exposures of sandstones and grits, the majority of which probably belong to the lower sandstone series. The presence of the "Muri sandstones" underneath the limestones of Arofu has not been demonstrated but, as already stated, the occurrence of brine springs at Akwana affords a considerable presumption in favour of their occurrence. Whether they outcrop in the unexplored Munchi country to the south and south-west of Akwana is, however, quite unknown.

In the north-eastern part of the province the "Muri sandstones" become more steeply folded, and in the Jenoe, Ligri, and Gateri hills appear to be underlaid by a series of very coarse-grained white felspathic grits or arkoses. The axes of the folds strike in a general way a few degrees north of east, and the folding appears to be most intense in the immediate neighbourhood of Gateri where the grits are much hardened, and in places set on end. On the western slopes of the Gateri hills are white and red grits

and sandstones, the representatives of the “Muri sandstones,” dipping generally westward and apparently underlying the fragmentary limestones of the Kudu valley. In the Gateri hills, moreover, there are indications, as at Awe, that the axes of the folds are not horizontal, but tilted gently to the west, and the distribution of the upper limestones in Muri province in elliptical areas or irregular centroclinals surrounded by the lower sandstones clearly suggests a later transverse corrugation of the whole series of gently folded rocks. It is probable, therefore, that the limestones of the Kudu valley, although originally occupying a much higher place in the series than the Gateri grits, have reached their present relatively low position, not so much as the result of faulting, but rather as the result of the irregular folding to which the whole series has been subjected. At the same time the occurrence of galena veins in the swamps to the west of Jebjeb indicates that the rocks of the Kudu valley, like those of Awe and Arofu, have undergone a certain amount of shattering and fissuring contemporaneously with the folding. Amongst the fissure-veins may also be included the broad dykes of limonitic ironstone which form conspicuous hillocks around Akiri in western Muri.

Yola Province. (A. L.)

The limestone-shale series runs up the plain of the Benue from Lau almost to Yola, and may even be found eventually farther east than Yola. This series occupies the lower ground and nowhere forms hills. In addition to it there are sandstones of various kinds, sometimes occupying the plain and sometimes building up hills above the plain. The sandstones of the hills are approximately horizontal and rest upon the lime-stone-shale series. They will be referred to as the “Upper Benue sandstones.” The question of their conformability or otherwise to the limestone-shales will be discussed later ([p. 188](#)).

The limestone-shale series, being soft and easily denuded, is very poorly exposed, and is generally hidden by the river alluvium. Calcareous sandstone and greenish yellow shale are seen near Lau, and south-west of Kunini large blocks and slabs of shelly limestone a few inches thick lie scattered about over the silt. An excavation at Kunini proved that the

limestone is in thin bands, 12 or 18 inches thick, intercalated between shales. The series is thrown into small anticlines and synclines. Of one small anticline noticed the strike of the axis was N. 10° E., one limb being inclined at 20°, the other at 60°. The shales are seldom seen but they give rise to a characteristic soil which is easily recognised and which leads to the discovery of limestone lumps and fragments on further search. Low-lying country of this character, with numerous slabs and lumps of limestone lying about on the surface stretches up the Benue to the vicinity of Namtari.

Similar areas of limestone-shales are found on the Gongola between Kiri and Shillen and between Mada and Kombo. The latter area runs up the plain of the Maio Hawal, but for what distance is unknown. In both areas the limestone is shelly and very fossiliferous and there are no sections showing the shales or giving the dips. The only natural section of this series which was met with in Yola province was at Numan. Just to the east of the village, white fine-grained sandstone crops out and is followed by still finer sandstones and mudstones on which the village itself is built. No limestone appears in the section and only a little shale. The dip is 10° towards N. 55° W. The mudstone is usually reddish, but the fresher parts are greenish and glauconitic-looking. Phosphatic nodules are found in this material and also in lumps of shelly limestone set up by the natives in Dong. A small inlier of Cretaceous limestone is exposed also between Jibaru and Murki in eastern Yola to the north of the Bagele hills.

The thick grits and sandstones which emerge from beneath the limestone-shale series in Muri province are probably responsible for the bold escarpment behind Wuduku, but in the absence of detailed examination it is impossible to say if these infra-limestone grits were ever deposited in Yola province, and if, being so deposited, they crop out between the limestone-shale series and the crystallines or are faulted out. The problem, a simple one in itself, is enormously complicated by the fact that the "Upper Benue Sandstones" rest upon the limestone-shales and conceal their boundary with the crystallines. Consequently, the relation of the Cretaceous rocks to the crystallines is at present unknown.

The Cretaceous rocks floor the trench of the upper Benue valley, while the plateau areas to the south and, east of the Gongola, to the north also, are

of crystallines. It is improbable from the characters of the limestone-shale series that these beds were formed in a narrow inlet running up into a land mass. On the contrary they are fine-grained marine sediments which were probably deposited in open water. Their present position, on the floor of the valley, at a much lower altitude than the crystalline plateaux north and south must be explained as the result of tectonic movements. The wrinkled character of the series is consistent with this explanation, and in this connection sections in the neighbourhood of Kombo are suggestive. There the broad channel of the Gongola, usually half a mile wide, becomes suddenly constricted to less than a hundred yards by a ridge of grits dipping at 50° towards S. 30° E. These grits pass under limestone-shales to the south-east, and must therefore be correlated with the Awe and Bomanda grits. When followed eastward, however, for only a few miles, this prominent ridge of grits disappears, and the limestone-shales are seen to approach to within at least a few yards of the granite. This strongly suggests that the limestone-shale belt running up the Maio Hawal is faulted down to abut against the granite. In a similar way, it is believed that the Upper Benue valley, to the east of its confluence with the Gongola, was initiated as a tectonic valley. (A. L.)

Bauchi and Bornu.

To the north as to the west of Gateri, a series of white sandstones intervenes between the Gateri grits and the limestones which outcrop in the swamps between Putu and Tangalto. These limestones are exposed in a very fragmentary manner, but the presence of a narrow belt of the lower white salt-bearing sandstones in the midst of the limestones half-way between Putu and Tangalto indicates here as elsewhere the occurrence of the usual succession and an irregular folding of the whole series. Approaching Tangalto the Cretaceous rocks run out upon the crystallines as white sandstones and quartzose conglomerates whose horizon probably coincides with those of the Ture grits to the east. In the hilly country to the east of Tangalto and on the plains to the north, the inclination of the Cretaceous rocks is in part due to a gentle folding, and in part to an extensive faulting which has in places on the Gongola plain brought up even the base of the

series and the underlying crystalline rocks to the same level as the limestones. Around Ture (Plate XIII, [Fig. 1](#)) and Awok, the varied dip of the well-bedded sandstones indicates a gentle curving into low domes and shallow basins, while a similar structure is very evident at Gulani, to the east of the Gongola, where the town is set within a circle of low hills which possess a well-marked centroclinal dip. On the other hand, the Wadi, Shinga, Bima and Gunna hills, are separated by faults, and owe their greater northerly inclination to the effect of a larger fault which is well seen at Gunna on the left bank of the Gongola, and on the track between Wadi and Wuyo, where the basal sandstones of the Bima hills with the underlying crystalline rocks are brought up to the surface of the plain. The faulting, like the folding, appears to be in two well-marked directions at right angles to each other and approximately N.N.W., and E.N.E., and it is probable that there is here, as in the neighbourhood of Awe, some intimate relation between the faults and folds, for whose complete demonstration much more detailed investigation would be required.

If the Gunna fault be taken as showing the base of the Cretaceous, the succession in the Gongola valley begins with a series of sandstones and grits at least 1,000 ft. thick. The basal beds are green and purple earthy grits, composed of the debris of crystalline rocks and containing large rounded fragments of granite and pegmatite. These are followed by alternating bands of red, white and purple grits, which pass into the white and red sandstones and grits of the Bima and Wadi hills. The marginal rocks, which overlie the crystallines between Bilaraba and Dusua are white, red, and purple grits, much broken and current bedded and probably belonging to the same horizon as the lower sandstones of the Bima hills. These "Bima sandstones" are succeeded after an unknown interval by the Cretaceous limestones which outcrop in many places upon the plains to the east and to the west of the Gongola, and are particularly frequent over the tract of country which lies between Dukul and Deba Habe to the west, and Wuyo and Gulani to the east of the river. In one locality, a mile to the south of the village of Gongila, between Nafada and Ashaka, the limestone is clearly shown resting upon the lower series of white and red sandstones and grits. As a rule, however, the rocks associated with the limestone, are badly exposed and obscured by drift, while the shales which presumably

accompany the limestones have nowhere been observed. At Gulani, the limestones pass conformably under a series of coarse white and red grits and white, yellow and brick red sandstones with intercalated bands of clay and flaggy ironstones. With these upper rocks the red and white sandstones, clays, and ironstones of Wuyo present much lithological resemblance and probably belong to the same horizon. Similarly at Reme in the extreme south-east of Bauchi province the limestones are overlaid by a well-stratified succession of white and red sandstones, grits, and flaggy ironstones, which are at least 800 ft. thick, and form the escarpments of the flat-topped hills of Waja. The whole series becomes markedly ferruginous towards the summit, and while the contact with the limestones below is not actually exposed, there is no reason to believe that it is other than conformable. These rocks in Waja represent the highest known members of the Cretaceous series, and it is they which are continued westward by Tula, Awok, and Ture, and run out upon the crystallines to the east of Chongwom. Around Awok and Ture these upper sandstones are distinctly and variously inclined, but in Tula and Waja the dips are so gentle as to be almost imperceptible. The white grits and sandstones are usually unfossiliferous, but in the neighbourhood of Ture and the Peak (Plate XIII, [Fig. 2](#)), they contain much silicified wood from the presence of which a subsidiary argument may perhaps be drawn in favour of their Cretaceous age. The Gongola valley to the east and north-east of Waja retains the faulted character which it exhibits to the north, while the country which lies between the escarpments of the Waja hills on the north, and the Yam Yam hills fronting the Benue on the south, is quite unknown and may contain representatives of both granitic and sedimentary rocks.

From the various exposures described, the Cretaceous series within the Protectorate may be tabulated as follows:

	Muri and Yola.	Bauchi and Bornu.
Upper grits and sandstones		Sandstones, grits, clays, and ironstones of Waja, Ture, Gulani and Wuyo.
Limestones	Turonian limestones and shales of the Benue valley	Turonian limestones of the Gongola valley.

Lower grits and sandstones	Passage beds (white sandstone and carbonaceous shales at Awe: calcareous grits and green calcareous marls at Bomanda). Muri sandstones (upper part salt-bearing). Gateri grits.	Bima sandstones and grits.
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No evidence of contemporaneous volcanic activity has been found within the Cretaceous rocks of the Protectorate. Intrusive sills of dolerite, however, have been observed within the Cretaceous sandstones in the neighbourhood of Awe, Bashar and Serakin Kudu in Muri province, but they, together with the numerous pipes and necks of basalt and trachyte which pierce the Cretaceous rocks in the valleys of the Benue and Gongola, are probably to be referred to some phase of Tertiary volcanic activity (see [Chapter VIII](#)).

In the French Sudan limestones of Turonian age have been described from the Damergou^[72] to the north of Zinder, from Bilma to the north of Lake Chad,^[73] and from Tabankort, Tabrichet and Mabrouka^[74] to the south and west of the Adr'ar' des Ifor'as. The occurrence in the Damergou of gypsiferous clays in association with Cretaceous limestones has not been paralleled in Northern Nigeria. On the other hand the presence of a salt-bearing series within the Cretaceous has not hitherto been recognised in the Sahara. Any correlation of the lower grits and sandstones of Northern Nigeria with the Lower Cretaceous of the Tegama, as established by M. Chudeau,^[75] would, however, be premature, in view of the absence of determinative fossils in both and the marked lithological differences between the two. It seems better for the present to avoid any definite subdivision other than that indicated above and to refer the whole series of grits, sandstones, clays, ironstones and limestones provisionally to the Upper Cretaceous. To the south of the Protectorate rocks of a similar age have been described by Mr. Parkinson^[76] and Dr. Esch^[77] from Southern Nigeria and the Kameruns respectively.

[70] See Chudeau, *Sah. Soud.*, 1909, pp. 47, 11, for pre-Cretaceous rocks of the Sahara.

- [71] See [Appendix II](#), *Palæontology*, p. 273.
- [72] De Lapparent, *C. Rd. Ac. Sc.*, 135, 1903, p. 1298; Chudeau, *Sah. Soud.*, 1909, p. 89.
- [73] De Lapparent, *C. Rd. Ac. Sc.*, 132, 1901, p. 388.
- [74] Chudeau, *Sah. Soud.*, 1909, p. 91.
- [75] *Ibid.*, p. 76.
- [76] Parkinson, "The Geology of the Oban Hills," *Q. J. G. S.*, 63, 1907, p. 314.
- [77] Esch, *Geologie von Kamerun*, 1904, p. 4.

CHAPTER V

EOCENE ROCKS

The Sokoto and Lokoja series: the nummulitic limestone of Sokoto; the upper ferruginous beds; the pisolitic clay of Giro; the white and pink clays of Kurukuru; the partial colouration of the lower sandstones and clays; the Lokoja sandstones and ironstones; the pisolitic ironstones of Mount Patti; outliers of the Lokoja series. The Duguri sandstones and grits: the clays of Pali, Kirifi and Deu; the Gombe sandstones; the ironstones of the Tagum hills; the grits and ironstones of Kerri Kerri; the age of the Duguri, Gombe and Kerri Kerri sandstones. The Upper Benue sandstones: their relation to the Cretaceous and crystalline rocks of Yola; age of the Upper Benue sandstones; outliers of Eocene sandstones in Muri province; Tertiary history of the Upper Benue valley. The Eocene rocks of the Upper Niger and the Kameruns. The stratigraphy of the Chad area.

THE dawn of the Eocene was marked by regional upheaval, by the folding and fracturing of the Cretaceous rocks and by subsequent intense erosion, as the result of which the underlying crystalline floor was again laid bare over the greater part of the Protectorate. A period of depression, however, ensued, during which the sandstones, clays, limestones and ironstones of the middle Eocene were extensively and unconformably accumulated both upon the crystalline rocks and upon the worn and denuded edges of the Cretaceous. The existence of this hitherto unsuspected unconformity between the Eocene and the Cretaceous introduces a new element of complication into the history of the Sudan and the southern Sahara.^[78] In the opinion of M. Chudeau^[79] deposition was continuous at least from the lower Turonian to the close of the Eocene, while the early Eocene was marked by a widespread transgression of the waters over the Niger valley. The presence of the unconformity, however, makes it evident that the known distribution of the Cretaceous cannot in any way be taken to indicate the former limits of the Cretaceous sea.

The Eocene rocks within the Protectorate may be conveniently classified into three groups:

1. The Sokoto and Lokoja series.
2. The Duguri, Gombe and Kerri Kerri sandstones.

3. The Upper Benue sandstones.

1. *The Sokoto and Lokoja series.*

The occurrence of fossiliferous limestones of Eocene age at Kalfu, Tamaske, Garadimi and Toronkawa to the north of Sokoto has been known since 1903.^[80] The limestones have been found, however, to extend much farther south and to be associated with sandstones, clays and ironstones which are continuous with those in the Niger valley from Ilo to Lokoja. One of the finest sections of the limestone series is found in the cliffs facing the river beyond Wurnu to the north-east of Sokoto where the following succession may be observed:

- 10 ft.—Dark red ferruginous grit on top.
- 10 „ Ferruginous oolite.
- 30 „ Soft, chalky, fossiliferous limestone with phosphatic nodules and rounded lumps of hard and somewhat phosphatised limestone.
- 10 „ Calcareous clay.
- 80 „ Blue shale, much obscured by landslips.
- 20 „ Compact yellow calcareous clay, with efflorescence of gypsum.
- 40 „ Red, yellow, green and black gypseous shales and clays.
- 50 „ Black carbonaceous shales with radiating crystals of gypsum alternating in thin bands with yellow ferruginous clays with knots of ironstone; efflorescences of alum and gypsum.

At Wurnu the same succession is represented by red sandstones above and white, yellow and blue clays and shales below, with fragments of limestone at the foot of the slope indicating a thin band of limestone intercalated probably near the summit of the shales. Around Sokoto there is much earthy ferruginous oolite on top passing downwards as a rule into well-bedded red sandstones. Below the Wurnu gate, however, there is an exposure of about thirty feet of blue and white calcareous shales and clays with thin bands of hard limestone 3–4 inches thick, numerous fossils and occasional fragments of gypsum. At Alkamu, near Sokoto, a chalky limestone with nodules and echinoids, associated with calcareous shales

and overlaid by earthy oolite, encloses rounded lumps of harder nummulitic and crinoidal limestone which look as if they had been originally washed into place. This appears to be the first record of the occurrence of nummulitic limestone in West Africa. A similar hard nummulitic limestone is found as far south as the neighbourhood of Jega and more detailed investigation would probably reveal its presence in many localities between Sokoto and Jega. The limestones, however, nowhere form thick and massive beds, but appear to occur always in thin bands or as nodules in calcareous shales or clays. The occurrence of gypseous shales below the limestones at Wurnu and Sokoto indicates the earlier presence of arid conditions, but whether or not the various exposures of limestone occur on one and the same horizon has not been determined.

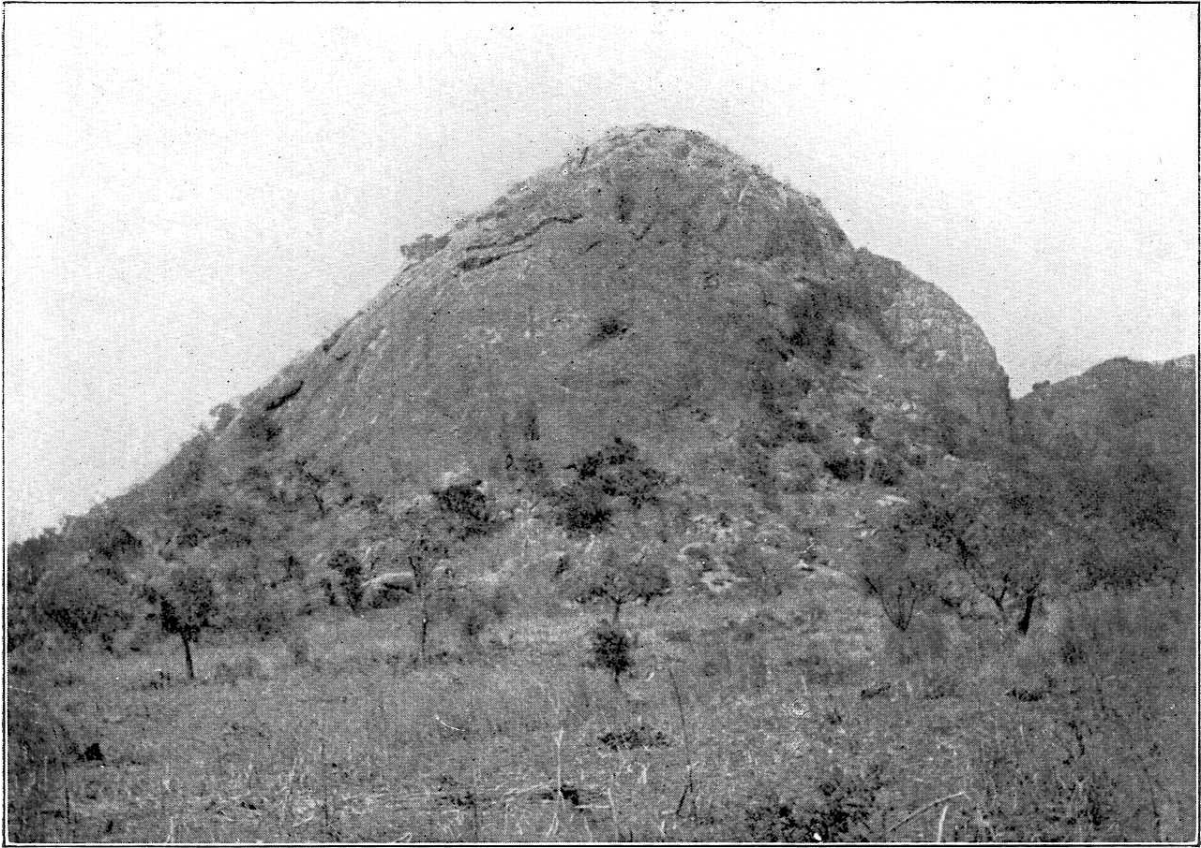
Neither limestones nor calcareous shales are known from the immediate neighbourhood of the Gulbin Kebbi below Sokoto.^[81] There are, however, numerous indications in the cliffs which bound the valley of a lower lighter-coloured earthy series overlaid by a succession of gritty and ferruginous beds. At Tozo white clays and earthy sandstones with thin bands of limonite are covered by oolitic ironstone which is overlaid in turn by hard red sandstone and pebbly ferruginous grit between Tozo and Kantami. The escarpments of Gulma are composed of soft white earthy grit below, becoming redder and more sandy towards the top and capped by a layer of hard ferruginous sandstone. At Birnin Kebbi the lower massive white and yellow earthy sandstones become mottled red and pass into dark red ferruginous sandstones on top. At Giro the lowest rock exposed is a yellowish pebbly grit with intersecting silicified ribs which passes in places into a friable white pebbly marl. This is followed by a thick (100 ft.) bed of white clay, frequently stained red along the joints and irregular cracks, almost every specimen of which shows a pisolitic or concretionary structure. Traced southward this bed becomes much silicified in its upper part, with retention of the original structure, and all stages may be traced between the normal pisolitic clay and a pure silicified and somewhat opalescent pisolite. Northward and eastward red and yellow sandy beds and friable pebbly grits become intercalated in the clay, while fragments of a fine-grained and very ferruginous sandstone afford indications of the former presence of ferruginous beds on top. Fragments of a similar silicified clay

are found as far south as Marake where the prevailing rock is a brick red earthy sandstone, in places spotted with feldspathic knots. At Kurukuru, near Ilo on the Niger, soft white and pink clays are again prominently exposed under a cap of ferruginous oolite. A pisolitic clay very similar to that of Giro has been described by M. Hubert^[82] from Gaya on the Niger.

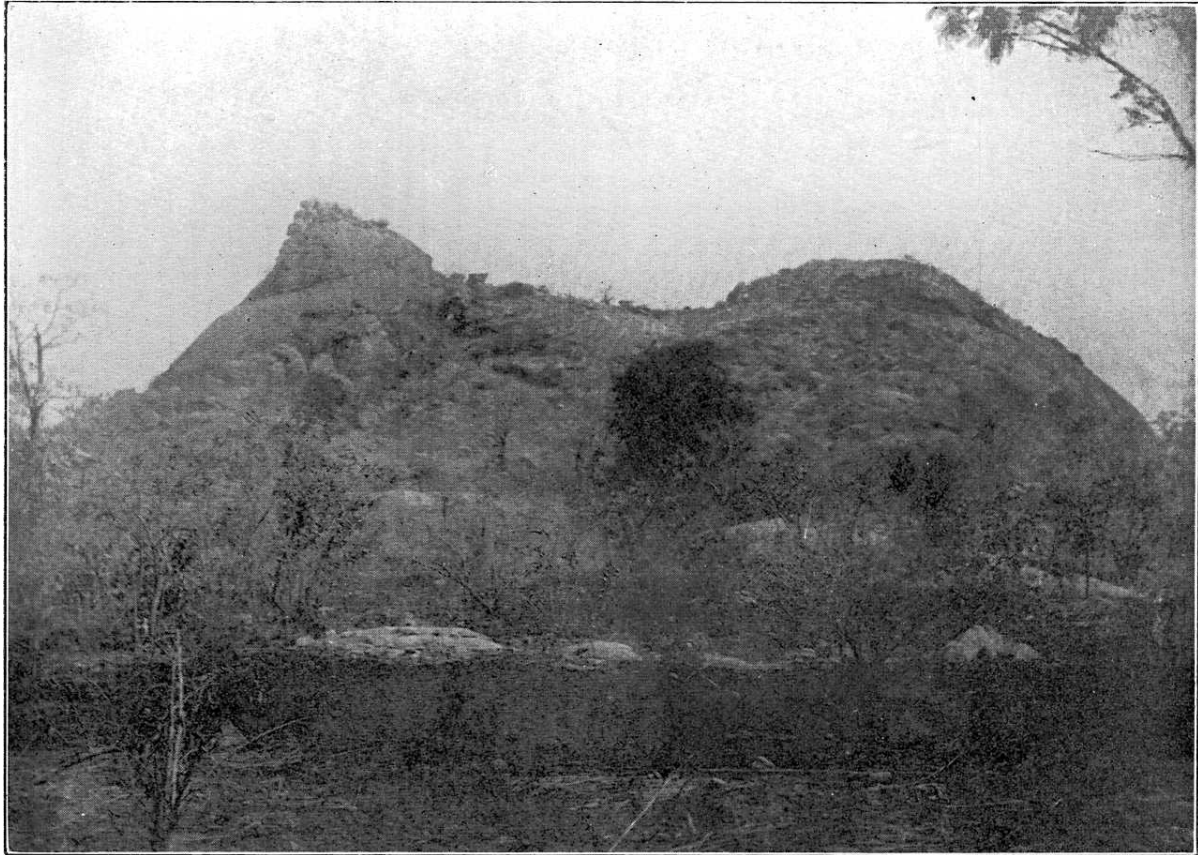
The sedimentary rocks run out upon the crystallines as friable pebbly yellow sandstones at Shunga to the east of the Niger, as a hard yellow silicified sandstone at Bakinrue in Borgu to the west of the Niger, and farther south as ferruginous sandstones and conglomerates at Konkwesso and Kabe, and soft white earthy sandstones on the Niger to the west of Warra. On the plateaux throughout Sokoto province the prevailing rock belongs to the upper ferruginous series of hard red sandstones, grits and ironstones, the softer and lighter-coloured rocks below being exposed for the most part only in the cliffs which bound the river valleys. The occurrence of outliers of still higher beds upon the surface of the plateaux indicates that the actual summit of the series has probably nowhere been observed.

There can be little doubt that the partial colouration of the lower beds, and to some extent also the hardening of the upper sandstones, is due to some considerable extent to the percolation of iron-bearing solutions from above downwards. There is equally little doubt that the oolitic ironstones are of primary and not of secondary or metasomatic origin and that the deposition of the ferruginous material took place during the latest stage in the history of Eocene sedimentation. All gradations indeed can be traced between a pure oolitic ironstone and a ferruginous sandstone with scattered oolitic grains of limonite. M. Chudeau^[83] insists that even where enclosed between thick beds of clay, the oolitic ironstones are in every case the product of the decalcification of limestone. He has himself, however, supplied a sufficient answer to his own argument in calling attention to the fact that the limestones where now exposed never show any trace of even a superficial alteration into ironstones and in admitting, as he does, the truth of M. Hubert's^[84] contention that the oolitic ironstones of the Niger valley are nowhere accompanied by or traceable into limestones of a similar character. The flints of Ansongo,^[85] moreover, like the nodules of Sokoto,

indicate nothing more than the local occurrence of limestone within the Eocene accumulations.



A GRANITE MOUNT NEAR BAUCHI.



A GRANITE MOUNT NEAR BAUCHI.

The sandstones of southern Kontagora and of the Niger valley below Jebba, which may be conveniently grouped as the "Lokoja series" from the locality of their greatest development, represent the southward continuation of the Sokoto sedimentary series and more particularly of the sandstones, grits, conglomerates and ironstones of the southern portion of the series, lying below the confluence of the Gulbin Kebbi and the Niger. The break in continuity at Warra is more apparent than real and is due merely to the fact that the Niger has entirely removed the sedimentary rocks from its channel and thus exposed the underlying crystallines. A well-marked difference in character may be traced between the Eocene rocks of Sokoto province and their southern representatives, a difference, however, which is entirely due to a dissimilarity in the earlier conditions of sedimentation. In the Sokoto series there is a well-marked succession of marine and deeper water accumulations by shallow water and lacustrine accumulations. Amongst the southern sandstones, however, there is no trace of a marine series, no trace of limestones or of carbonaceous or calcareous shales and even clays are

conspicuous by the rarity of their occurrence. The series throughout, which attains at Lokoja a thickness of 1,000 feet, is a shallow water accumulation of sandstones, grits, conglomerates and ironstones, comparable to some extent with the upper shallow water portion alone of the Sokoto series. The resemblance indeed between the uppermost members of both series is heightened by the prevalence in both of oolitic and pisolitic ironstone, and it seems probable that, in spite of their earlier dissimilarity, the conditions towards the close of sedimentation in Eocene times were identical over the whole area from Sokoto to Lokoja. A further resemblance between the Lokoja series and the rocks of the lower valley of the Gulbin Kebbi is the frequent staining of the lower members of the series by descending ferruginous solutions. Wherever the lowest rocks of the Lokoja series are exposed they are almost invariably light-coloured sandstones and conglomerates. The colourisation begins along the joints and fissures and spreads inward as an irregular pink or reddish mottling. Traced upward the rocks gradually become uniformly coloured and the tints gradually deepen until the rocks assume the characteristic dull dark red or chocolate colour of the uppermost members of the series. The staining in places, owing probably to local differences in texture, has reached a greater depth than in others with the result that at certain points within the series the lower lighter-coloured rocks appear to attain a less development than in others. There is, however, to be observed in places towards the upper portion of the series an alternation of more and less ferruginous beds underneath the richly ferruginous cap of chocolate-coloured sandstones and ironstones, while here and there intercalations of ferruginous beds are to be found even within the lower lighter-coloured members of the series. In northern Borgu, moreover, it is common to find the basal conglomeratic beds richly ferruginous, a fact which is probably to be explained as due to the gradual extension of the marginal swamps during the advent of shallow water conditions.

The outliers of the Lokoja series which lie to the north and north-east of Kontagora belong to the uppermost portion of the series and consist for the most part of oolitic and pisolitic ironstones. Immediately south of Kontagora, soft white sandstones and earthy grits are exposed in the gorge of a stream at Kagara. Again at Sonko in a narrow canyon valley a fine-

grained hard ferruginous sandstone is intercalated between beds of yellow gritty clay. Between Sonko and Jebba, there is a great development of ferruginous sandstones, ironstones and grits which extend also over a large portion of western Nupe. In western Kontagora, yellow grits and sandstones occur round Auna, while at Shaffini yellow and reddish pebbly sandstones form the base of the hills of ferruginous grit which rise above the plain. At Kumboji the rocks are red pebbly sandstones and ferruginous grits, while at Chika a lower bed of yellow sandstone is succeeded by red grit and ferruginous conglomerate.

On the Niger below Jebba, ferruginous sandstones and gritty ironstones are exposed at Rabba on the left bank, and red sandstones and pebbly grits at Shonga on the right. At Share white pebbly sandstones and conglomerates with thin bands of clay occur below the ferruginous grits which form the summit of the tabular hills. At Padda, to the east of Pategi, the basal beds are white conglomerates and sandstones, knotted with fragments of felspar and passing upward into the alternating series of white and reddish sandstones which rise into the pinnacle peaks of Egbom. On the banks of the Kaduna above Wuyo, white earthy sandstones are frequently exposed passing upward in the isolated hills through alternations of white and red sandstones into hard ferruginous grit and ironstone. Throughout southern and western Nupe, the rocks, where exposed, are invariably red sandstones or ferruginous grits. Between Kacha and Baro, however, the lower beds are white or mottled red and pass upward into a succession of red and white sandstones and grits. A similar alternation occurs in the base of the outlying Jakpara hills in northern Kabba which are capped by red ferruginous grit.

From Baro southward along the Niger and eastward into Nassarawa and Bassa, oolitic ironstones and earthy limonites associated with ferruginous sandstones and grits attain a greater development while the colouration of the lower beds is in general much more complete. At Baro a white or reddish sandy clay with pebbles of quartz and kaolinised felspar is exposed at the base of Mt. Gidda. This is followed by a thick mass of oolitic ironstone covering the greater part of the slope, in places much decayed and ribbed, banded and knotted with concretionary limonite. The upper portion of the hill is composed of red sandstone and ferruginous grit, the uppermost

beds being well banded with coarser material, and enclosing pebbles of quartz and felspar. At Lokoja the lowest rock exposed at the base of Mt. Patti is a soft white felspathic sandstone with pebbles of quartz, kaolinised felspar and pegmatite which occasionally accumulate into conglomeratic bands. The greater part of the hill, however, is composed of white sandstones and grits indistinctly bedded and stained irregularly, and in places to a considerable depth, with iron ore. In the upper part of the hill the white sandstones give place to ferruginous sandstones with concretionary knots of earthy limonite and intercalated bands and irregular lenticular patches of oolitic ironstone. The uppermost 150 feet are made up of oolitic, pisolitic and earthy ironstones enclosing irregular masses of hard lustrous concretionary limonite, while the summit is capped by a pebbly ironstone composed of small rounded and elliptical pebbles of compact ironstone cemented by earthy limonite containing a few grains of quartz. ^[86]

The prolongation of the Lokoja series into Kabba, Nassarawa and Bassa is accompanied by much variation in the extent and thickness of the lower lighter-coloured rocks, and in the proportions of ferruginous sandstone and oolitic ironstone in the upper part of the series. There are also indications to the north and east of Lokoja of the ironstones occupying in places a much lower horizon than they do in Mt. Patti, as for example in Mt. Gidda where the ironstones are found only a little way above the river level and at Amara on the Benue where a ferruginous conglomerate very similar to that on the summit of Mt. Patti is exposed in the bank of the river. On the Benue, the Lokoja series runs out unconformably and presumably upon Cretaceous rocks in the Doma hills behind Udeni which present the usual succession of white sandstones below and red sandstones above. In Bassa the Lokoja series, although not actually traced, is believed to be continuous with the white and red sandstones and clays with plant remains exposed in the neighbourhood of Idah and Igore on the borders of Southern Nigeria. A small outlier of red and white sandstones and earthy limonite, near Ojerami and Akwi in southern Kabba, bears witness moreover to the former extent of the Lokoja series to the west of the Niger.

(2). *The Duguri, Gombe, and Kerri Kerri sandstones.*

The post-Cretaceous rocks of eastern Bauchi and Bornu, while more or less continuous from south to north, may be conveniently divided into three groups, the Duguri sandstones and grits, the Gombe grits and clays, and the Kerri Kerri grits and ironstones. The series throughout is a well-bedded succession of grits, sandstones, clays and ironstones, usually lighter-coloured below, and dark red and ferruginous above. Although everywhere unfossiliferous the series presents many analogies with the Eocene rocks of the Niger valley, and it is largely upon the strength of those analogies that these eastern rocks are assumed to be also of Eocene age. Their stratification throughout is practically horizontal, though in a few localities a gentle dip may be observed. Their junction with the underlying Cretaceous is believed to be unconformable, but the actual contact has nowhere been observed. Their total thickness, from the level of the Gongola at Nafada to the summit of the Kerri Kerri plateau, is about 1,000 ft., and there is no reason to believe that they elsewhere exceed this figure to any great extent.

In Duguri, the basal beds are not exposed, but in the cliffs behind Tongolan there is an excellent section of 400 feet of white and purple grits capped by hard red ferruginous sandstone, and banked against the crystalline rocks to the west. Between Tongolan and Yuli there is an alternation of white and red sandstones with occasional beds of gritty clay, while at Samia a series of white sandstones and clays overlaid by red sandstones and grits is exposed in the cliffs which bound the valley of the Yuli river. The boundary between the Cretaceous and Eocene is very approximate, and has been drawn provisionally a little east of Samia, where the horizontal sandstones appear to give place to sandstones possessing a distinct inclination to the west and probably belonging to the inclined Cretaceous series. Southward, the white grits of Duguri appear to be largely replaced by red ferruginous sandstones, which are well exposed around Tukkur to the east of Kantana. On the plateau at Kantana, a thin series of red sandstones with bands and streaks of flaggy limonite, underlain by white sandstones, pebbly grits, and conglomerates, rests directly upon the crystalline rocks, while the presence of outliers of hard red sandstone on the summits of the hills around Kanna affords some indication of the originally greater westward extension of the series. The correlation of the various

rocks in Duguri and the Kanna district is rendered difficult both by the rapid lateral variation in character of individual beds and by the occurrence of much faulting along the eastern margin of the plateau. The distinction of a lower and lighter-coloured series from an upper ferruginous series is less marked in Duguri than elsewhere, the lower series being apparently in great part an irregular succession of white and red sandstones and grits.

Over the central plains of eastern Bauchi the prevailing rock is a soft reddish grit in places quite pebbly, in others passing into white and purple sandstones. Exposures are found mostly in the deep and narrow channels which the streams have cut below the surface of the plain. At Pali bands of white gritty clay in every respect similar to the clays of Kurukuru on the Niger ([p. 168](#)) are intercalated amongst the sandstones, some of them containing knots and lenticular patches of purer kaolin. At Panguru rounded masses of earthy and oolitic ironstone are found in the surface alluvium, but it is doubtful whether they are concretions in situ or rolled fragments of an earlier deposit. A thin series of soft white and red sandstones and earthy limonites capped by hard red grit and possessing a dip of 10° S. and belonging probably to the upper part of the series is exposed immediately south of Putu. At Kirifi and Deu on the Gongola, thick beds of white clay similar to the clays of Pali are exposed below harder beds of red sandstone and red ferruginous grit. Similarly at Keffin Serakin Yaki, to the west of the Gongola, banks of white clay occur as large lenticular patches in reddish grit. Between Deu and Gombe white sandstones and clays are frequently exposed in the lower part of the cliffs which bound the river valley, the upper part being invariably red sandstone or hard ferruginous grit. At Gombe a characteristic section is exposed in the hill behind the Residency. A coarse white clay with gritty and sandy bands in its upper part and somewhat stained with iron forms the lower slopes of the hill, while on top are dark red sandstones and pebbly grits passing in places into flaggy or oolitic ironstones. The Gombe grits and clays are continuous within the bend of the Gongola with the grits and ironstones of the escarpments behind Tilde and Dukul which bound the narrow plain of the Gongola in the extreme east of Bauchi province. These eastern escarpments, however, are much higher than the escarpments of the west and the upper ferruginous series attains in them a much greater development. At Dukul the thinly

bedded soft white sandstones which outcrop at the base of the escarpment are covered by a thick series of hard red grits with bands of flaggy limonite. Around Tilde and Ribadu the ferruginous grits alone are exposed, but at Birri to the north the lower series is again exposed as thinly bedded white sandstones under the usual ferruginous cap. To the east of the Gongola, the Gombe series is continued in the low hills of ferruginous grit, conglomerate, and ironstone around Ligdar and Buni, and farther east in the sandstones and ironstones of the broken hills which stretch from Ndufa to Lua in southern Bornu. Around Ndufa yellow and brick-red sandstones are exposed below with ferruginous grits above, containing in places scattered knots of hæmatite and streaks of lustrous pisolite. At Lua the Tagum hills are entirely composed of ferruginous grit enclosing in the upper layers large knots, lenticles, and irregular masses of hard pisolitic limonite similar in every respect to the lustrous pisolite of Mt. Patti at Lokoja. Immediately south of Kojé the lower rocks run out upon the crystallines in alternating bands of yellow sandstone, coarse grit, and fine white clay somewhat stained with iron ore.

In the Kerri Kerri country in western Bornu to the north of the Gongola, the post-Cretaceous rocks are magnificently exposed in the southern escarpments of the Kerri Kerri plateau. The stepped cliffs of Kadi consist of two thick and imperfectly stratified beds of soft white earthy grit, passing in places into a coarser pebbly grit and separated by a thin bed of red ferruginous sandstone with streaks of limonite which forms the floor of the step. The summit of the upper bed is reddened and capped by a thin layer of concretionary ironstone. The detached tabular hills which fringe the plateau are either of the same height as the plateau or cut down to the level of the intercalated bed of hard red sandstone (Plate XIV, [Fig. 1](#)). In many cases the detached hills are stepped like the escarpment of the plateau, and at times the upper grits are represented merely by a conical peak set upon a platform floored by the ferruginous band which separates the upper and lower grits. Farther east the prominent tabular hill behind Fika has a very similar structure. The base is formed of white and brick-red grits upon which is set a thick bed of white earthy grit vertically jointed and bounded by precipitous cliffs. A narrow ledge of hard red sandstone separates this lower grit from an upper bed of similar character somewhat stained with iron,

whose gentler slopes are covered with broken fragments of the red ferruginous grit which caps the summit. Beyond Fika the Potiskum road leads as far as Boza through the trench-like intersecting valleys so characteristic of the margin of the plateau with the rocks well exposed in the vertical cliffs on either side (Plate XIV, [Fig. 2](#)). Almost invariably soft reddish grits are found below and white earthy grits with bands and streaks of white and pink clay above, the whole being capped at Boza by a series of coarse red ferruginous grits. The stratification which is indistinct in the lower part becomes more marked towards the summit of the series and affords indications in places of a very gentle inclination towards the north. The upper ferruginous grits frequently rise into low hills and mounds on the surface of the plateau between Farsawa and Potiskum and near Gojin they contain abundant pisolitic knots of limonite, whose presence renders the grits very similar in appearance to those of the Tagum hills in southern Bornu.

To the west, the Kerri Kerri grits run out upon the crystalline rocks in the neighbourhood of Hardawa, as hard white conglomerates and grits, while to the north and east they are gradually lost under the recent accumulations of blown sand and earthy drift. Between Potiskum and Gujba, the surface rocks belong mostly to the upper ferruginous series, the lower red and white grits being exposed only in the base of the ironstone hills of Durua. The ironstones are very variable in character and include richly ferruginous grits with streaks of limonite, gritty pisolitic ironstones, pure lustrous pisolitic ores and flaggy and earthy limonites, the latter containing in places much decayed and unrecognisable plant remains. East of Abakri outcrops become rare and the last trace of the Kerri Kerri rocks is found at Girboa in some scanty exposures and broken fragments of ferruginous grit. In central Bornu the only known occurrence of rock of any kind is the outcrop of sandstone recorded by Barth in the banks of the Maidugari river.^[87]

It should be remembered that the assumption of an Eocene age for the Duguri, Gombe, and Kerri Kerri grits, clays, and ironstones rests entirely upon analogy. They exhibit frequent resemblances to the Eocene rocks of the Niger valley and more particularly to the Lokoja series and to the southern portion of the Sokoto series. They are throughout shallow water

accumulations, for the most part distinctly bedded, still practically horizontal and presenting, like the rocks of the Niger valley, a succession of a lower lighter-coloured series by an upper and more richly ferruginous series. The lithological resemblances are also particularly striking. The earthy grits of Kerri Kerri can be exactly paralleled at many points in the Niger valley and in the valley of the Gulbin Kebbi, although in Kerri Kerri they attain a much more extensive development. The clays of Pali, Kirifi and Deu are similar in every respect to the clays of Kurukuru on the Niger, while the peculiar pisolitic ironstones and lustrous limonites of Kerri Kerri and the Tagum hills are exactly similar to those of Mt. Patti at Lokoja.

The assumption of an unconformable contact of the Duguri, Gombe, and Kerri Kerri rocks with the underlying Cretaceous rests upon the practically constant horizontality of the former and the practically constant folding and faulting of the latter. Certain broad resemblances, however, can be detected between the uppermost and least inclined Cretaceous rocks of Ture and Waja and the sedimentary series of Duguri, Gombe, and Kerri Kerri. Like the latter the rocks of Waja are of shallow water origin and show a gradual transition from lighter-coloured sandstones below to hard red ferruginous sandstones and ironstones above. They are, however, as a rule more distinctly bedded and inclined and much harder and more compact than the Gombe and Kerri Kerri series, while in lithological character they differ in the absence of intercalated clays in the lower part and in the absence in the upper part of the oolitic and pisolitic ironstones so characteristic of the ferruginous series in the Kerri Kerri and the Tagum Hills.

There is little doubt that in structural and lithological characters the rocks of Duguri, Gombe, and Kerri Kerri present a greater resemblance to the Eocene rocks of the Niger than to the Cretaceous rocks of Waja. The unfortunate absence of palæontological evidence and of any decisive contacts with known Cretaceous rocks is however to be greatly regretted. On the whole it seems improbable that rocks so distinctly inclined and so evidently undulating as the Cretaceous of the Gongola valley should give place so rapidly in the west and north to such an extensive series of horizontal sandstones and ironstones as is to be found in eastern Bauchi and Bornu. The entire absence from the latter moreover of the pipes and necks

of basalt, so abundant in the Cretaceous of the Gongola valley, is subsidiary evidence in favour of a difference of age. It would seem indeed that, as in the case of the Lokoja sandstones and the Cretaceous rocks of the Benue, all the facts are much more readily explicable on the assumption of an unconformity between the Cretaceous and the Eocene in Bauchi and Bornu.

The possibility of contemporaneous volcanic activity during the deposition of the Eocene sandstones and ironstones is discussed in [Chapter VIII](#).

3. The Upper Benue sandstones. (A. L.)

Upon the Cretaceous rocks which occupy the Benue plain in Yola province there rests a thick series of sandstones which will be referred to as the “Upper Benue Sandstones.” The relation between these sandstones and their basement is extremely interesting and yields valuable information as to the history of this part of Nigeria.

The southern boundary of the sandstones is a very irregular one. Thus on passing from Gidan Jauro to Bamga the path proceeds over sandstones for a time, then passes on to granite for a short distance, then on to sandstone (an arkose) again and finally on to the crystallines. North of Bamga the sandstone partly wraps round a small hillock of granite. But in the cliff-like sides of the Yendam Hills the evidence that the sandstones have buried up an old and hilly land-surface is very clear. Thus in [Plate XVI](#) a hill some hundred feet high is seen to be composite. The bulk of the hill (off the plate to the left) is granite, but its base is hidden by conglomerates, which pass upwards into pebbly sandstones. The latter are banked up against the granite and overtop it, thus forming a tabular hill. Many other sections in the district show a similar relation between sandstones and crystallines. (Plate VI, [Fig. 1](#).) There is here no question of a faulted margin, but evidence on all sides that an older and hilly surface has been buried up in sandstones. If now one stands amongst these hills, there is the Benue plain to the north, floored by the Cretaceous limestone-shales, with the “Upper Benue Sandstones” resting upon them; around are the composite hills with the sandstone banked up against the granite, while to the south is

the crystalline plateau of southern Yola, with the highest of its hills capped by sandstone. The accompanying section^[88] is approximately correct, though the maps will not permit of great accuracy.

Passing on towards Yola, there are crystallines at Bikola and again beyond Jelima, but in the lower ground between the two are pebbly sandstones and conglomerates running up from the plain into the crystalline area as a tongue. Here the present stream course seems to occupy the same position as an older drainage channel existing before the deposition of the detrital rocks. The mass of the Gumba hills is almost all of granite. The southern part, entirely of granite, rises to 1,886 ft. above sea-level, but the cone of the northern peak, 1,727 ft. high, is of sandstone which, as in the case of the Yendam hills, must have been banked against the granite at the time of deposition.

The observations made may be summarised by saying that if the “Upper Benue Sandstones” were stripped off, so as to lay bare the underlying basement, no great change in the physical geography would result. Some detached hills and outliers would disappear, but the essential features of the Benue plain with its Cretaceous rocks and the crystalline plateau of southern Yola would still remain. The sandstones evidently accumulated upon lower ground to the north, at first abutting against the granite hills of the south but eventually rising above them. The physical features were buried up and have been exhumed by later denudation with great modification in detail but with the essential characters persisting throughout.^[89]

The sandstones are found to run up the Benue plain past Yola to Garua and they also extend some little way north and north-west of Garua. The main spread, however, appears to turn southward, and according to German writers,^[90] they are found between Djerim and Rei-Buba on the uppermost Benue, and also on the hilly area to the west.

Bearing in mind this distribution of the sandstones and their field-relations to the underlying basement, there is seen to be an intimate connection between the sandstones and the river, and it appears fairly certain that the sandstones were accumulated in the valley or valley-plain of

an “Older Benue.” The outliers of the Yendam Hills and the Bagele Hills at Yola, which rise a thousand feet or so above the river, bear witness to the former thickness of these accumulations, now so largely removed.

The Upper Benue Sandstones cannot yet be subdivided even locally. No clay or shale bands have been found which would serve as horizons. The material does not vary much. The bulk is very felspathic and has evidently been formed out of granitic debris. At Bamga it is not always easy to distinguish between the arkoses of the sandstone series and the compacted granite sand at the base of the hills. The base of the series may become conglomeratic and the sandstone is liable to be pebbly even in the highest layers. Most of the rocks are white or slightly stained with iron oxide, but there are also intercalated layers of very ferruginous sandstone which occur as definite bands a few inches thick, traceable across the entire section exposed. The ferruginous character of these bands was determined at the time of deposition, possibly by local concentration of the heavier minerals. In addition to such bands there are in places much thicker masses of highly ferruginous grit occurring at what is locally the base of the series.

Much of the sandstone is massive, without bedding planes, and it tends to form caps to the hills with vertical sides, fifty feet or more in height. Cross bedding is beautifully exhibited in places in the lower part of the series, especially in the Bagele Hills, around Tagumbilli, and at Lamurde. The laminæ succeed each other in a perfectly regular manner through considerable thicknesses of material, one set of laminæ being truncated and overlaid by another set with a different inclination. Cross bedding is frequently met with in the Cretaceous grits of the Protectorate, where the laminæ are oblique to the major bedding planes. The cross-bedded sandstones of the Upper Benue, however, are built up of sets of laminæ, indistinctly marked off from those above and below, and without any true bedding planes at all. The structure resembles in every way the cross bedding of the Benue sandbanks, and in size of grain, in the presence of occasional pebbles and even of pebbly layers, and in the material itself, a coarse felspathic sand, the sandstones closely resemble the sandbanks of the present day ([p. 15](#)). The oblique lamination has been likened to that of dunes, but the coarse size of grain and the presence of pebbles indicate running water rather than wind as the immediate cause.

The flat tops of the sandstone hills and the terrace-like features on their sides show that the Upper Benue sandstones have not been folded. They rest on the Cretaceous limestone-shale series, which is folded in Yola province as it is in Muri. This suggests that the Upper Benue sandstones rest unconformably on the Cretaceous and are therefore post-Cretaceous in age. The actual boundary between the sandstones and the Cretaceous was only met with in one place, between Murki and Jibaru, about twenty-five miles north of Yola. There a coarse, dark red grit rests directly on Cretaceous limestone without any transition beds between the two. Unfortunately the dip of the limestone is not known, and therefore the evidence of the contact is not conclusive. The immediate change from limestone to unbedded ferruginous grit is easily understood if the overlying series is unconformable to the lower one, but it is not easy to account for it on the view of a continuous sequence. This section is regarded as strongly supporting the argument drawn from the unfolded character of the sandstones.

Though the reliance to be placed on lithological character is uncertain, so far as it goes the evidence is all in favour of correlating the Upper Benue sandstones with the Lokoja series. It is true that the oolitic ironstones so frequently met with in that series are not known from the Upper Benue area, but the soft friable sandstones and the ferruginous character of much of the material strongly remind one of the Lokoja series.

The above conclusions, and certainly the map of this area, will require modification in details, for the following reason. Two groups of sandstone are known in the Benue area, one below the limestone-shales and therefore Cretaceous, the other resting upon the limestone-shales, in all probability unconformably, and therefore post-Cretaceous. In the absence of fossils from either group of sandstones, the only method of determining the age of any particular exposure is to prove continuity either with the sandstones passing below the limestone-shales or with the sandstones resting upon the limestone-shales. This would require careful work under the most favourable conditions, and is out of the question in a preliminary survey. Thus in the Yola province it may ultimately prove that Cretaceous (infra-limestone) sandstones have in places been mapped with the Upper Benue sandstones.

In a few places in Muri Province there are sandstones having lithological resemblances to the post-Cretaceous series. Near Mutum Biu, for example, there is a porous, very ferruginous grit, breaking up into small fragments, each coated with a skin of limonite—characters exactly paralleled, in many localities, by the Lokoja series. Again at Loko, sixteen miles north of Katsena Allah, there is a soft unbedded pebbly sandstone which is much more like the Upper Benue sandstones than the Cretaceous. It may yet prove that on the widespread Cretaceous rocks of Muri there are here and there patches of post-Cretaceous sandstone, serving to link up the Upper Benue sandstones with the Lokoja series.

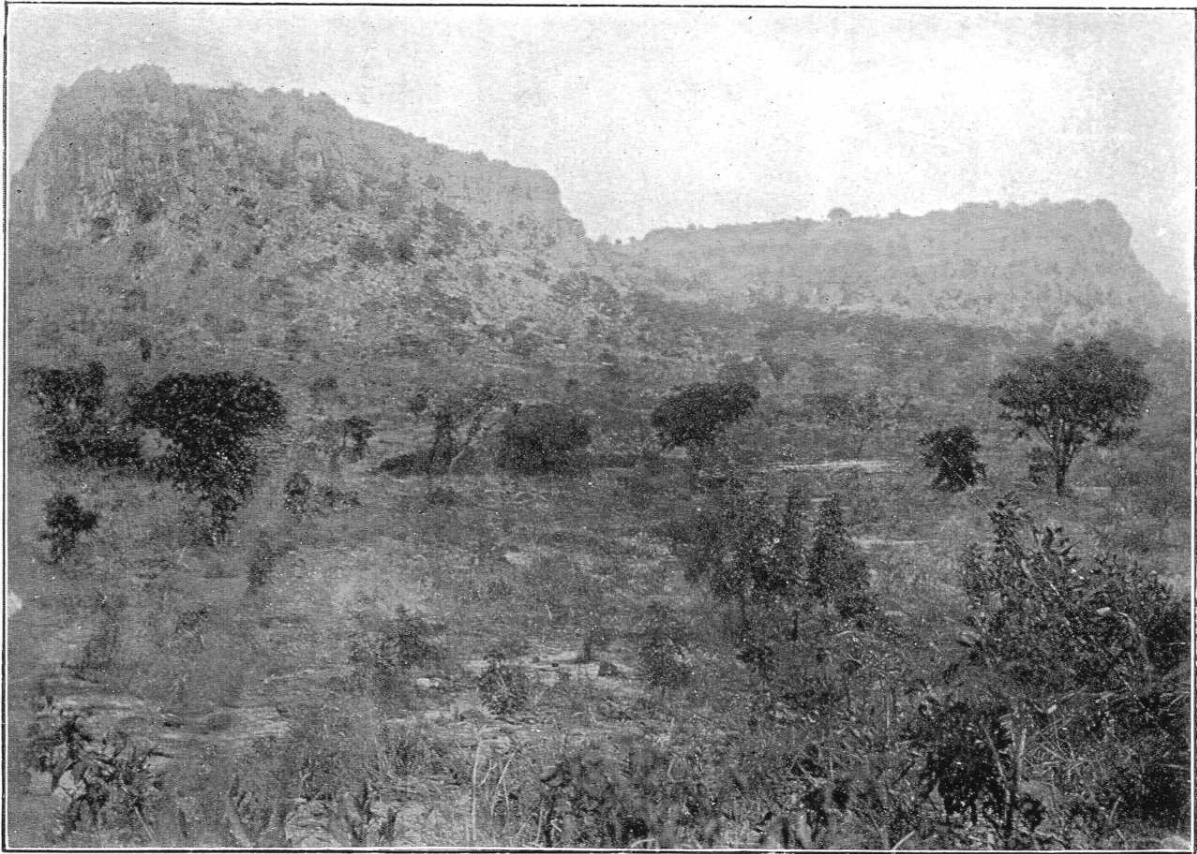
In spite of such future modifications, it is believed that the history of the Upper Benue can be already sketched out as follows:

1. Crustal movements (folding and perhaps faulting) dropped down the Cretaceous rocks below the crystalline basement along the Lau-Yola line.

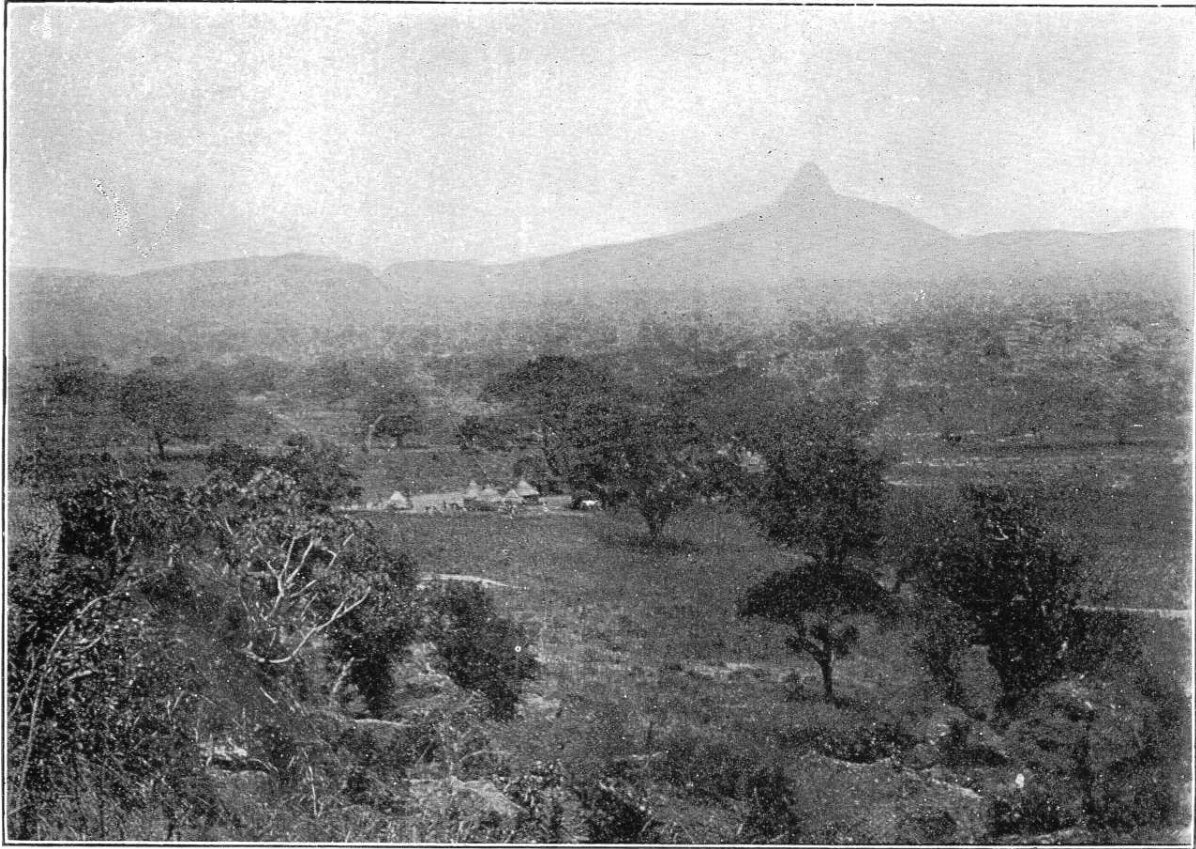
2. The soft character of the Cretaceous rocks marked them out for rapid denudation, and an “Older Benue” was established along this line with crystalline plateaux north and south.

3. For some reason the detritus carried by the river accumulated, and on consolidation it formed the Upper Benue sandstones. The unstratified character of the material and the absence of alluvial mud or clayey bands indicates a rapid accumulation, probably under somewhat different conditions from those now prevailing. The sands were banked up against the granites of the south and buried up at least the foot hills. [\[91\]](#)

4. Later denudation has largely removed the sandstone and has exposed the Cretaceous floor again in many places. (A. L.)



THE SANDSTONE HILLS OF TURE.



TURE OR TANGALE PEAK.

The great development of rocks of Eocene age in the region of the Upper Niger has been sufficiently emphasised by M. Chudeau.^[92] The Niger sandstones of Dahomey^[93] are undoubtedly continuous with the western members of the Sokoto series, and are properly ascribed by M. Chudeau to the Eocene. The sandstones of the Upper Benue valley in the northern territories of the Kameruns described by Dr. Passarge,^[94] are likewise continuous with the Upper Benue sandstones of N. Nigeria, and therefore most probably also of Eocene age. From the Southern Kameruns Dr. Esch^[95] has described Cretaceous and Eocene rocks whose mutual relations are very obscure. The locally folded character of the Cretaceous, however, suggests an unconformity at the base of the Eocene. In Southern Nigeria Mr. Parkinson^[96] has indicated the occurrence of both Cretaceous and post-Cretaceous rocks whose relations are at present unknown. The clays and lignites of Asaba^[97] may possibly, however, have been

accumulated contemporaneously with the upper shallow-water portion of the Eocene of Northern Nigeria.

M. Chudeau's^[98] admirable attempt at the construction of a map of the central Sahara is somewhat marred by his admittedly audacious method of deducing the solid geology of certain areas from the average depth of the wells. A striking example of the inaccuracy thus introduced is the relegation by M. Chudeau of the Eocene rocks of Sokoto to the Lower Cretaceous. Another fruitful source of error is the failure of M. Chudeau to recognise, doubtless through the want of sufficiently suggestive exposures, the possibility of an unconformity between the Cretaceous and the Eocene in the Sudan. There is every reason to believe that the erosion which followed the post-Cretaceous elevation was active over the whole of the Sudan, and that the greater part of the hummocky and irregular surface thus produced was again under water during the period of the middle Eocene depression. Moreover, the deposits then accumulated were in turn subjected to erosion during a subsequent movement of elevation, with the result that the early Eocene surface of crystalline and Cretaceous rocks was again laid bare over considerable areas. In view, therefore, of the former probable extension of the soft sandstones and ironstones of Kerri Kerri and the Tagum Hills over the plains of Bornu, it is permissible to doubt the accuracy of much of M. Chudeau's suggested stratigraphy of the Chad area. It seems more reasonable to regard the sandstones of Kanem and Manga as Eocene rather than Lower Cretaceous, and to look upon them as having been deposited, as in the case of the Eocene rocks of the Gongola valley, upon an earlier surface of Upper Cretaceous and crystalline rocks now buried beneath the superficial accumulations of Bornu. The rocks of Koutous and Alakhos, Ouame and Tirminy^[99] present, moreover, characteristics which ally them more with the Eocene rocks of the south than with the Turonian of the Damergou, and it is possible that even the southern escarpment of the Tegama itself^[100] where it overlooks the Damergou should be referred to the Eocene. The red, white, and yellow sandstones of Ennedi and of Snoussi on the Upper Shari are probably also of the same age, and in no way related (as suggested by M. Courtet^[101]) to the Karroo beds of South Africa. (J. D. F.)

- [78] Cf. Suess, *The Face of the Earth*, Vol. IV, Oxford Trans., 1909, p. 89.
- [79] Chudeau, *Sah. Soud.*, 1909, p. 104.
- [80] De Lapparent, *C. Rd. Ac. Sc.*, 136, 1903, pp. 1, 118; Bather, *Geol. Mag.*, [5], 1, 1904, p. 292.

The Eocene fossils from Sokoto and Jega are similar to those collected by Col. Elliot and Capt. Lelean from the Sokoto frontier, of which accounts have been published by Dr. F. A. Bather (*Geol. Mag.*, 1904, p. 292, Plate XI) and by Mr. R. Bullen Newton (*G. J.*, XXIV, 1904, p. 522), who regard them as of Middle Eocene age. The fossils consist principally of echinoids and internal casts of mollusca. The latter are not sufficiently well preserved for exact determination.

From Sokoto numerous examples of *Hemiaster sudanensis* (Bather were obtained, and also specimens of *Lucina* and gasteropods. At Jega *Hemiaster sudanensis* is also abundant, and a portion of *Plesiolampas* (probably *P. Sahara*, Bather) was obtained. The mollusca belong to the genera *Nautilus*, *Turritella*, *Voluta*, *Cardium*, *Lucina*, *Ostrea*, etc.

(H. W.)

- [81] Fossiliferous limestone has, however, been reported from Yelu at the confluence of the Dallul Mauri and the Niger valley.
- [82] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 384.
- [83] Chudeau, *Sah. Soud.*, 1909, p. 274.
- [84] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 112.
- [85] Chudeau, *op. cit.*, p. 275.
- [86] Cf. Gurich, *Zeit. Deut. Geol. Gesell.*, XXXIX, 1887, p. 122.
- [87] Barth, *Central Africa*, 1857, II, p. 368.
- [88] See Section IV on margin of [coloured map](#).
- [89] The crystalline rocks at the base of the Bagele hills, mentioned by Barth (*Central Africa*, Vol. II, p. 478) may be taken to represent the outcrop of a portion of the earlier hummocky floor which was buried under the sandstones of Yola.
- [90] Passarge, *Adamaua*, 1895, p. 363.; Edlinger, *Niger-Tsad Expedition*, 1904, p. 154.
- [91] Cf. Passarge, *Adamaua*, 1895, p. 390. Passarge did not recognise the presence of Cretaceous rocks in the Upper Benue valley.
- [92] Chudeau, *Sah. Soud.*, 1909, p. 97.
- [93] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 399.
- [94] Passarge, *Adamaua*, 1895, p. 363.
- [95] Esch, *Geologie von Kamerun*, 1904, p. 11.
- [96] Parkinson, "The post-Cretaceous Stratigraphy of S. Nigeria," *Q. J. G. S.*, 63, 1907, p. 309.
- [97] Parkinson, *op. cit.*

- [\[98\]](#) Chudeau, *Sah. Soud.*, 1909, p. 85; and map.
- [\[99\]](#) Chudeau, *Sah. Soud.*, 1909, p. 81 *et seq.*
- [\[100\]](#) Barth, *Central Africa*, 1857, Vol. I, p. 531.
- [\[101\]](#) Courtet, *C. Rd. Ac. Sc.*, 140, 1905, p. 162.

CHAPTER VI

SUPERFICIAL ACCUMULATIONS

The general absence of a deeply decomposed crust upon the rocks of the Sahara and the Sudan. The drift of Hausaland and Borgu; kankar; origin of the plains of Hausaland; the surface ironstones; origin of the drift; Mounts Dala and Kogon Dutsi; the surface drift of the Bauchi plateau; conditions of formation of the surface ironstones; periods of maximum and minimum erosion; periods of elevation and depression of the crust; the recent arid period. The drift of Bornu and Katagum; the absence of surface ironstones; the “firki”-land of eastern Bornu; the shelly marls of Kukawa; the conditions of formation of the drift of Bornu; the early existence of swampy and lacustrine conditions in Bornu; regional oscillations of level in the Chad area; erosion of the floor of the present lake; the cuvettes of Manga. The drift of the Gongola valley. The drift of the Benue valley. The ferruginous crusts of the flat-topped hills of the Lower Niger and Benue.

No fossiliferous deposits of later age than Eocene have been discovered within the Protectorate. M. Chudeau^[102] quotes the occurrence of Miocene shales at Boutoutou^[103] to the north of Gober and concludes that the sea did not finally quit Central Africa until the close of the Miocene. If, however, any deposits of similar age were ever accumulated within the Protectorate, they appear to have been subsequently removed and replaced by unconsolidated accumulations, most probably of late Pliocene age, which now in places cover Eocene, Cretaceous, and crystalline rocks alike. These accumulations, which vary considerably in composition in different parts of the Protectorate, are of continental and in part of lacustrine origin and represent the products of weathering and disintegration of the crystalline and sedimentary rocks, sorted and rearranged by the action of rain, river, and wind. Their superficial portion now forms the surface soil which except in swampy localities is characterised by the small quantity of humus which it contains. M. Chudeau^[104] has remarked upon the absence of any deep decomposition of the rocks in situ in the Sahara and Sudan to the north of lat. 11°, and his remarks may with justice be extended to the greater part of Northern Nigeria, where the original products of decomposition have rarely remained in place but have travelled outwards for a variable distance from the parent rocks and in places accumulated to very considerable depths. This drifted or travelled character of the surface alluvium may be typically

observed round the margins of the larger groups of granite hills. The disintegrated debris rapidly accumulated during torrential rains at the mouths of the narrow valleys or at the base of the weathered slopes is gradually broken up and spread out upon the plain during succeeding periods of less intense fluvial action.

1. *Hausaland and Borgu.*

The red sandy loam which covers the greater part of the crystalline plains of central Hausaland and Borgu, while predominantly of local derivation, possesses for the most part a drifted or travelled character, and in composition presents relatively little affinity with the rocks which anywhere immediately underlie it. A rude stratification may at times be observed within the drift, but for the most part it belongs to the category of ill-defined alluvium and surface wash.^[105] Red earth and subangular quartz grains in varying proportions naturally make up the bulk of the deposit, but it contains also more or less worn and weathered crystals of all the common rock-forming minerals together with pebbles and fragments of quartz, quartzite, granite, gneiss, amphibolite, and other rocks which are naturally most abundant where the drift is thinnest. Calcareous concretions similar to the kankar of India are common in places where the drift is predominantly earthy. They have, however, nowhere been observed forming any continuous sheet of sub-surface limestone, while, contrary to M. Hubert's suggestion,^[106] their distribution appears to have little or no relation to the occurrence of crystalline limestones within the schistose series. This composite drift, while evidently resulting from the rearrangement of weathered and disintegrated material, can frequently be seen covering up and wrapping round smooth and little weathered surfaces of rock in the midst of the open plain. The smaller inselberge moreover appear to have little influence upon the composition of the drift, which preserves its heterogeneous character up to the base of the dome, and even encloses on the margin sub-angular blocks and boulders of granite detached at an earlier period from the surface of the dome itself. In places, as on the high plains of Zaria and Bauchi, a very considerable thickness of drift has accumulated, and where locally it has been in part removed the rocky floor below is

hummocky and irregular and as a rule remarkably free from any evident decomposition. It is highly probable, indeed, that the sandy loam with its heterogeneous composition and its subangular pebbles of various rocks was slowly accumulated upon an irregular surface and spread out and levelled by wind and rain.^[107] That it was accumulated, moreover, under very different conditions from those which now prevail is evident from the fact that it is nowhere still in process of deposition upon the open plains of Hausaland. On the contrary it is now undergoing erosion, rearrangement, and gradual removal by the rivers and streams which in many parts of their course have not yet succeeded in exposing the rock below. Its presence also to a remarkable depth upon the watersheds and the loftiest plains makes it highly improbable that such regional accumulation could have taken place at the present elevation of the central provinces. It would seem rather that there has been within comparatively recent times a rearrangement of the drainage system of the Protectorate, brought about by a crustal movement of sufficient intensity to transform the plains of Hausaland from low-lying areas of accumulation into high-lying areas of erosion.

Such an hypothesis serves also to explain the present typical aspect of the plains themselves when it is remembered that their gently undulating character is due not so much to the smooth and even surface of the underlying rocks as to the covering of drift which has obscured all the minor irregularities of the crystalline floor below. There can be little doubt that the plains of Hausaland arose primarily as approximately base-levelled plains of erosion. There is, however, no reason to believe that the actual surface of the underlying rocks coincides in more than a general way with that of the open rolling plains of to-day. All the indications indeed point to quite the opposite conclusion. The base-levelled plains appear to have suffered some slight regional elevation and erosion, and the consequent hummocky and irregular surface has been in turn levelled up and buried in drift. The plains as we know them, therefore, while primarily plains of erosion, are in their more obvious characteristics to some considerable extent plains of accumulation.^[108] The processes of erosion and accumulation moreover, to which they owe their present aspect, are for the most part the same as those which are now active in these regions. The isolated hills and the bare surfaces of rock frequently show evidence of

abundant pot-holing and of extensive fluvial action. Messrs. Andrews and Bailey^[109] have suggested that a sub-aerial plain of erosion may in time, by means of dry weathering and run-off, replace a tract of hilly country at an altitude of even more than 3,000 ft., without the help of river erosion or planation. Whatever local influence these processes may have, however, the possibility of such a result upon a regional scale is exceedingly doubtful. The efficacy of dry weathering indeed, as an agent of extensive erosion, is, in spite of the able championship of Professor Walther,^[110] becoming more and more discounted with the progress of research in the Sahara.^[111]

A characteristic feature of the red sandy loam on the high plains of Borgu, Zaria, and Bauchi is the occurrence in places of a ferruginous cap or crust, which frequently forms a continuous floor many acres in extent. The ferruginous cap has arisen through the cementation of the quartz grains and pebbles into a hard resistant mass by means of hydrated ores of iron originally deposited in the subsoil above ground water level. The characteristic cellular or vesicular character and hard lustrous surface of the ironstone are due to the leaching out of the uncemented material and the rearrangement and partial dehydration of the iron ores upon exposure at the surface. Wherever, therefore, the ferruginous cap is now exposed, it may be assumed that the soil originally overlying it has been removed in the course of denudation. In southern Bauchi and Zaria^[112] the vesicular ironstone may at times be found forming low platforms raised a few feet above the general level of the plain, which is itself in places provided with a similar floor of ironstone. At other times where erosion has been more intense it caps low flat-topped hillocks of drift whose slopes are littered with fallen blocks and boulders of gritty ironstone. Rolled pebbles of concretionary limonite and subangular fragments of an earlier ferruginous crust are frequently also characteristic constituents of the surface soil. It is evident, therefore, that in addition to the period of formation of the ferruginous crust which now in places forms the floor of the plain, there has been at least one earlier period of formation of gritty ironstone within the drift. It may therefore be presumed that the drift was originally of sufficient thickness to cover the highest platform of ironstone of which traces still remain, and that the ironstone platforms of lower levels were formed during subsequent intervals in the later erosion of the drift. In places fragments or masses of

pebbly ironstone of heterogeneous composition can be seen adhering to bare and unweathered surfaces of granite, sandstone, and other rocks. Such occurrences are to be regarded as relics of earlier sheets of ironstone formed, it may have been only locally, when the drift was thicker and the ground-water at a higher level than it is at present.

The clue to the origin of the drift is found within the walls of Kano. The two flat-topped hills, Dala and Kogon Dutsi, each 150 feet in height, are composed almost entirely of soft and thoroughly decomposed rock, stained red towards the summit and capped by a layer of vesicular ironstone ([p. 90](#)). The presence of rolled fragments of an earlier ironstone in the ferruginous cap indicates that the flattened summits once formed the floor of the plain and were themselves probably covered by an earlier sheet of drifted alluvium. These hills bear witness, however, not only to the extensive decomposition *in situ* to which the crystalline rocks were once exposed, but also to the intensity of the subsequent erosion which has left them as solitary relics of the earlier surface of the plain. It is possible, indeed, that the formation of such a sheet of weathered rock possessed a regional character, and that in its later erosion is to be found the explanation of the general absence of any deep decomposition of the rocks *in situ* not only within the Protectorate but also over the whole surface of the central Sahara and Sudan. To the erosion of this weathered crust is also to be ascribed the formation of the hummocky and irregular surface upon which the drift was afterwards spread out, and it is probable that the greater part of the drift itself was derived from the further erosion of those portions of the original weathered surface which still remained above the level of accumulation. No facts have been ascertained with regard to the occurrence of true lateritic material either in the original sheet of weathered rock or in the drifted alluvium. There is some reason to believe, however, that the surface ironstones in places contain considerable quantities of free alumina.

It is possible that before accumulation began the weathered rock had not been entirely removed from the surface of the plains, and that portions of the original sheet of decomposed material may therefore be preserved in places underneath the drift. Moreover worn surfaces of the original weathered rock may have been left practically devoid of any covering of drift, and these surfaces, as around Aribi and Gantam on the Nassarawa

tableland, may have been in course of time provided with a typical crust of gritty and vesicular ironstone. The floor of the plain of Kano around the two prominent hills probably represents in this manner a platform cut in the earlier sheet of weathered rock and covered with a ferruginous crust. A similar origin should perhaps be ascribed to the plains of Rauni which are characterised by the great depth of their surface ironstones and the presence of hillocks of similar composition which recall the buttes of Djougou in Dahomey.^[113] Between Kano and Katsina, moreover, the amount of drift varies much from place to place, and appears to have been locally very largely removed and replaced by recent products of weathering and considerable accumulations of blown sand and clay which have probably arisen to some extent through the rearrangement of the earlier drift under the influence of the winds. These recent accumulations are sometimes found resting upon older surfaces of cellular ironstone, but are never themselves covered with a ferruginous crust.

The red earthy drift which forms the detached flat-topped hills upon the summit of the Bauchi plateau (Plate XI, [Figs. 1](#) and [2](#)), while located for the most part upon a crystalline floor, may possibly have originated to a considerable extent through the disintegration of the fragmental material of the volcanic cones in the neighbourhood, and through the rearrangement of the original weathering products which may be presumed to have formed upon the lavas of the Morrua plateau. There are, however, few remaining traces of any earlier weathering of the crystalline and igneous rocks preserved underneath the superficial accumulations. For the most part the latter rest upon smooth and comparatively unweathered surfaces of rock (Plate XV, [Fig. 2](#)), while their travelled origin is indicated not only by the presence of quartz grains and granitic debris in the ferruginous cap, but by the occurrence of tinstone in the drift throughout the whole area. The tinstone is naturally most abundant around Ngell in the neighbourhood of its centre of dispersion, but traces of it are to be found even on the hills of Rukuba from which the drift itself has now been almost entirely removed. There is good reason to believe, indeed, that the Bauchi plateau up to a comparatively recent date was continuous with the central plains of Hausaland, and that it participated in all the oscillations of conditions to which the latter were subjected. It is probable, therefore, that the plateau

was at first covered with a sheet of decomposed rock of variable thickness, by the subsequent erosion of which the irregular surface was produced upon which the drift was accumulated and spread out. It is obvious, however, that the drift itself with its crust of compact or vesicular ironstone is now undergoing erosion, and that the present tin-bearing alluvium is derived for the most part from the rearrangement of the drift and only to a small extent from the denudation of the underlying rocks which is now again in process. It is probable that, during the earlier stages in the erosion of the plateau drift, accumulation continued upon the plains of Zaria and Nassarawa round the base of the plateau, but any such action is practically negligible under the present *régime*.

M. Hubert^[114] in Dahomey, and Dr. Passarge^[115] in Adamaua, assign only a very subordinate place to the formation of surface ironstones upon drifted alluvium. They believe that the ironstones have almost always formed within material which has arisen as the result of the decomposition *in situ* of the rocks immediately underlying.^[116] M. Hubert's insistence upon this point of view has led him to suggest that the crystalline rocks of the plains of Borgu were levelled before being decomposed and covered with surface ironstones, and that the plains themselves have preserved their present character and elevation since the close of the Eocene. It may be granted that the crystalline rocks of Nigeria were at one time deeply decomposed, but Mounts Dala and Kogon Dutsi bear unmistakable testimony to the subsequent intense erosion of this former crust of weathered rock. In southern Bauchi and Zaria, moreover, extensive sheets of surface ironstone can frequently be seen capping typical gritty alluvium, and there can be little doubt that within the Protectorate as a whole, the formation of surface ironstone upon material actually weathered *in situ* is the rarer phenomenon. For the most part the ironstones rest upon and have been formed within drifted and travelled superficial material which has been accumulated, near or at base-level, upon an irregular and hummocky surface, and spread out in such a manner as to obscure all the minor irregularities and to produce the present smooth and even character of the plains of Hausaland. M. Chautard^[117] has noted the similar extensive formation of surface ironstones upon the ancient alluvial plains of Upper Guinea and the Futa Djalon.

The actual conditions of formation of the surface ironstones are very problematical. Their association with the older alluvium affords some reason for believing that their extensive formation in the past has been accomplished under somewhat different conditions from those which now prevail. It seems probable also that the presence of surface ironstone as a ferruginous cap upon drifted alluvium is an analogous phenomenon to its occurrence upon material which has arisen, as in Mounts Dala and Kogon Dutsi, through the decomposition *in situ* of the rocks below. Drs. Hatch and Corstorphine^[118] believe that the surface ironstones of South Africa have been formed very largely as bog iron ores by the oxidation of ferruginous solutions or by the deposition of the iron salts during the evaporation of pools of stagnant water. Captain Freydenberg^[119] has noted the accumulation of ferruginous clay on the northern shores of Chad, but extensive sheets of ferruginous material comparable to the platforms of surface ironstone have nowhere been observed in process of formation upon the surface of recent alluvium or of æolian accumulations under the present *régime* of periodic rains. The surface waters accumulated during the summer rains are for the most part rapidly and completely drained, and any probability of the formation at high altitudes within the Protectorate of pan or bog iron ores upon an impervious subsoil would appear to be prohibited by the general open and porous character of the drift. Moreover wherever true swamps, subject to partial or complete annual desiccation, are to be found, surface ironstones are typically absent, a waterlogged condition of the subsoil being apparently unfavourable to the deposition of the iron ores. M. Lacoïn^[120] believes that in alluvial deposits such as those of the Shari and Oubangui an enrichment in iron takes place in the lower beds at the expense of the upper, and that the banks of ferruginous material, formed by sub-surface evaporation and oxidation after the manner of the “*alios*” of Landes,^[121] are afterwards exposed by the removal of the overlying accumulations. Such an hypothesis, however, is equally open to the objection that the surface ironstones of Nigeria are most typically found where the sub-surface drainage is most perfect. Professor Walther^[122] believes that the surface ironstones are entirely the product of arid conditions, and that their origin is similar to that of the protective crust of the Libyan desert. While, however, there can be little doubt that the

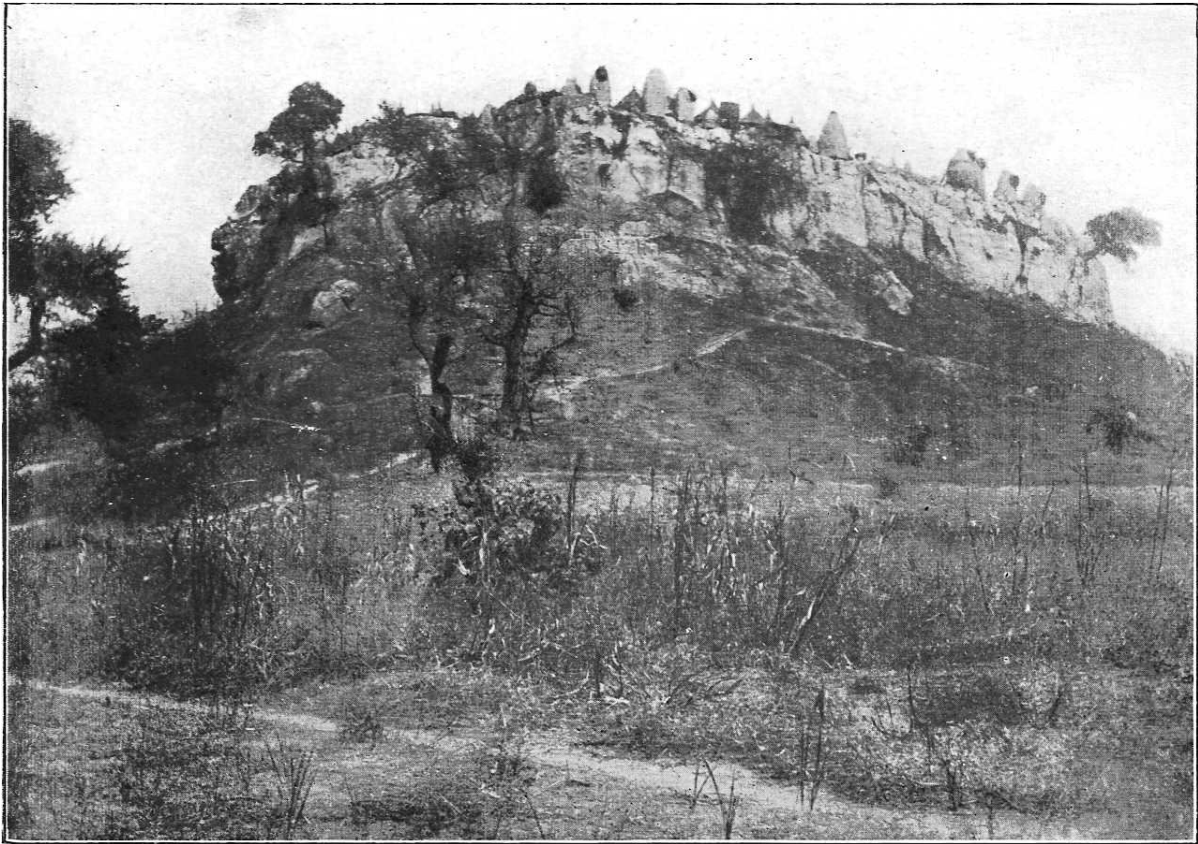
hardening of the surface of the ironstone and the production of the characteristic varnish is the result of the partial dehydration of the iron ores upon exposure to conditions of excessive evaporation, it is difficult to believe that the occasional rains and dews of an arid climate could suffice to produce any extensive superficial accumulation of the iron ores. MM. Chautard^[123] and Hubert,^[124] in view of the sporadic distribution of the ironstones, would extend to their formation Sir T. H. Holland's hypothesis^[125] of the bacterial production of laterite under humid conditions, an hypothesis, however, which still awaits experimental verification. Dr. Passarge^[126] is of opinion that surface ironstones are formed under semi-arid conditions where the aridity causes the rapid evaporation of the occasional rains, while the high temperature promotes rapid chemical weathering of the rocks, and the excess of ozone and oxygen compounds in the rains, together with the absence of reducing substances in the soil, brings about the rapid oxidation and precipitation of the iron ores. Mr. Maufe^[127] believes that the formation of surface ironstones (muram) in East Africa is an immediate consequence of deforestation. There is some reason to believe, however, that even underneath a covering of vegetation the slow precipitation of iron ores in the subsoil may give rise to the formation of ferruginous concretions which may in time coincide to form a continuous layer. Where the streams have trenched the deeply decomposed surface of certain belts of schists in Kabba and Nassarawa which support an abundant vegetation, it may be observed that the iron shows a decided tendency to accumulate in the upper portion. The solution of the iron is brought about by the descent of water charged with carbon dioxide and with organic acids, arising primarily from the decomposition of the superficial vegetation, while the subsequent rise and redeposition of the iron ores above ground water level is accomplished during periods of less relative humidity. It is conceivable that a similar action may take place within alluvial deposits when covered with a rich vegetation, and that a subsequent change in conditions may bring about the more rapid evaporation of the ferruginous solutions, the removal of the vegetation and of the surface soil, and the hardening and polishing of the crust into a typical surface ironstone. The present irregular distribution of the ironstones may have been conditioned by the amount of subsequent erosion as well as by the original

composition of the drift, the content in iron of the circulating surface waters, the abundance of the vegetation and the character of the drainage, the free access of oxygen to the subsoil being a necessary condition of their formation.

Dr. J. M. Maclaren,^[128] Mr. F. P. Mennell^[129] and Mr. J. M. Campbell^[130] have suggested an analogous origin for certain laterites and ironstones of India, Rhodesia and Upper Guinea respectively, and postulate for their formation an alternation of wet and dry seasons similar to the present *régime*. Messrs. Mennell and Campbell are of opinion, moreover, that the concentration of the ores upon definite horizons to form the floors of successive platforms or terraces may be explained as the result of a spasmodic lowering of the ground water level under climatic conditions similar to those which now prevail. While, however, the erosion of the platforms themselves within the drift may certainly be ascribed to this cause, it is much more probable that, at least in the case of Nigeria, the spasmodic activity in the deposition of the iron ores has been brought about by an alternation of humid and arid conditions, the formation of ironstone being most marked during periods of gradual desiccation. It is well known that the present *régime* of periodic rains within the Sudan represents a return to more humid conditions after a period of aridity, and there is some reason to believe that there have been many such climatic oscillations in the past. Nothing is more probable, therefore, than that certain periods of desiccation should immediately follow the formation of certain platforms of erosion within the drift, and that to the incidence of these periods of desiccation is to be ascribed not only the maximum deposition of the iron ores but also the removal of the overlying alluvium and the hardening of the surface of the ironstones. It is probably correct to assume that an annual alternation of rains and droughts similar to the present *régime* is favourable to the deposition of iron ores in the subsoil, and there is every reason to believe that their accumulation is still going on where conditions favourable to their formation are to be found. Under present conditions, however, the amount of iron salts dissolved in the ground water must be relatively less than it would be at the close of a period of maximum humidity, and it may therefore be presumed that in Northern Nigeria the present is a period of comparatively slow accumulation of iron ores. There is indeed, as already

remarked, little or no evidence of any extensive formation of surface ironstone, since the beginning of the present *régime*.

The ironstones must be exposed actually at the surface before they can assume their characteristic lustrous carapace, and the removal of the vegetation and of the surface soil can be accomplished only by a climatic change. Mr. Campbell^[131] in Upper Guinea, and Mr. Maufe^[132] in East Africa, would assign the deforestation very largely to the influence of man, and the subsequent removal of the soil to the agency of the tropical rains.^[133] It is difficult to believe, however, that the characteristic ironstone platforms of Hausaland can possess such a comparatively recent origin, and there can be little doubt that the highest platforms of which traces still remain should be referred to at least the later Tertiary period. Once formed, the hard resistant crust may be covered by later alluvial or æolian accumulations or by the products of disintegration and re-arrangement of the ironstone itself, upon the return of more humid conditions. Wherever covered by later deposits, however, the ironstone is softer and lacks its characteristic varnish. Moreover, where a scattered vegetation is now found creeping over a surface of lustrous ironstone, it is frequently a difficult matter to decide how much of the scanty soil is a recent accumulation, and how much of it represents the original covering of alluvium which had not been entirely removed.



DOZI HILL, KERRI-KERRI.



A TRENCH-LIKE VALLEY NEAR LELE, IN KERRI-KERRI.

The formation of the surface ironstones of the plains of Hausaland is thus by no means readily explicable. Their origin is probably due, in part, to many different causes, and their fragmental occurrence at various levels, sometimes upon weathered rock, sometimes upon unweathered rock, and sometimes upon drifted alluvium, points to repeated formation and repeated erosion at several different periods. It is believed that, if thoroughly investigated, their distribution and the distribution of the surface drift would afford some clue to those repeated climatic and crustal changes to which within comparatively recent times the surface of the central Sudan has been subjected. The evidence already detailed appears to be sufficient to establish the following succession of events upon the plains of Hausaland:—

(1) A period of minimum erosion during which the superficial rocks were deeply decomposed to a variable depth and covered with a ferruginous cap.

(2) A period of maximum erosion during which the earlier sheet of decomposed rock was largely removed, with the exception of such relics as Mounts Dala and Kogon Dutsi.

(3) A period of minimum erosion and accumulation during which the surface drift was spread out upon the plains.

(4) The present period of erosion of the drift during which, in places, successive platforms have been cut in the drift and floored with concretionary ironstone.

The cause of the alternation of these periods of accumulation and erosion is probably to be sought in crustal changes of a regional character. If a condition of low relief be granted to begin with, a general depression of the surface to the neighbourhood of base level may be postulated during the periods of accumulation, and a corresponding elevation of the surface above base level during the periods of erosion. The various periods indicated above may therefore be termed respectively:—

- (1) The first period of depression.
- (2) The first period of elevation.
- (3) The second period of depression.
- (4) The second period of elevation.

The history of the earlier periods is somewhat obscure and it is possible that there may have been many minor oscillations of the crust during the first period of depression and the first period of elevation. It is possible also that the first period of depression may have been represented merely by a general base levelling of the surface through erosion, and that the original sheet of weathered rock may have been formed in part underneath sedimentary rocks which were afterwards removed. The flat-topped hillocks of consolidated granitic grit and clay between Kano and Kazaure may perhaps represent contemporaneous accumulations during the first period of erosion, which were themselves subsequently exposed to denudation. There is some reason to believe, also, that there has been within the various periods a minor oscillation in climatic conditions. These minor oscillations would naturally be most clearly recognisable within the second period of

elevation which is continuous with and includes the present. There is little doubt that the formation of the surface ironstones was favoured by the successive occurrence of periods of maximum evaporation, and it is highly probable that these periods assumed at times a typically desert character, and that the partially eroded surface of the drift was covered up at intervals by æolian accumulations. One such period at least immediately preceded the present more humid conditions, and is clearly recognisable in the worn and rounded dunes of the dead *erg* of northern Hausaland. The dunes are now grassed over and covered with scattered trees, but their origin is still sufficiently indicated in their composition and general outlines as well as in the characteristic banking of the sand against rocky exposures. The surface soil and the recent alluvium of the plains of northern Hausaland probably contain much wind-blown material, and the swampy character of much of the surface of the plains during the rainy season is probably to be in part explained as the result of the interference of the accumulations of this recent arid period with the earlier system of drainage. At the same time, however, it should be remembered that the erosion of the drift has in most places not yet become very intense and that the average slope of the plains of Hausaland is as a rule so gentle that a considerable volume of water must accumulate before a perceptible flow can be established. Much swampland is found on the high plains around and to the east of Zaria, where Dr. Baikie^[134] first recognised the drifted character of the alluvium.

2. Bornu and Katagum.

A striking feature of the plains of Bornu and Katagum and of the Gongola valley between Nafada and the Yola border is the absence of any superficial crust of vesicular ironstone. The drift which may reach in Bornu a depth of more than 300 feet differs considerably in composition from the characteristic drift of Hausaland. Upon the open plains of Bornu the drift is composed for the most part of a mixture of white or yellow sand and light or dark grey clay. A red ferruginous staining of the drift is comparatively rare to the east and north-east of Gujba. The clays in places yield irregular concretions of kankar and small nodules of concretionary ironstone are dug from the surface sands in the neighbourhood of Kukawa, but no evidence

has been anywhere obtained of the formation or earlier existence of any continuous sheet of vesicular ironstone. Well sections exhibit below the surface alternations of sand and clay in beds of very varying thickness.^[135] The sand varies much in coarseness of grain and is in part angular and of drifted fluvial origin and in part well rounded and of æolian origin, the latter apparently predominating largely towards the north-east. The clays are finely divided and flocculent in water and mixed with minute grains of blown sand. In places they become darker coloured from the inclusion of organic matter and pass into the “firki” (p. 52) or so-called black cotton soil of Chad and the lower Shari. Where exposed at the surface, the clay becomes plastic and very tenacious in the rains, but hard and brittle and much cracked and fissured in the dry season. These clays are rich in azote which contributes much to their fertility.^[136] This advantage, however, is counterbalanced by the almost complete absence of lime, potash, magnesia, and phosphoric acid, and by the presence of large quantities of silicate of alumina and of salts of soda. Fragments of broken shells of land snails and lacustrine molluscs are common in the clays of eastern Bornu, while thin beds of calcareous marl^[137] with abundant remains of recent molluscs are intercalated among the sands and clays of Mongonu and Kukawa. It is evident that the conditions of accumulation of the drift in Bornu were different from those under which the drift of Hausaland was accumulated. The latter was preeminently a terrestrial or sub-aerial accumulation while the former in the finer-grained character of the sand, the abundance of clay and the absence of pebbles and of any intense ferruginous staining affords indications of having been deposited to a great extent under water in a shallow lacustrine or swampy region. It is probable, however, that the accumulation of the drift of Bornu was, at least in the earlier stages, contemporaneous with the accumulation of the drift of the plains of Hausaland and may therefore be referred to a considerable extent to the second period of depression. No traces are left, however, upon the plains of Bornu of the earlier periods of depression and elevation, and it may therefore be presumed that the surface upon which the drift was deposited was entirely cleared of earlier accumulations during the first period of erosion and that its minor irregularities have been completely obscured by the later drift. The second period of elevation which brought about the

erosion of the drift in Hausaland appears to have resulted in comparatively little erosive action upon the drift of Bornu. It is highly probable indeed that Bornu retained its swampy and lacustrine and therefore receptive character to a comparatively recent period and that the oscillations of less and more humid conditions which probably prevailed during the accumulation and erosion of the drift of Hausaland are reflected in Bornu in the intermingling of æolian and fluvatile and lacustrine material. The local abundance of alkaline salts in the drift of Bornu may also be taken as indicating an occasional and local desiccation.

The presence of swampy and lacustrine conditions in Bornu dates, therefore, from the second period of depression and there is no reason to believe that there has been any marked break in continuity between these later Tertiary swamps and the present Lake Chad. There is, however, no evidence that the present lake or the former more extensive swamps in any way represent the shrunken relics of the early Tertiary seas. The later swamps indeed, with which the lake is more or less continuous in point of time, appear never to have reached any great depth and to have been liable to repeated desiccation. Desert periods may have changed them partially or wholly into sand fields, but the return of humid conditions constantly re-established their former character.^[138] It may be presumed also that during periods of exceptional rainfall the swamps became united into a shallow lake of considerable extent which may even at times have covered the greater part of central and Eastern Bornu. There is no reason to believe, however, that this was to any extent a normal condition of things, or that the drift was primarily accumulated in a large permanent lake. So far as known, marly beds occur in Bornu only in the neighbourhood of the present lake, and whatever the extent of the marginal swamps may have been, the deepest and clearest water seems always to have been confined to eastern Bornu. The thinness and local character of the marls would seem to imply, however, that the requisite conditions for their formation were by no means common or universal. Captain Freydenberg has described in sufficient detail the former extension of the swamps to the north, east, and south of Chad.

The character and thickness of the drift deposits appear to indicate that Bornu has in the past formed part of an area of predominant depression. The

shelly marl of Kukawa which lies at least fifty feet above the present level of Chad, undoubtedly indicates that the floor of the lake was at one time at a higher level than it is at present. The long-continued accumulation was probably conditioned at first by the second period of depression and later by the gradual uprise of the Congo watershed during the latest period of elevation. In the neighbourhood of Kukawa, however, the low hillocks of drift show evident signs of erosion, and it may therefore be presumed that the present floor of the lake has been excavated out of the earlier lacustrine and drift deposits. This erosion which M. Garde^[139] would presumably ascribe entirely to æolian activity, was probably in great part conditioned by the continued relative subsidence of the Bodele depression, the consequent eastward draining of the lacustrine region and the establishment of the Bahr el Ghazal as a waterway from west to east. The lower relative level of Bodele has now been definitely established, and M. Chudeau's^[140] arguments in favour of the eastward flow of the Bahr el Ghazal may be provisionally accepted until fuller information has confirmed or contradicted them. Desert conditions, however, followed close upon erosion. The worn and rounded hillocks of drift around Kukawa were partially buried in blown sand and the Bahr el Ghazal was blocked by the action of the wind. On the establishment of the present conditions the lake as we now know it was formed and limited towards the west by the line of sand dunes, now covered with abundant vegetation, which stretches from Kukawa to the river Yo along the margin of the lake. There is no reason, however, to postulate the presence of an earlier lake actually on the site of the present Chad at a period immediately anterior to the coming of the desert.

It may be presumed that during the period of erosion which preceded the advent of the desert, shallow watercourses were cut in the drift of Bornu and that they were in part dismembered and filled up again during the succeeding period of desiccation. One such watercourse may be clearly traced from Murguba to Maigumeri in central Bornu. The present channels of the Yo, the Maidugari river, the Yedseram, and the Shari were probably also established at the same period. The cuvettes of Manga probably originated at the same epoch and represent portions of earlier watercourses detached and isolated by æolian accumulations. The sites of the cuvettes

may possibly have been determined in part by the presence of springs in the earlier river beds. While, therefore, there is little reason to believe with Captain Freydenberg^[141] that the bottoms of the cuvettes represent detached portions of the original floor of an extensive lake almost entirely filled up with æolian accumulations, there is equally little reason to believe with M. Chudeau^[142] that the cuvettes have arisen through the local subsidence of the underlying rocks brought about by the dissolution of beds of soluble salts, hypothetically contained within the sandstones of the Tegama below. M. Chudeau's objection that an approximately level surface such as that of the plains of Bornu cannot be produced under the influence of the wind is clearly untenable in view of the known occurrence of extensive level sandfields in modern deserts. Moreover, in Katagum solitary dunes or isolated hummocks of blown sand formed during the recent arid period may be found alone in the midst of an open undulating plain. There is some reason also to believe that the winds which never attained any great intensity varied in direction in different parts of the Protectorate. In Bornu the dunes are elongated in a N.-S. direction while in Katagum they are more generally E.-W. and it is possible that a successive variation in the direction of the winds over the same area may have been responsible for much of the levelling of the present surface of the plain.

The lacustrine conditions which prevailed over central and eastern Bornu during the deposition of the drift extended also for a considerable distance westward through Katagum. The drift of Katagum is very similar in character to that of Bornu and the abundance in it of wind-blown material, as in the drift of Bornu, bears witness to the constant struggle between humid and desert conditions during the period of its accumulation. The predominance of the latter at a period immediately preceding the present is indicated as in Bornu by worn and rounded dunes and banks of sand. There is every reason to believe that Bornu and Katagum have been for long on the margin of the desert and that the history of the repeated advance and retreat of the desert can be clearly read in the varied character and composition of the drift.

3. The Gongola Valley.

To the west of Gujba, thick accumulations of red sandy loam effectually obscure the irregularities of the topography of eastern Kerri Kerri. The drift has evidently been largely derived from the disintegration of the red sandstones and ironstones of Kerri Kerri, but the surface of the original deposits has been in great part rearranged by subsequent æolian action. Upon the plateau itself the æolian accumulations reach in places a very considerable thickness. In the valley of the middle Gongola to the south of Kerri Kerri there is as in Bornu and Katagum, a remarkable absence of surface ironstone. Around Nafada and southward upon the narrow plain to the west of the Gongola the drift is predominantly a red sandy loam. To the east of Ashaka, however, the drift consists very largely of a stiff tenacious black or dark grey clay, very similar in appearance to the “firki” of eastern Bornu. This material also covers large areas upon the central plain of Bauchi to the south and south-east of Deba Habe. It is clear from the manner in which the red loam and the dark coloured clay obscure the irregularities of the surface and creep up the flanks of the sandstone hills that the superficial accumulations are here also, as on the plains of Hausaland, of drifted or travelled origin and not primarily of the nature of material weathered and disintegrated in situ. The local origin of the drift may however be presumed. The red sandy loam of the northern and western plains has been derived very largely from the disintegration of the Kerri Kerri and Gombe sandstones, while the clays of the east and south have probably arisen in great part from the decomposition of the lavas of the Burra plateau. The original depth of the drift is unknown, but must have been very considerable. The Gongola in places has not yet succeeded in cutting its way through the drift to the rock below, while the highest portion of the plain which is still buried in drift lies about 300 feet above the present river bed. The total thickness, however, may well have been much greater and there is every reason to believe that what is now the valley of the middle Gongola between Nafada and Gasi formed, along with that portion of the central plain of Bauchi which lies between Deba Habe and Putu, an area of special accumulation throughout the second period of depression and that the activity of the river and its tributaries during the

most recent period of elevation has been directed mainly towards the removal of the drift which had accumulated in the earlier valley.

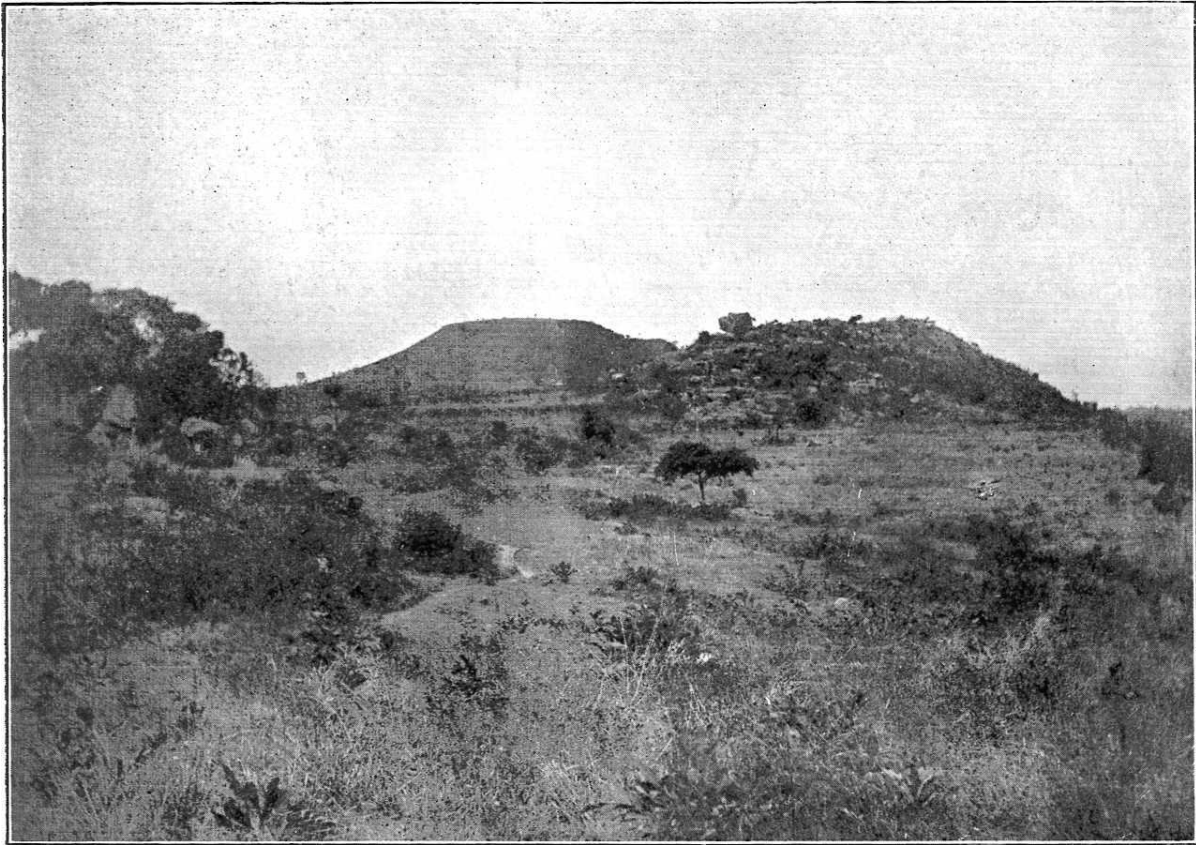
4. The Benue Valley.

The alternation of the major periods of accumulation and of erosion as established for the plains of Hausaland was probably to some extent of a regional character, although the relative amount of respective depression or elevation may have varied much in different parts of the Protectorate. The irregular movements to which the Benue valley as a whole was subjected during the latest period of elevation renders the correlation of the superficial deposits with those of the northern States a matter of considerable difficulty. There is good reason to believe, however, that the lower Benue valley and the greater part of the provinces of Kabba and Ilorin were exposed during the same period to very intense erosion, in consequence of which the original drift deposits have been almost entirely removed and the underlying Eocene sandstones deeply trenched and worn. The middle Benue, on the other hand, appears to have been long a region of relative depression and accumulation, and although platforms of surface ironstone may be found in places even on the plains of Muri,^[143] the requisite conditions for its formation appear to have been for the most part wanting. It is possible that its absence may be due, as in Bornu and the Gongola valley, to a prolonged water-logged condition of the subsoil. Wherever, on the other hand, the surface of a hillock of red sandstone was exposed above the drift, there seems to have been a rearrangement of the ferruginous material and a concentration of the iron ores in the upper part which has given rise upon exposure to a typical honeycombed limonitic cap. The ferruginous crusts of the flat-topped hills and plateaux of the lower Benue and the Niger have a somewhat similar origin and have arisen in part through the rearrangement and deposition in the subsoil of the original iron of the sandstones and oolites and in part, like the ferruginous conglomerate on the summit of Mount Patti, through the cementation of travelled material of a similar composition. Where the summits are bare and lustrous it may be presumed, as in the case of the ironstone platforms of Hausaland, that the surface soil, originally overlying, has been removed in the course of

denudation. At Awe, on the middle Benue, a sheet of sintery calcareous material appears to have been deposited upon the upturned edges of the Cretaceous before the accumulation of the drift.



CRETACEOUS SANDSTONES AND SHALES AT AWE.



A FLAT-TOPPED HILLOCK OF DRIFT ABOVE NARAGUTA.

Attention has been already directed ([pp. 27, 77](#)) to the occurrence in Kabba, Nassarawa, and Southern Zaria of belts of felspathic muscovite magnetite schists whose deeply decomposed crusts support a luxuriant vegetation. Where partially eroded, it may be observed that the iron ores have a tendency to accumulate in the subsoil, and it is possible that if the overlying soil and vegetation were removed, the subsoil might harden into a typical vesicular ironstone. It is difficult to decide, however, to what extent the decomposed crusts may be considered the result of weathering under present conditions. It seems probable in view of the absence of any deep decomposition of the adjoining rocks that they may to a large extent represent the irregular base of the earlier sheet of decomposed rock to which the presence of Mounts Dala and Kogon Dutsi bears witness upon the plains of Hausaland and which may possibly have extended at the same time over the remainder of the Protectorate. In some places, moreover, as at Kanna, there is evidence that the crystalline floor had been locally and deeply decomposed as the result of differential weathering underneath the

Eocene sandstones. A similar occurrence has been noted by M. Hubert^[144] at Youri in the Niger Valley.

Little consolidated accumulations of late Tertiary age, frequently provided with a ferruginous cap like the drift of Northern Nigeria, have been described under the designation of “Benin Sands” by Mr. Parkinson^[145] from Southern Nigeria, of “Ossa Sandstones” by Dr. Esch^[146] from the Kameruns, of “terre de barre” by M. Hubert^[147] from Dahomey, and of *reg* from the Northern Sahara by numerous French authorities.^[148] It is yet premature, however, to attempt any correlation of these deposits with the drift of Northern Nigeria.

- [102] Chudeau, *Sah. Soud.*, 1909, p. 97.
- [103] De Lapparent, *C. Rd. Ac. Sc.*, 26 Dec., 1904.
- [104] Chudeau, *op. cit.*, p. 273.
- [105] See Chamberlain and Salisbury's *Geology*, 1905, p. 174.
- [106] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 102.
- [107] Cf. Lenz, *G. M.*, 1879, p. 172.
- [108] Cf. Passarge, Rumpfffläche und Inselberge (Kordofan-type), *Zeit. Deut. Geol. Gesell.*, 56, 1904, p. 193.
- [109] Andrews and Bailey, *Q. J. G. S.*, Vol. LXVI, 1910, p. 228.
- [110] Walther, *Das Gesetz der Wüstenbildung*, 1900.
- [111] Cf. Chudeau, *Sah. Soud.*, 1909, p. 285.
- [112] Rohlfs evidently mistook the surface ironstones of Bauchi and Zaria for rocks of sedimentary origin. *Pet. Mitt. Erg.*, VII, 1872, p. 64.
- [113] Hubert, *Miss. Sci. au Dahomey.*, p. 160, Plate VIII.
- [114] Hubert, *Miss. Sci. au Dahomey*, 1908, pp. 109, 123.
- [115] Passarge, *Adamaua*, 1895, p. 400.
- [116] Cf. also Mennell, *G. M.*, VI, 1909, p. 350.
- [117] Chautard, *Le Fouta-Djallon*, 1905, p. 126.
- [118] Hatch and Corstorphine, *The Geology of South Africa*, 2nd Ed., 1909, p. 330. Cf. also Hatch, *Mineral Resources of Natal*, 1910, p. 73; Rogers and Du Toit, *Geology of Cape Colony*, 1909, p. 390.
- [119] Freydenberg, *Tchad et Chari*, 1908, p. 21.
- [120] Lacoïn, *Bull. Soc. Geol. Fr.*, 1903, p. 484. Cf. Foureau, *Miss. Sah.*, p. 674 et seq.
- [121] De Lapparent, *Traité de Géologie*, 4th Ed., p. 333.

- [122] Walther, *Das Gesetz der Wüstenbildung*, 1900, p. 19.
- [123] Chautard, *Le Fouta Djallon*, 1905, p. 137; *C. Rd. Ac. Sc.*, 146, p. 239.
- [124] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 199.
- [125] Holland, *Geol. Mag.*, X, 1903, p. 59.
- [126] Passarge, *Adamaua*, 1895, p. 396.
- [127] Maufe, *The Geology of East Africa*, Col. Reps. (Miscell.), No. 45, 1908, p. 53.
- [128] Maclaren, *Geol. Mag.*, 3, 1906, p. 536.
- [129] Mennell, *Geol. Mag.*, 6, 1909, p. 350.
- [130] Campbell, *Tr. Inst. M.M.* 19, 1910, p. 432.
- [131] Campbell, *op. cit.*, p. 209.
- [132] Maufe, *op. cit.*, p. 208.
- [133] Cf. also Mennell, *op. cit.*, p. 209.
- [134] Baikie, *A Journey from Bida to Kano*, 1866.
- [135] Cf. Freydenberg, *Tchad et Chari*, 1908, p. 1.
- [136] Courtet, *C. Rd. Ac. Sc.*, 140, p. 163.
- [137] Cf. Courtet, *C. Rd. Ac. Sc.*, 140, p. 160; Garde, *C. Rd. Ac. Sc.*, 148, pp. 1616, 1698.
- [138] Freydenberg, *Tchad et Chari*, 1908, p. 63.
- [139] Garde, *C. Rd. Ac. Sc.*, 148, pp. 1698, 1616.
- [140] Chudeau, *Sah. Soud.*, 1909, p. 232.
- [141] Freydenberg, *Tchad et Chari*, 1908, p. 61.
- [142] Chudeau, *Sah. Soud.*, p. 84.
- [143] Passarge, *Adamaua*, 1895, p. 400.
- [144] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 368.
- [145] Parkinson, *Q. J. G. S.*, 63, 1907, p. 309.
- [146] Esch, *Geol. von Kamerun*, 1904, p. 14.
- [147] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 219.
- [148] See Gautier, *Sahara Algerien*, 1908.

CHAPTER VII

TERTIARY CRUSTAL MOVEMENTS AND DRAINAGE CHANGES

Post-Cretaceous erosion in the Sudan; the Middle Eocene depression; western and eastern basins of sedimentation; crustal oscillations of the Middle Tertiary period; later Tertiary oscillations; the first period of depression; the first period of elevation; early drainage system of the Protectorate; the second period of depression; lacustrine conditions in Bornu; the second period of elevation; flexuring of the crust; α - and β -flexures; differential elevation of the Nassarawa tableland and the Bauchi plateau; deflection of the Gongola; depression of the Chad area. Peculiarities of the watersheds and river valleys of the Protectorate: explanation of the characteristic features of the valleys of the Niger, Benue, and Gongola; the origin of inselberge; M. Hubert's explanation of the hydrography of Dahomey; the period of formation of the "dallols" of Sokoto; correlation of the Tertiary crustal movements in the Sudan with those of the Mediterranean area.

THE close of the Cretaceous was marked by regional upheaval, accompanied by fracturing and folding of the crust and by subsequent intense erosion. The Cretaceous sandstones, shales, and limestones, which may be presumed to have at one time covered the greater part of the Protectorate, were largely removed from the underlying hummocky and irregular surface of crystalline rocks. The plateau of southern Yola, as described by Mr. Longbottom ([p. 32](#)), and the eastern cliffs of the central tableland of Kabba ([p. 16](#)), would appear to have originated at this period. The final character of the post-Cretaceous drainage system and of the early Eocene land surface is, however, very imperfectly known. Mr. Longbottom believes that there were earlier river valleys in the place of the Upper Benue and Lower Gongola of the present day ([p. 190](#)), and there is some reason to believe that there was also an earlier river in the present valley of the Lower Niger. At all events it is in these localities that the greatest thickness of Eocene sediments is now preserved, although there is little doubt that the later crustal movements have also, to some extent, affected their present distribution.

The movement of depression in the Middle Eocene which resulted in the establishment of open sea in the north-west and in the deposition of the marine beds of Sokoto province, brought about also the gradual submergence of the land surface and the predominance of shallow water

conditions over the greater part of the Protectorate. In the earlier river valleys the greatest thickness of sediments would naturally be deposited, while only the ferruginous sandstones and ironstones of the latest lacustrine phase would be deposited over the higher lying portions of the early Eocene land surface. The formation of the lava fields of Bauchi and Burra was probably contemporaneous with the latest phase of sedimentation. There is some reason to believe, moreover, that during the greater part of the period occupied by the movements of depression, there were two major regions or basins of sedimentation, a western and an eastern, separated by a stretch of higher ground which extended from Zinder and the northern frontier of Hausaland to the southern border of the Benue valley, and of which the Murchison Range formed the only striking physical feature. The thinning out of the Eocene sandstones to the east of the Niger valley, on the middle Benue, and to the west of the Gongola, lends some support to this hypothesis. The accumulations within the western basin, which was probably continuous westward with the Atlantic, would then include the lacustrine series of Southern Nigeria, the shallow water deposits of the middle Niger, the marine limestones of Sokoto and the sandstones of Dahomey and the Futa Djallon, while the Duguri, Gombe, and Kerri Kerri sandstones, the Upper Benue sandstones and their prolongation into the northern Kameruns, together with the ferruginous sandstones of the Upper Shari and the Congo, would represent the accumulations within the eastern basin. The outliers of Eocene upon the Kanna hills and the relics of ferruginous sandstones upon the plains of Muri indicate, however, that the western and eastern basins became finally united, and it seems probable that towards the close of Eocene sedimentation, shallow water and lacustrine conditions prevailed over the greater part of the southern Sudan. Free communication may even have been established at this time over Central Africa, ^[149] between the western seas of the Upper Niger and the eastern seas of Lybia and Egypt. There is no reason to believe, however, that the lower marine facies of Sokoto ever extended eastward much beyond its present limits.

There can be little doubt that the close of the Eocene was marked by the cessation of the predominant movement of depression and the beginning of those crustal oscillations which culminated in late Pliocene or early

Pleistocene times in the formation of the Bauchi plateau and the establishment of the present river system. No deposits of later recognisable age than Eocene have been discovered within the Protectorate and it may therefore be presumed that until the close of the Miocene, when the sea is believed to have finally retired from the Sahara ([p. 194](#)), the surface of the Protectorate remained near or at the level of the sea. The predominant movement, however, was probably that of elevation, with the result that the uppermost members of the Eocene suffered some considerable erosion and were entirely removed from the greater part of the surface of the central provinces. The later Tertiary oscillations, which may be conveniently grouped as Pliocene, began with a negative or downward movement, which has already been termed the “first period of depression” ([p. 213](#)), and during which the crystalline rocks of the central provinces were deeply decomposed and probably also covered with a layer of drift. The “first period of elevation” which followed was marked by a somewhat irregular upheaval of the surface of the Protectorate. The maximum movement appears to have been attained in the east, where the Eocene and underlying rocks were elevated in a low extended arch whose axis formed the primary watershed and ran approximately from Eastern Yola to Zinder and Air. As a result of the erosion which followed the movement of elevation the greater part of the sheet of decomposed rock formed during the first period of depression was removed from the surface of the central plain. The crystalline areas assumed very largely their present physical features and characteristic scenery. The plains of Hausaland, however, were continuous with the plains of Nassarawa and Muri across the site of the Bauchi plateau and the Nassarawa tableland. The Murchison Range, the Wadai Hills, and the Ningi and Kagoro massifs remained then as now relics of a former erosion, and beyond acting as secondary watersheds exerted comparatively little influence upon the general system of drainage. The relics of higher beds occasionally found on the sandstone plateaux of the Niger may, however, be referred to this period of erosion. The larger rivers probably drained south-westward into Southern Nigeria and Dahomey across the present valley of the lower Benue and the Niger. The gorges of the Niger above Lokoja and between Jebba and Yelwa clearly indicate the much more recent origin of the present river valley. In the east, however, along the watershed the sedimentary rocks were deeply trenched and worn on either

side by the headwaters of the rivers. The valleys of the middle Gongola and of the Upper Benue present when compared with the Niger Valley all the characteristics of a more ancient topography. It seems probable indeed that towards the close of erosion the rivers which drained westward over the plains of Bauchi had already established the escarpments of Kerri Kerri, the narrow plains of the middle Gongola, and the cliffs of Waja and Tangale. An earlier river, moreover, flowed westward from the hills of Yola with its valley limited as now by the highlands of Wurkum and Waja, which for a time interposed a continuous barrier between the rivers of Bauchi and the headwaters of the Benue. The establishment of the gap at Kombo was probably accomplished by a tributary of the Maio Haul towards the close of erosion. To the east of the watershed the plains of eastern and central Bornu were likewise established at this period by rivers which probably drained eastward and southward over the site of Chad and the Shari valley into the Congo basin.

There followed, however, the “second period of depression,” during which the drift was accumulated upon the irregular surface of the plain. The maximum movement would appear to have taken place in the east in the region of the former elevation. It seems probable that not only was the earlier watershed depressed to such an extent that the lateral valleys in Bauchi and Bornu were filled with drift to a depth of 400 to 500 feet, but that lacustrine conditions prevailed farther east over the plains of Bornu and Chad and the present valleys of the Logone and the Shari. Another area of special depression was apparently situated in southern Zaria and western Bauchi, where the remains of a covering of drift 200 to 300 feet in thickness forms a series of flat-topped hills on the highest portion of the plateau. The older Benue gravels, as described by Passarge^[150] and Edlinger,^[151] were probably also accumulated at this period. The drainage in the earlier stages may be presumed to have been, in general, towards the areas of maximum depression, while the close of the period was marked by the prevalence of conditions of accumulation and of uncertain drainage over the whole surface. The inland seas of the Upper Niger, as described by MM. Chevalier,^[152] Gautier and Chudeau,^[153] should probably also be referred to this period, but it should be remembered that, although they may have been at times connected with the open sea to the west, there is, as in the

case of Lake Chad ([p. 217](#)), no reason to postulate any kind of continuity between them and the earlier Miocene or Eocene seas. [\[154\]](#)

Radical changes ensued upon the cessation of the movement of depression and the return of a positive movement of elevation in late Pliocene or early Pleistocene times. It should be remembered, however, that the irregular upheaval of the surface during the “second period of elevation” and the consequent changes of drainage were in no sense catastrophic. On the contrary, the movements and changes were gradual, and possibly to some extent successive and spread over a considerable interval of time. The regional elevation of the Protectorate, with the exception of the middle Benue valley, to an average height of 1,000 feet was accompanied and complicated by a gentle flexuring of the crust into low arches and shallow troughs along axes which ran approximately W.S.W.–E.N.E. and N.N.W.–S.S.E. (see [map, p. 236](#)). The two sets of flexures may be termed respectively the α -flexures and the β -flexures, and it may be presumed that they were produced for the most part simultaneously and only to a small extent successively by crustal movements, which were themselves contemporaneously functional. The irregular elevation and depression of the surface which gave rise to the present system of drainage may therefore be looked upon as the result of the interference of the two sets of flexures. Two axial arches of the α -series, a northern (α_1) and a southern (α_2), are readily distinguishable within the Protectorate, the southern being the more important and of the greater elevation. The northern (α_1) runs from Borgu over the high plains of Kontagora, Sokoto, and Katsina towards Machina and Zinder, and forms the watershed between the Gulbin Kebbi and its tributaries on the one hand, and the Kaduna and the Yo on the other. These rivers may be presumed to occupy the shallow troughs on either side of the axial ridge. The axis of the southern arch (α_2) runs from central Kabba through Nassarawa and southern Bauchi into northern Yola, and the Benue may be presumed to occupy the trough to the south. The two α -arches are not strictly parallel, but diverge somewhat towards the east. Of the β -flexures two axial arches are also distinguishable within the Protectorate, a central (β_1) and a western (β_2), the latter being the more important of the two. The central arch (β_1) is of a very partial character, and is recognisable only in the sinuous watershed which runs northward between the two α -

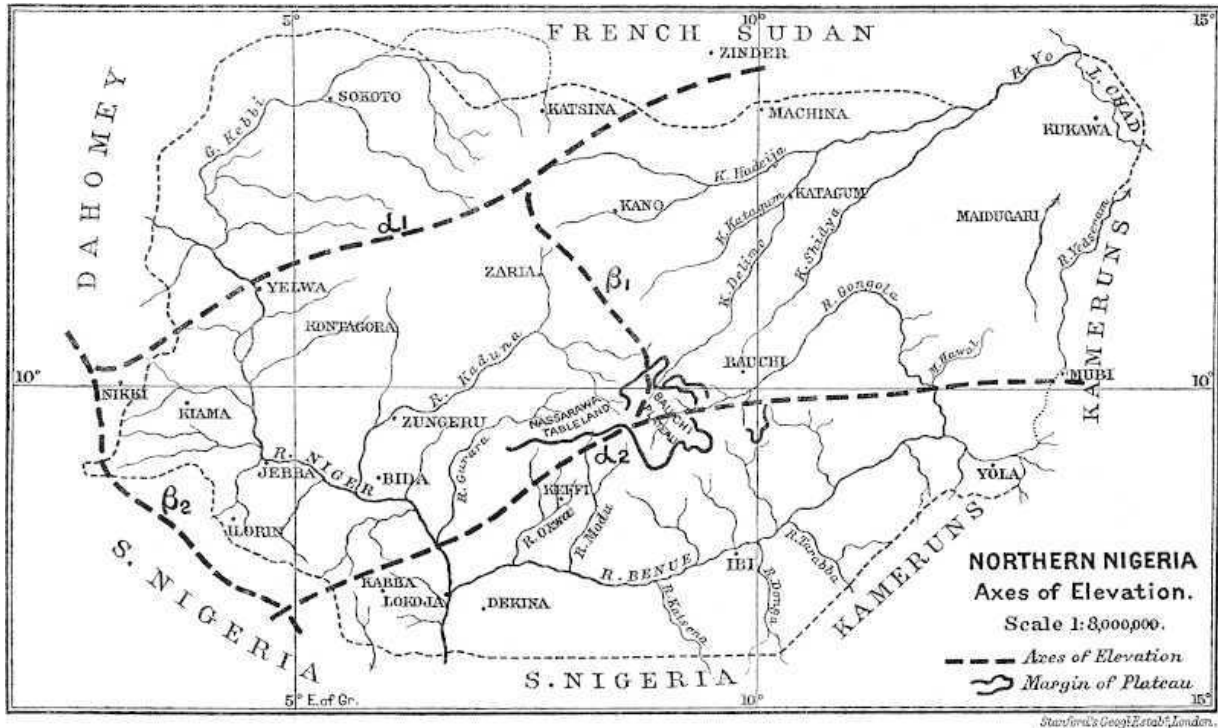
arches from the summit of the plateau towards the Sokoto-Kano frontier and separates the headwaters of the Kaduna from those of the Yo. The western arch (β_2) forms, in part, the western watershed of the Niger, and runs northward from the high plains to the south of Ilorin approximately along the frontier of the Protectorate to the neighbourhood of Nikki in Dahomey.

The arches throughout are asymmetrical in character, and vary somewhat in elevation from point to point. The sinuous character of the crests is probably due to the mutual influence of the complementary forces which produced the flexuring at right angles. The intensity of the flexuring appears also to have varied, and in the case of two intersecting flexures one or other usually predominates. In the central region, for example, the α -arches clearly predominate at their intersection with the β -arch (β_1), while in the west the β -arch (β_2) has evidently been the determining factor in the establishment of the present course of the Niger. The interference of the β -arch (β_2) with the western extension of the α -flexures is clearly reflected in the repeated bending of the river. Where the α -arches were crossed by the β -arch, dome-shaped elevations of a regional character were formed, whose effect may still be traced in the westwardly concave curves which the Niger describes at Yelwa and above Lokoja. Where the intervening α -trough, however, was crossed by the β -arch, the resulting curve was concave eastward as at Jebba. The asymmetrical position of the Niger valley indicates a shorter radius for the western portion of the β -arch, while the fact that the river was able to pursue a continuous course across the α -arches and troughs makes it sufficiently obvious that the β -arch (β_2) must have reached originally a somewhat greater elevation than the western portion of the α -arches. The influence of the formation of the western β -trough, now occupied by the Niger valley, may also be traced in the southward deflection of the lower courses of the Kebbi, Kaduna and Gurara. It is obvious also from the height of the sandstone plateaux above Lokoja that the greatest and perhaps most prolonged movement of elevation in the Niger valley took place in the region of the confluence of the Niger and the Benue. This movement, moreover, was evidently sufficiently slow to allow of the Niger cutting a continuous trench across the rising arch, a result, however, which was rendered possible only by the presence of

sedimentary rocks in the river bed. There is no reason to believe with Gurich^[155] that, during the formation of their trench-like valleys in the region of the confluence, the middle courses of the Niger and the Benue were ever converted into a series of detached basins or inland seas. It should be remembered also that the relative elevation of the arches was for the most part insufficient to produce any marked divergence from the apparent horizontality of the sandstones in the Niger valley.

NORTHERN NIGERIA Axes of Elevation.

MAP N^o. 3.



London. Macmillan & Co., Ltd.

Face p. 236.

The symmetrical elevation of the southern α -arch and the southward prolongation of the central β -arch were modified and interrupted by the formation of lines of weakness longitudinally along the former and in the region of the intersection of the two arches. The relative influence of the forces which produced the flexures can be traced in the varying direction and sinuous character of the lines of weakness. Along these lines differential movements took place which resulted in the disproportionate elevation of the Nassarawa tableland and of the Bauchi plateau with reference to the middle Benue valley and the northern plains of Bauchi. On the north the fracturing of the crust, as already described ([p. 39](#)), was confined only to the central region, while to the west and east the tableland and the plateau merged respectively into the general slope of the arch. On

the south, however, a continuous line of fracture extended from western Nassarawa by way of Darroro and Assab to Sura and Angass in southern Bauchi. There is no reason to believe that the southern α -arch as a whole ever reached the elevation of the Bauchi plateau or that the plateau originated in any sense as a “horst,”^[156] separated from the adjoining tableland and the plains of Nassarawa and Muri by the later differential subsidence of the latter. On the contrary, the movement appears to have been in one direction only, and characterised by the differential elevation of the plateau and the tableland and the lagging behind of the lower plains of Nassarawa, Muri and northern Bauchi. The trough of the Benue above the confluence of the Niger affords clear evidence indeed both of the comparatively stationary character of the middle Benue valley and of the gradual relative elevation of the Lokoja sandstones in the region of the confluence.

In the east the prolongation of the northern α -arch through Manga and the region to the north of Chad extends beyond the limits of the Protectorate. The southern arch, on the other hand, is prolonged as a simple flexure of varying elevation through eastern Bauchi into Yola province. In the trough between the northern and southern arches, the river Yo occupies the same relative position to the east of the central β -arch as the Kaduna does to the west. The presence in the east, however, of the former watershed of the first period of elevation has introduced complications into the otherwise simple relations of the α -axes and troughs. It may be presumed that on the formation of the southern α -axis and the differential elevation of the Bauchi plateau, the river Gongola, like the Shidya and the Delime, would have followed a general north-easterly course and ultimately joined the Yo but for the intervention of the higher ground along the earlier watershed which diverted the river to the south. It should be remembered that the whole of this region had been deeply trenched and eroded during the first period of elevation (p. 231) and that the earlier valleys had been largely filled with drift during the second period of depression. It seems probable, therefore, that on the subsequent elevation of the southern α -arch the Gongola on leaving the high plains of Bauchi was able to make its way for long distances through the earlier river valleys and stream courses and ultimately to escape into the Benue valley through the gap at Kombo. The

trench of the Gongola above Gombe may possibly be to some extent the product of recent erosion, but there is little doubt that the plain of the Gongola below Nafada was excavated at an earlier period and is in no way to be regarded as the result of the activity of the present river, whose energy has been almost entirely directed to the removal of the drift, which formerly obscured the hummocky surface of the plain.

The prolongation of the southern α -axis into eastern Bauchi appears to have been characterised by the formation of a flexure of a flattened monoclinal type, which resulted in the elevation of the plains of eastern Bauchi and the middle Gongola to a height of 500 feet above the valley of the Benue. The actual summit of the flexure is probably situated in the vicinity of the rapids of Kombo. The hydrography of northern Yola is very imperfectly known and complicated by the presence of the earlier watershed. It is probable, however, that the present watershed between the Benue and Lake Chad on the eastern frontier corresponds approximately with the prolongation of the α -axis, which here resumes its normal character of an asymmetrical arch. The last trace of the flexure may be recognised to the east of the Mandara hills in the swampy watershed between the Benue and the Logone. The configuration of southern Yola suggests an ancient topography, and its physical character is probably very similar to that which prevailed at the close of the first period of elevation. The Chad area to the north remained a region of relative depression during all the later flexures, and it seems probable that accumulation continued and the escape southward of the surplus waters remained possible until the final elevation of the Congo watershed along a parallel flexure to the south. The contemporaneous formation of the Bodele depression, however, brought about the establishment of the effluent of Chad through the Bahr el Ghazal and the partial erosion of the drift and lake deposits of Bornu. The subsequent advent of arid conditions during the period immediately preceding the present resulted, as already indicated ([p. 219](#)), in the diminution of the water supply, the closure of the Bahr el Ghazal, and the restriction of the lake to its present limits.

This attempted reconstruction of the more recent hydrographical changes affords an explanation of many of the more striking peculiarities in the physical geography of the Protectorate. Attention has already been

directed to the fact that the larger tributaries of the Niger and the Benue offer throughout their courses a reversal of the normal sequence of mountain tract, valley tract and plain tract. The Kebbi, the Kaduna, and the Gongola rise upon lofty plains and flow for long distances in broad sandy channels bounded by imperceptible slopes until they enter, in their lower courses, well-defined trench-like valleys bounded on either hand by steep walls of rock. The watersheds are marked by no ranges of jagged hills. On the contrary, the lofty plains on which the rivers rise are frequently more deeply buried in drift than the flood plains of their lower courses. The rivers flow peacefully around and between the lofty granite domes which rise abruptly from the upper plains, and even such larger groups of hills as those of Ningi form merely local centres of dispersion for the smaller streams. Moreover, the hydrographical centre of the Protectorate is not a rugged mountain mass, but an open treeless plain, a plateau 4,000 feet in height, bounded by steep cliffs on the south and west and provided with worn and flattened hillocks of drifted alluvium perched in places upon the very margin of the cliffs. A complete reorganisation and rejuvenation of the drainage system is clearly indicated, and it is possible only in some such manner as that already described. The crust has been flexured and fractured during a recent regional elevation, and from the crests of the arches the consequent streams have been thrown off to pick a course for themselves over the drift-covered plains and between the solitary hills and finally to unite to form the major rivers in the bottoms of the shallow troughs. Dr. Passarge^[157] has frequently called attention to the youthful age of the rivers of Africa, and his suggestion would appear to be amply confirmed by the whole character of the present river-system of Nigeria.

The sinuous course of the Niger within the Protectorate still represents in a general way the original adaptation of the drainage system to the tectonic conditions. The diversity of scenery in the present valley is, however, largely due to the influence of the lithological character of the underlying rocks. There is little doubt that at first the Niger flowed throughout upon a sandstone surface, relics of which now lie from 500–1,000 feet above its present level, and that it eroded its channel originally as a continuous, deep and narrow trench. Complications first ensued when the river succeeded in cutting through the overlying sandstones between Yelwa

and Jebba, and in reaching the crystalline rocks below. The greater thickness of sandstones below Jebba conditioned the cutting back of the gorge between Jebba and Yelwa with which the river is still occupied. The resistant character of the crystalline rocks brought about the formation of falls and rapids, while the consequent lateral erosion at the head of the gorge gave rise to the broad sandy plains between Yelwa and Ilo. Below Jebba the prolonged elevation of the southern α -arch brought about the continued trenching of the sandstones above Lokoja and the lateral erosion of the banks in Nupe and Ilorin, where the level of the river remained stationary for considerable intervals of time. The irregular degradation of the channel of the Niger conditioned also the formation of the broad, flat-bottomed valleys (dallols) of Sokoto and the narrow canyon valleys of southern Kontagora, the dissection of the sandstone plateaux of Nupe, the removal of the Eocene rocks from northern Kabba and the cutting of the lower gorge of the Gurara. Everything, indeed, points to the recent erosion of the Niger valley, while the rapids and falls of the Kaduna and the Gurara beyond the limit of the sandstones and the ill-defined valleys of their upper and middle courses clearly indicate that they have only just begun to cut their way backwards and downwards into the central crystalline plains.

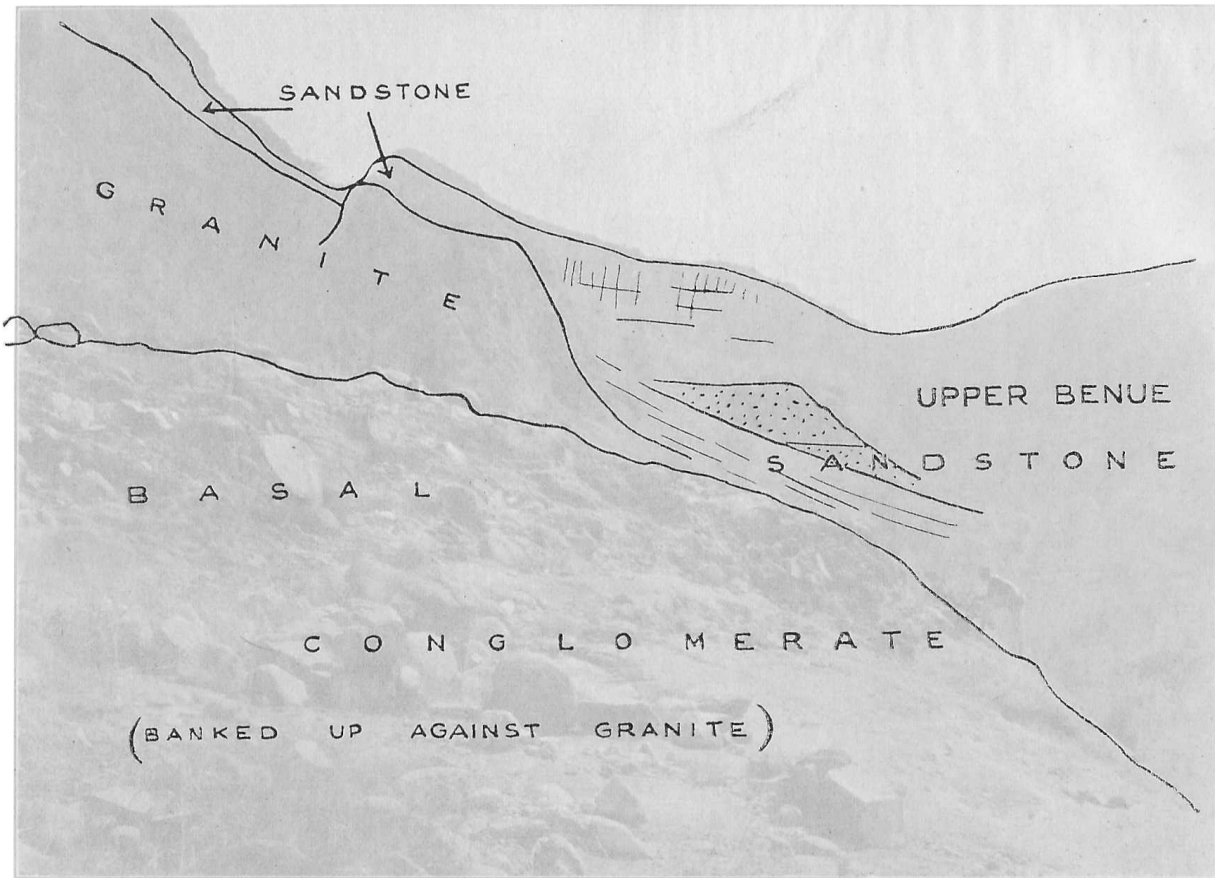
Compared with the Niger valley, the plains of the middle and upper Benue present all the characteristics of a more ancient topography, and it is probable that they retain to a great extent the configuration which they possessed at the close of the first period of erosion. Only in its lower course has the Benue cut a trench in the Eocene sandstones at all comparable with the gorge of the Niger above Lokoja, and this has probably been accomplished, as already explained ([p. 238](#)), during the slow differential elevation of the region of the confluence. There seems reason to believe, also, that during the earlier stages of the more recent erosion the Benue was provided with a greater volume of water than the Niger, and that to this influence is to be ascribed the greater imperfection of the trench of the lower Benue and the removal of the sandstones from the hummocky crystalline plain of eastern Kabba. This increase of volume, however, was probably only relative and contemporaneous with the diminution of the Niger during the recent arid period in the Sudan. The return of more humid conditions, however, together with the capture of the upper Niger by a

tributary of the Taffassasset, as described by M. Chudeau,^[158] restored the activity of the Niger and enabled it to cut its present channel in the surface of the crystalline plains above and below Lokoja.

The northern tributaries of the Benue, with the exception of the Gongola, take their rise on the southern border of the Nassarawa tableland and the Bauchi plateau. Their headwaters alone, however, exhibit evidence of any recent intense erosive activity. Behind Jagindi and in the neighbourhood of Assab the headwaters of the Modu river have trenched and worn the margin of the plateau. The Wase river has laid bare in Yergum and Angass the hummocky and irregular slopes which separate the upper from the lower plains. The headwaters of the Kudu river have cut picturesque steep-sided valleys in the rocks of Duguri, and narrow canyon-like trenches in the central plains of Bauchi. In the Gongola valley, on the other hand, the only traces of recent erosion are to be found in the rapids of Baddera and Kombo and, in part, in the imperfect trench which the river has cut in the sandstones between Gombe and Deu. From its source on the plateau until it enters the plains to the east of Gombe, the topography of the river valley recalls in every detail the characteristic features of the valleys of the Kaduna and the Gurara. Nevertheless, although of similar age and origin, the Gongola flows from Nafada southward through a valley which presents all the features of a more intense erosion than the Kaduna or the Gurara or even the Niger itself has been able to accomplish in its lower course. It is difficult to escape the conclusion that upon the rearrangement of the drainage system and the establishment of the present upper Gongola, the river entered to the east of Gombe, an older river valley which had been excavated during some former period of erosion. The interlacing valleys of the margin of the Kerri Kerri plateau may possibly have arisen during the latest period of erosion, but if so, the streams which excavated them have never regained the activity which they possessed before the last appearance of the desert in the Sudan.



UPPER BENUE SANDSTONES AND CONGLOMERATES UPON A HUMMOCKY GRANITIC FLOOR.



The isolated domes and kopjes ([Plates III., XII.](#)) and detached groups of hills which form the most striking feature of the higher plains are characteristic also of all the older gneissic areas of Africa, and the question of their origin has given rise to considerable difference of opinion. The particular type of scenery to which they give rise has been termed by German writers the “Inselberglandschaft” (Plate XVII., [Figs. 1, 2](#)). Passarge^[159] sees in them the product of arid conditions, and concludes that the surfaces which they characterise assumed their present character for the most part in early Mesozoic times when the greater part of Africa was a barren desert. Bornhardt,^[160] on the other hand, believes that the inselberge have arisen as the result of the superposition of several cycles of erosion. There is, however, no reason to believe that this peculiar topography cannot arise simply through an alternation of periods of weathering and periods of erosion, brought about either by a gentle oscillation of the crust or by a repeated base levelling of the plains and rejuvenation of the drainage system. A plane surface of granite and gneiss subjected to long-continued

weathering at base level would be decomposed to unequal depths, mainly according to the composition and texture of the various rocks. When elevation and erosion ensued, the weathered crust would be removed, and an irregular surface would be produced from which the more resistant rocks would project. Those rocks which had offered the greatest resistance to chemical weathering beneath the surface would upon exposure naturally assume that configuration of surface which afforded the least scope for the activity of the agents of denudation. In this way would arise the characteristic domes and turtle-backs which suffer further denudation only through insolation and exfoliation.^[161] Their general elliptical outlines, which Merrill^[162] would ascribe very largely to the influence of crustal stress and strain, are probably in great part due simply to the modification by weathering of original phacolitic intrusions. A slight subsequent movement of depression would suffice to cover with drift large areas of the irregular surface between the domes and to produce the characteristic topography of isolated hills rising abruptly from the level surface of the plain (Passarge's Kordofan Type). That such indeed has been the mode of formation of the kopjes of Hausaland can be demonstrated within the walls of Kano. Kogon Dutsi, the larger of the two flat-topped hills of diorite, although deeply decomposed, still preserves in its lower part detached boulders or cores of unweathered rock. If the subsequent erosion had continued until the weathered material had been entirely removed, the flattened hill would have been replaced by a typical kopje of loose boulders resting upon a smooth and rounded surface of rock below. On the plains around ample evidence can be found of the levelling of the surface by means of the accumulation of drift. A repetition of these conditions of formation would give rise to an accentuation of the earlier domes and kopjes and the formation of others of lower level. A similar origin to that of the domes may be ascribed also to the detached groups of hills and even to the larger granite massifs like those of the Ningi, Ruruma, Kagoro, and Vere Hills. There is no reason to believe with Passarge^[163] that the latter partake in any way of the nature of "horsts," which have assumed their present position through the differential subsidence of the surrounding plains. Their flattened though hummocky summits evidently bear witness to an earlier regional level of erosion, but their characteristic mode of occurrence is to be

ascribed not to marginal faulting, but to a succession of periods of decomposition and accumulation and periods of erosion which has brought about the relative lowering of the surrounding and less resistant gneisses and schists.

The pre-Cretaceous land surface, as exposed around Tangalto, was marked like the present with domes and inselberge. The pre-Eocene surface was somewhat broken and irregular, but probably also characterised, as in the Niger valley below Lokoja, by detached and solitary hills. There is no reason, however, to suppose that the actual pre-Cretaceous configuration of the plains of Hausaland is still retained. It is much more probable that domes and solitary hills have been formed in the manner indicated at many different periods, and are a characteristic product of sub-aerial tropical weathering of a crystalline surface when exposed to oscillations of level.

Dr. Passarge,^[164] in explanation of the structure of Adamawa, has assumed the existence of two tectonic directions, an E.–W. or Benue line, and a S.S.W.–N.N.E. or Kamerun line, along both of which extensive faulting has taken place at various periods and in such a manner as to leave a numerous series of isolated hills and ranges of hills standing out as horsts upon the lower plains. According to him, also, the Bagele and other hills of Yola have assumed their present position through the subsidence of the surrounding plains, while the Benue valley between Yola and Garua is of tectonic origin, and bounded by faulted margins to the north and south. While, however, there is some reason to believe that the Cretaceous rocks of Yola have been faulted down against the crystallines to the north and south, and that in pre-Eocene times an earlier river occupied a tectonic valley in the region of the upper Benue (p. 190), there is no evidence to show that there has been a recurrence of faulting on the upper Benue in post-Eocene times, or that the slopes of any of the hills of “Upper Benue Sandstone” have been defined by later Tertiary or recent movements. Passarge’s hypothesis is applicable only to pre-Eocene topography, and cannot be extended in explanation of those features of the upper Benue valley which are of more recent origin.

M. Hubert^[165] has suggested in explanation of the hydrography of Dahomey that the courses of the present streams bear a direct relationship

to the strike of the gneisses and schists, and that the surface of the colony has remained stationary since the close of the Eocene. His hypothesis, however, is admittedly inadequate to explain the transverse character of the Niger valley. In Nigeria, moreover, all the principal rivers without exception flow across the strike of the gneisses, while the drift-covered character of the lofty plains and of the watersheds and the nature of the whole hydrographical system clearly indicate the recent elevation of the present surface of the Protectorate. It is very probable indeed that Dahomey has participated in all the recent crustal movements which have affected Nigeria, and that the central watershed of the colony corresponds with the westward prolongation of the northern α -arch and the western β -arch of the Protectorate. A recent rejuvenation of the drainage system is postulated, moreover, by M. Chudeau^[166] in the French Sudan, by Mr. Parkinson^[167] in Southern Nigeria, and by Dr. Esch^[168] in the Kameruns, and it is improbable that Dahomey should have escaped the general movement of elevation.

M. Chudeau^[169] has recognised in the Sudan the recent period of arid conditions of which traces are still to be found within Northern Nigeria. He believes, however, that the dallols were excavated during the period of desert conditions by rivers which drew their supplies from the mountains of Air. It is possible that there may have been repeated alternations of desert and humid conditions within the latest period of erosion; but it is clear in Nigeria that the dallols of Sokoto were already excavated before the advent of the latest arid period, and that the rivers which formed them have not regained the activity which formerly characterised them. Capt. Meynier's suggestion, quoted by M. Chudeau, that the rivers of Hausaland turned their backs to the desert, though picturesque, is little in harmony with the facts. The courses they follow within the Protectorate were defined long before the advent of the desert, and not by any adaptation to climatic conditions, but by the direct influence of the early Pleistocene crustal movements which brought about the rejuvenation of the drainage system of Nigeria and the formation of the Bauchi plateau. These movements were probably also responsible for the establishment of the dallols of Tahoua and the quaternary wadis of the northern Sahara, as well as for the elevation of the plateaux of Air and Adr'ar' des Ifor'as. These movements, moreover, were

probably contemporaneous with the later Tertiary movements in the Mediterranean basin and in East Africa.

M. Chudeau^[170] has pointed out the analogy between the pre-Devonian and the post-Carboniferous movements to the south of the Atlas and the Caledonian and hercynian movements to the north of the Alps. In Nigeria and the Sudan some analogy may perhaps be permitted between the general meridional strike and vertical folding of the gneisses and schists and the Caledonian folding of Scotland and Scandinavia. Of hercynian movements, however, there is little or no trace in the Sudan, while the presence of an unconformability at the base of the Eocene and the absence of any later folding makes the post-Cretaceous and early Tertiary movements in the Sudan more nearly parallel with the sequence of events in northern Europe than with the sequence in the Alps and the Atlas. The period of maximum movement in the Alps would appear to be represented in Nigeria and the Sudan only by minor post-Eocene oscillations of the crust, while the major movements in the Sudan which gave birth to the present drainage system appear to be contemporaneous with the later Mediterranean stages of the Alpine movements.^[171] At the same time, however, it should be emphasised that, while the movements may have been contemporaneous, there was probably little or no genetic connection between them. The Tertiary period was everywhere a period of great instability of the crust, and the movements would appear to have varied both in intensity and in kind, and to have been localised now in one place and now in another, at different periods of their history.

[149] Cf. De Lapparent, *C. Rd. Ac. Sc.*, 136, p. 1118.

[150] Passarge, *Adamaua*, 1895, p. 385.

[151] Edlinger, *Niger-Tsad Expedition*, 1904, p. 157.

[152] Chevalier, *C. Rd. Ac. Sc.*, 15 April, 1901, p. 926.

[153] Gautier and Chudeau, *Sah. Soud.*, 1909.

[154] Cf. Suess, *The Face of the Earth*, Vol. IV., Oxf. Trans., p. 92.

[155] Gurich, *Zeit. Deut. Geol. Gesell.*, xxxix, 1887, p. 126.

[156] Cf. Passarge, *Adamaua*, 1895, p. 377.

[157] Passarge, *Die Kalahari*, 1904, p. 637.

[158] Chudeau, *Sah. Soud.*, 1909, p. 222.

- [159] Passarge, *Rumpffläche und Inselberge*, Z. D. G. G., 56, 1904, p. 193.
- [160] Bornhardt, *Deutsch Ost Afrika*, Bd. VII, 1900, p. 37.
- [161] Cf. Harrison, J. B., *Handbook of British Guiana*, p. 87.
- [162] Merrill, *Rocks and Soils*, 1897, p. 245.
- [163] Passarge, *Adamaua*, 1895, p. 377. Cf. also Suess, *The Face of the Earth*, Vol. IV, Oxford Trans., 1909, p. 283.
- [164] Passarge, *Adamaua*, 1895, p. 387. See also Suess, *The Face of the Earth*, Vol. IV, 1909, p. 280.
- [165] Hubert, *Miss. Sc. au Dahomey*, 1908, p. 443.
- [166] Chudeau, *Sah. Soud.*, 1909, p. 220.
- [167] Parkinson, G. J., XXIX., 1907, p. 56.
- [168] Esch, *Geologie von Kamerun*, 1904.
- [169] Chudeau, *Sah. Soud.*, 1909, p. 220.
- [170] Chudeau, *Sah. Soud.*, 1909, p. 1.
- [171] Suess, *The Face of the Earth*, Vol. I., Oxf. Trans., 1904, p. 279.

CHAPTER VIII

TERTIARY VOLCANIC ACTION

Absence of pre-Tertiary volcanic rocks within the Protectorate: two periods of Tertiary volcanic activity. The early Tertiary period: the Maradu vent; the nepheline basalts of Wase and Tangalto; the phonolites of Wase Rock, Tangale Peak and Bantaji; the plateau lavas of southern Bornu, Bauchi and Nassarawa; petrographical character of the lavas; early Tertiary lavas of the Sahara, Sudan and Adamawa. The later Tertiary volcanic period: the puyes of northern Yola, the middle Benue valley and the Bauchi plateau; petrographical character of the lavas; petrographical provinces.

It may be presumed that the intrusion of the alkali granites and dyke rocks throughout the Protectorate was accompanied in pre-Cretaceous times by considerable superficial volcanic activity (p. 142). M. Chautard^[172] has, moreover, described the intercalation of volcanic rocks in the Upper Cretaceous of Senegal. All trace of pre-Tertiary effusive rocks, however, within the Protectorate appears to have been removed in the course of denudation. The fracturing and folding of the crust at the close of the Cretaceous was apparently accompanied in places by the formation of mineral veins within the fissures, but no contemporaneous igneous activity has been definitely identified within the Protectorate (p. 140). Within the Tertiary era itself two separate periods of volcanic activity may be recognised, one in the middle Eocene and probably to some extent contemporaneous with the accumulation of the Eocene sandstones within the Protectorate, and one in late Pliocene or early Pleistocene times and contemporaneous with the later Tertiary oscillations of the crust. For the most part the eruptions were of an isolated character and gave rise to the formation of detached pipes, cones and puyes of igneous material. More rarely, as in southern Bornu and on the borders of Bauchi and Nassarawa, the extruded lavas united to form a continuous sheet of volcanic rock, which covered a considerable area of the underlying sedimentary and crystalline rocks.

The earliest outburst of volcanic activity appears to have taken place towards the close of the post-Cretaceous erosion or possibly in the middle

Eocene at a period contemporaneous with the accumulation of the lower marine beds of Sokoto, and is represented now by the extensive neck or vent of fragmental material near Maradu in Sokoto province whose levelling was accomplished during the later extension of the middle Eocene sea. The greater number of the boulders in the agglomerate are composed of a microgranitic quartz porphyry with phenocrysts of orthoclase, biotite and brown hornblende in addition to quartz (1323). To the same period is probably also to be referred the line of well-worn stumps of basalt, phonolite and trachyte which stretches from the Gongola valley westward through Tangale and Angass to Namu in northern Muri. The Tangale Peak (Pl. XIII, [Fig. 2](#)) and the Wase Rock (Pl. V, [Fig. 2](#)), which still retain a portion of the original flow attached to the rock in the vent, were probably among the latest eruptions of the series. The intrusive dolerites of Bashar (41) and Awe (1014) may be ascribed to the same phase of igneous activity, while the Rock of Bantaji in Southern Muri may be taken to represent contemporaneous volcanic action to the south of the Benue.

In petrographical character the basaltic pipe rocks are always olivine-bearing and usually porphyritic with olivine and augite. Felspar is found only in the ground-mass which at times contains a little intersertal glassy base. The basalt of Wase Topa is remarkable for the number and size of the nodules of olivine which it encloses. A worn vent immediately south of the town of Wase is filled with a fine-grained nepheline basalt (32) containing glomeroporphyritic aggregates of augite crystals and well-formed crystals of olivine. Felspar is entirely wanting, the ground-mass being composed of small rods of augite and granules of magnetite set in a brown glassy base studded with minute crystals of nepheline. A somewhat similar but coarser-grained rock from Tangalto (1661) contains large crystals of decomposed olivine and purple and well-zoned augite in a holocrystalline ground-mass of rods and granules of augite, magnetite grains, flakes of biotite and abundant nepheline in well-formed and irregular crystals. The nepheline was evidently the last to crystallise, and to some extent takes the place of matrix, being in part moulded upon the other constituents of the ground-mass.

The phonolite of the Wase Rock (31) is of a trachytoid type, containing columnar phenocrysts of sanidine, ragged crystals of ægirine augite and

scattered pseudomorphs after nepheline in a ground-mass consisting predominantly of minute felspar prisms with a well-marked fluxional arrangement, mingled with magnetite grains and rods and granules of green pyroxene. The phonolite of Tangale Peak (1665–6) contains phenocrysts of sanidine and soda pyroxene in a ground-mass of a more orthophyric type, composed of minute idiomorphic crystals of nepheline, imperfect prisms of ægirine and ægirine augite and elongated microlites of felspar which pass into an ill-defined feldspathic matrix. Traces of brown hornblende in irregular porous growths may also be observed in the ground-mass. A similar rock from the immediate neighbourhood of Tangalto (1663) contains soda pyroxene, but no evident nepheline in the ground-mass, and may be grouped provisionally as a trachyte. The phonolite of Bantaji Rock is of a more nephelinitoid type, with the ground-mass enclosed in a somewhat intersertal manner between the abundant but much decomposed phenocrysts of felspar. With the felspar are associated as phenocrysts hypidiomorphic crystals of ægirine and ægirine augite frequently in zonal growths and smaller idiomorphic and abundant crystals of somewhat weathered nepheline. The ground-mass is composed of lath-shaped felspars, granular pyroxene and irregular patches of analcime probably of secondary origin. The association of phonolite and nepheline basalt, observed in the neighbourhood of Wase and the Peak, has not been recognised around Bantaji.

The period of formation of the lava fields in southern Bornu and on the borders of Bauchi and Nassarawa may be provisionally referred to the close of the middle Eocene, on the strength of the apparent intercalation of a friable red sandstone amongst the lavas in the escarpment of the Barbur plateau behind Dukshi. The extensive erosion which the plateau lavas themselves have undergone indicates, moreover, that they are of no very recent age. The eastern lava field is the more extensive of the two and perhaps somewhat younger, so far as one can judge from the less denuded character of the cones. The rich content in iron of the upper part of the Eocene accumulations may possibly have been conditioned to a considerable extent by the distribution of much fragmental basic igneous material upon the surface of the waters.

In petrographical character the lavas are basaltic throughout and rich in olivine, and vary from felspathic doleritic types to limburgitic varieties in which felspar is greatly reduced in amount.^[173] Nepheline-bearing types have nowhere been recognised, nor any effusive rocks more acid than basalt. MM. Chudeau and Gentil^[174] have, however, indicated the occurrence of ægirine rhyolites and phonolites, as well as of basalts, tephrites and limburgites, amongst the older lavas of Ahaggar and Air, which are probably of similar age to those of Nigeria. Dr. Passarge^[175] has described the occurrence of basalt, augite andesite and phonolite on the summit of the Shebshi hills on the borders of Nigeria and the Kameruns. MM. Gentil and Freydenberg^[176] believe that the ægirine porphyries of Mounio and of Hadj el Hamis on the southern shore of Chad are of volcanic origin. M. Garde,^[177] however, is of opinion that they possess an intrusive character and belong to the hypabyssal phase of the alkaline magma which gave rise to the Mounio and Melfi granites.

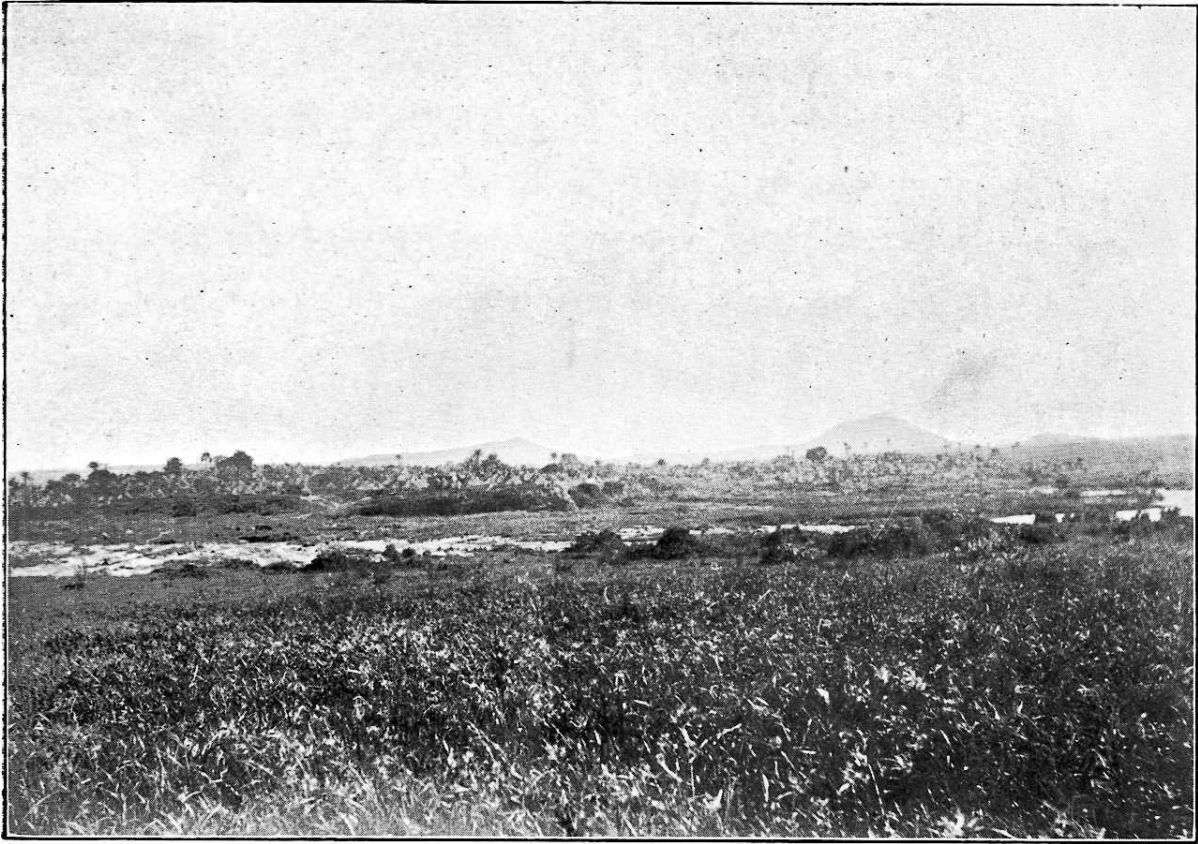
The plutonic rocks of this early Tertiary volcanic period are nowhere exposed within the Protectorate, and hypabyssal types are almost unknown. The doleritic sills of Awe and Bashar were probably injected during one of the earlier phases of igneous activity. Dr. Passarge^[178] has indicated the occurrence of basaltic dykes in the sandstones of Adamawa and a boss of nepheline syenite between Yola and Garua on the upper Benue. The latter, however, more probably belongs to the underlying crystalline floor. M. Chudeau^[179] believes that the alkali granites of Zinder and Mounio have been intruded since the close of the Cretaceous; but while there is every reason to believe that the nepheline basalts and ægirine rhyolites of Tertiary age drew their origin from a magma rich in soda, there is, as already stated ([p. 140](#)), no evidence within the Protectorate upon which any immediate connection between the younger soda granites and the Tertiary volcanic rocks can be established. On the contrary, there would appear to have been in the middle Eocene within the Protectorate merely a renewal of volcanic activity from the same alkaline source from which the pre-Cretaceous granites and dyke rocks originated. If, however, further investigation should confirm M. Chudeau's suggestion of the Tertiary age of the granites and porphyries of Zinder, Mounio and Goure, these rocks may then be taken to

represent the plutonic and hypabyssal phases of this middle Eocene period of igneous activity.

To the later Tertiary period of volcanic activity, which may be referred to the close of the Pliocene and which perhaps also extended into the early Pleistocene, are to be assigned the puy of Kereng (Plate X, [Fig. 2](#)) on the southern margin of the Bauchi plateau, the puy of Song and Mboi (Plates VI, [Fig. 2](#); VII, [Figs. 1, 2](#)) in northern Yola, and somewhat doubtfully also those of the Awe district on the middle Benue (Plate V, [Fig. 1](#)). The puy of the Kereng group are remarkably perfect and exhibit steep crater walls of fragmental material breached by a single short flow of olivine basalt. Elsewhere on the plateau also, as between Vom and Bukuru, there are small puy with breached craters which are probably of a similar recent age. From the cliffs of Assab, moreover, later flows of lava may in places be seen on the worn slopes of the plateau edge, and there is every reason to believe that the erosion of the drift and of the margin of the plateau was considerably advanced before the final close of volcanic activity. The presence of craters and the local character of the associated flows serve to distinguish these later products of igneous activity from the earlier lavas of the plateau. A similar superposition of later volcanic material upon earlier lavas has been observed by M. Chudeau^[180] in the Ahaggar and in Air. Unlike the central Sahara, however, there is in Nigeria no reason to believe that there has been any actual continuity in volcanic action between the earlier and the later periods.

In Yola province the Mboi Hills are crossed by a line of small craters four or five miles long, running roughly in a N.W.–S.E. direction. To the south-east between the Mboi Hills and Song the line is continued by a number of hillocks of basalt for a few more miles, but how far it extends to the N.W. is not known. Only a relatively small amount of basalt was erupted from these vents. In the case of the largest crater with a crater floor of 250 yards in diameter the volcanic material was not sufficient to conceal the granite through which it came. The granite was broken away during the eruption so as to form a cliff extending one-third round the crater, while the other two-thirds is formed of agglomerate and tuff with very many fragments of granite. The crater thus resembles a swallow's nest attached to its support. With one exception, where the wall is entire, all the craters were

breached by a lava flow. They are in excellent preservation, and the cliff-like walls, with insignificant talus heaps at their bases, indicate a very recent origin. (A. L.)



THE SALT-FIELDS OF AWE; VOLCANIC HILLS IN THE BACKGROUND.



AKIRI SALTINGS.

The vents on the plain to the south-east are not marked by craters or by fragmental materials, but by mounds of basalt less than fifty feet high spreading out into lava flows, which cover most of the surface near by. Here the eruptions seem to have consisted in a much quieter extrusion of lava alone. The basalts are all vesicular and very commonly quite light and cellular. (A. L.)

The basalts of the Benue plain likewise form low hillocks or mounds with no tuff or fragmental material to be seen. Generally there is no lava flow proceeding from them, and the extrusion of quite a small quantity of lava seems to have sufficed to relieve the pressure. Kaddemi Hill is an exception; an old basalt flow leads from it for two to three miles to the south-east. (A. L.)

All the basalt flows around Awe are raised considerably above the level of the plain, and protect the limestone shales below from denudation, though elsewhere the shales are very badly exposed. In consequence the

flows rest on “pedestals” from thirty to fifty feet in height. Assuming that the lava chose the level surface of the plain rather than ridges upon it, it follows that since the eruptions around Awe there has been a general lowering of the surface by that amount. From this it appears that the Awe eruptions were not quite contemporaneous with those of Mboi, for at the latter place only a small amount of talus has been broken off from the crater cliffs. (A. L.)

The ash cones and puy of the middle Benue are much more imperfectly preserved than those of Kereng or Mboi and are certainly considerably older. Their comparatively recent age is presumed, however, as in the case of the later volcanic period in Air and Tibesti,^[181] from the presence of thermal springs in the immediate neighbourhood. It seems probable that they were established as the result of the crustal movements anterior to the accumulation of the drift, while the puy of Kereng and Mboi owe their origin to the differential movements of the latest period of elevation.

In petrographical character these later rocks are very similar to the earlier Tertiary lavas. All are basalts rich in olivine and augite and varying in the same locality from types with abundant felspar in the ground-mass to thoroughly limburgitic^[182] varieties (12) from which felspar is entirely wanting. The ash of the Kereng puy is remarkable for the number and size of the rounded augite crystals and olivine nodules which it contains. Recent flows of olivine basalt and limburgite are known also from Ahaggar and Air^[183] and from the Kameruns.^[184] M. Chudeau^[185] has described rhyolites from Inzize to which he assigns a later Tertiary age, Dr. Passarge^[186] has noted a nepheline basalt from an isolated cone of recent origin in the Benue valley between Yola and Garua, and Dr. Esch^[187] has described nephelinites and leucitites from the Kamerun Mt. which is still periodically in active eruption. No rocks of similar composition, however, have been recognised amongst the products of the most recent volcanic period within the Protectorate.

MM. Chudeau^[188] and Chautard,^[189] following M. Lacroix,^[190] have suggested that the alkaline rocks of North Africa may be grouped into three petrographical provinces, a western province including the Azores, Canaries, Cape Verde and Senegal, a central province including Ahaggar,

Inzize, Air, Zinder and Mounio, Chad and the Shari valley, the Kameruns, and Dahomey, and an eastern province including Abyssinia, Arabia and Somaliland. Upon this hypothesis the alkaline rocks of Northern Nigeria belong to and complete the central province of Lacroix, as extended by M. Chudeau. It is probable, however, that this central province includes intrusive and volcanic rocks of many different ages, and that alkaline types have been a characteristic product of differentiation from the earliest times to the present day, ([p. 142](#)). There is, moreover, every reason to believe that the progress of investigation will reveal many connecting links between the various provinces, which are themselves somewhat too extensive and comprehensive to warrant their recognition as such. A petrographical province in the stricter sense of the term may perhaps be set up to include the alkaline plutonic rocks of Bauchi province and their associated hypabyssal types, but there can be little doubt that the general alkaline character of the igneous rocks of northern Africa, to which Lacroix has given expression, may be quite sufficiently indicated by emphasising their natural classification within the Atlantic branch of igneous rocks. [\[191\]](#)

[\[172\]](#) Chautard, *Bull. Soc. Geol. Fr.*, 4th Series, VII., 1907, p. 427.

[\[173\]](#) These rocks present in microscopical structure a decided resemblance to the Carboniferous lavas of the Bathgate Hills (Falconer, *Trans. Roy. Soc. Edin.*, Vol. XLV., 1906, p. 133) in Scotland. With the exception that the doleritic types are somewhat less weathered, the description and classification of the Bathgate rocks may be applied almost word for word to these Tertiary lavas of Nigeria. Doleritic rocks (Falconer, *op. cit.*, Plate I., Fig. 2) with phenocrysts of augite and olivine, and more rarely of feldspar are common on the Morrua plateau in eastern Nassarawa and along the base of the escarpment between Darroro and Assab (928, 1573, 1630). Feldspar is abundant in the ground-mass and the augite occurs in granules, granulitic aggregates, small ophitic plates, or minute idiomorphic crystals between the feldspars. The basalts of Zammagan (1578) and Buratai (1640) show a typical fluxional arrangement of the feldspars (Falconer, *op. cit.*, Pl. I., Fig. 3). The basalt of Kwoia (1654) exhibits an intimate mixture of feldspar and augite in the ground-mass (Falconer, *op. cit.*, Pl. I., Fig. 4). The basalt of Assab (934) contains scattered feldspars with abundant minute granular aggregates of augite, while a similar rock from Buratai and Dukshi (1641) contains some residual brown glassy material and passes into limburgitic varieties with phenocrysts of augite and olivine in an ill-defined microlitic base.

[\[174\]](#) Chudeau, *Sah. Soud.*, 1909, p. 258; Gentil, *Doc. Sc. Miss. Sah.*, 1905, p. 726.

[\[175\]](#) Passarge, *Adamaua*, 1895, p. 374. See also Downes, *Geol. Rep. Yola-Cross River Boundary Commission*, 1910, p. 35.

- [176] Gentil and Freydenberg, *Bull. Soc. Geol. Fr.*, 4th Ser., VIII., 1908, p. 35; Lacoïn, *C. Rd. Ac. Sc.*, 22 June, 1903.
- [177] Garde, *C. Rd. Ac. Sc.*, 149, 5 July, 1909, p. 43.
- [178] Passarge, *Adamaua*, 1895, pp. 363, 384.
- [179] Chudeau, *Sah. Soud.*, 1909, p. 266.
- [180] Chudeau, *Sah. Soud.*, 1909, p. 361.
- [181] Chudeau, *Sah. Soud.*, 1909, p. 264.
- [182] Cf. Falconer, *Trans. Roy. Soc. Ed.*, XLV., 1906, Pl. I, Fig. 6.
- [183] Chudeau, *Sah. Soud.*, 1909, p. 361.
- [184] Esch, *Geologie von Kamerun*, p. 40.
- [185] Chudeau, *op. cit.*, p. 259.
- [186] Passarge, *Adamaua*, 1895, p. 384.
- [187] Esch, *Sitz. Ber. Ak. Wiss. Berlin*, 1901, p. 277.
- [188] Chudeau, *op. cit.*, p. 268.
- [189] Chautard, *Bull. Soc. Geol. Fr.*, 4th Ser., VII., 1907, p. 427.
- [190] Lacroix, *C. Rd. Ac. Sc.*, 24 May, 1899, p. 1353; 1905, 140, p. 22. *Nouv. Arch. Mus.*, 4th Ser., IV., p. 156; V., p. 246.
- [191] Cf. Prior, *Min. Mag.*, 1903, XIII., p. 228; Harker, *Nat. Hist. of Igneous Rocks*, 1910, p. 91.

APPENDIX I

MINERAL RESOURCES

Salt — Iron — Gold — Monazite — Galena — Tinstone.

DETAILED information respecting the principal mineral resources of Northern Nigeria will be found in Professor Dunstan's Reports on the Results of the Mineral Survey.^[192] The following is a brief summary of the more important facts.

(1). *Salt.*

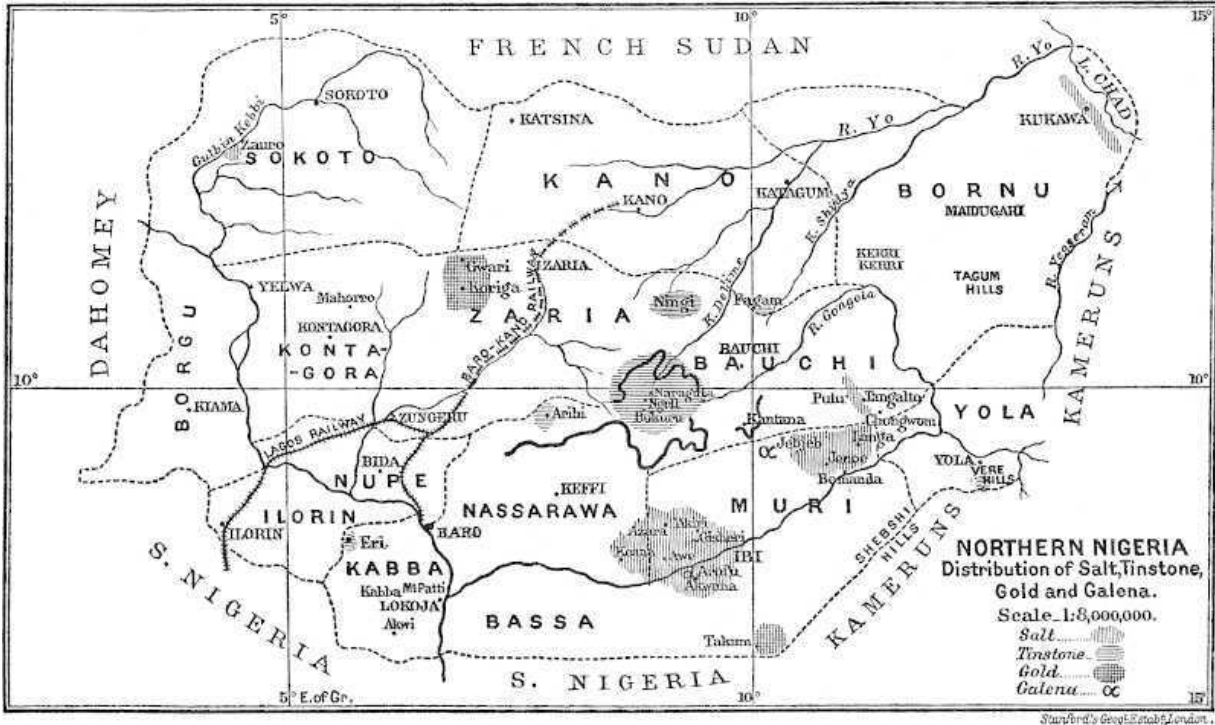
Salt has long been a staple article of inter-tribal commerce on the middle Benue, and its manufacture has in the past been extensively carried on in the province of Muri. The introduction of European salt has, however, adversely affected the native industry, and salt-making is now confined only to those localities in which the natural conditions are most favourable. In Muri province the salt is prepared by the concentration and evaporation of the waters of weak brine springs which are distributed over a western and an eastern area. In the western area the town of Awe is the centre of the industry, while Akiri and Azara are subordinate in both production and population. (Plate XVIII, [Figs. 1, 2.](#)) Keana on the Nassarawa border forms the western limit of the salt-producing area, while Akwana to the south of the Benue marks its southernmost extension. In addition to these towns which still manufacture a surplus for export, salt-making for local purposes is carried on at Kanji, Doya, Abuni, Ribi and Arofu within the same area. At a former period also salt was made at Lafin Gisheri, between Tunga and Ibi, and this locality may be taken to mark the eastern limit of the salt area. In eastern Muri the salt-producing district is of less importance than that in the west. The centres of the industry are Bomanda and Jenoe, but small quantities of salt are also prepared at Jebjeb, Jebero and Langa.

With regard to geological occurrence it is believed, as already indicated, that the weak brine from which the salt is made in Muri province is located within the upper part of the "Muri sandstones" and immediately underneath the passage beds and the Turonian limestone. Mr. Longbottom found that at Awe the springs supplied from the sandstones of the passage beds are entirely free from salt and cease flowing before the end of the dry season, while the neighbouring brine springs which rise from rocks lying below the surface drainage maintain a constant flow throughout the year. Moreover, wherever the shaly bands within the passage beds were pierced by boreholes, the brine rose with a steady flow under considerable pressure. The sandstones and grits underneath the passage beds appear to be saturated with weak brine which is retained, as in a reservoir, by the overlying shales. It is believed that the western and eastern salt-bearing districts form detached and irregular synclinal areas affected by a subordinate folding. The brine springs at Awe, Azara and Bomanda are especially associated with the subordinate anticlinal axes along which the salt-bearing beds within the synclinals approach the surface. Some of the minor salt towns, such as Ribí, Keana and Jenoe, are probably located on or near the outcrop of the salt-bearing beds. Wherever the brine springs are found, bubbles of a colourless gas, probably carbon dioxide, rise with the water. Calcareous incrustations round the orifices of the springs are found at Awe and Jebero. Some of the springs of Awe are, moreover, slightly thermal and possess a temperature a few degrees above sun heat. This is probably to be correlated with the recent volcanic action in the neighbourhood.

NORTHERN NIGERIA

Distribution of Salt, Tinstone, Gold and Galena.

MAP N^o. 4.



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Face p. 266.

The present springs rise in stream courses and are unavailable during the rainy season. Boring, however, within the synclinal areas would afford a copious and continuous supply. Professor Dunstan has analysed the salt of various localities and discussed the possibilities of increasing the native output. The general weakness of the brine, however, would appear to be unfavourable to European exploitation.

Salt is won from brine pools in south-eastern Bauchi over an extensive belt of country which stretches northward from the Pero Hills and passes to the east of Chongwom and Tangalto. The pools are presumably fed by springs and the salt is said to be abundant and easily prepared. The whole area, however, is very imperfectly known and largely unexplored. Scanty

incrustations of natron, utilised by the local herdsmen, are known from the Gongola valley in the neighbourhood of Nafada.

In Bornu salt is made from the ash of the siwak tree (*Salvadora persica*) which grows abundantly on the ancient dunes which fringe the western shore of Chad. The slaves of Mongonu and Musara prepare salt also from the ash of three varieties of grass, "pagam, kalaslim and kanido," which are found in the immediate neighbourhood of the lake. Over the whole of central Bornu the waters of the wells are charged with salts of soda in very varying amount. The ancient alluvium of Bornu is evidently impregnated with soda, and incrustations are found in places upon the surface of desiccated swamps. Natron pools, similar to those of Manga and presumably fed by springs, are unknown in central Bornu.

In Sokoto province incrustations of salt are found in the dry season at Zauro in the valley of the Gulbin Kebbi, and natron (kungwa) occurs in a similar manner between Gulma and Argungu. Knots and streaks of gypsum are common in the shales and clays around Sokoto, while efflorescences of gypsum and alum are found on the cliffs of Wurnu to the north-east. Scanty incrustations of natron are found upon desiccated swamps in many localities in the provinces of Sokoto, Kano, Zaria and Ilorin, but never in sufficient quantity to give rise to even a native industry. Hausaland has been in the past supplied with salt and natron from Asben, Bilma and Manga in the French Sudan. European salt, however, is rapidly replacing the native article, and many caravans now bring only natron from the north to the markets of Kano.

(2). *Iron.*

As in the case of salt, so in the case of iron, the native manufacture has been very largely superseded by the importation of European bar-iron. Only in the more inaccessible districts or in those places where the ore is abundant do the natives still smelt the local ironstones in their primitive fashion. The surface ironstones were at one time generally utilised throughout the Protectorate as a source of iron. The magnetite sands are still washed and smelted by the pagans of the Mumuye hills in eastern Muri,

while in northern Nassarawa the Koro tribe around Kao is still noted for its proficiency in the manufacture of iron from the local limonitic ores which occur within the schists in veins of varying thickness.^[193] Similar ores occur in veins in the sandstones around Akiri in western Muri, but are no longer utilised by the natives. The oolitic and earthy ironstones of sedimentary origin and of Eocene age give rise to an extensive native industry around Akwi and Ojerami in central Kabba, around Mahorro in eastern Kontagora, around Aliero in southern Sokoto, and around Kantana in southern Bauchi, as well as in the Kerri Kerri country in western Bornu and in the Tagum hills in southern Bornu. So far as European exploitation is concerned, the sedimentary ores of Mt. Patti and the lower Niger are most favourably situated for transport. The ores, however, while abundant, are of a somewhat low grade character and relatively rich in phosphoric acid. Magnetite and hæmatite schists are extensively developed in eastern Kabba, but their composition would render necessary costly crushing and separating processes before the ores could be available for export.^[194]

(3). *Gold.*

Traces of gold have been found in some of the quartz veins which are so widely distributed throughout the Protectorate, but no payable reefs have yet been discovered. Small quantities of free gold occur in the streams between Koriga and Birnin Gwari in northern Zaria and around Tagum in southern Muri. In Zaria the gold is evidently located within the northern prolongation of the Zungeru belt of slates and schists, and further exploration may possibly lead to valuable discoveries. All the slaty belts already described within the Protectorate are, indeed, worthy of careful prospecting for gold. Small quantities of gold are also found associated with cassiterite in some parts of Bauchi province.

(4). *Monazite.*

Monazite is widely distributed throughout the Protectorate, and appears to be a fairly constant constituent of the gneisses and schist. Small

quantities are contained in almost every river and stream within the crystalline areas. It has been found most abundantly in central Kabba and in northern Nassarawa, but, so far as known, it occurs nowhere in sufficient quantity to afford any hope of profitable European exploitation.

(5). *Galena*.

A slightly argentiferous galena occurs in veins within the Cretaceous rocks around Arofu and Akwana in western Muri, and in the Kudu valley to the west of Jebjeb in northern Muri. In the Arofu district the galena is associated with fluorspar and calcite, and the formation of the veins has been accompanied as already described by much silicification of the adjoining limestones. The Arofu galena is excavated by the natives for use as a cosmetic, and from time to time small pockets of silver are discovered within the veinstuff. To the west of Jebjeb the galena is no longer dug by the natives, and the old workings are now levelled and grassed over. The galena appears to have been associated with quartz, hæmatite and blende, but there is no evidence of silicification of the country rock.

(6). *Cassiterite (Tinstone)*.

Tinstone is the most valuable asset of the Protectorate from the point of view of mineral resources. Its occurrence in western Bauchi was known to the natives long before the advent of Europeans. Recent investigation, however, by the Mineral Survey has proved its presence over considerable areas in widely separated localities within the Protectorate. The most extensive area over which tinstone is found in the surface alluvium occurs in western Bauchi, and includes the summit of the plateau and the greater part of its northern margin. The alluvial deposits of Naraguta, which lie within this area, were originally developed by the Niger Co., but have now been acquired by the Consolidated Gold Fields, Ltd. Elsewhere, however, tinstone becomes equally abundant, and there is little doubt that further investigation will reveal many localities within this extensive area in which alluvial workings may be profitably carried on. Several syndicates indeed are now engaged in exploiting certain portions of the tinfield. In the

neighbourhood of Ngell the tinstone, accompanied by sulphide ores of copper, zinc and lead, occurs, as already described, in part as a pneumatolytic modification of the Bukuru granite, and in part as a true lode or pegmatitic formation. This particular occurrence is being developed by the Niger Co., but there is every reason to believe that similar occurrences will be found throughout the area as investigation proceeds.

Apart from this extensive tin-field in the centre of the Protectorate, cassiterite is also found in Bauchi province in many places throughout the Ningi and Burra hills, and detailed prospecting will probably here also reveal several payable localities. The completion of the Baro-Kano railway^[195] within the next two years will afford an easy means of access to the tin-producing areas of Bauchi, and will at the same time largely overcome the difficulty of transport which is at present the greatest obstacle in the way of the development of the tin fields.

Beyond the limits of Bauchi province small quantities of tinstone have been found in the neighbourhood of Gadama and Fagam on the borders of Bauchi and Kano, around Gantam and Aribi on the Nassarawa tableland as well as on the eastern slopes of the Vere hills in Yola, the western slopes of the Shebshi hills in southern Muri, and in the neighbourhood of Eri in eastern Ilorin. In the latter locality the tinstone was found to be derived from a richly stanniferous pegmatite vein, but the actual extent and possibilities of the occurrence are as yet unknown. The same may be said also of the other localities indicated, and there can be little doubt that all of them are worthy of further investigation. Tinstone has been reported to occur abundantly in the Mada country in eastern Nassarawa, but this discovery likewise awaits confirmation.

[192] Dunstan, *Col. Reps. (Miscell.)*, Nos. 32 (1906), 46 (1908), 47 (1908). A fourth report on the results of the Mineral Survey will shortly be issued.

[193] Sir William Wallace has reported an extensive native iron industry at Fawa in Katsina, where the ore appears to occur in a similar manner to that in the neighbourhood of Kao.—*G. J.*, VIII., 1896.

[194] In *The Iron Ore Resources of the World*, Vol. II., 1910, p. 1029, Dr. J. W. Evans has given a *résumé* of the results of the investigations of the Mineral Survey into the extent and value of the ferruginous deposits of Northern Nigeria.

[195] And the branch line to Leri in Zaria province.

APPENDIX II

THE PALÆONTOLOGY OF THE UPPER CRETACEOUS DEPOSITS OF NORTHERN NIGERIA

By HENRY WOODS, M.A., *University Lecturer in Palæozoology, Cambridge*

THE fossils collected by Dr. J. D. Falconer and Mr. A. Longbottom from the Upper Cretaceous deposits of Northern Nigeria include Fishes, Ammonites, Gasteropods, and Lamellibranchs. The localities from which these fossils were obtained are: Awe, Deba Habe, Jibaru, Gongila,^[196] Gulani, Kumberi, Kunini, and Reme. At present the number of species which can be identified is comparatively small, but the collection is sufficient to show that under more favourable conditions a large and important fauna could probably be obtained from the deposits of this region.

Teeth, vertebræ, &c., of fishes were collected at Kumberi and Jibaru, and have been examined by Dr. A. S. Woodward who identifies them as *Gigantichthys*, a genus which is known from the Chalk of Egypt and the Sahara.^[197] Ammonites have been found at Gongila, Reme, Kunini, and Kumberi, and they show that the deposits are of Turonian age—probably Lower Turonian. The Ammonites belong to the genera *Vascoceras*, *Mammites* (*Pseudaspidoceras*), and *Hoplitoides*. *Vascoceras* is confined to the Turonian. *Pseudaspidoceras* is mainly of Turonian age. The species of *Hoplitoides* belongs to the section of that genus which occurs in Turonian deposits. The species of *Vascoceras* and *Mammites* (*Pseudaspidoceras*) found in Nigeria are closely allied to forms which occur in the Lower Turonian of Northern Africa and the Iberian peninsula.

The larger number of the Gasteropods and Lamellibranchs were found at Awe, but no Ammonites were obtained at that locality; these fossils, so far as they can be identified, do not give any definite evidence that the deposit at Awe is of earlier age than the Turonian.

At Deba Habe *Ostrea* (*Exogyra*) *olisiponensis*, Sharpe, is abundant and is the only fossil obtained at that place. In Portugal that species occurs in both Turonian and Cenomanian deposits. In Nigeria it has also been found

at Gongila, where the Turonian *Hoplitoides* occurs. So that it is probable that the deposit at Deba Habe is of the same age as the one at Gongila.

At Gulani *Ostrea praelonga*, Sharpe, is abundant, but no other fossil was found at that locality. In Portugal *O. praelonga* occurs in the Bellasian—an horizon which is believed to come between the Aptian and the Cenomanian. The same species is found at a similar level in Tunis and Algeria. The occurrence of this one species can, I think, hardly be taken as sufficient evidence that the deposit at Gulani is of Bellasian age, since it is not unlikely that, as is the case with *O. (Exogyra) olisiponensis*, this form may recur at later horizons when a favourable facies of deposit again appears.

At present, then, we can state that the deposits in which Ammonites occur are of Turonian age, and that there is not sufficient evidence to show that beds of earlier age are represented.

The fauna of the Turonian deposits of Northern Nigeria resembles closely that found in beds of similar age in Portugal, Tunis, and Algeria. This is shown especially by the Ammonites and oysters. At present the genus *Vascoceras* is known only from Portugal, Spain, Algeria, Tunis, Egypt, Nigeria and Brazil. *Hoplitoides* occurs in Algeria, Tunis, Nigeria, the Kameruns, and probably in Egypt. The species of *Pseudaspidoceras* found in Nigeria is allied to forms found in Portugal, Tunis and Brazil. The oysters likewise belong to species characteristic of Northern Africa and South-western Europe, and one of them has also been found as far south as Angola. The Ammonites also show, as Pervinquière has already pointed out is the case in Tunis, that the fauna is related to that found in the Upper Cretaceous of Brazil.

Amongst other places in West Africa, Cretaceous deposits have been found in the Kameruns, and are regarded by Solger as of Upper Turonian and Lower Senonian age, but by Harbort^[198] they are referred to the Emscherian (Lower Senonian). They certainly appear to be of later date than the Nigerian deposits with which they are connected chiefly by the presence of *Hoplitoides*.

Until the fauna of the Nigerian Cretaceous is more completely known any further discussion of its relationship to the faunas of other areas must be

postponed.

The list on the following page shows the distribution of the species found in Nigeria, and is followed by palæontological notes and descriptions of some of the species.

	Awe.	Deba Habe.	Gongila.	Gulani.	Kumberi.	Kunini.	Reme.
<i>Nuculana</i> , sp.					*		
<i>Trigonoarca</i> , sp.	*						
<i>Trigonia</i> , sp.	*						
<i>Modiola</i> , sp.	*						
<i>Pecten (Neithea) quinquecostatus</i> , Sow.	*						
<i>Lima (Mantellum)</i> , sp.	*						
<i>Ostrea praelonga</i> , Sharpe				*			
„ (<i>Exogyra</i>) <i>olisiponensis</i> , Sharpe		*	*				
<i>Gervillia (Pseudoptera)</i> , sp.	*						
<i>Astarte awensis</i> , sp. nov.	*						
<i>Lucina</i> , sp.					*		
<i>Cardium</i> , sp.	*						
<i>Corbula</i> , sp.	*						
<i>Pseudomelania</i> , sp.	*						
<i>Turritella</i> , sp.					*		
<i>Semifusus africanus</i> , sp. nov.	*						
<i>Vascoceras nigeriensis</i> , sp. nov.						*	*
„ <i>gongilensis</i> , sp. nov.			*				
<i>Mammites (Pseudaspidoceras)</i> , sp.			*				
<i>Hoplitoides nigeriensis</i> , sp. nov.			*				
„ sp.							*

LAMELLIBRANCHIA.

NUCULANA, sp.

[Plate XIX.](#), Fig. 4.

An internal cast resembles in form *N. Mariæ* (d'Orbigny), but in the absence of the shell the species cannot be identified.

Locality.—Kumberi.

TRIGONOARCA, sp.

[Plate XIX.](#), Figs. 1–3.

Several specimens of *Trigonoarca* were found, but are not sufficiently well-preserved for specific identification. In general form they resemble some examples of *T. trichinopolitensis* (Forbes)^[199] from the Trichinopoli Beds of Southern India.

Locality.—Awe.

TRIGONIA, sp.

Two specimens of a species belonging to the Scabræ section of *Trigonia* are in the collection, but are not sufficiently well-preserved for identification.

Locality.—Awe.

MODIOLA, sp.

Examples of a small form of *Modiola* occur at Awe.

PECTEN (NEITHEA) QUINQUECOSTATUS, Sowerby.

Three specimens with the shell preserved were found at Awe.

LIMA (MANTELLUM), sp.

An imperfect specimen from Awe resembles *Lima* (*Mantellum*) *gaultina*, Woods. [\[200\]](#)

OSTREA PRÆLONGA, Sharpe.

[Plate XIX.](#), Figs. 5–8.

1850. *Ostræa prælonga*, D. Sharpe, *Quart. Journ. Geol. Soc.*, VI., p. 187, Pl. XX., Fig. 4.
1869. *Ostrea prælonga*, H. Coquand, *Mon. Ostrea. Terr. Crét.*, p. 169, Pl. LXVI., Figs. 1, 2.
1890. *Ostrea prælonga*, A. Peron, *Moll. Foss. Terr. Crét. Tunisie*, p. 110, Pl. XXIII., Figs. 1–6.

Remarks.—About twenty more or less perfect specimens of an *Ostrea*, agreeing closely with *O. prælonga*, Sharpe, were found. *O. pentagruelis*, Coquand,^[201] is considered by Peron to be only a very old individual of *O. prælonga*.

This species occurs in the Bellasian of Portugal. In Algeria and Tunis it is characteristic of beds which occur below the Cenomanian, and are referred by Peron to the Upper Albian.

Locality.—Gulani.

OSTREA (EXOZYRA) OLISIPONENSIS, Sharpe.

[Plate XX.](#), Figs. 1–3.

1850. *Exogyra olisiponensis*, D. Sharpe, *Quart. Journ. Geol. Soc.*, VI., p. 185, Pl. XIX., Figs. 1, 2.
1862. *Ostrea Overwegi*, H. Coquand, *Géol. pal. rég. sud. prov. Constantine*, p. 226, Pl. XIX., Figs. 1–6.
1862. *Ostrea Coquandi*, Julien, in Coquand, *ibid.*, p. 324, Pl. XXXIII., Figs. 10–12.
1869. *Ostrea olisiponensis*, H. Coquand, *Mon. Ostrea. Terr. Crét.*, p. 125, Pl. XLV., Figs. 1–7.
1869. *Ostrea Overwegi*, Coquand, *ibid.*, p. 140, Pl. XLIV., Figs. 1–9; Pl. XLVI., Figs. 14, 15.
1873. *Ostrea olisiponensis*, L. Lartet, *Ann. Sci. géol.*, Vol. III., p. 59, Pl. XI., Fig. 1.

1882. *Exogyra olisiponensis*, G. Seguenza, *Atti d. R. Accad. dei Lincei*, Ser. 3; *Mem. Scien. Fis. Math.*, Vol. XII., p. 180, Pl. XVII., Fig. 2.
1882. *Exogyra oxyntas*, Seguenza, *ibid.*, p. 178, Pl. XVIII., Fig. 1.
1888. *Ostrea olisiponensis*, P. Choffat, *Mém. Soc. Phys. et d'hist. nat. de Genève*, XXX., 2, p. 95.
1891. *Ostrea olisiponensis*, A. Peron, *Moll. Foss. Terr. Crét. Tunisie*, p. 114, Pl. XXIII., Figs. 14–18.
1902. *Ostrea (Exogyra) olisiponensis*, P. Choffat, *Faune Crét. Portugal*, Vol. I., Ser. 4, p. 166, Pl. VI., Figs. 17–19.
1903. *Ostrea olisiponensis*, R. Fortau, *Bull. Instit. Égypt.*, Ser. 4, Vol. IV., p. 283, Figs. 3–5.
1905. *Ostrea (Exogyra) olisiponensis*, P. Choffat, *Géol. Col. Portug. d'Afrique. II, Nouvelles données sur la zone littorale d'Angola*, p. 44, Pl. I., Figs. 4, 5.

Remarks.—This species is represented by numerous examples, in a few of which the surfaces of the shell are well-preserved and show a few coarse ribs. Sharpe and other writers have noted the great variability in the ribbing of this species. The Nigerian specimens agree with the varieties which possess only a few ribs. Peron, Choffat, and Fortau have discussed fully the affinities and synonymy of *O. (Exogyra) olisiponensis*. This species is common in Northern Africa and Portugal; it occurs in the Cenomanian of Algeria, Tunis, Egypt, and Palestine; in Portugal it is found in both the Upper and Lower Turonian, as well as in the Cenomanian, wherever the deposits are of the marly facies. South of Nigeria this species has been discovered in the province of Angola.

Localities.—Deba Habe and Gongila.

GERVILLIA (PSEUDOPTERA), sp.

[Plate XX.](#), Figs. 4–6.

There are several internal casts resembling in form *G. (Pseudoptera) anomala* (Sowerby),^[202] but there is no evidence to show whether radial ribs were present on the shell or not, and the species cannot be determined at present.

Specimens from the Cenomanian of the neighbourhood of Tebessa (Algeria) have been identified with Sowerby's species by Coquand,^[203] and Peron^[204] states that it is abundant and well-preserved at that locality.

Locality.—Awe.

ASTARTE AWENSIS, sp. nov.

[Plate XX.](#), Fig. 7.

Description.—Shell rather small, moderately convex, with the postero-dorsal part somewhat compressed, outline subquadrate or subtriangular, very inequilateral. Anterior margin slightly concave; ventral margin convex; posterior margin short, truncated, forming an obtuse angle with the long, slightly convex postero-dorsal margin. Umbones pointed, rather prominent. Lunule large, ovate, smooth, with a sharp edge. Escutcheon large, deep, with a sharp edge. Ornamentation consists of about fourteen broad, prominent concentric ribs, which are marked by fine concentric lines. Length, 8·25; height, 7 mm.

Affinities.—In the character of its ornamentation *A. awensis* resembles several small species of *Astarte* found in the Cretaceous rocks. It is more inequilateral, more elongate, and has the posterior margin more distinctly truncated than in *A. subsimilis*, J. Böhm.^[205] In similar respects it differs from *A. griesbachi*, Woods,^[206] from Pondoland.

Remarks.—Numerous casts of this species occur in ferruginous nodules.

Locality.—Awe.

LUCINA, sp.

[Plate XX.](#), Figs. 8, 9.

Description.—Shell nearly orbicular, convex, slightly inequivalve; height and length nearly equal. Anterior and ventral margins rounded; posterior margin truncated, forming an obtuse angle with the straight or slightly convex postero-dorsal margin. Umbones rather prominent. Ornamentation consists of regular, concentric lamellar ribs separated by broad, flat interspaces which are marked by fine, concentric lines. Length, 11 mm.; height, 11·5 mm.

Affinities.—In general form, and in the character of its ornamentation, this species resembles one of the varieties of *Lucina fallax*, Forbes, described by Stoliczka^[207] from the Utatúr Group, but the umbones are more prominent, and the postero-dorsal angle is more distinct. It also resembles *L. saharica*, Quaas,^[208] and some forms of *L. subnummismalis*, d'Orbigny.^[209]

Locality.—Kumberi.

CARDIUM, sp.

[Plate XX.](#), Fig. 10.

In the possession of numerous radial ribs and in the general form of the shell, some internal casts resemble *Cardium cenomanense*, d'Orbigny,^[210] but the valves are less convex.

Locality.—Awe.

CORBULA, sp.

Specimens of *Corbula* resemble *C. elegans*, Sowerby,^[211] in the character of their ornamentation, but the posterior part of the shell is more produced.

Locality.—Awe.

GASTEROPODA.

PSEUDOMELANIA, sp.

[Plate XXI.](#), Fig. 1.

Several specimens of a small gasteropod resemble *Pseudomelania Pallaryi*, Peron and Fortau,^[212] from the Cenomanian of Egypt, but they are of much smaller size, being from 8 to 13 mm. long. The surface of the shell is smooth, except for slightly curved growth-lines, and, on the base, spiral grooves.

Locality.—Awe.

TURRITELLA, sp.

[Plate XXI.](#), Fig. 2.

A species of *Turritella* resembles *T. Choffati*, Thomas and Peron,^[213] from the Turonian of Tunis. The anterior part of each whorl is bent at an angle with the large flattened posterior part, and the ornamentation consists of four strong spiral ribs, one of which occurs at the angle, giving the appearance of a carina. On the later whorls the ribs become granular or tubercular. The ribs are much coarser than in *T. Choffati*.

Locality.—Kumberi.

SEMIFUSUS AFRICANUS, sp. nov.

[Plate XXI](#), Figs. 3–5.

Description.—Shell subfusiform. Spire shorter than the last whorl, consisting of about five whorls. Whorls carinate, the carina being more or less distinctly nodular. Posterior part of the whorls flattened, much larger than the anterior part in the spire. Anterior part of last whorl convex. Ornamentation consists of narrow spiral ribs with flat interspaces crossed by growth-lines; on the later whorls the ribs become divided up into rows of elongated tubercles.

Remarks.—This species, which is represented by several imperfect specimens, shows some resemblance to *Lagena nodulosa*, Stoliczka,^[214] and *L. secans*, Stoliczka,^[215] from the Ariyalūr Group, and also to *Serrifusus dakotensis* var. *vancouverensis*, Whiteaves,^[216] but possesses one carina only.

Locality.—Awe.

CEPHALOPODA.

Genus.—VASCOCERAS, P. Choffat.

(*Faune Crét. Portugal*, I., 1898, p. 51.)

VASCOCERAS NIGERIENSIS, sp. nov.

[Plate XXI](#), Fig. 6; [Plate XXII](#), Figs. 2, 3.

Description.—Shell inflated, rounded, subglobose. Whorls thick, largely embracing; umbilicus deep, rather more than a third of the diameter of the shell, with nearly vertical walls, the umbilical margin subcarinate, without tubercles.

Flanks curving gradually from the umbilical margin and not limited from the rounded periphery. Surface of shell with growth-rings.

Sutures: Saddles large, broad, and low; external saddle unsymmetrically divided and with small minor divisions; first lateral smaller than the external saddle, more rounded, less divided, with the outer margin sloping and the inner margin nearly perpendicular; second lateral saddle broad, only slightly divided. External lobe deeper than the first lateral lobe; the latter is small with one main division; second lateral lobe smaller, with two points.

	(1)	(2)
Diameter	88	93 mm.
Thickness	58	54
Height of last whorl	37	34
Width of umbilicus	30	33

(1) Reme. (2) Kunini.

Affinities.—This species resembles the thicker forms of *V. durandi* (Peron and Thomas),^[217] but the walls of the umbilicus are steeper, the umbilical margin is without tubercles, and the external and first lateral saddles are broader and not so high. In general form, and in the absence of tubercles at the umbilical margin, *V. nigeriensis* resembles *V. amierensis*, Choffat,^[218] but the whorls embrace to a larger extent, and their sides are less flattened; there are also some differences in the suture. In the form of the umbilicus, and in the breadth of the saddles, this species resembles *V. harttiformis*, Choffat,^[219] but the shell is much less thick. Another similar form but with a thicker shell, is *V. hartti*, Hyatt,^[220] from Brazil.

Localities.—Reme and Kunini.

VASCOCERAS GONGILENSIS, sp. nov.

[Plate XXI.](#), Fig. 7; [Plate XXII.](#), Fig. 1.

Description.—Shell moderately inflated. Umbilicus less than a quarter of the diameter of the shell, with vertical walls and subcarinate margin. Flanks somewhat compressed. Periphery flattened. Inner whorls with strong ribs on the flanks and prominent tubercles at the umbilical margin and at the limit of the

flanks and periphery; on the last whorl the ribs and tubercles gradually become obsolete.

Sutures: External saddle broad, subquadrate, with small divisions; first lateral saddle broad, rounded, divided unsymmetrically by a small lobe; second lateral saddle subquadrate; one small auxiliary saddle on the flanks. External lobe deep and larger than the first lateral lobe, which has only small divisions; second lateral lobe small, its lower margin with pointed divisions.

Diameter	108 mm.
Thickness	57 „
Height of last whorl	52 „
Width of umbilicus	24 „

Affinities.—This species resembles *V. subconciliatum*, Choffat,^[221] from the Turonian of Portugal, but the ribs and tubercles become obsolete at an earlier stage, the umbilicus is smaller, the divisions in the external saddle are less deep, the first and second lateral saddles are more rounded, and the first lateral lobe is narrower. *V. gongilensis* also resembles some forms of *V. cauvini*, Chudeau.^[222]

Locality.—Gongila.

Genus—MAMMITES, Laube and Bruder.

(*Palæontographica*, XXXIII., 1887, p. 229.)

Sub-genus—PSEUDASPIDOCERAS, Hyatt.

(“Pseudoceratites of the Cretaceous,” *U. S. Geol. Surv. Mon.*, XLIV., 1903, p. 106.)

MAMMITES (PSEUDASPIDOCERAS), sp.

[Plate XXIII.](#), Figs. 1, 2.

In the collection there is one specimen of *Mammites* (*Pseudaspidoceras*); it resembles the form figured by Choffat^[223] under the name *Acanthoceras* (?), cf. *Footeanus* (Stol.), from the Turonian of Portugal, but differs in having a smaller umbilicus and less distinct siphonal tubercles. The sutures are not shown in the specimen from Nigeria.

The Portuguese form differs from *A. footeanus*, Stoliczka,^[224] by the absence of secondary ribs on the siphonal region of the inner whorls, and by the forward inclination of the ribs between the lateral tubercles and the siphonal tubercles.

Other allied species are *M. (Pseudaspidoceras) salmuriensis* (Courtillet) var. *byzacenica*, Pervinquièrè,^[225] from the Lower Turonian of Tunis, and *M. (Pseudaspidoceras) Pedroanus* (White),^[226] from Brazil.

Locality.—Gongila.

Genus—HOPLITOIDES, A. von Kœnen.

(“Nachtrag Foss. Unter. Kreide Mungo in Kamerun,” *Abhandl. d. k. Gesellsch. d. Wiss. Göttingen, Math.-Phys. Kl., N.F. Band 1*, 1898, p. 53. Emended F. Solger, *Foss. Mungokreide in Kamerun*, 1904, p. 127; L. Pervinquièrè, *Études Pal. Tunis, I. Ceph.*, 1907, p. 215.)

HOPLITOIDES NIGERIENSIS, sp. nov.

[Plate XXIII](#), Fig. 3; [Plate XXIV](#), Figs. 1–5. [Text-figure 1](#).

Description.—Shell discoidal, compressed, with truncated periphery. Whorls largely embracing, with the flanks slightly or moderately convex, the greatest convexity being at a short distance from the umbilical margin. Umbilicus small, or very small, about one-seventh of the diameter of the shell, its sides nearly perpendicular. Periphery usually rather broad, bounded on each side by a prominent rounded carina; the prominence of the siphuncle gives rise to a median rounded ridge between which and the carinas the shell is concave.

In fully-grown individuals the shell is smooth, but in smaller examples the flanks bear a few strong, rounded ribs, which curve slightly forwards near the periphery; the larger ribs start from the umbilical margin, and one or two smaller ribs may be intercalated on the flanks. Where the ribs join the carina the latter may become more or less distinctly tuberculate. Usually the ribs become obsolete when the shell reaches a diameter of about 50 mm., but occasionally they persist on larger individuals.

Sutures: External saddle broad, rather low, with a variable number of small divisions. First and second lateral saddles similar to, but distinctly smaller than the external saddle; each has usually some small divisions, but in some cases

these are absent, and the saddle is then rounded. One or two auxiliary saddles may occur. External lobe deep, sometimes nearly as wide as the space between the two carinæ. First lateral lobe usually rather narrower and not so deep as the external lobe, with more or less numerous small divisions. The second lateral lobe is similar to, but smaller than the last.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Diameter	110	95	84	67	68	62	57	44 mm.
Thickness	47	42	34	28	28	27	28	20 „
Height of last whorl	61	49	45	37	38	38	34	25 „
Width of umbilicus	15	15	10	10	7	6·5	8	6 „

(1–8) From Gongila.

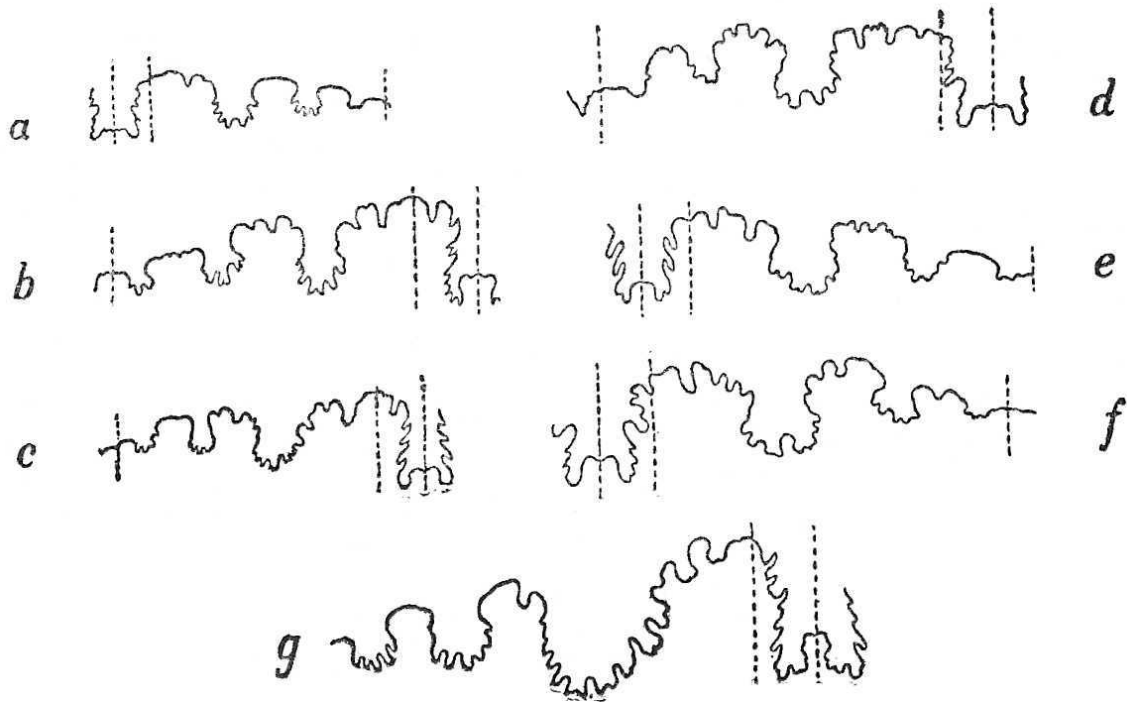


FIG. 1.—*Hoplitoides nigeriensis*, sp. nov. Gongila. Sutures of six specimens. Natural size.

Affinities.—This species appears to belong to the bicarinate group of *Hoplitoides*. In general form it resembles *H. munieri*, Pervinquière,^[227] from the Lower Turonian of Tunis, but the whorls, are thicker, the flattened periphery is broader and shows a distinct siphuncular ridge. The sutures are similar to those of *H. munieri*, but are symmetrically placed, and the first lateral lobe is

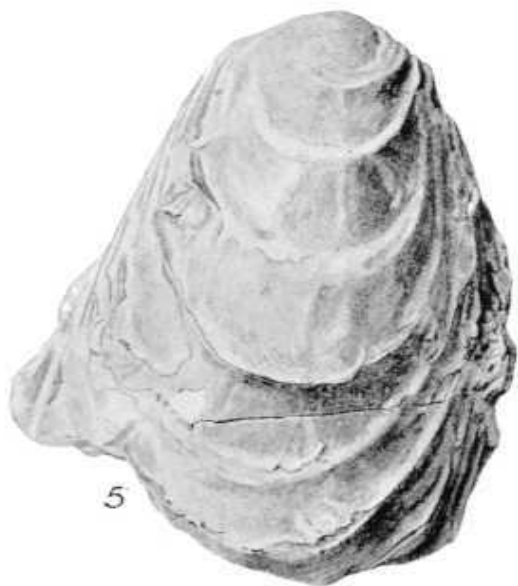
relatively narrow. The ribbing of the smaller specimens is similar to that of *Hoplites ingens*, v. Kœnen em. Solger.^[228]

In general form, *H. nigeriensis* is similar to *Heterotissotia neoceratoides*, Peron,^[229] from the Lower Senonian of Algeria and the Upper Turonian of Tunis. Both possess a broad, flattened periphery with a keel on each side and a prominent siphuncular ridge, but in *H. neoceratoides* the ribs are retained to a later stage. The figures given of the suture of Peron's species are not satisfactory, and I am indebted to Dr. L. Pervinqui re for comparing figures of the sutures of *Hoplitoides nigeriensis* with the type of *Heterotissotia neoceratoides* which is now in the Sorbonne. Dr. Pervinqui re states that the differences in the suture are too great to allow of these two species being referred to the same genus.

Remarks.—The sutures of *Hoplitoides nigeriensis* vary considerably in appearance on account of the difference in the number and size of the small divisions of the saddles. A somewhat similar variation is seen in the sutures of *H. ingens*, v. Kœnen, em. Solger, and *H. munieri*, Solger. Variations also occur in the thickness of the whorls, the breadth of the periphery, the size of the umbilicus, and the period at which the ribs become obsolete. In one specimen, which is less thick than the other, the siphuncular ridge is absent.

This species is represented by numerous specimens, all of which are internal casts or have only a small portion of the shell preserved, and in nearly all cases the sutures are clearly exposed.

Locality.—Gongila.^[230]



CRETACEOUS MOLLUSCA.

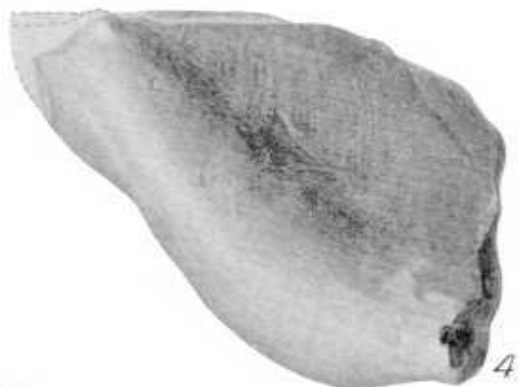
T. A. Brock, del.

(The figures are of natural size unless the amount of enlargement is stated).

FIGS. 1–3. *Trigonoarca*, sp. Awe. 1, right valve; 2, dorsal view of the same; 3, hinge of left valve. ([Page 276.](#))

„ 4. *Nuculana*, sp. Kumberi. Internal cast of left valve. $\times 1\frac{1}{2}$. ([Page 276.](#))

„ 5–8. *Ostrea praelonga*, Sharpe. Gulani. 5, right valve; 6, 8, right valves with umbo of left valves; 7, left valve. ([Page 277.](#))



CRETACEOUS MOLLUSCA.

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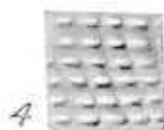
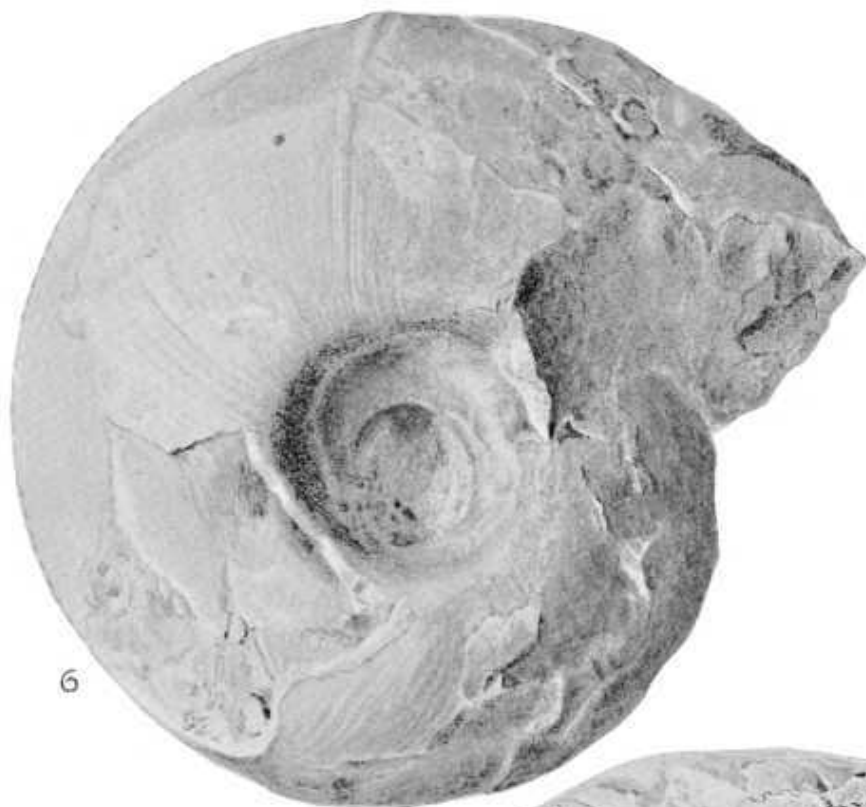
FIGS. 1–3. *Ostrea (Exogyra) olisiponensis*, Sharpe. Deba Habe. Left valves. ([Page 277.](#))

„ 4–6. *Gervillia (Pseudoptera)*, sp. Awe. Internal casts. 4, 5, left valves; 6, right valve. ([Page 278.](#))

„ 7. *Astarte awensis*, sp. nov. Awe. Right valve, drawn from a wax impression of an external cast, × 1½. ([Page 279.](#))

„ 8, 9. *Lucina*, sp. Kumberi. 8, right valve; 9, dorsal view of the same, × 1½. ([Page 279.](#))

„ 10. *Cardium*, sp. Awe. Internal cast of right valve. The marginal part of the valve is imperfect. ([Page 280.](#))



CRETACEOUS MOLLUSCA.

T. A. Brock, del.

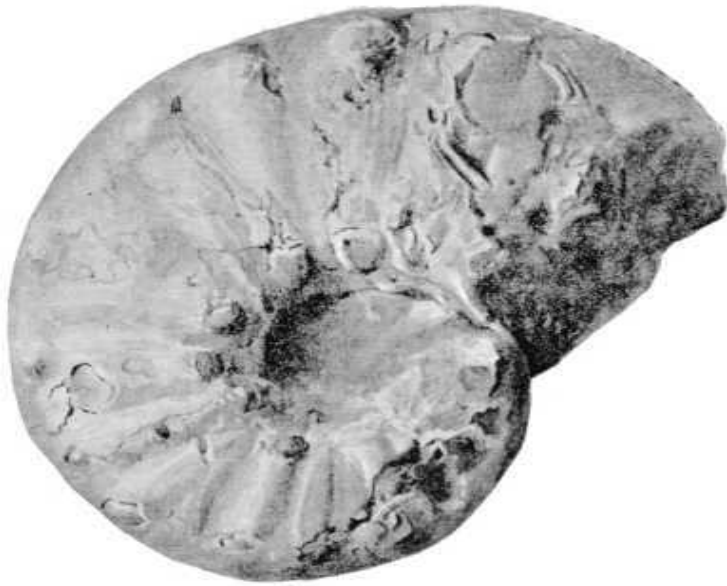
- FIGS. 1. *Pseudomelania*, sp. Awe. $\times 1\frac{1}{2}$. ([Page 280.](#))
„ 2. *Turritella*, sp. Kumberi. $\times 1\frac{1}{2}$. ([Page 281.](#))
„ 3–5. *Semifusus africanus*, sp. nov. Awe. 3, drawn from a wax impression of an external cast; 4, portion of the ornamentation of 5, $\times 6$. ([Page 281.](#))
„ 6. *Vascoceras nigeriensis*, sp. nov. Reme. ([Page 281.](#))
„ 7. *Vascoceras gongilensis*, sp. nov. Gongila. Internal cast. ([Page 282.](#))



CRETACEOUS MOLLUSCA.

T. A. Brock, del.

- FIGS. 1. *Vascoceras gongilensis*, sp. nov. Gongila. Siphonal view of Fig. 7, [Plate XXI](#). ([Page 282](#)).
„ 2, 3. *Vascoceras nigeriensis*, sp. nov. Kunini. Two views of the same specimen. ([Page 281](#).)



1



2



3

CRETACEOUS MOLLUSCA.

T. A. Brock, del.

FIGS. 1, 2. *Mammites* (*Pseudaspidoceras*), sp. Gongila. Two views of the same specimen. ([Page 283.](#))

„ 3. *Hoplitoides nigeriensis*, sp. nov. Gongila. ([Page 284.](#))



CRETACEOUS MOLLUSCA.

T. A. Brock, del.

FIGS. 1–5. *Hoplitoides nigeriensis*, sp. nov. Gongila. 1, siphonal view of Fig. 3, [Plate XXIII](#); 4, siphonal view of Fig. 3. ([Page 284](#).)

- [196] The occurrence of Turonian fossils at this locality has been recorded by G. C. Crick, *Geogr. Journ.*, Vol. XXIV., 1904, p. 522, footnote 1.
- [197] Dames, *Sitzungsber. Ges. nat. Freunde, Berlin*, 1887, pp. 69, 137.
- [198] Guillemain and Harbort, “Profil der Kreideschicht. am Mungo,” *Abhandl. d. k. preuss. geol. Landesanst. N.F.*, Heft 62, 1909, p. 405.
- [199] Stoliczka, *Cret. Fauna S. India*, 1871, p. 353, Pl. XVIII., Figs. 12, 14; Pl. XIX., Figs. 2, 3; Pl. XX., Figs. 2, 8, 9, 10.
- [200] *Mon. Cret. Lamell. England*, Vol. II., 1904, p. 31, Pl. V., Figs. 16–20.
- [201] *Mon. Aptien de l’Espagne*, 1865, p. 166, Pl. XXVI., Figs. 1, 2.
- [202] For figures see Woods, *Mon. Cret. Lamell. England*, Vol. II., 1905, p. 64, Pl. IX., Figs. 2–4. Compare also d’Orbigny, *Pal. Franç. Terr. Crét.*, Vol. III., 1846, p. 478, Pl. CCCXCII., Figs. 1–3. When describing this species I overlooked Guéranger’s statement that he had seen specimens showing ligament pits; from the presence of those pits it is evident that this species cannot belong to *Pteria*. It seems probable that *Pseudoptera* should be regarded as a sub-genus of *Gervillia*. Guéranger, *Album Paléont. de la Sarthe*, 1867, pp. 17, 20, Pl. XXII., Figs. 9, 10; Pl. XXV., Figs. 10, 11.
- [203] *Géol. et Pal. de Constantine*, 1862, p. 292.
- [204] Peron, *Moll. Foss. Terr. Crét. Tunisie*, 1890, p. 238.
- [205] “Kreidebild. des Fürbergs u. Sulzbergs,” *Palæontographica*, XXXVIII., 1891, p. 74, Pl. III., Fig. 8.
- [206] *Ann. S. African Mus.*, IV., 1906, p. 300, Pl. XXXV., Figs. 18, 19.
- [207] *Palæont. Indica, Cret. Fauna S. India*, III., 1871, p. 256, Pl. XIV., Fig. 7.
- [208] *Palæontographica*, XXX., 2, 1902, p. 214, Pl. XXIV., Fig. 7.
- [209] Ravn, *Mollusk. Danmarks Kridtafl. I. Lamellibr.*, 1902, p. 129, Pl. IV., Fig. 21.
- [210] *Pal. Franç. Terr. Crét.*, III., 1844, p. 37, Pl. CCXLIX, Figs. 5–9.
- [211] Woods, *Mon. Cret. Lamell. England*, Vol. II., 1908, p. 216, Pl. XXXIV., Figs. 23–28.
- [212] *Bull. Institut Égypt.*, Ser. 4, Vol. VI., 1903, p. 270, Pl. I., Fig. 22.
- [213] *Moll. Foss. Crét. Tunisie*, 1889, p. 47, Pl. XIX., Figs. 13, 14.
- [214] *Cret. Fauna S. India*, Vol. II., 1867, p. 137, Pl. XI., Fig. 18.
- [215] *Ibid.*, p. 138, Pl. XI., Figs. 19, 20.
- [216] *Mesozoic Fossils*, Vol. I., 1879, p. 119, Pl. XV., Fig. 5. Compare also Woods, *Ann. S. African Mus.*, Vol. IV., 1906, p. 324, Pl. XL. Figs. 5, 6. Newton, *Trans. Roy. Soc. S. Africa*, Vol. I., 1909, p. 28, Pl. IX., Fig. 6.
- [217] *Moll. Foss. Crét. Tunisie*, 1889, p. 27, Pl. XVIII., Figs. 5–8. Peron, *Amm. Crét. supér. Algérie*, 1896–7, p. 44, Pl. IV., Fig. 1; Pl. V., Fig. 1; Pl. XVII., Fig. 5. Pervinquière,

Paléont. Tunisienne, Ceph. secondaires, 1907, p. 332, Pl. XXI., Fig. 1. V. Douvillei, Choffat, *Faune crét. Portugal*, Vol. I., 1898, p. 59, Pl. X., Fig. 6; Pl. XI., Figs. 2–5; Pl. XXI., Figs. 13–16.

- [218] Choffat, *ibid.*, p. 61, Pl. XII., Figs. 1, 2; Pl. XIII., Figs. 1, 2; Pl. XXI., Figs. 17–21.
- [219] *Ibid.*, p. 61, Pl. XII., Fig. 3; Pl. XIII., Figs. 3–6; Pl. XXI., Figs. 22–24.
- [220] “Contrib. Paleont. Brazil,” *Archiv. Mus. Nac. Rio de Janeiro*, VII., 1887, p. 226, Pl. XIX., Figs. 1, 2; Pl. XX., Fig. 3. Hyatt, “Pseudoceratites of the Cretaceous,” *U. S. Geol. Survey, Mon.*, XLIV., 1903, p. 103, Pl. XIV., Fig. 16.
- [221] *Faune crét. Portugal*, Vol. I., 1898, p. 64, Pl. XV., Figs. 1–3; Pl. XVI., Fig. 4; Pl. XXII., Figs. 28–31.
- [222] “Ammonites du Damergou (Sahara meridional),” *Bull. Soc. géol. de France* (4), IX., 1909, p. 68, Pls. I., II., III.
- [223] *Faune crét. Portugal*, Vol. I., 1898, p. 66, Pl. XVI., Figs. 9, 10; Pl. XXII., Figs. 3, 4.
- [224] *Palæont. Indica, Ceph. Cret. S. India*, 1864, p. 101, Pl. LII., Figs. 1, 2. Kossmat, “Südindische Kreideformat,” *Beitr. Pal. Geol. Osterr-ungarns u. d. Orients*, XI., 1897, p. 20. Petrascheck, “Amm. sächsich. Kreideformat., *ibid.*, XIV., 1902, p. 144. Pl. IX., Fig. 1.
- [225] *Études Paléont. Tunisienne, I. Ceph. Second.*, 1907, p. 315, Pl. XIX., Fig. 1.
- [226] “Contrib. Paleont. Brazil,” *Archiv. Mus. Nac. Rio de Janeiro*, VII., 1887, p. 212, Pl. XXII., Figs. 1, 2.
- [227] *Études Pal. Tunis, I, Ceph.*, 1907, p. 217, Fig. 83, Pl. X., Figs. 1, 2.
- [228] Solger, *Foss. Mungokreide in Kamerun*, 1904, p. 137, Pl. V., Figs. 8, 9.
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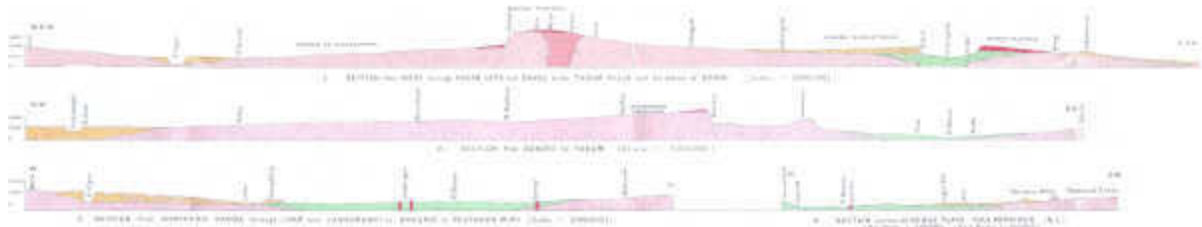
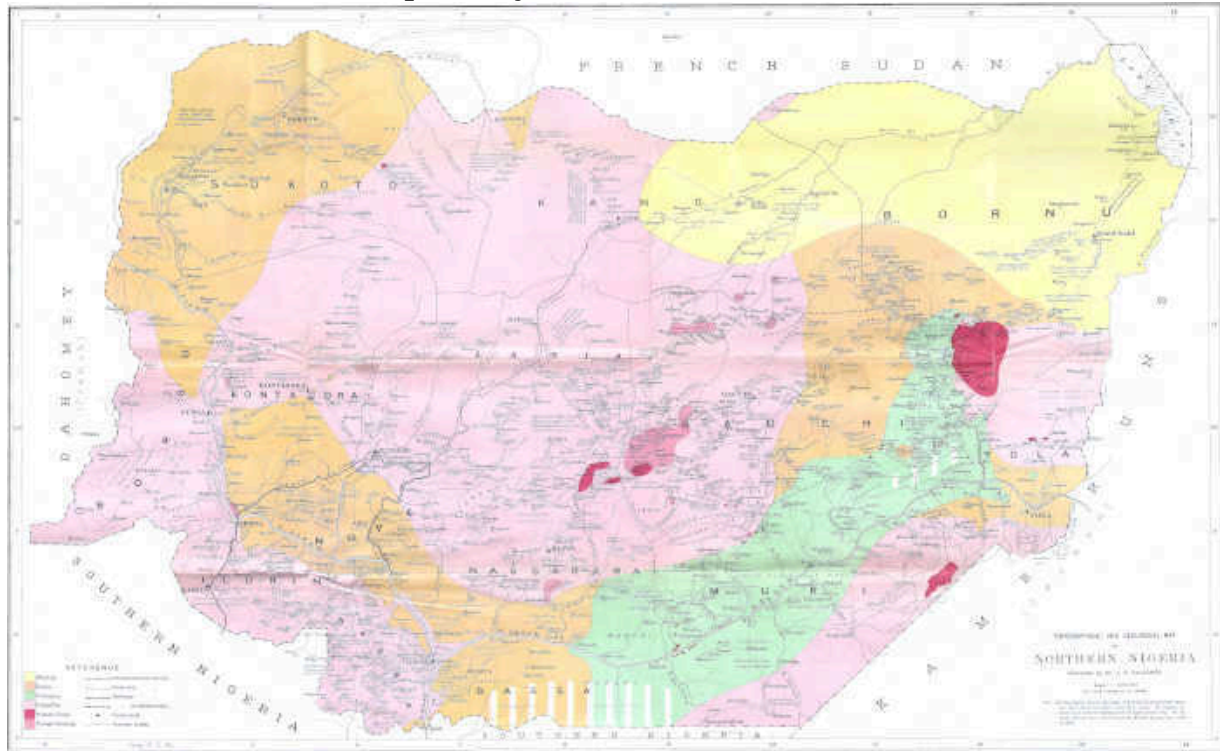
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ERRATUM.

[p. 203](#), line 13, for “larvas” *read* “lavas.”

TOPOGRAPHICAL AND GEOLOGICAL MAP
OF
NORTHERN NIGERIA
Compiled by Dr. J. D. FALCONER.



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Transcriber's note:

- The change in the [ERRATUM](#) has been made.
- Page [6](#), Changed: "down on either si de, a n from" to: "side, and"
- Page [45](#), One ref. to footnote missing. Ref. "2" treated as ref. to [ftn. 13](#), and a ref. to [ftn. 14](#) placed after "the plains of Yergum."
- Page [49](#), Changed: "those which form the the western boundary" to: "form the western"
- Page [115](#), Changed: "may be recognised all the familar stages" to: "familiar"
- Page [122](#), Changed: "comparision which M. Chudeau" to: "comparison"
- Page [123](#), Changed: "the epidorites, amphibolites, and granulites" to: "epidiorites"
- Page [151](#), Changed: "Gateri but the the associated rocks" to: "but the associated"
- Page [154](#), Changed: "corrugation ofthea whole series of gently" to: "of the whole"
- Page [154](#), Changed: "limestones of the Kudu vlley" to: "valley"
- Page [167](#), Changed: "limestones, however, no here form thick" to: "nowhere"
- Page [207](#), footnote [121](#), Changed: "De Lapparent, *Traité des Geologie*" to: "*Traité de Géologie*"
- Page [232](#), Changed: "that lacustrune conditions prevailed" to: "lacustrine"
- Page [233](#), footnote [150](#), Changed: "Passarge, *Adamana*, 1895, p. 385." to: "*Adamaua*"
- Page [290](#), Changed: "Stoliczka" to: "Stoliczka"
- Minor changes in punctuation have been done silently.
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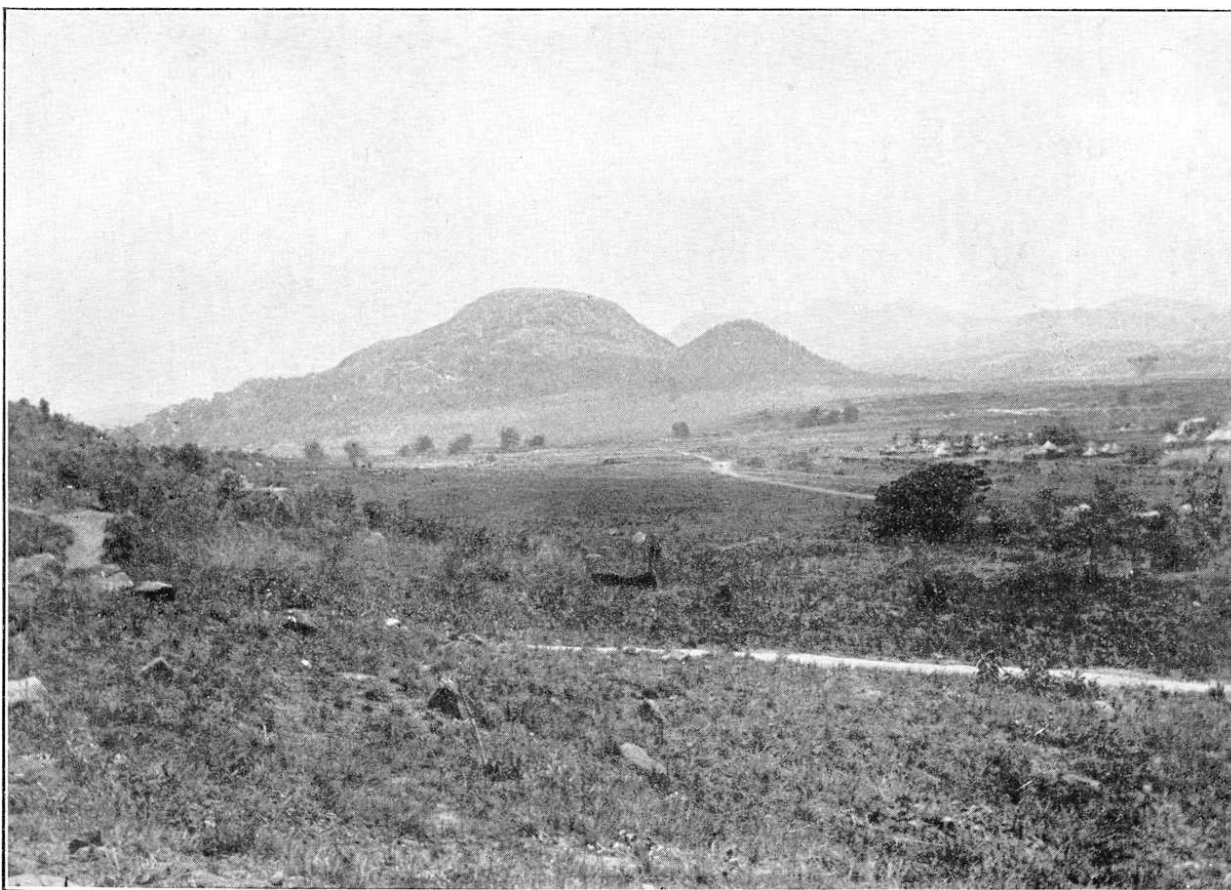
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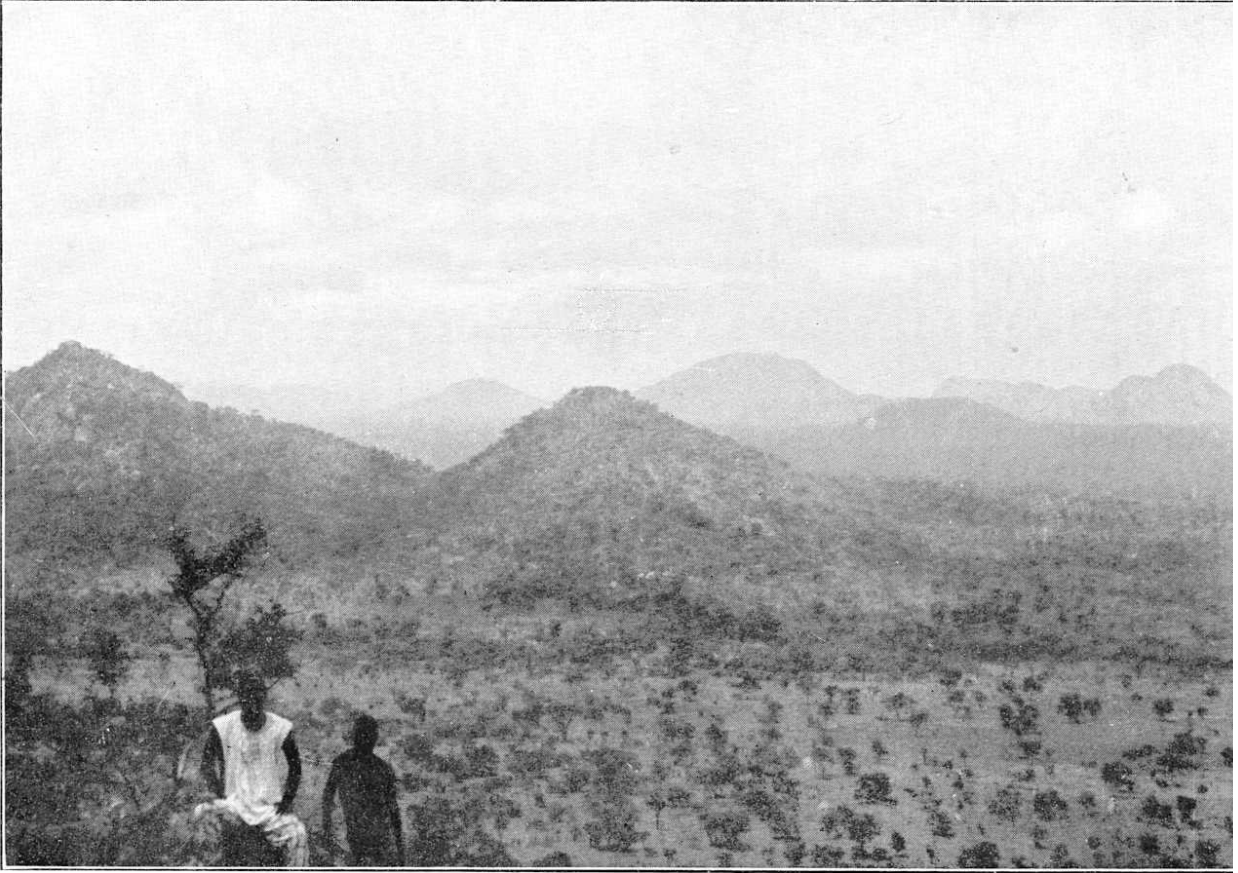
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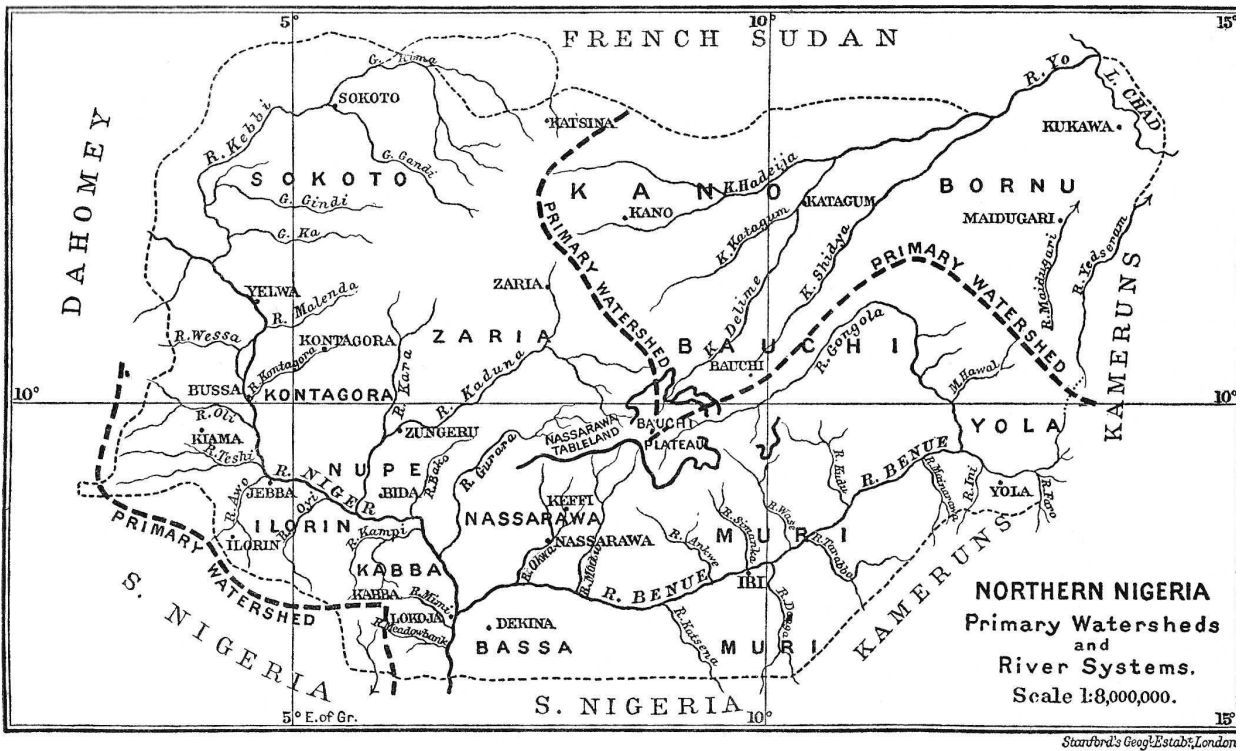
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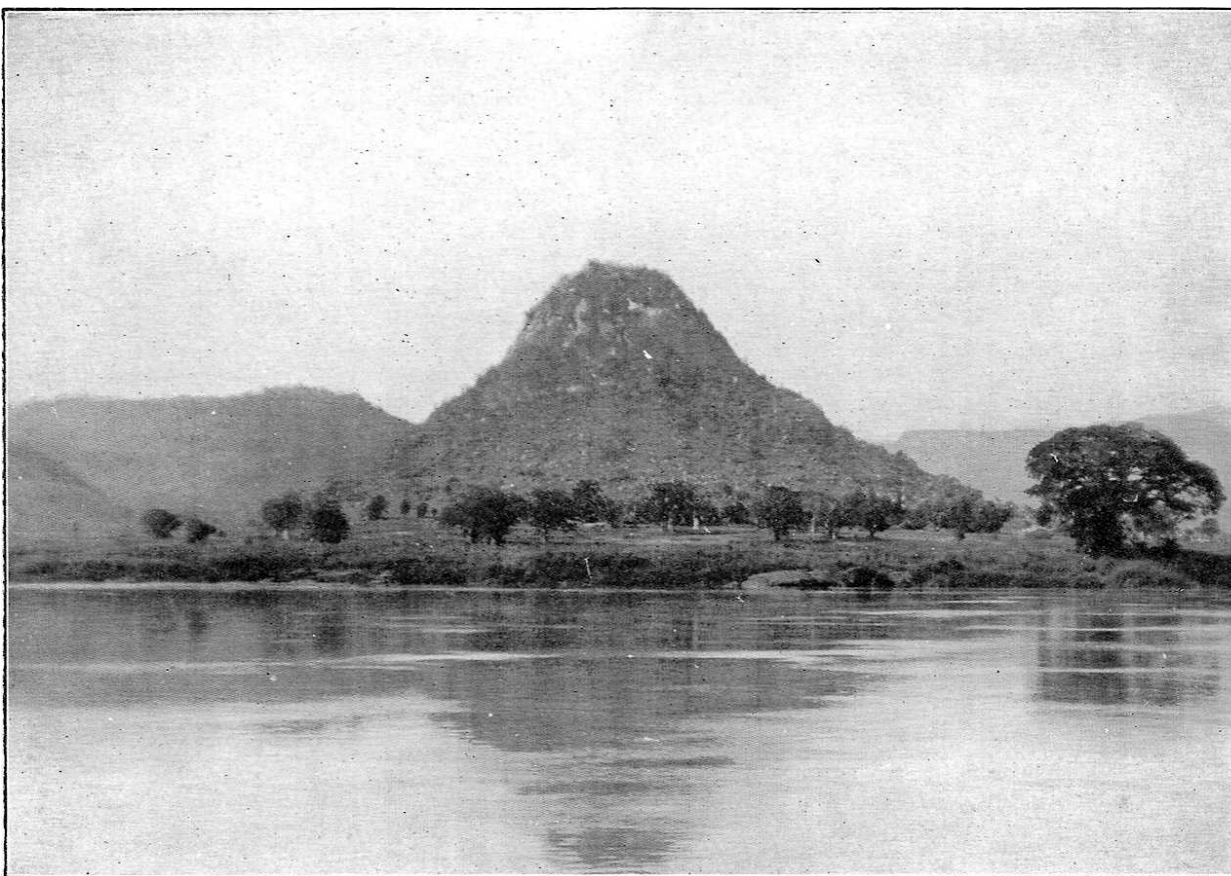


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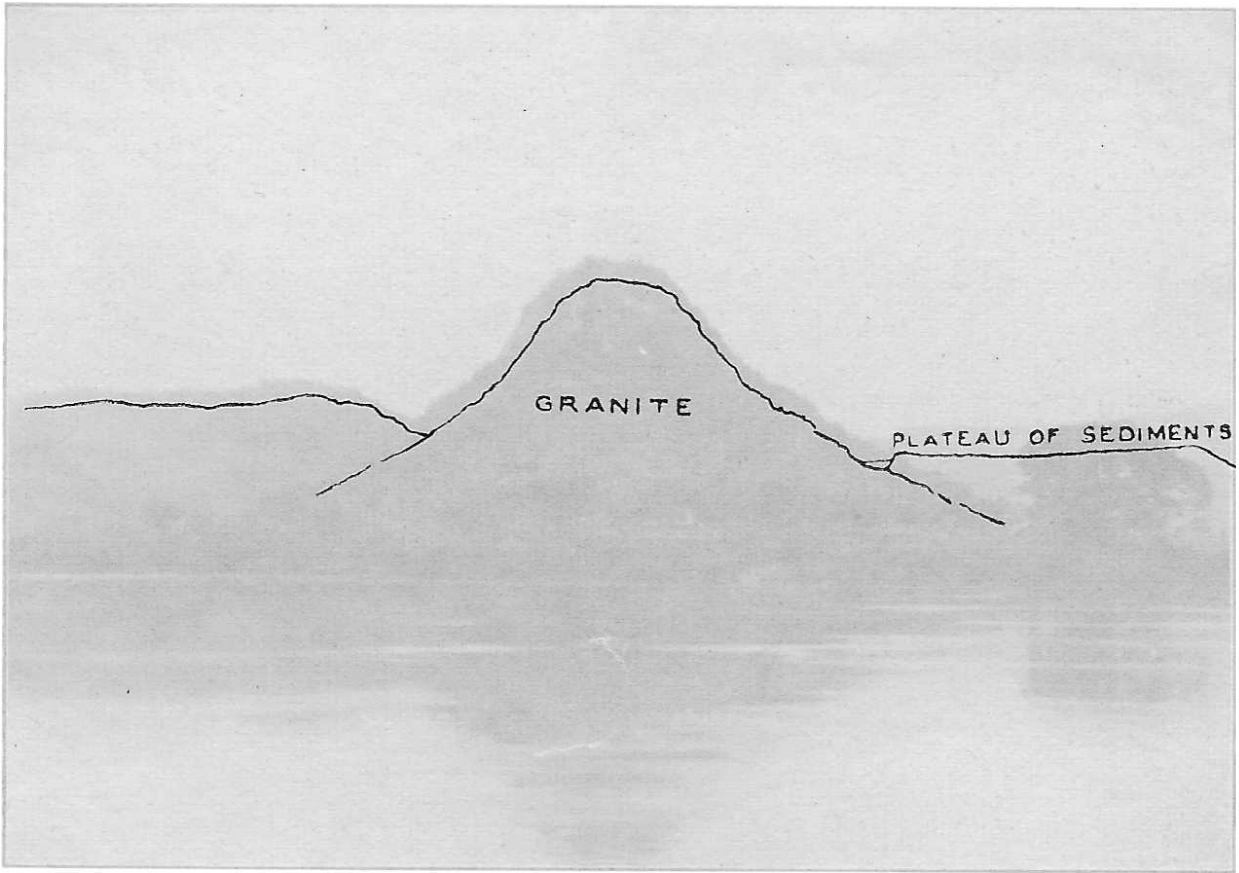


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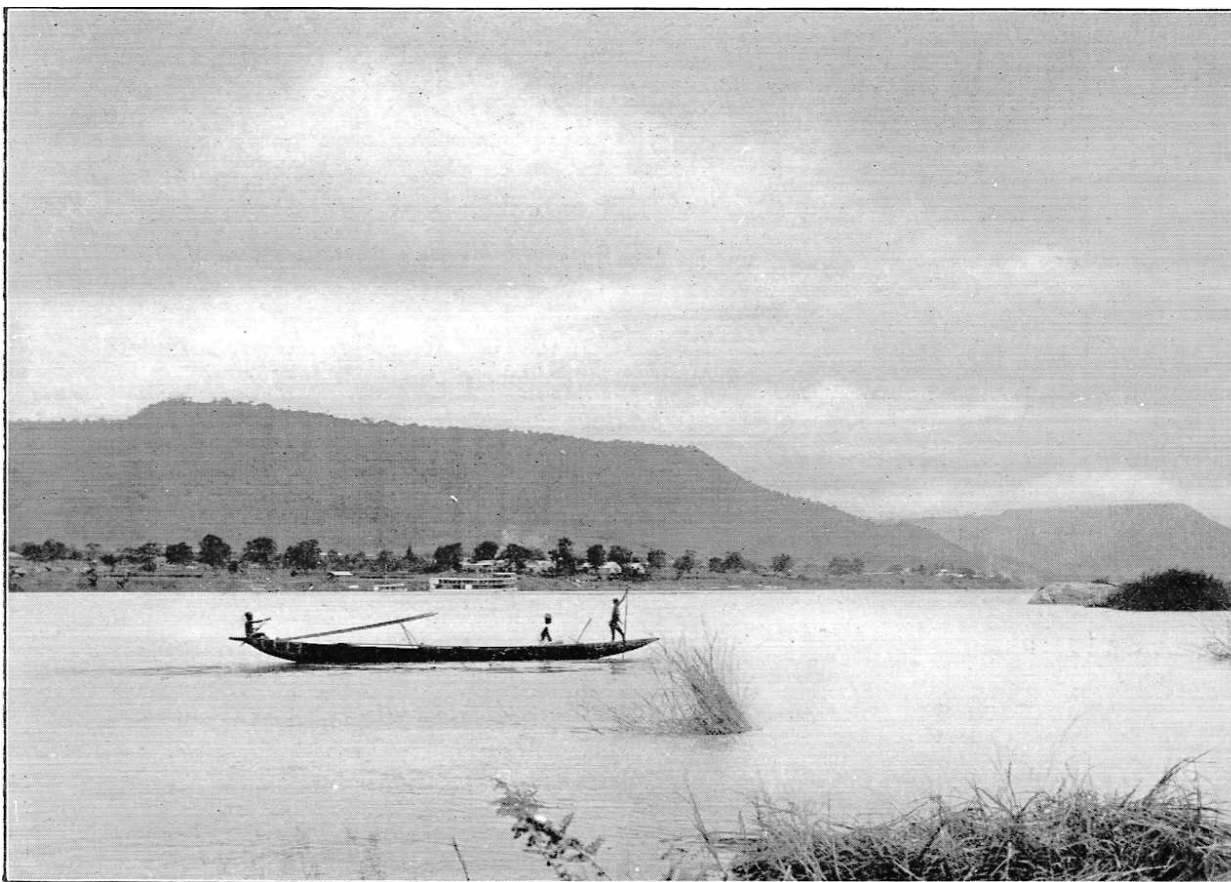




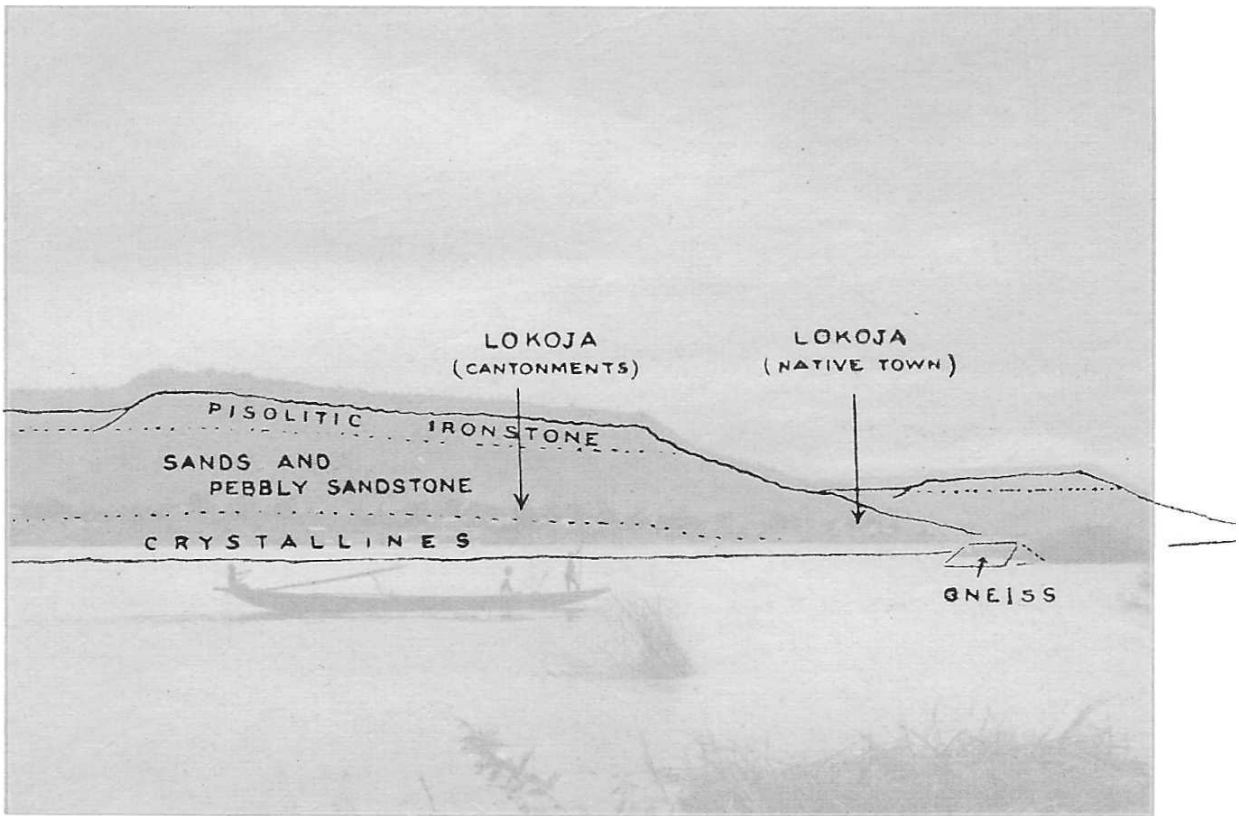
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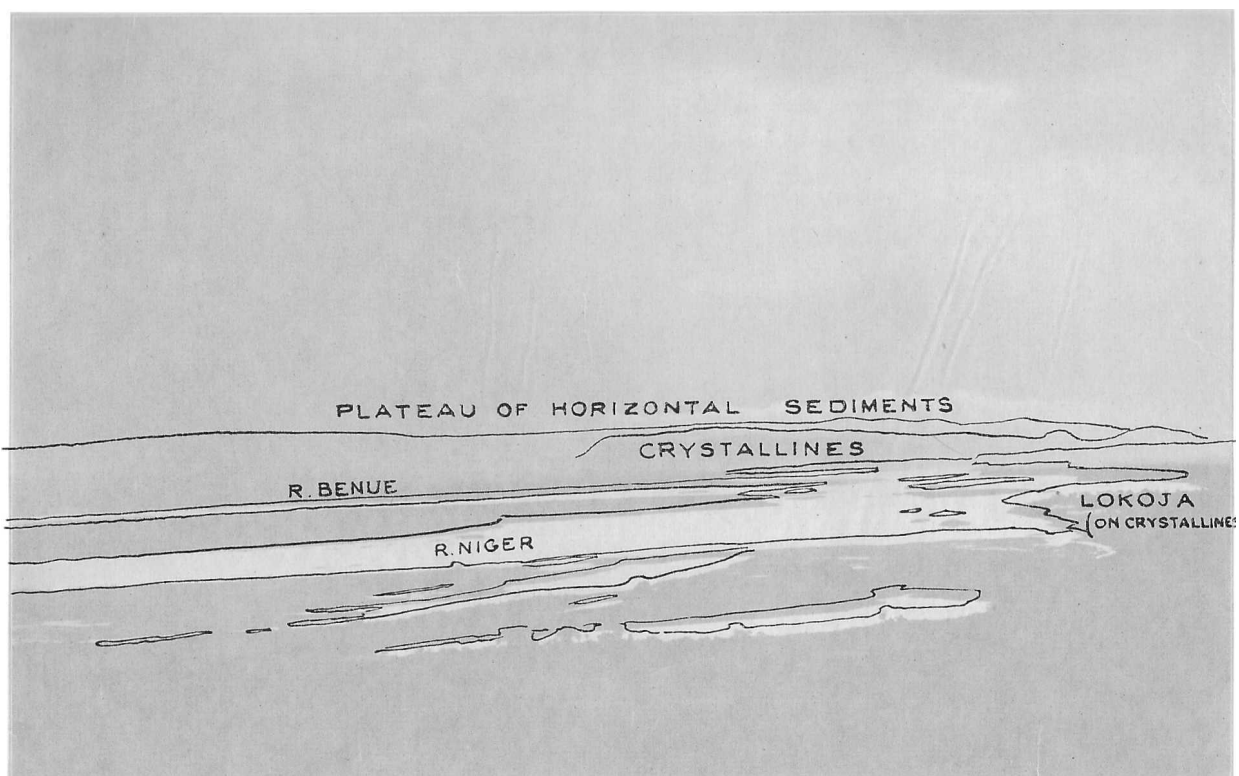
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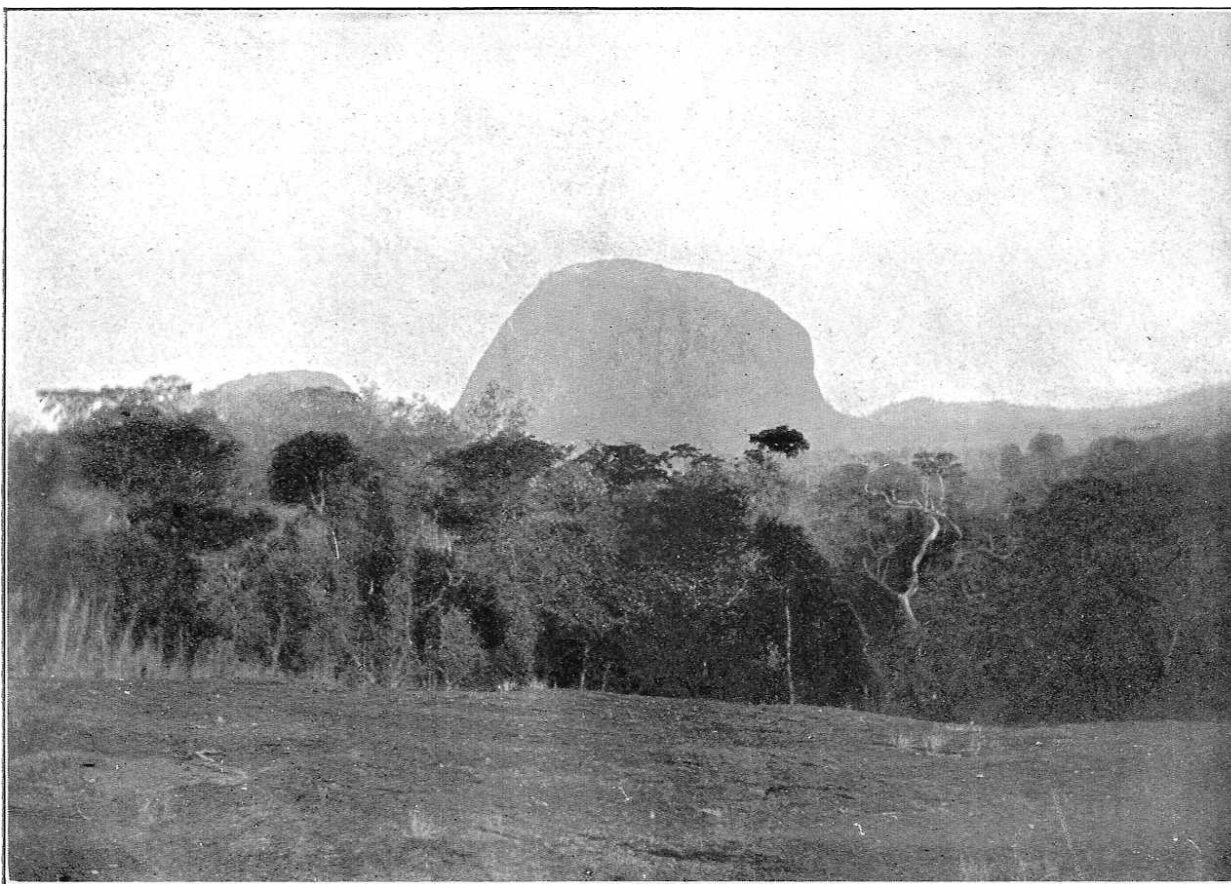
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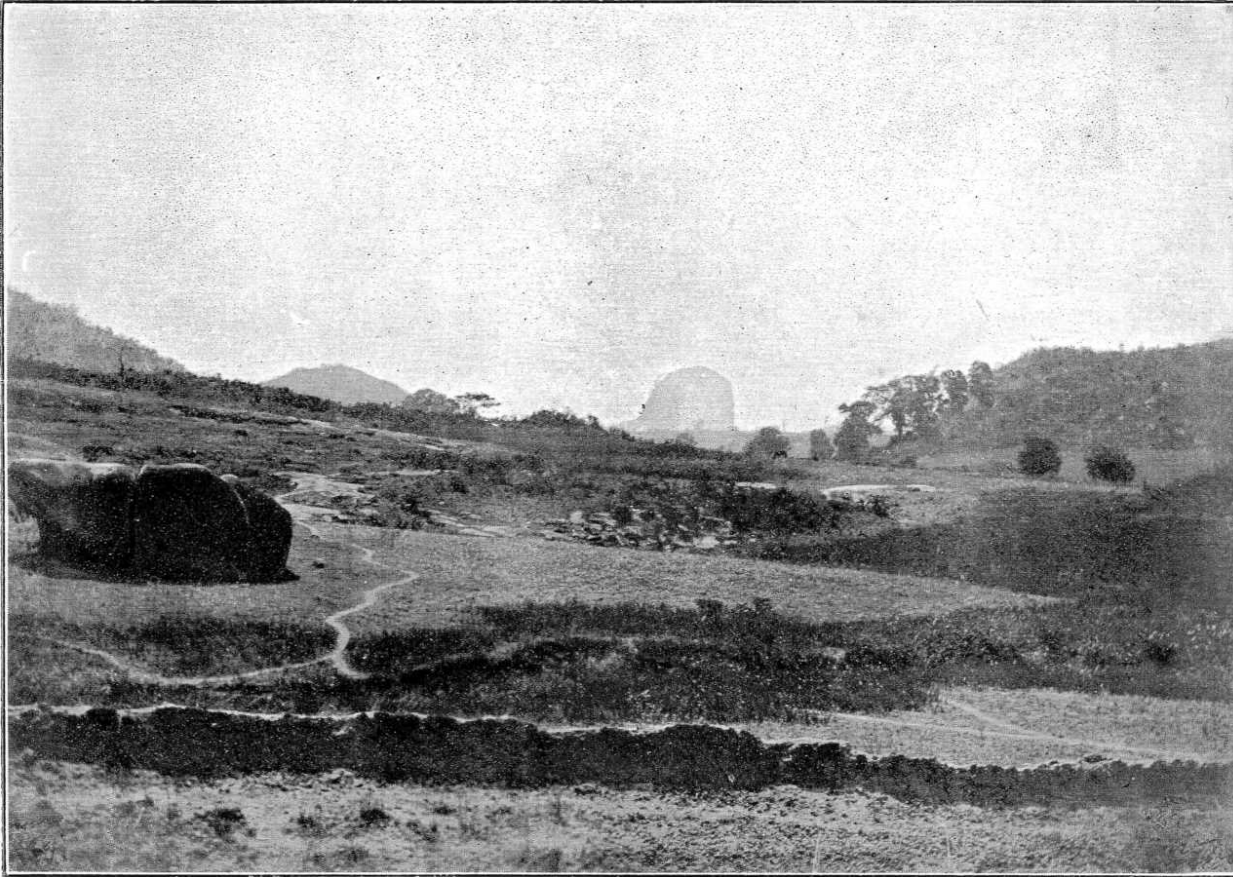
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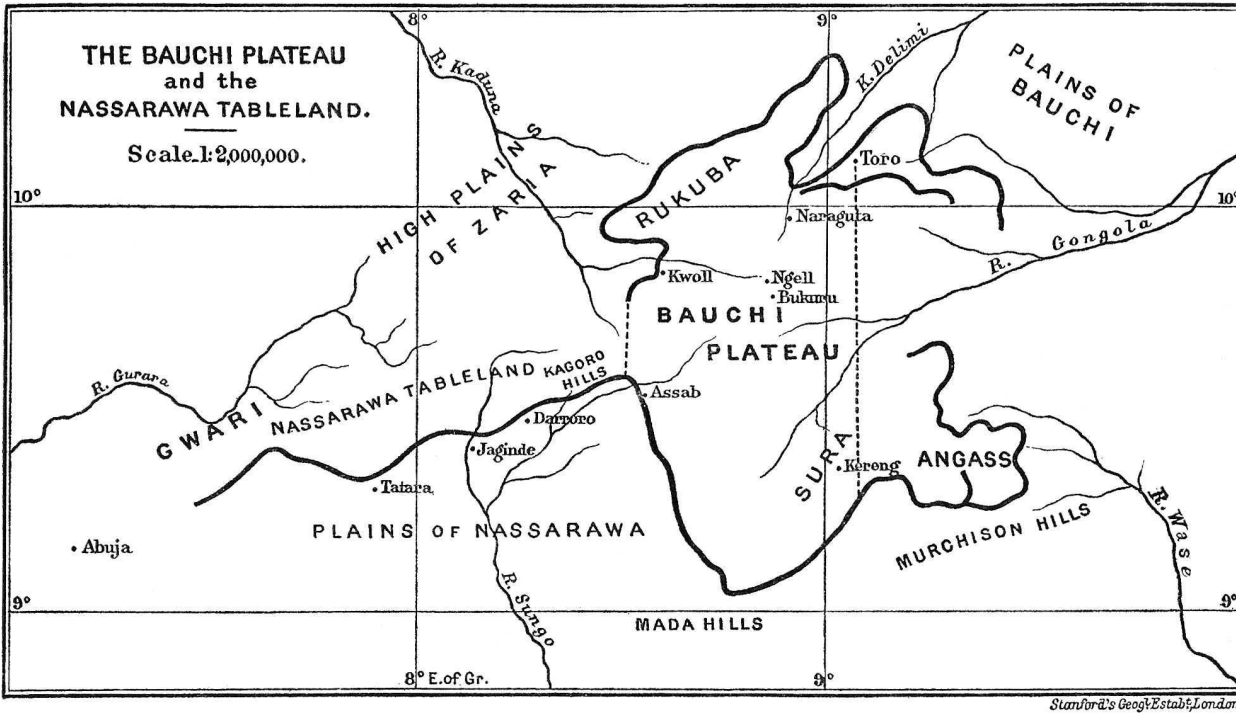
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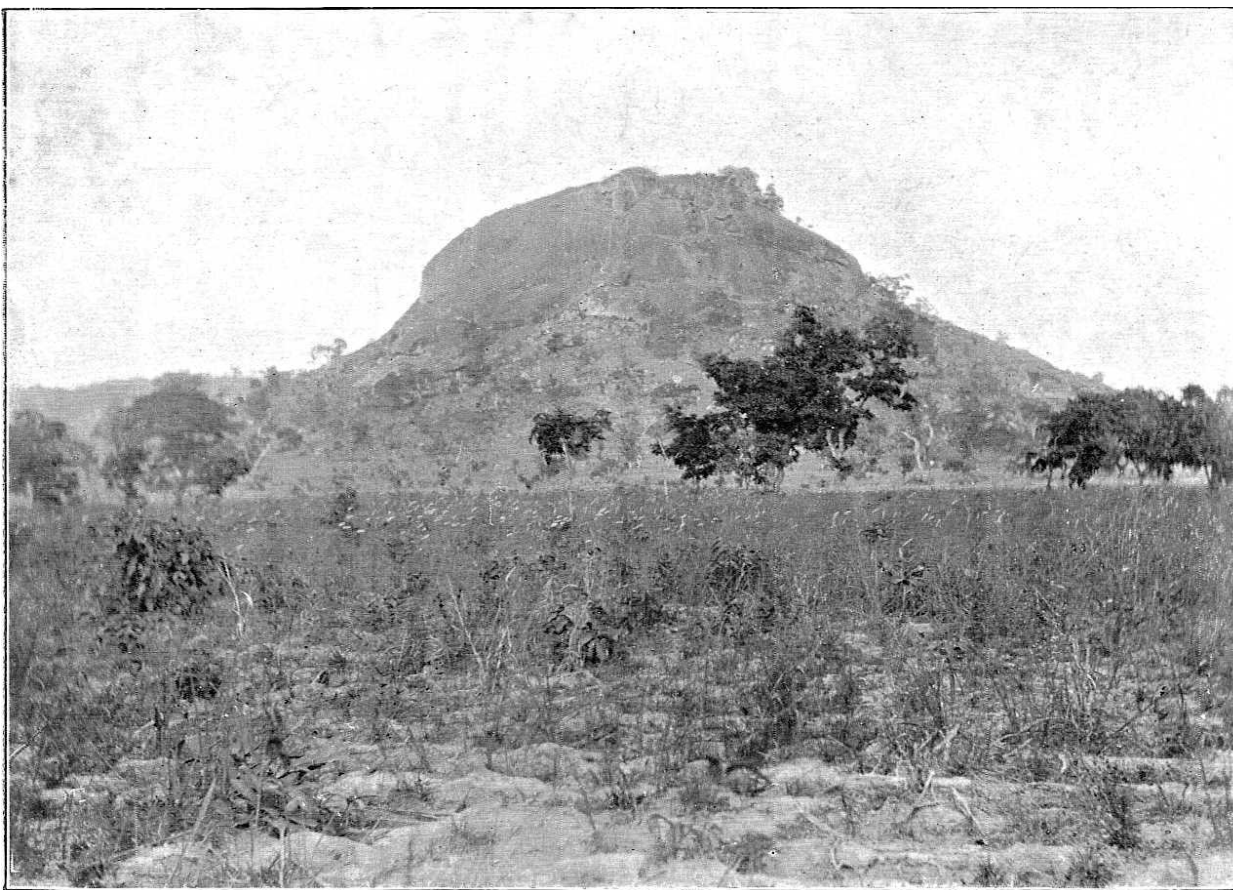
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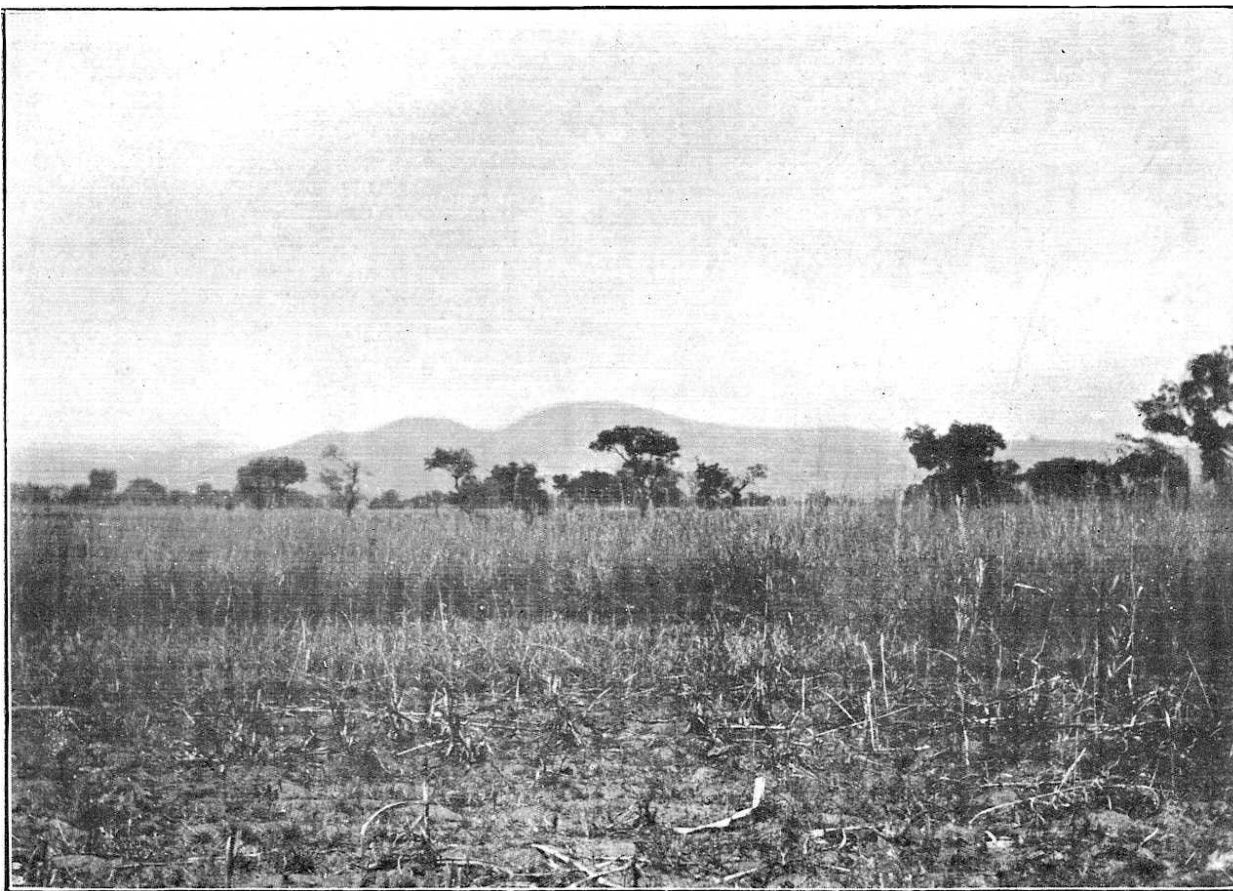
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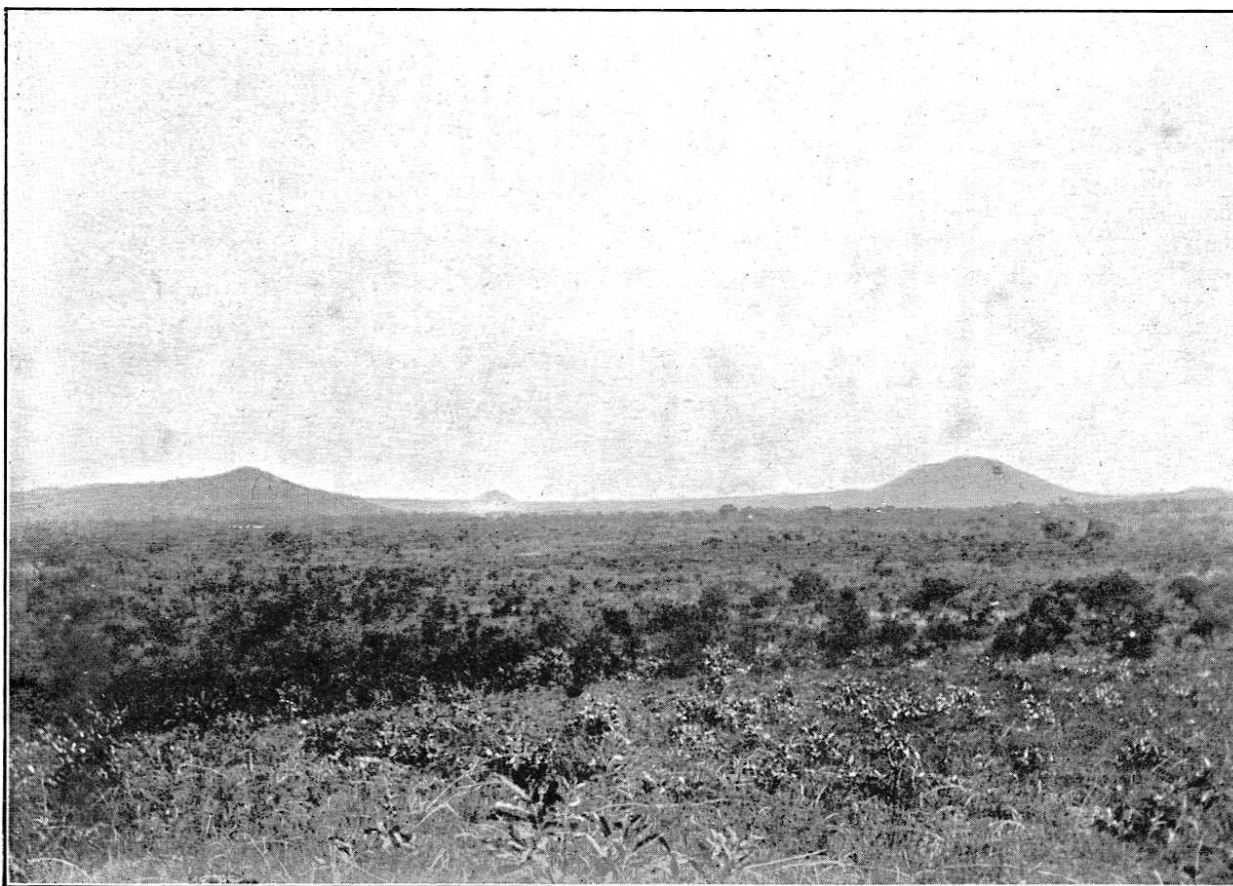
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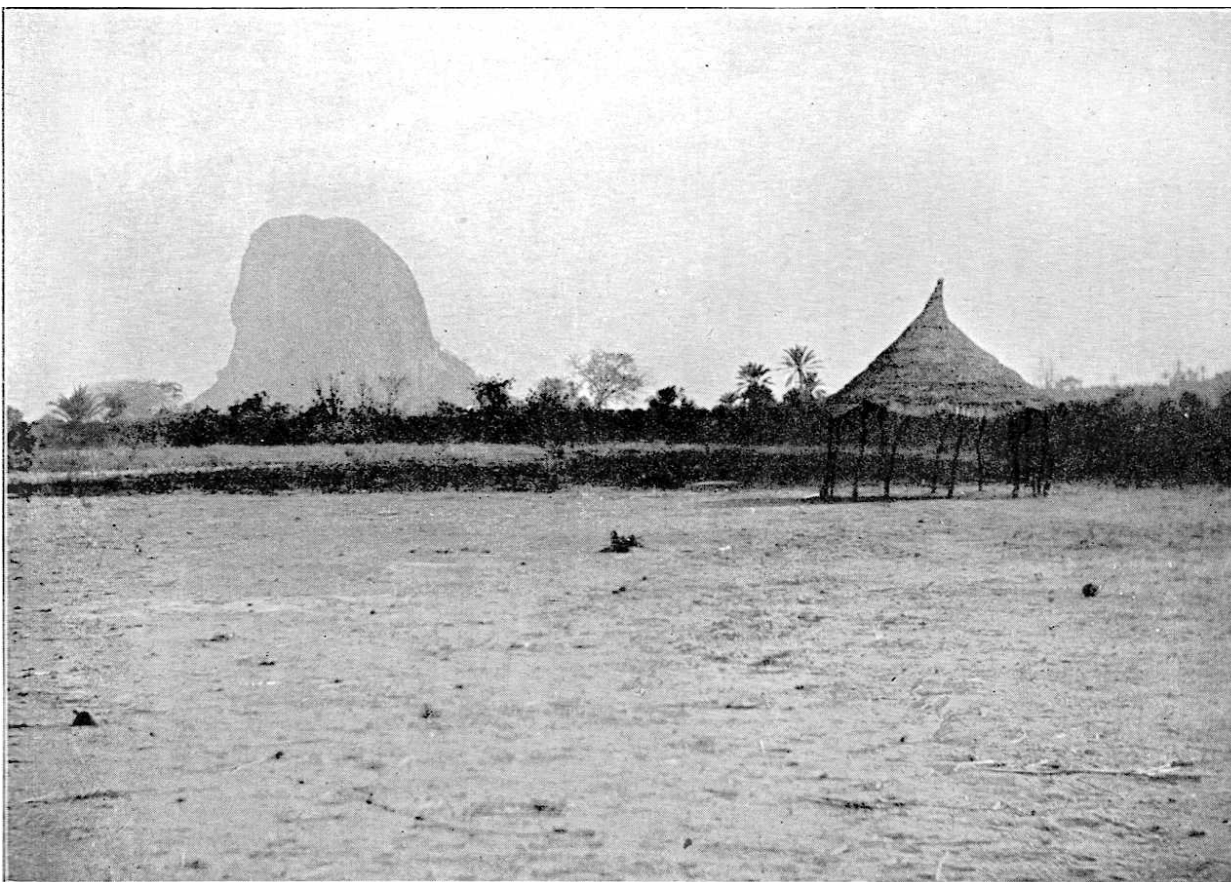
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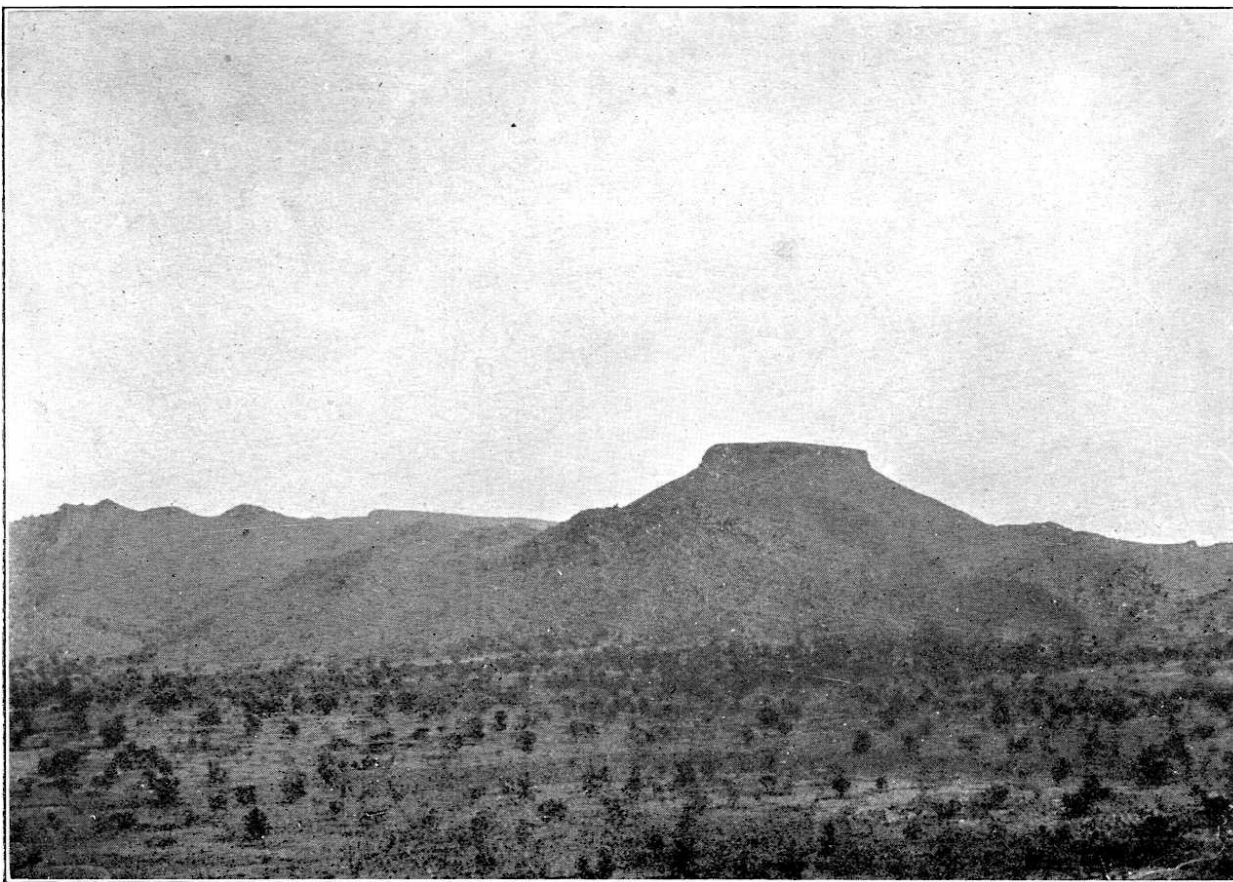
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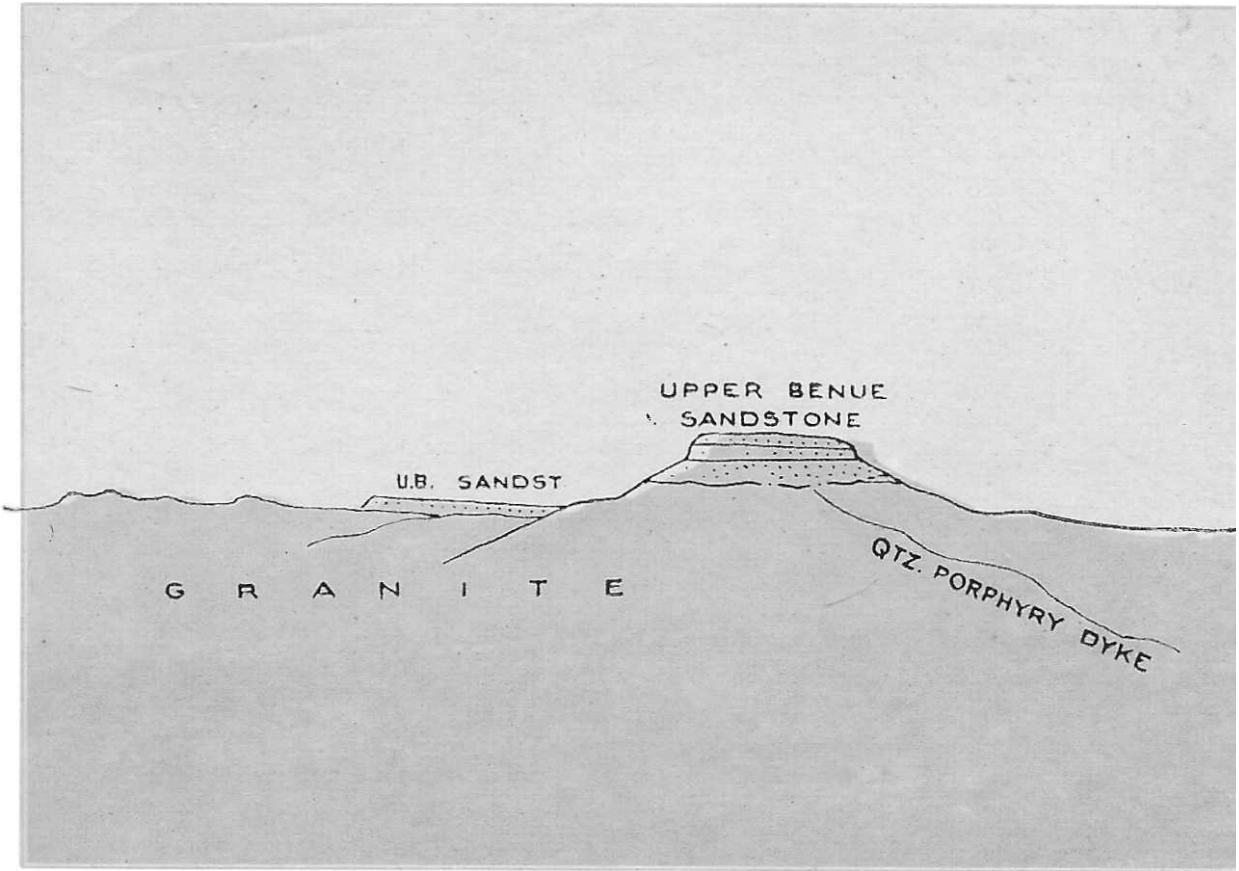
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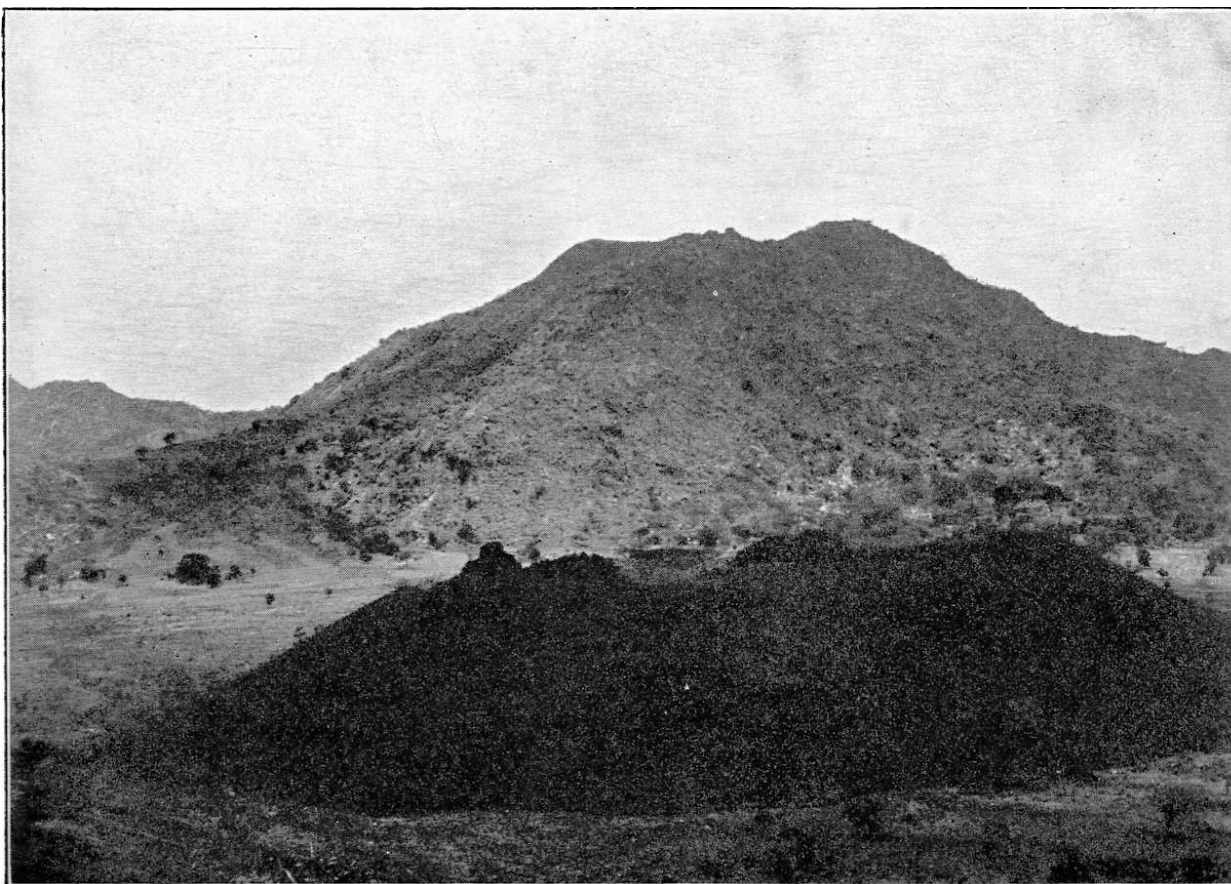
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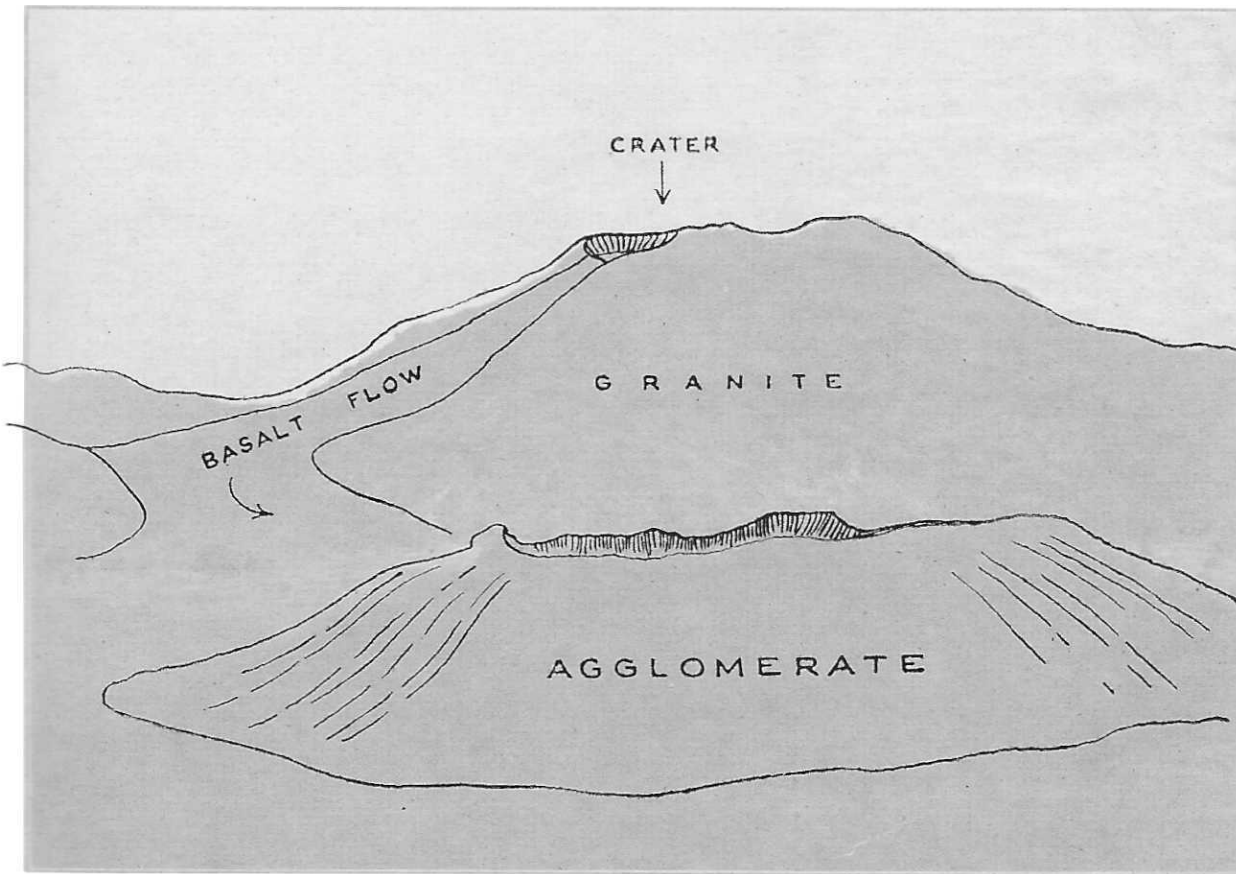
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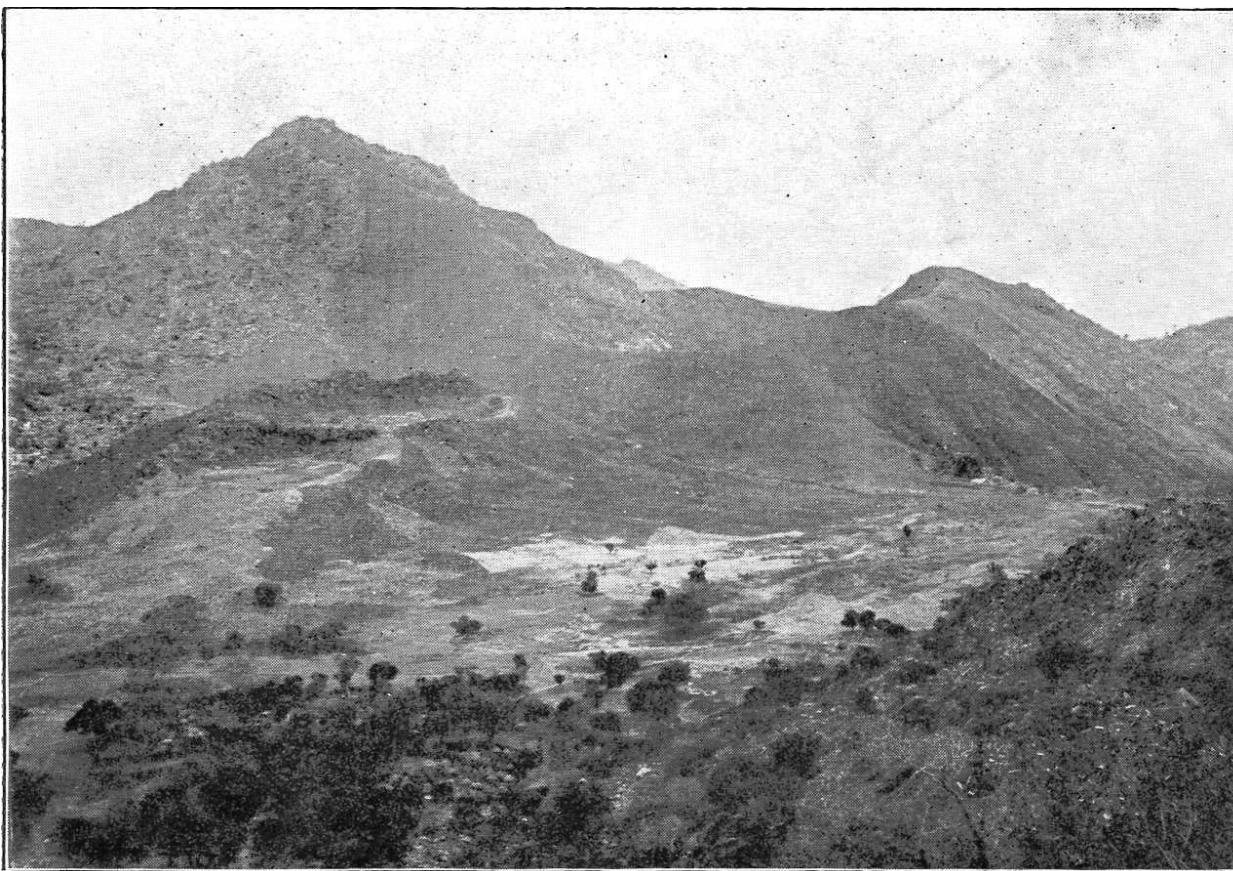
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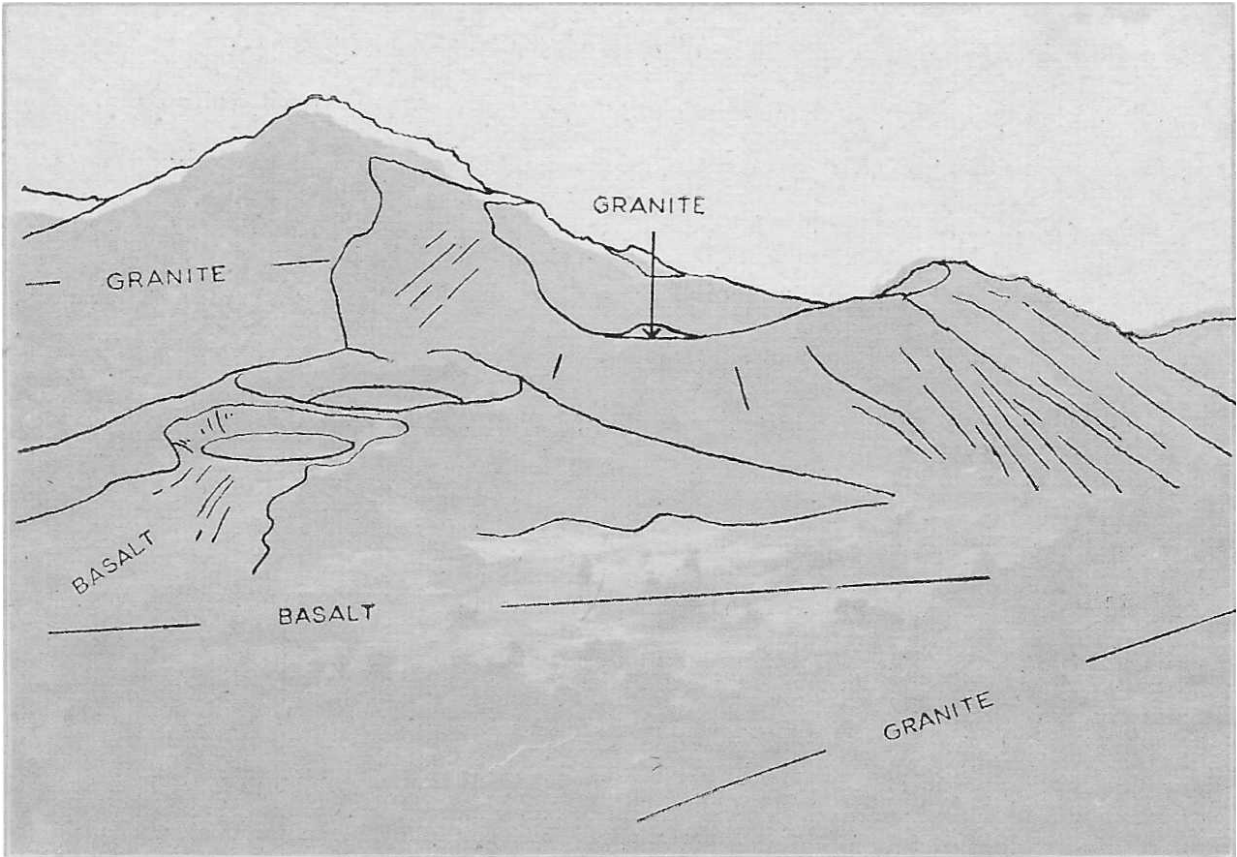
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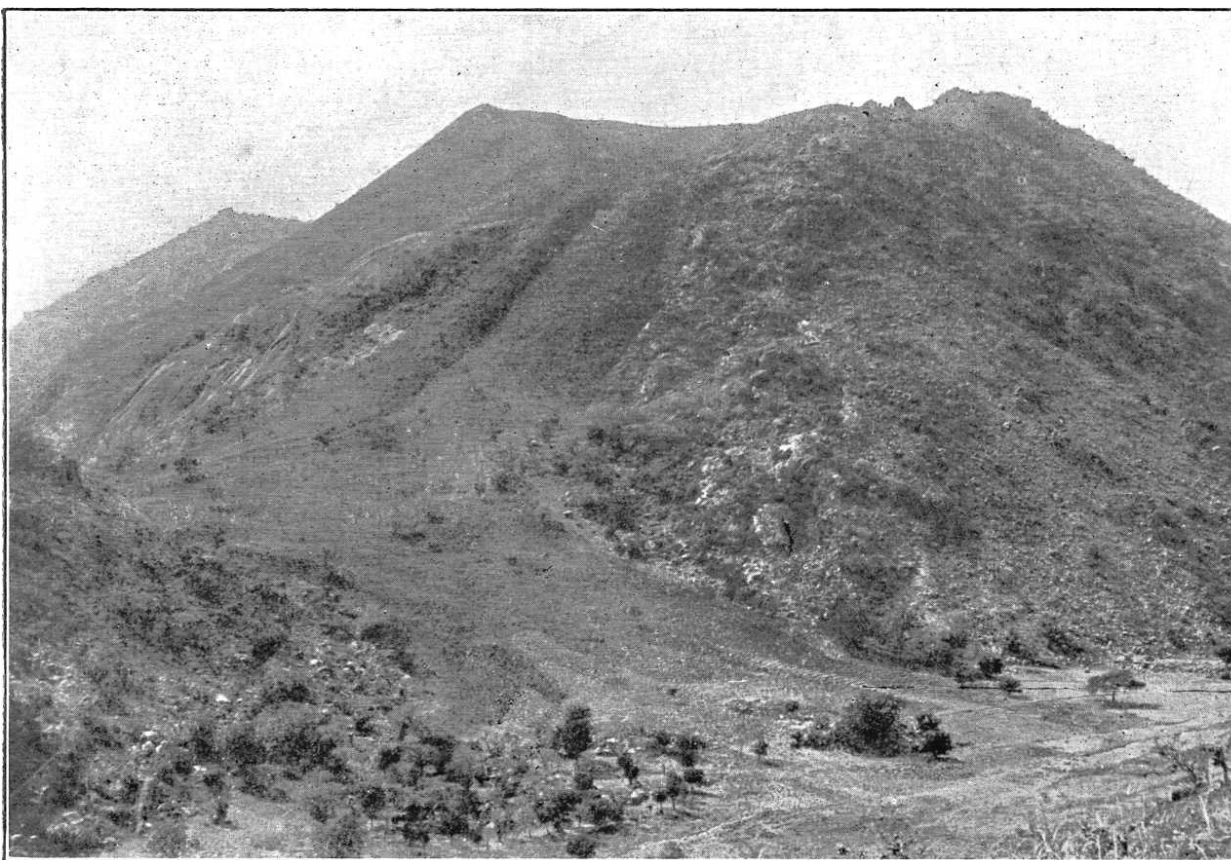
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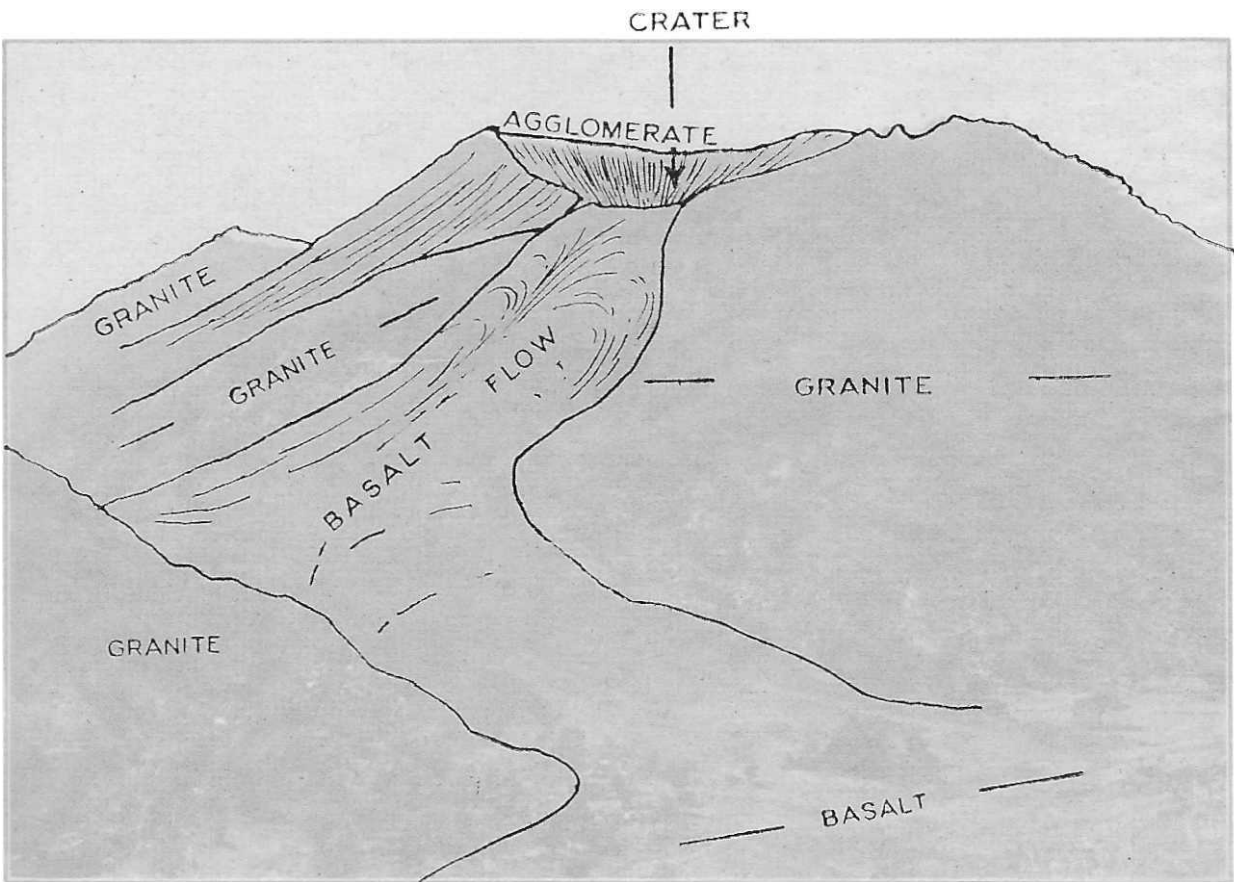
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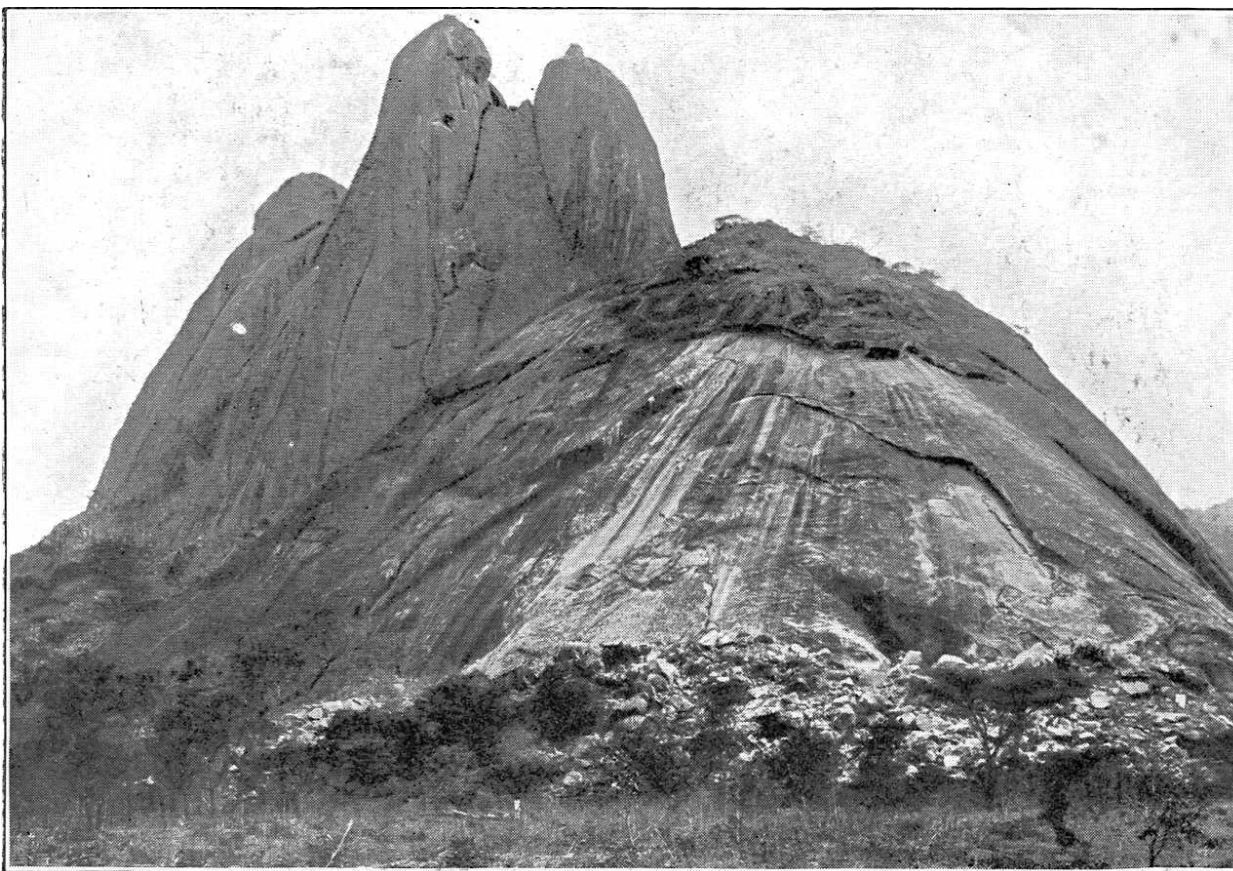
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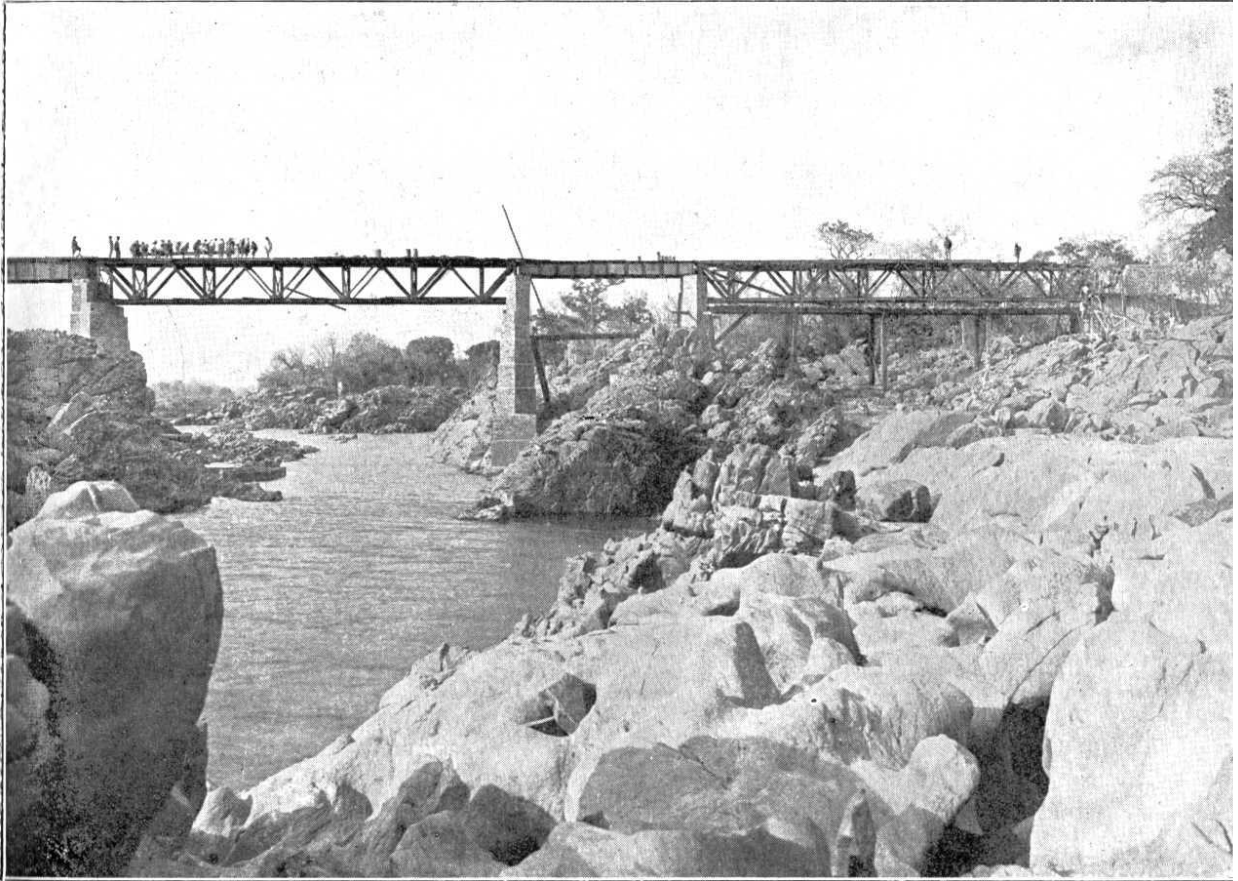
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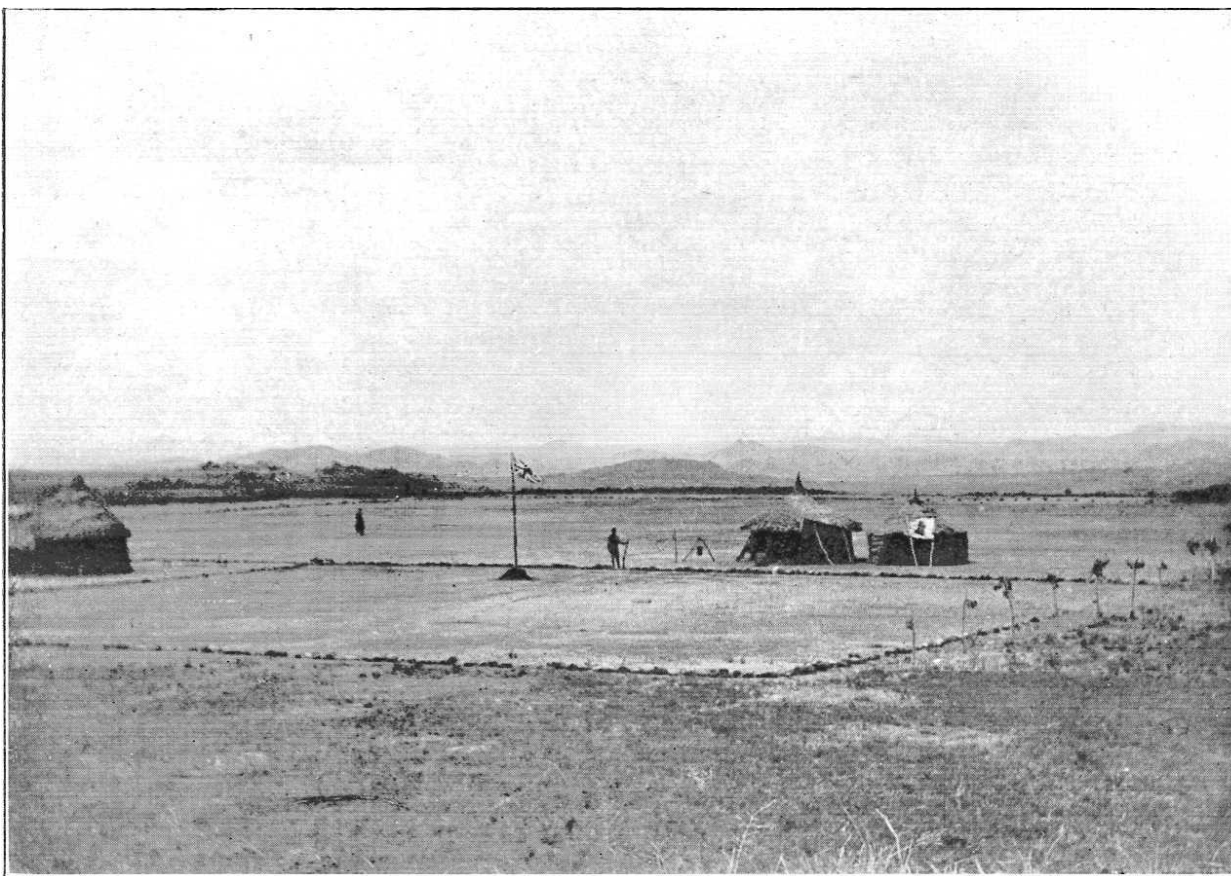
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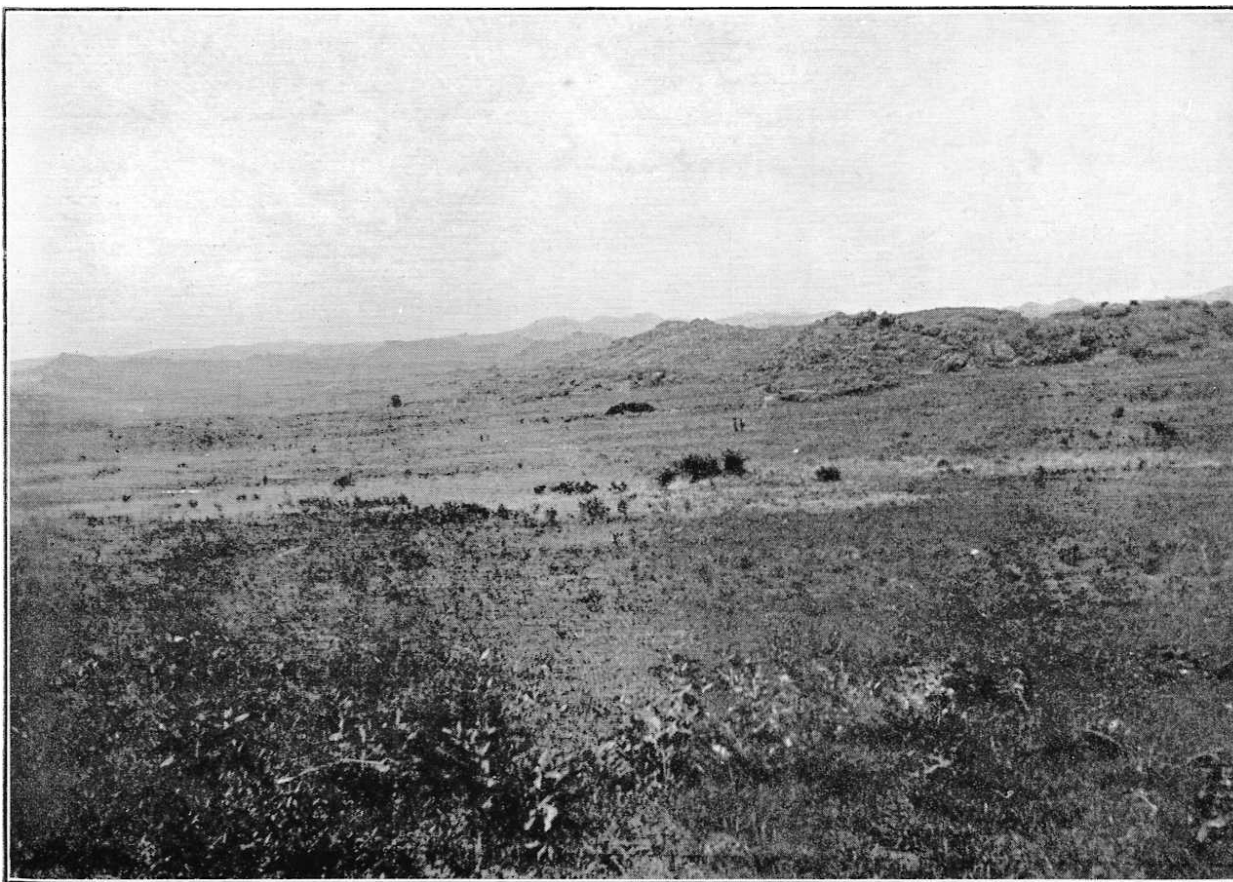
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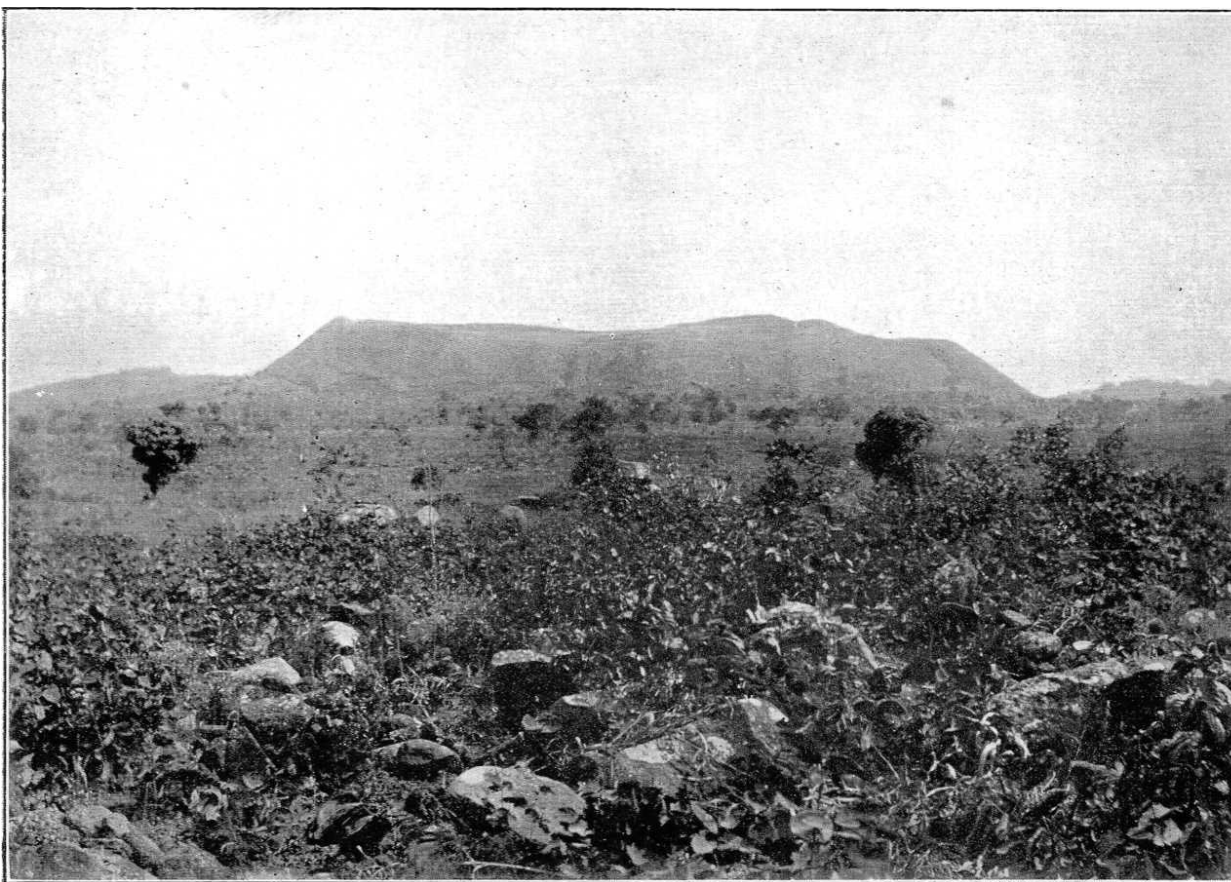
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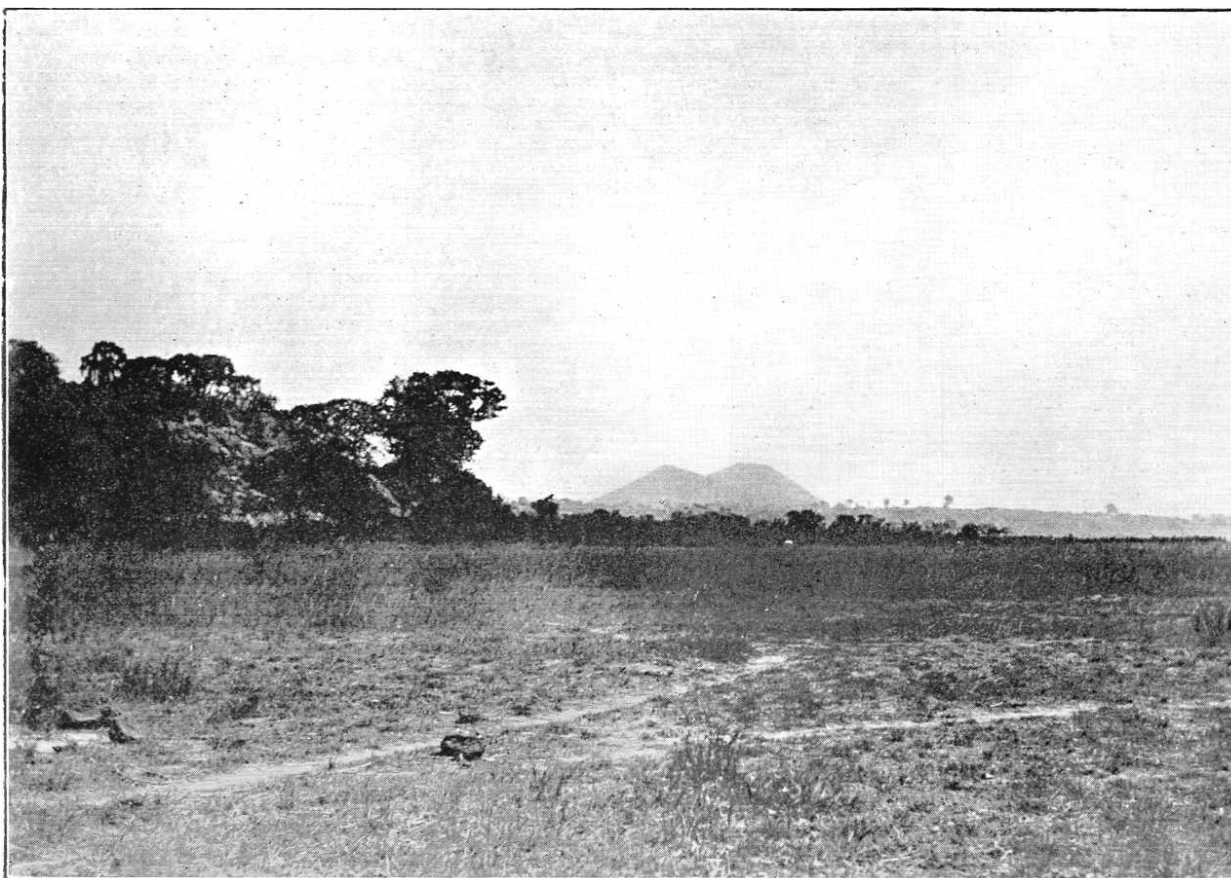
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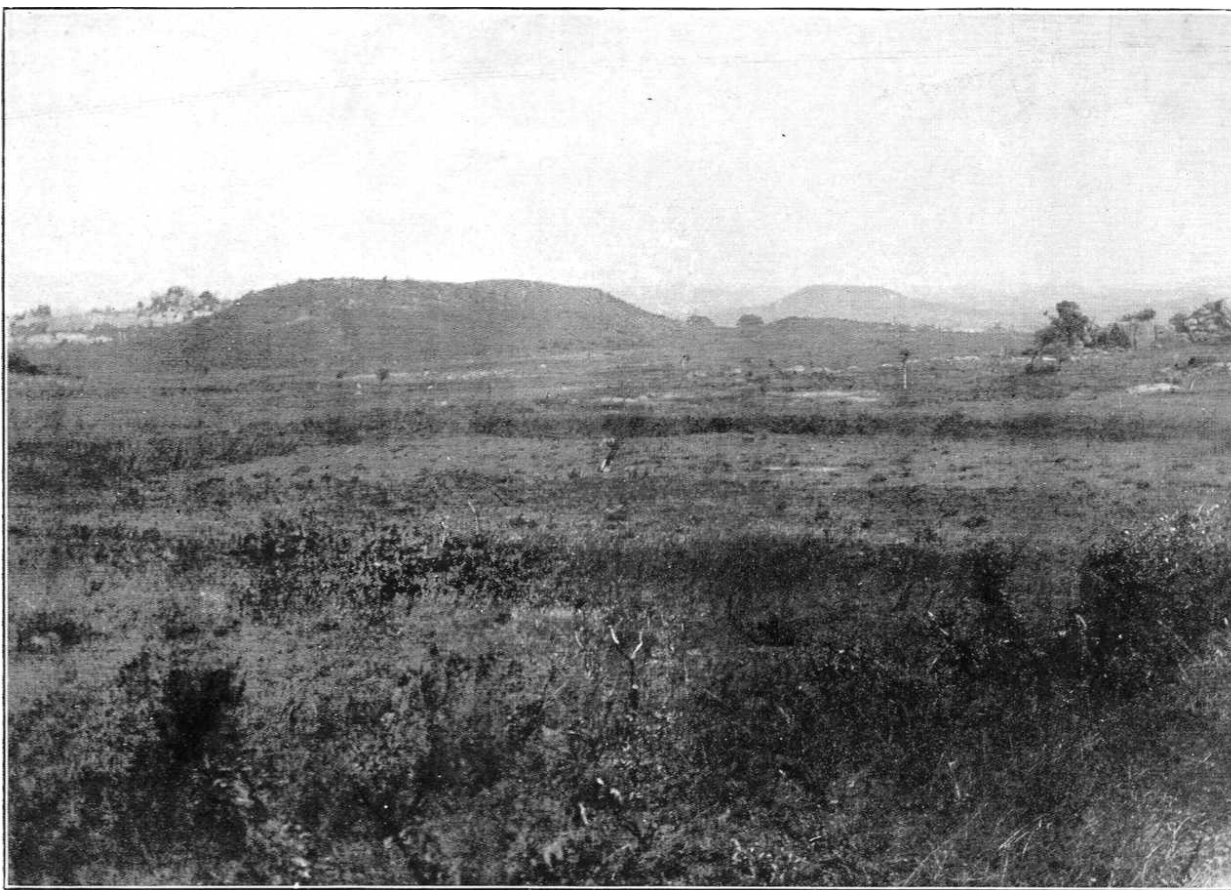
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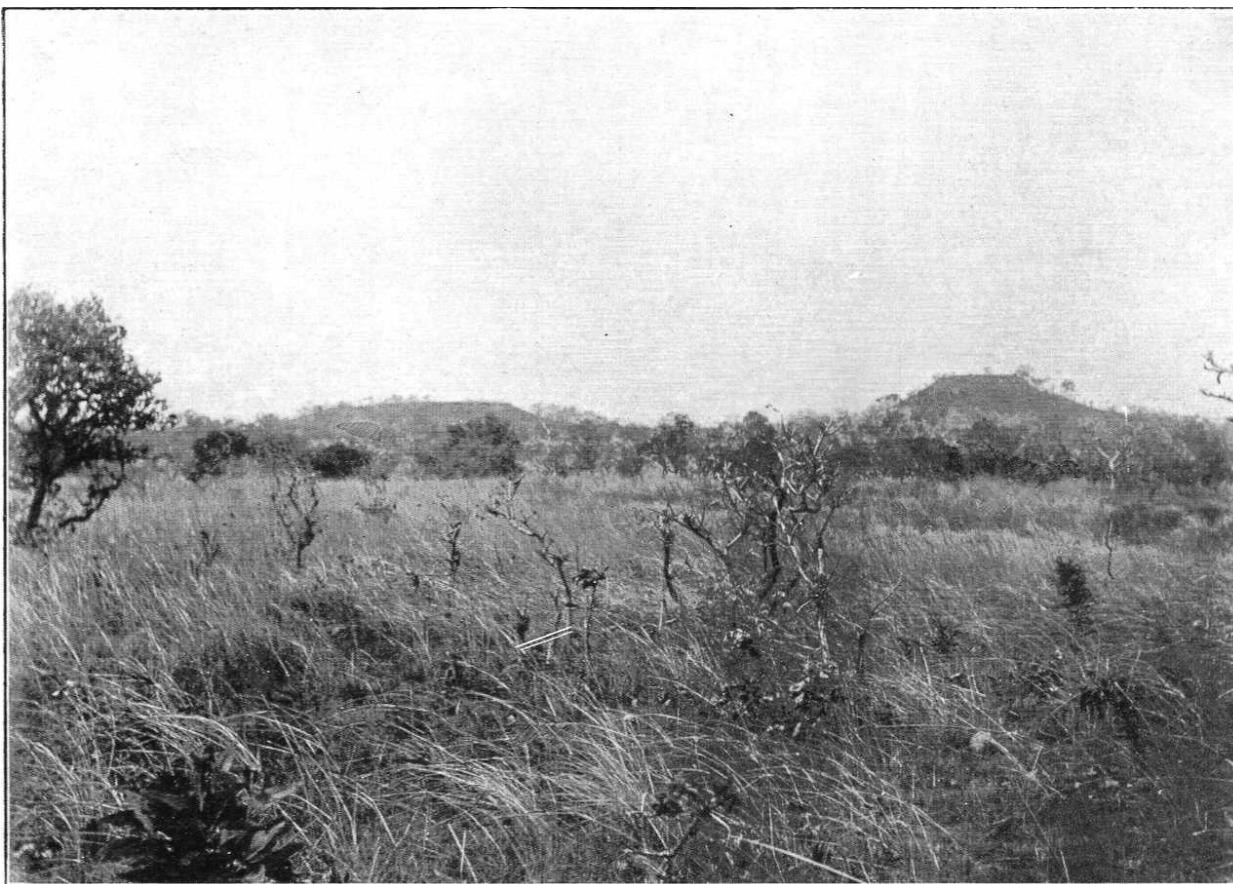
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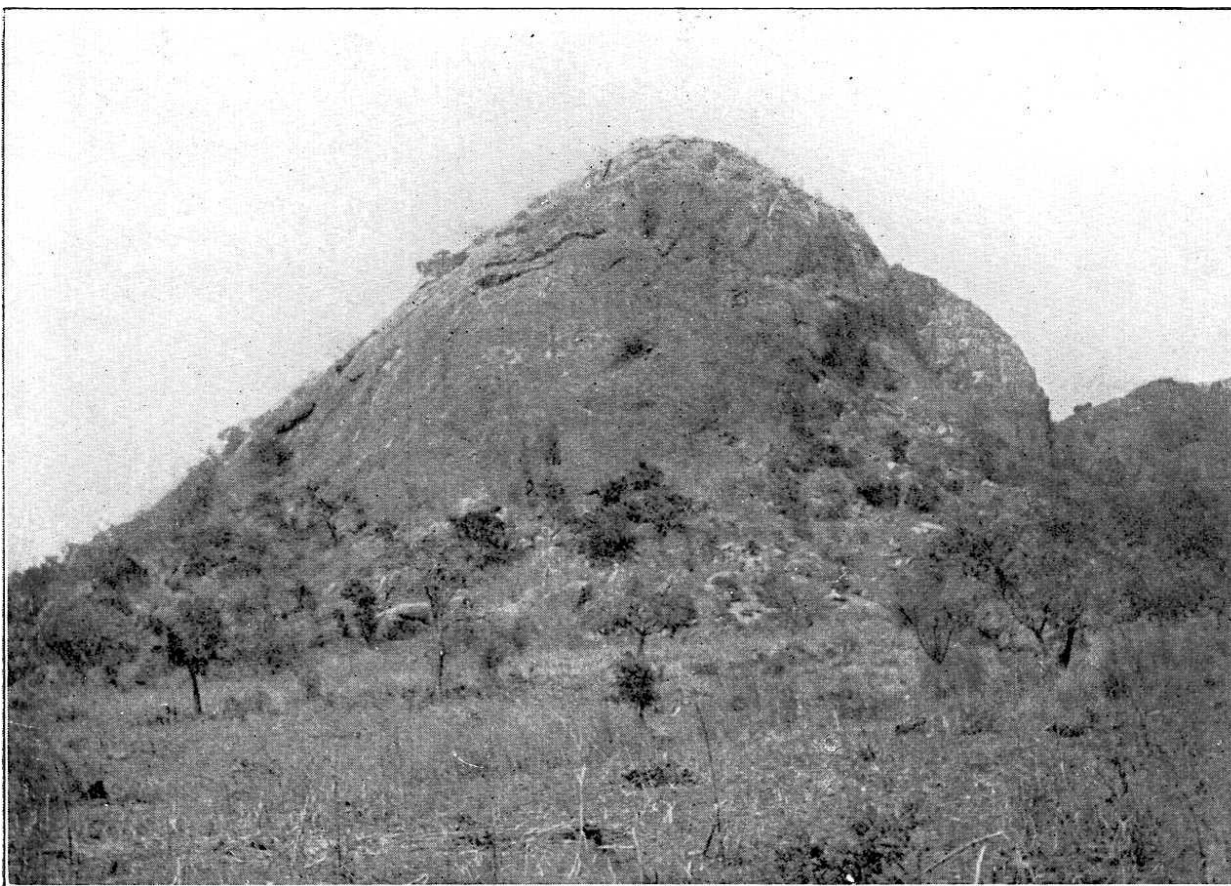
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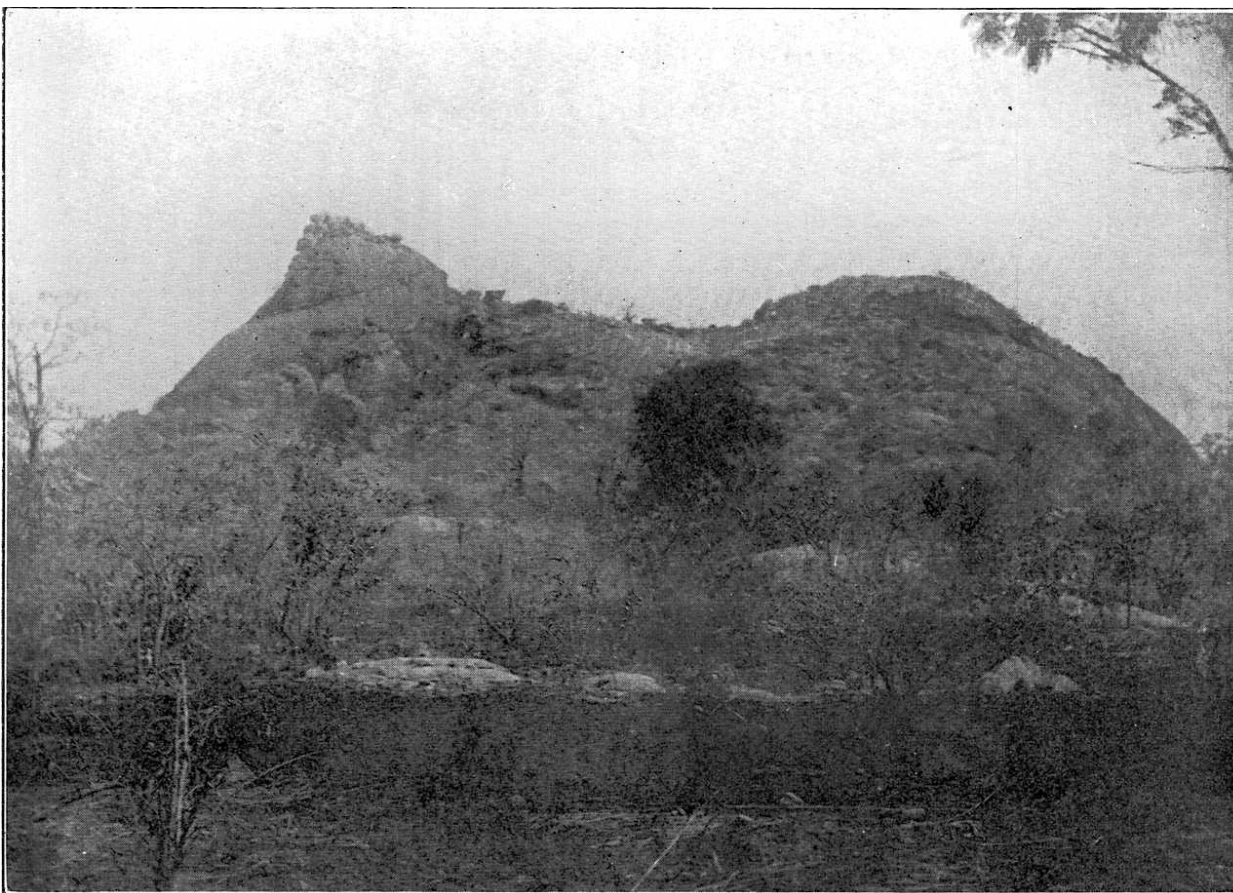
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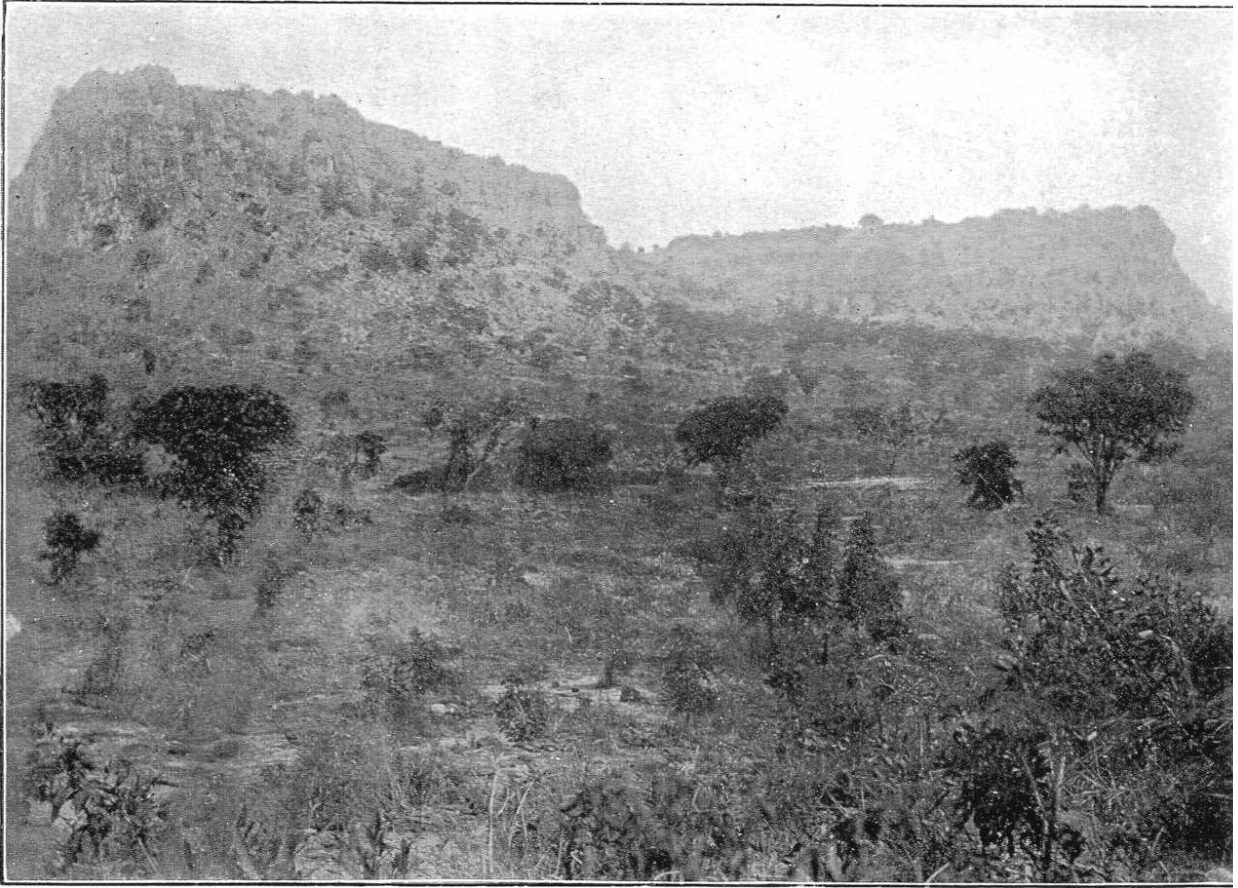
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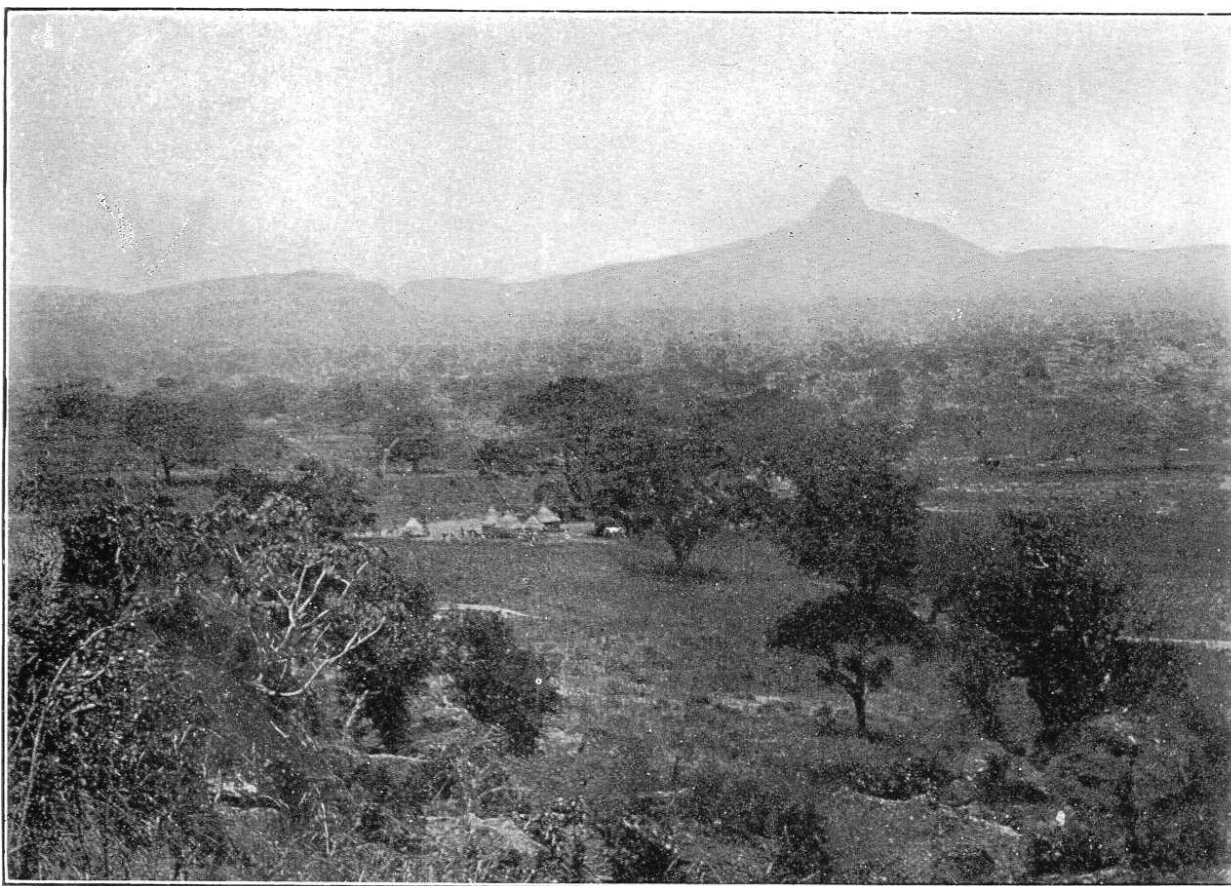
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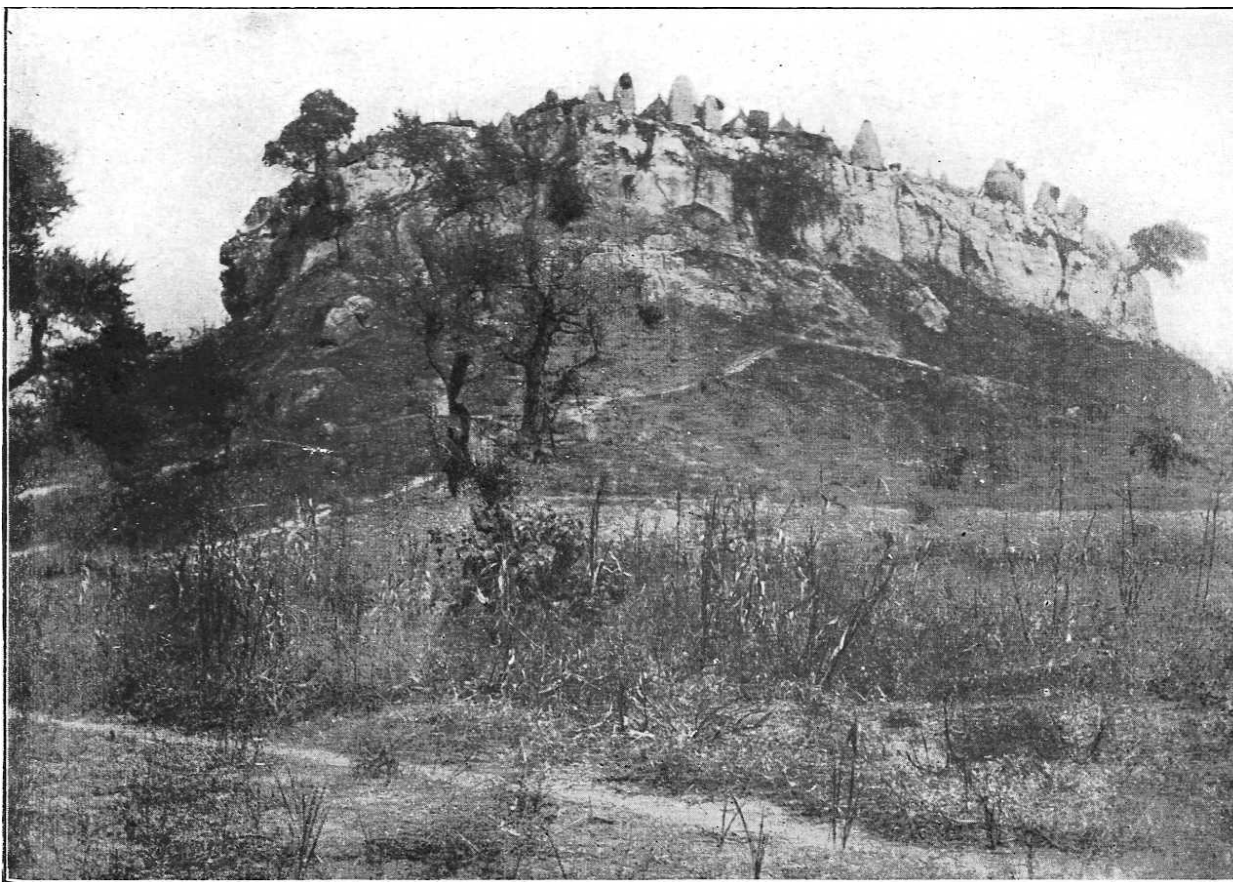
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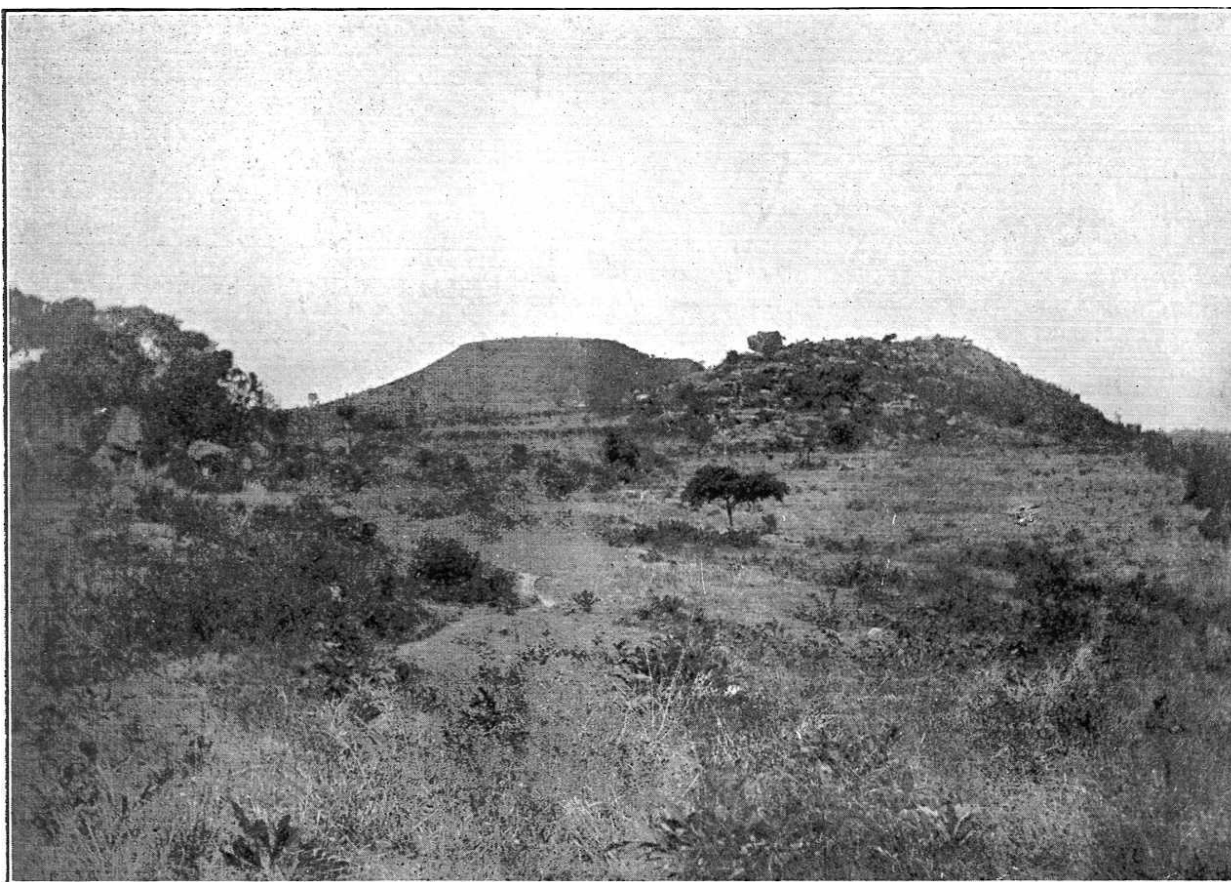
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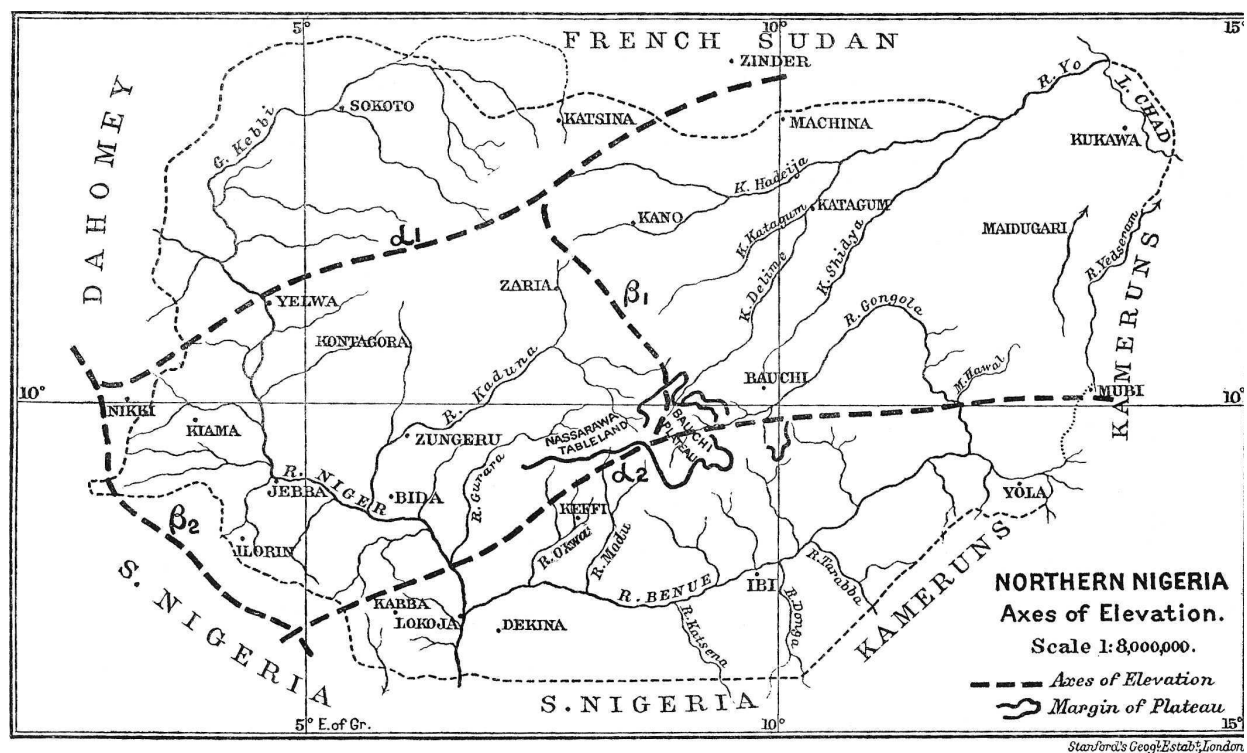
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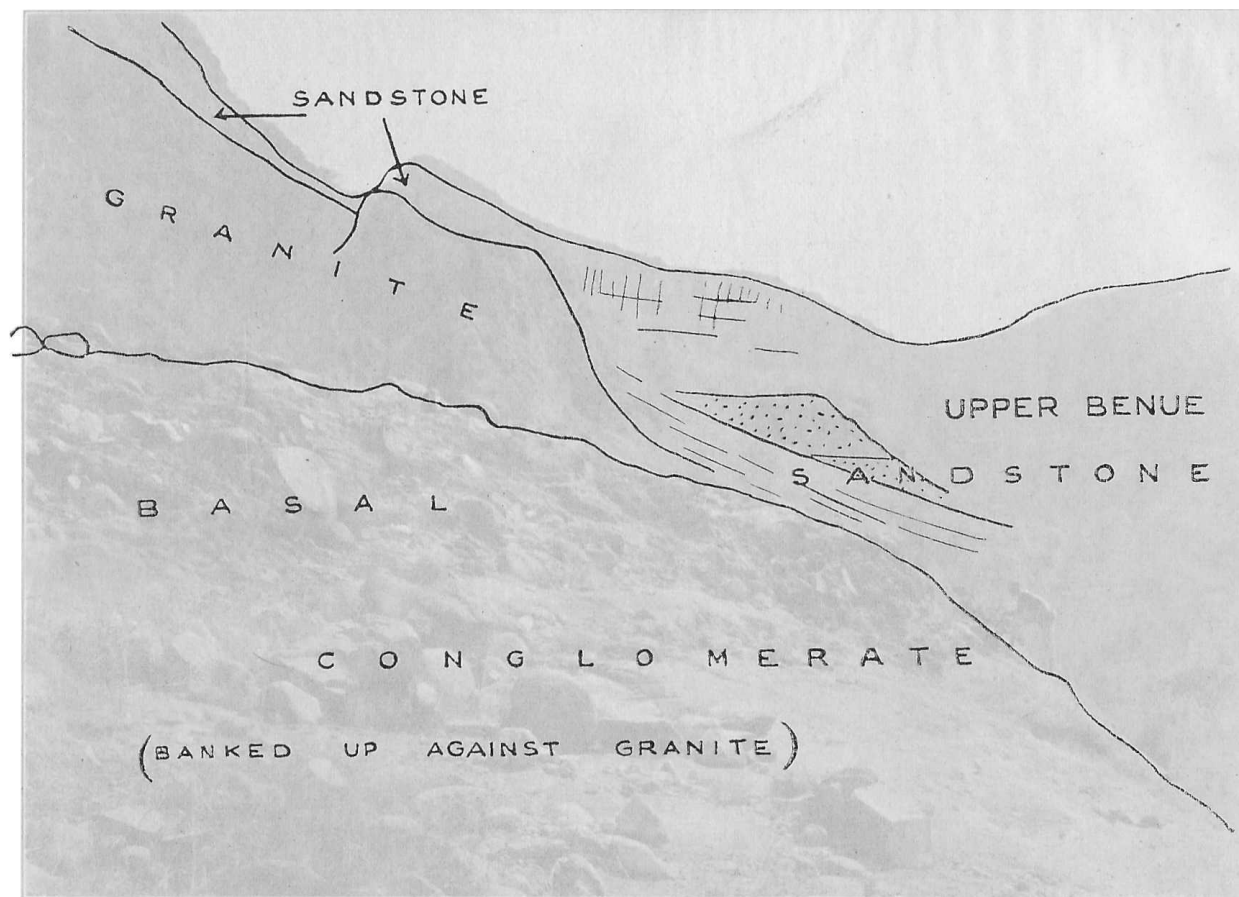


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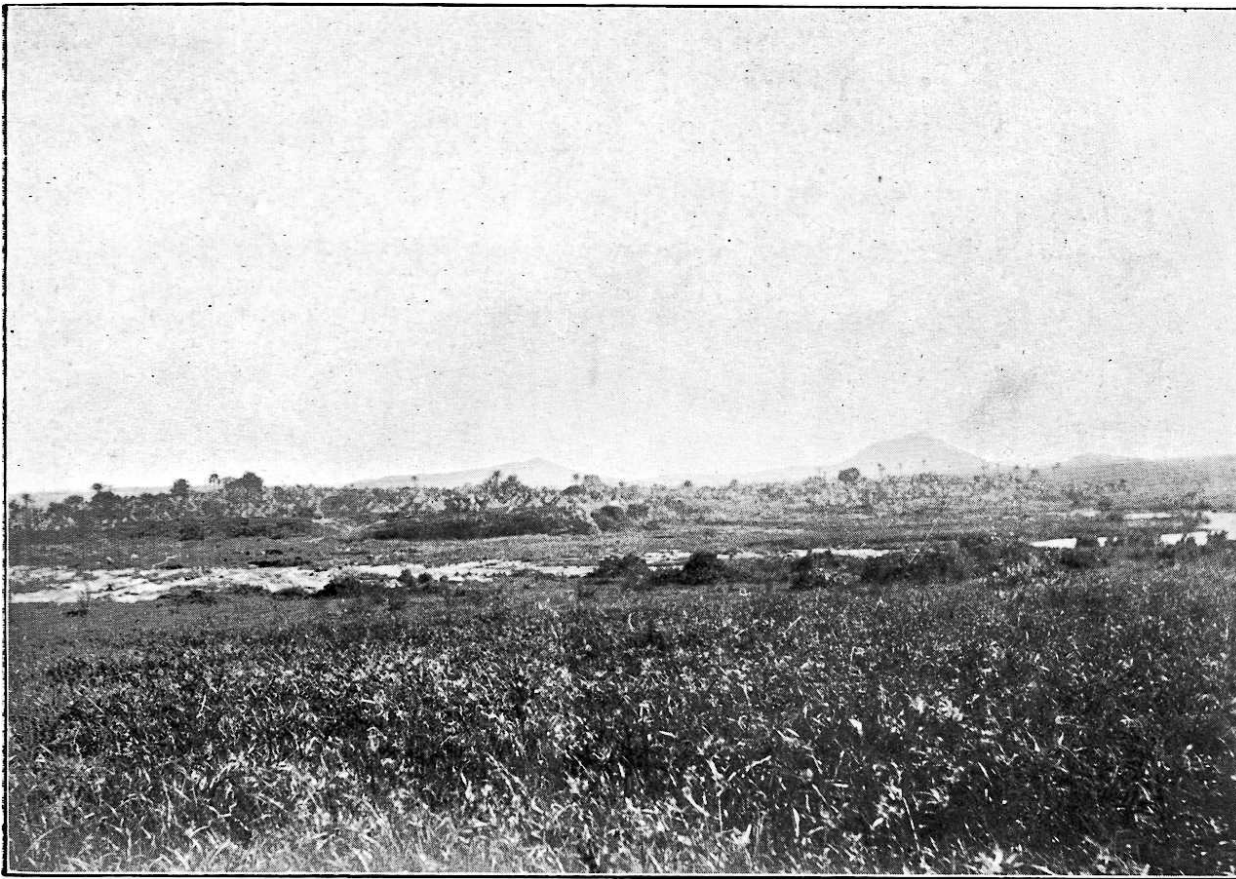




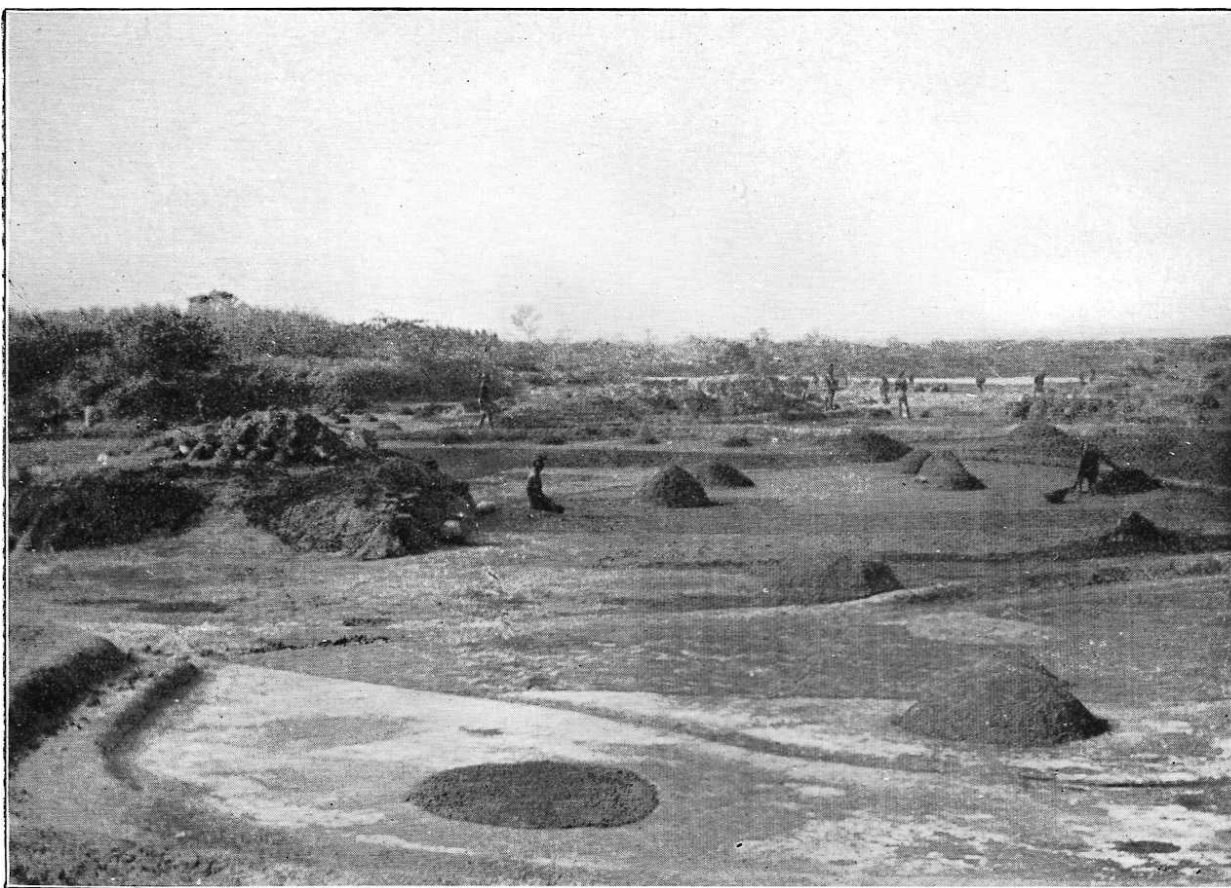
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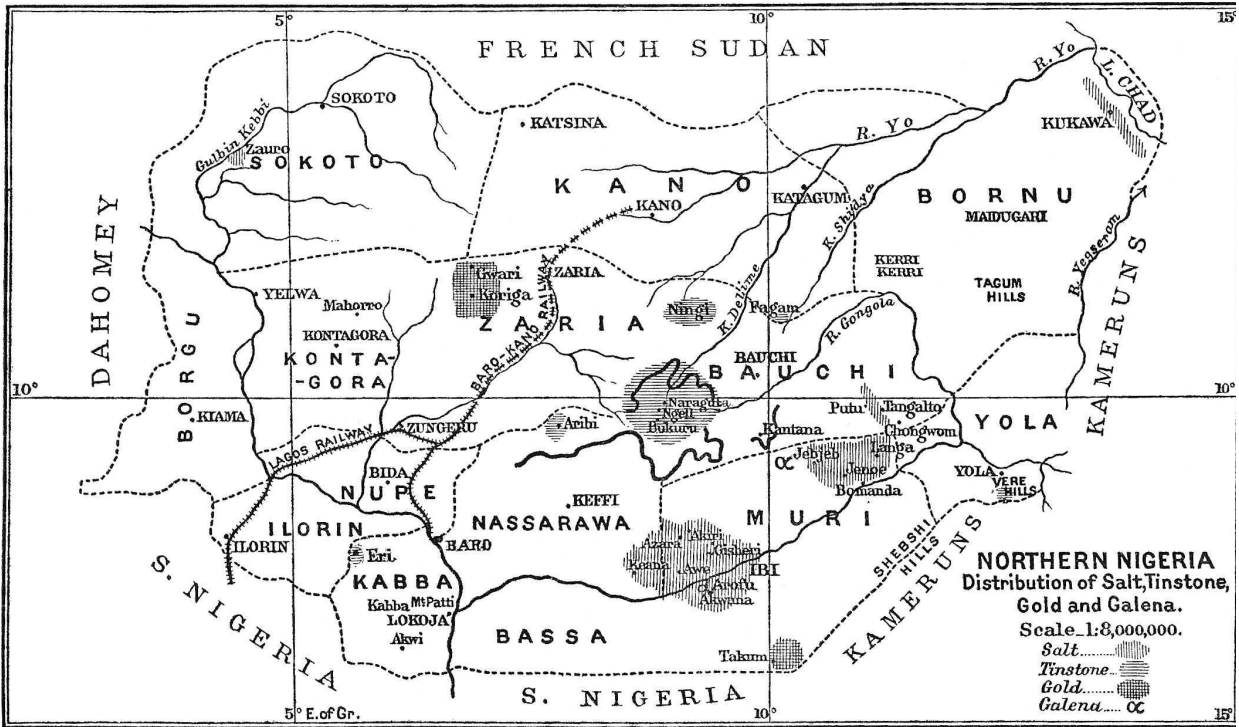
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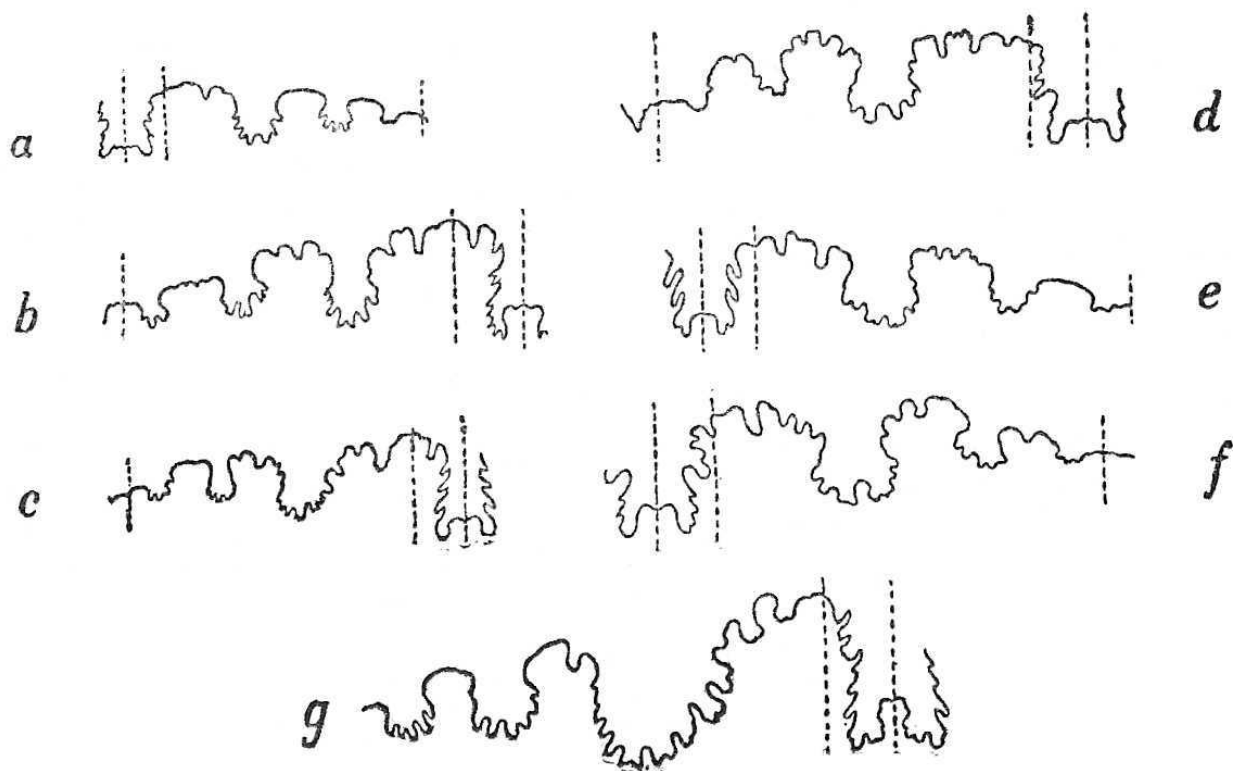


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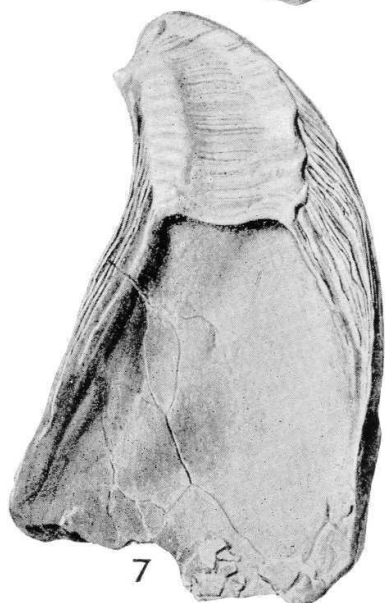
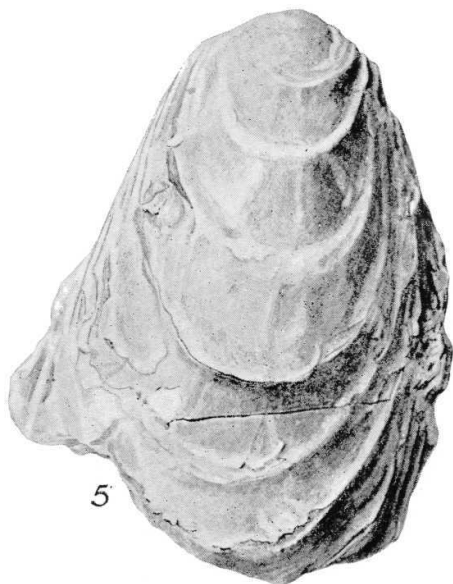
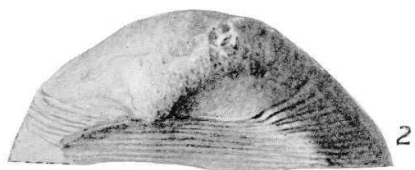


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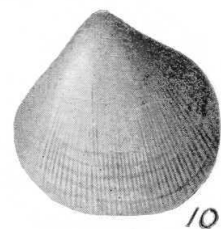
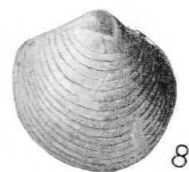
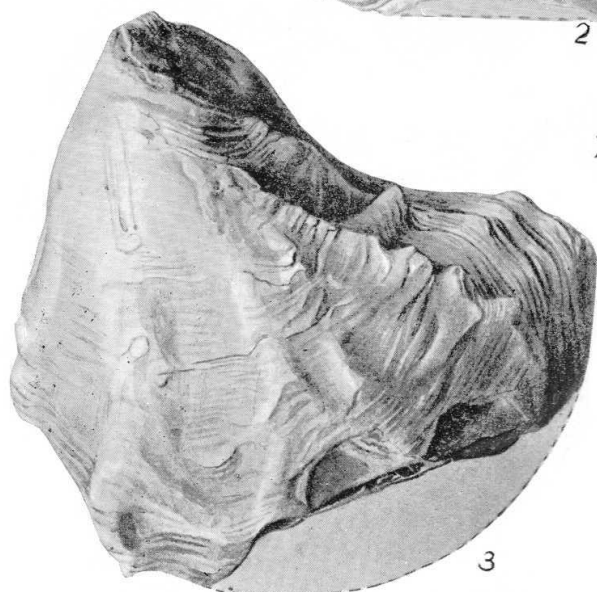
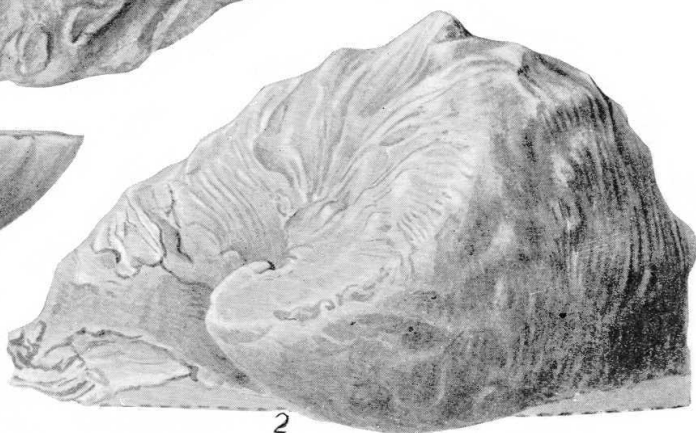
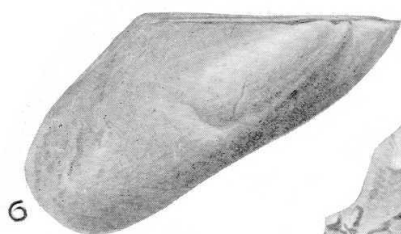
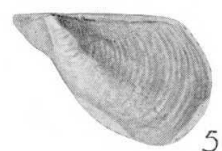
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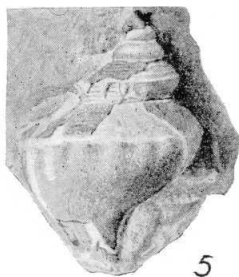
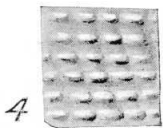
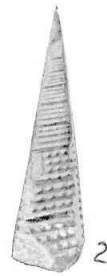
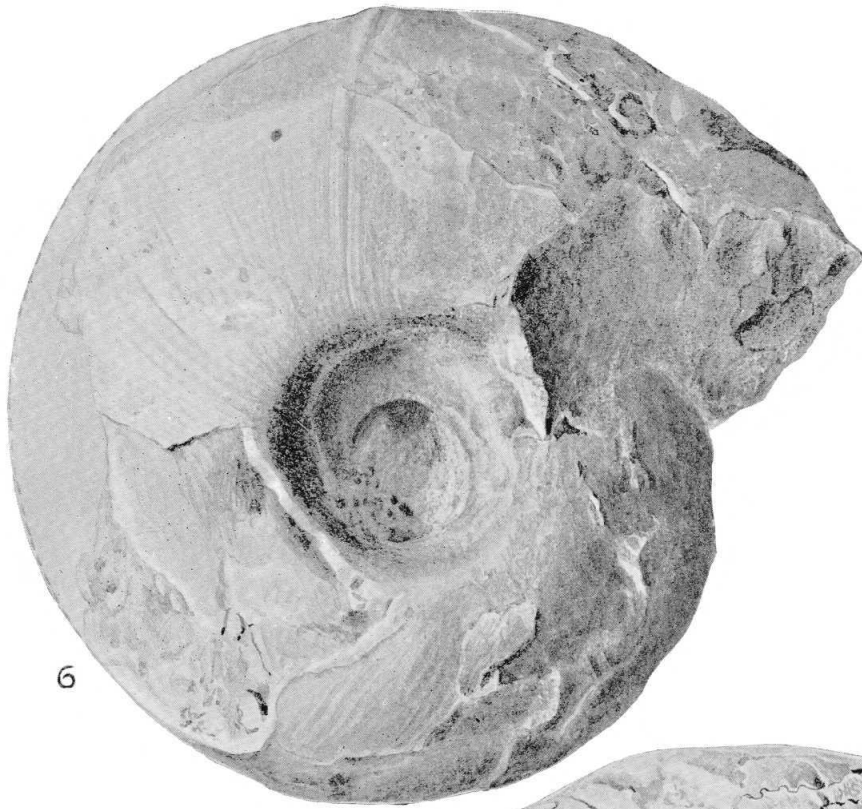
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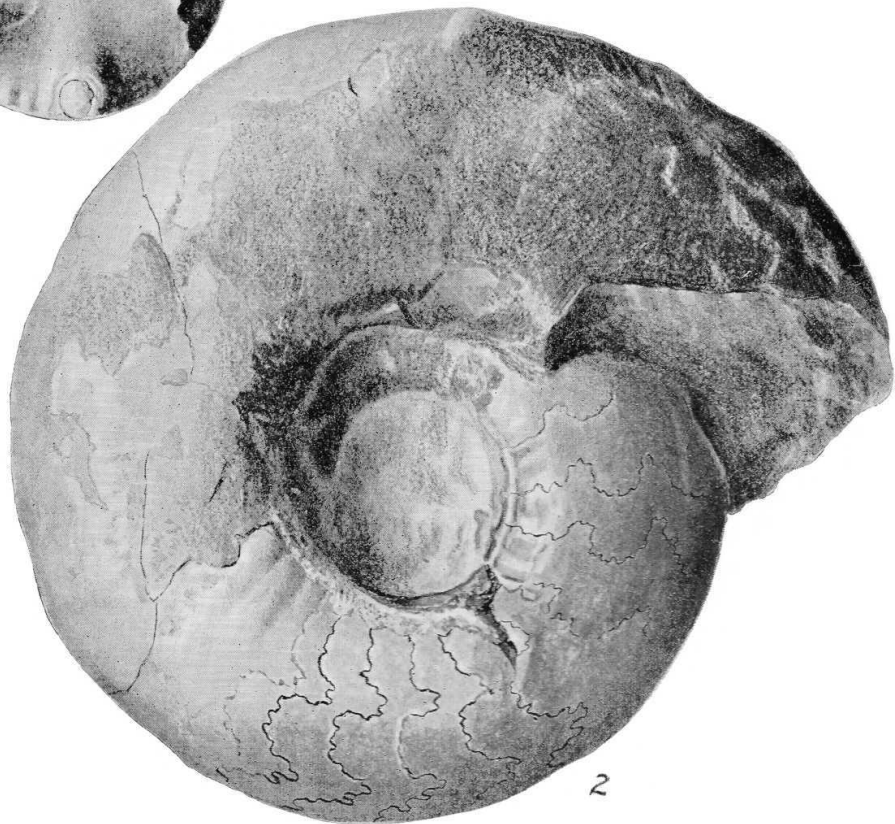
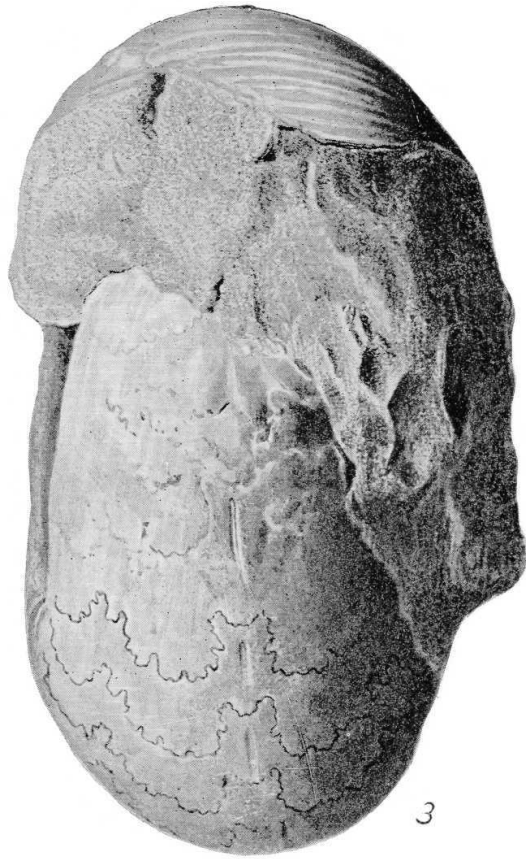
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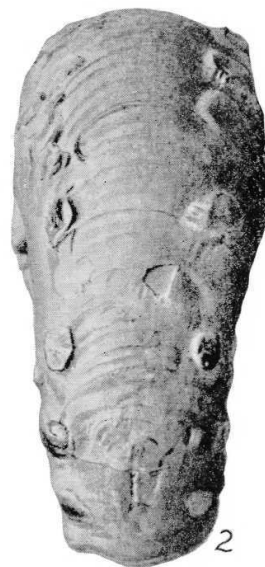
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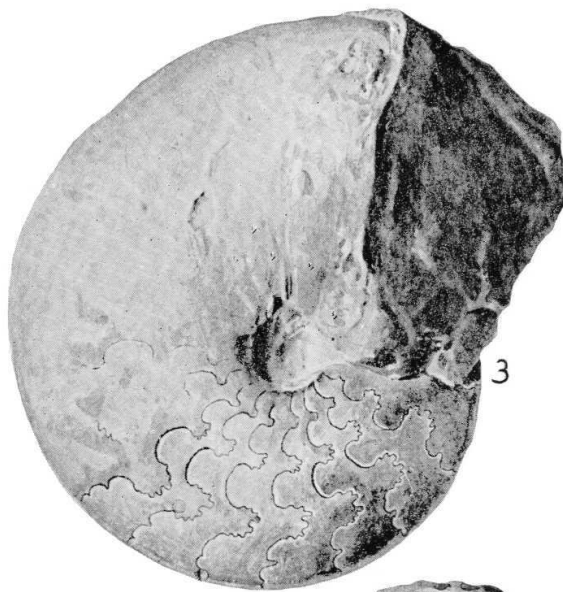


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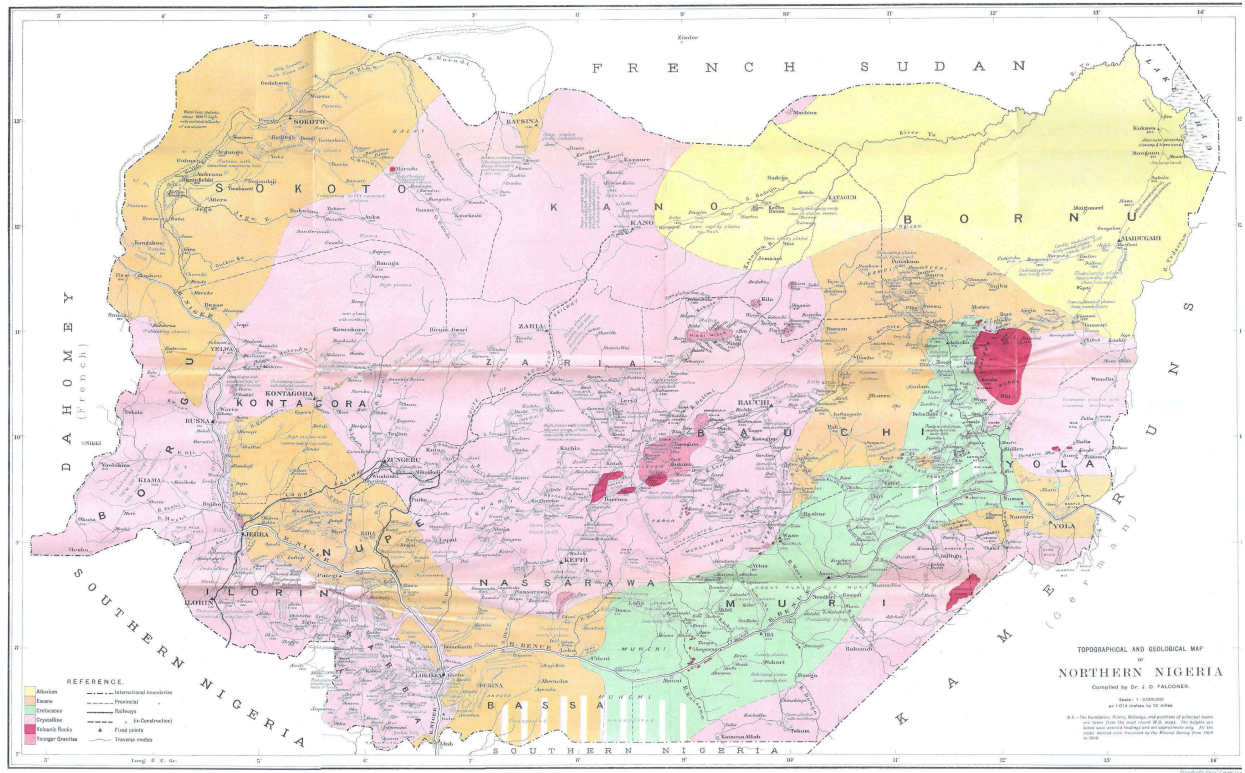


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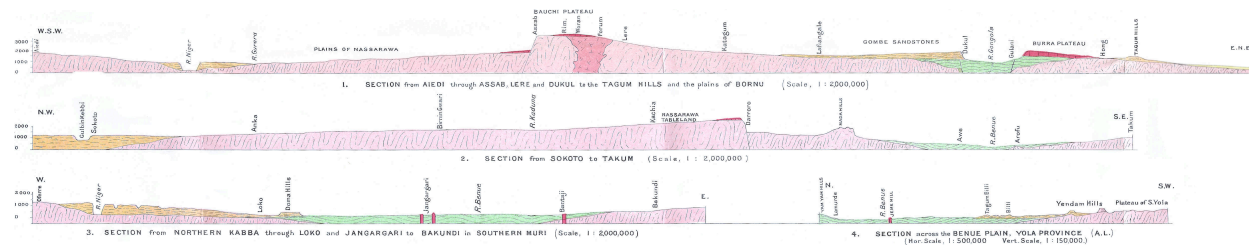
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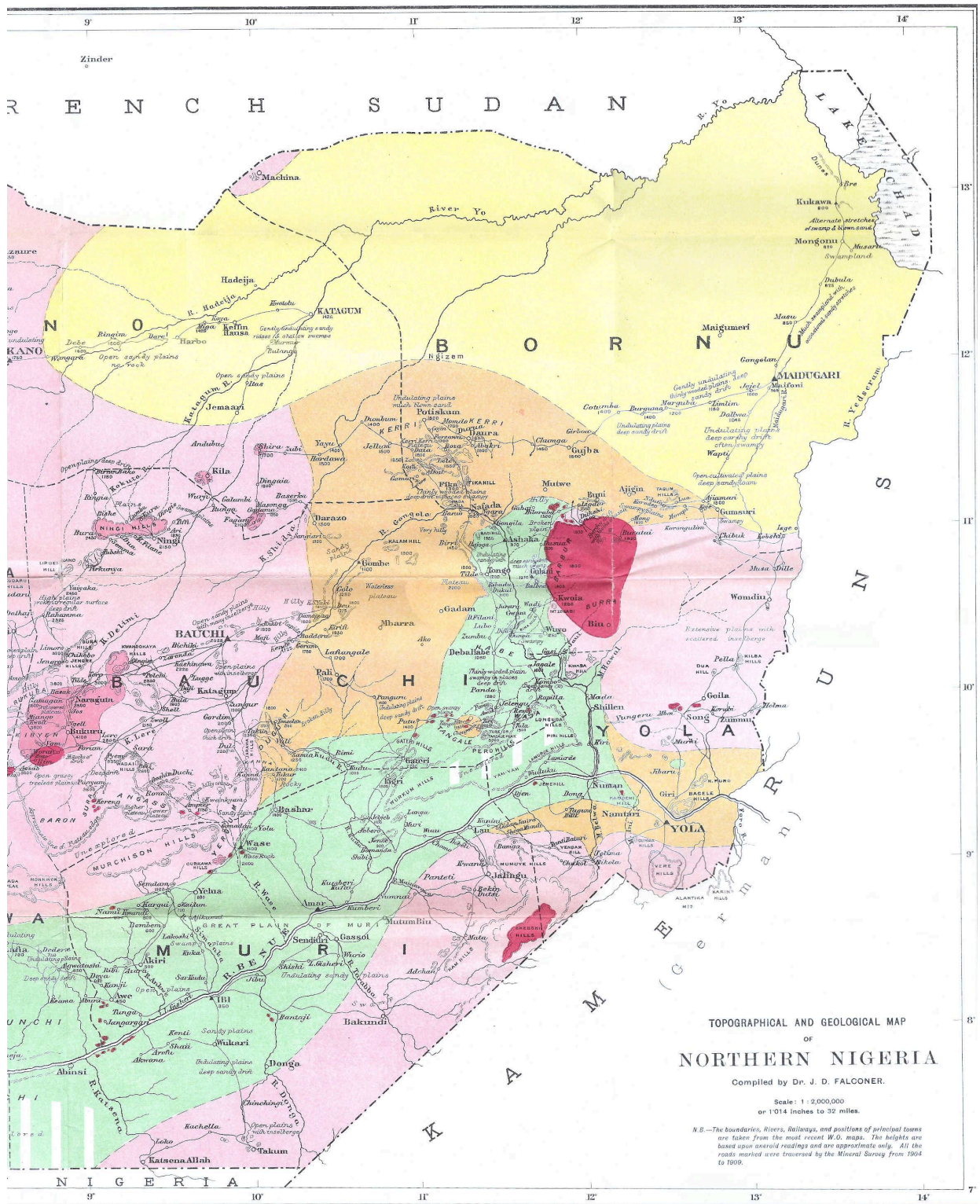
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